

从 C++ 到 AI 计算：第三册实践与代码卷

本地量化推理引擎、完整源码、KV Cache、后端优化与验收

CPU Performance Study

本卷是《从 C++ 到 AI 计算：第三册》的配套实践与代码卷，围绕 `linux_cpu_inference` 贯穿项目展开：模型资产、tokenizer、张量布局、int8 kernel、reference decoder、KV cache、trace、7B 账本、服务调度、CPU/GPU 后端边界和工程验收都要落到可构建代码、完整源码、测试和报告。

前言

第三册原理卷把 AI 推理系统拆成模型资产、typed graph、张量布局、量化、KV cache、后端、服务运行时和验收体系。本实践与代码卷只做一件事：把这些概念压到同一个可运行项目里，并把完整源码阅读路径并入同一目录，让读者知道每个概念对应哪段代码、哪条命令、哪种输入、哪份报告和哪条失败路径。

贯穿项目是 [books/cpu-volume-3/labs/linux_cpu_inference](https://github.com/QuintusLiu/books/cpu-volume-3/labs/linux_cpu_inference)。它不是玩具目录，而是第三册已经持续维护的本地 CPU 量化推理切片：包含教学模型格式、tokenizer、safetensors manifest、gate、int8 linear、packed/AVX2 路径、tiny reference decoder、GPT-2 reference path、KV cache trace、7B compute ledger、serving SLO probe、自研 GPT-2 smoke、真实 small 开源模型 smoke、CLI、benchmark 和 CTest。本卷不维护第二份实现；源码章节通过 [lstinputlisting](#) 直接读取同一份工程文件，所有练习、讲解和完整代码都回到同一份源码。

本卷的学习顺序是从小证据到大系统：先让一个 tiny MLP 算对，再解释张量地址流和 int8 累加；再让 reference decoder、sampler 和 KV cache trace 固定语义；然后接模型导入、shape verifier、memory planner 和 IR lowering；最后用 CPU/GPU 后端边界、工业专项和回归门禁收束。代码走读不再被放到最后当附录，而是插入到相同主题部分中：讲完项目入口就读构建文件和调用链，讲完量化就读公共合同、tiny MLP 和 int8 kernel，讲到真实模型时读 tokenizer、safetensors 和 GPT-2，收束验收时读工具、benchmark、smoke 和测试。每章都要求读者交付代码运行结果或报告字段，而不是只抄概念定义。

核心思想

第三册实践的目标不是“写一个看起来像大模型的 demo”，而是建立一条能被复查的推理证据链：资产能校验，shape 能验证，kernel 能对照，trace 能定位，性能能降级，服务指标能解释，回归门禁能长期维护。

常见缩写和术语先导

缩写	全称	本卷中的含义
AI	Artificial Intelligence	这里特指本地推理工程，不讨论泛泛应用口号。
LLM	Large Language Model	decoder-only 文本生成模型是本卷主线。
KV Cache	Key/Value Cache	attention 历史 K/V 状态，属于请求生命周期。
GEMM	General Matrix Multiply	矩阵乘，是预填充和训练常见核心算子。
GEMV	General Matrix Vector Multiply	矩阵向量乘，是单 token decode 的核心形态。
SIMD	Single Instruction Multiple Data	一条指令处理多个 lane 的 CPU 执行方式。
AVX2	Advanced Vector Extensions 2	x86-64 上的 256-bit SIMD 指令集之一。
FMA	Fused Multiply-Add	乘加融合指令，常出现在浮点 kernel 中。
SLO	Service Level Objective	服务目标，例如 TTFT、TPOT、p95/p99。
TTFT	Time To First Token	从请求进入到首 token 输出的时间。
TPOT	Time Per Output Token	decode 阶段每个输出 token 的平均或尾部时间。
IR	Intermediate Representation	编译器内部表示，用于 pass、lowering 和计划生成。
RAG	Retrieval-Augmented Generation	检索增强生成，需要把外部文档和 prompt 编排起来。
MoE	Mixture of Experts	多专家模型，会改变路由、负载和内存账本。
LoRA	Low-Rank Adaptation	低秩适配权重，常用于微调和多租户加载。

这些缩写不要死记。每个缩写在正文里都要落到一个代码对象或报告字段：KV cache 对应 trace 里的 slot 和 position，SLO 对应 serving probe 的 CSV，IR 对应 executable plan，SIMD/AVX2 对应 kernel 反汇编和 benchmark。

本卷结构

本卷按第三册原书的四个大块组织，但目录不再采用“前面做实践、后面贴源码”的割裂方式。每个大块都把实践任务、代码走读和验收证据放在同一条主线里：[linux_cpu_inference](#)。

第一部分从模型资产到量化 kernel。你会运行 tiny MLP，先读构建入口、项目边界和调用链总图，再检查 tokenizer 和 safetensors manifest，手算张量地址，比较 scalar、packed 和 AVX2 路径，并在公共合同、tiny MLP 与 int8 kernel 源码里解释 scale、accumulator 和 requantize 的边界。

第二部分进入 Transformer、KV cache、sampler 和服务运行时。你会跑 reference decoder trace，解释每个 checkpoint 的 shape、stride、dtype 和 layout，构造 KV cache 状态表，分析 admission、queue wait、TTFT、TPOT、取消和拒绝。

第三部分处理模型导入、真实模型加载闭环、memory planner 和 IR lowering。你会把 manifest、shape verifier、tensor role、state edge、workspace、liveness 和 executable plan 写成可检查报告，并在同一部分中读 reference decoder、tokenizer、safetensors 和 GPT-2 reference path 的完整实现，显式运行 LCQI 自研 GPT-2 safetensors smoke 和一个 small 级别开源 GGUF 模型 smoke，而不是只停留在 tiny fixture。

第四部分收束到后端优化和验收。你会比较 CPU kernel 证据、GPU 后端边界、工业专项能力和回归门禁，并直接读工具、smoke、benchmark、ledger、serving probe 和测试源码。最终交付不是“跑通一次”，而是可复查的工程档案：命令、输入、输出、trace、benchmark、不可验证边界和回归规则。

目录

前言	ii
常见缩写和术语先导	iii
本卷结构	iv
第一部分 推理链路、张量账本与量化	
第 1 章实践：端到端推理链路和项目总入口	2
一、对应原书问题：概念必须落到对象和命令	2
二、从特例推到规律：先抓一个能手算的切片	2
三、实践任务：把观察固定成报告字段	2
四、验收和递进大作业	2
代码走读：构建入口、项目边界与调用链总图	3
源码阅读覆盖范围	3
根构建入口	3
LCQI 目标定义	6
项目说明与模型资产	10
为什么要先讲调用链	18
一、全工程入口：哪些文件决定能不能复现	18
二、Tiny MLP 调用链：最短闭环怎样跑通	19
三、Int8 Kernel 调用链：reference、packed、AVX2 和 benchmark 怎样对照	20
四、Tokenizer、Safetensors 与 Reference Decoder：真实模型前的三道门	21
五、GPT-2 调用链：从 HuggingFace 资产到生成 token	22
六、报告、服务探针、外部对标和测试	23
第 2 章实践：张量布局、地址流与第一个 MatVec	24
一、对应原书问题：概念必须落到对象和命令	24
二、从特例推到规律：先抓一个能手算的切片	24
三、实践任务：把观察固定成报告字段	24
四、验收和递进大作业	24
第 3 章实践：GEMM、packing 与 SIMD 证据	25
一、对应原书问题：概念必须落到对象和命令	25
二、从特例推到规律：先抓一个能手算的切片	25
三、实践任务：把观察固定成报告字段	25
四、验收和递进大作业	25
第 4 章实践：int8 量化、累加器与重新量化	26
一、对应原书问题：概念必须落到对象和命令	26
二、从特例推到规律：先抓一个能手算的切片	26
三、实践任务：把观察固定成报告字段	26
四、验收和递进大作业	26
代码走读：公共合同、Tiny MLP 与 Int8 Kernel	27
为什么先读头文件	27
Tiny MLP 与推理入口	27
Int8 Kernel、Tokenizer 与 Safetensors	28
GGUF 与 Q4_K 合同	33
Reference Decoder 与 GPT-2 合同	38
最短闭环：从模型文件到 logits	47
模型加载与基础推理	47
Int8 Kernel 基础实现	54
第二部分 Transformer、KV Cache 与服务运行时	
第 5 章实践：Transformer 切片与 KV Cache 状态	65
一、对应原书问题：概念必须落到对象和命令	65
二、从特例推到规律：先抓一个能手算的切片	65
三、实践任务：把观察固定成报告字段	65
四、验收和递进大作业	65

第 6 章实践：Reference Decoder、logits 与采样	66
一、对应原书问题：概念必须落到对象和命令	66
二、从特例推到规律：先抓一个能手算的切片	66
三、实践任务：把观察固定成报告字段	66
四、验收和递进大作业	66
第 7 章实践：服务调度、SLO 与资源预算	67
一、对应原书问题：概念必须落到对象和命令	67
二、从特例推到规律：先抓一个能手算的切片	67
三、实践任务：把观察固定成报告字段	67
四、验收和递进大作业	67
第 8 章实践：RAG、Agent 与状态图边界	68
一、对应原书问题：概念必须落到对象和命令	68
二、从特例推到规律：先抓一个能手算的切片	68
三、实践任务：把观察固定成报告字段	68
四、验收和递进大作业	68
第三部分 模型导入、AI 编译器与内存计划	
第 9 章实践：模型导入、manifest 与 Shape Verifier	70
一、对应原书问题：概念必须落到对象和命令	70
二、从特例推到规律：先抓一个能手算的切片	70
三、实践任务：把观察固定成报告字段	70
四、验收和递进大作业	70
第 10 章实践：真实模型加载闭环与首 Token Trace	71
一、对应原书问题：概念必须落到对象和命令	71
二、从特例推到规律：先抓一个能手算的切片	71
三、自研 GPT-2 reference smoke：先把标准 safetensors 跑通	71
四、真实 small 模型 smoke：不用 tiny，先让开源模型实际跑起来	72
五、实践任务：把观察固定成报告字段	73
六、验收和递进大作业	73
代码走读：Reference Decoder、Tokenizer、Safetensors 与 GPT-2	74
从 tiny MLP 走向 decoder-only 模型	74
Tiny Reference Decoder	74
Tokenizer 与 Safetensors 实现	91
GPT-2 Reference Path	107
一次可审查的 cached-KV 优化	107
第 11 章实践：运行时内存计划、liveness 与 Workspace	180
一、对应原书问题：概念必须落到对象和命令	180
二、从特例推到规律：先抓一个能手算的切片	180
三、实践任务：把观察固定成报告字段	180
四、验收和递进大作业	180
第 12 章实践：IR、Pass、Lowering 与 Executable Plan	181
一、对应原书问题：概念必须落到对象和命令	181
二、从特例推到规律：先抓一个能手算的切片	181
三、实践任务：把观察固定成报告字段	181
四、验收和递进大作业	181
第四部分 后端优化、工业专项与验收	
第 13 章实践：CPU 后端、微内核与性能证据	183
一、对应原书问题：概念必须落到对象和命令	183
二、从特例推到规律：先抓一个能手算的切片	183
三、实践任务：把观察固定成报告字段	183
四、验收和递进大作业	184
第 14 章实践：GPU 后端边界、copy 与同步成本	185
一、对应原书问题：概念必须落到对象和命令	185
二、从特例推到规律：先抓一个能手算的切片	185
三、实践任务：把观察固定成报告字段	185
四、验收和递进大作业	185
第 15 章实践：工业 LLM 专项能力对照	186
一、对应原书问题：概念必须落到对象和命令	186
二、从特例推到规律：先抓一个能手算的切片	186
三、实践任务：把观察固定成报告字段	186
四、验收和递进大作业	186

第 16 章实践：工程验收、回归门禁与最终项目	187
一、对应原书问题：概念必须落到对象和命令	187
二、从特例推到规律：先抓一个能手算的切片	187
三、实践任务：把观察固定成报告字段	187
四、验收和递进大作业	187
代码走读：工具、Smoke、Benchmark 与回归测试	188
工具不是附属品	188
Benchmark、Trace、Ledger 与 Serving Probe	188
外部模型 Smoke 与对比脚本	239
完整测试	285
实践与代码总索引	320
一、实践主题到代码走读的合并位置	320
二、最小复现命令	320
术语表	321

第零部分

推理链路、张量账本与量化

第 1 章实践：端到端推理链路和项目总入口

本章对应原书中的第三册“端到端链路：从模型文件到下一个 token”。不要把它当作概念复述，本章要把模型目录、manifest、tokenizer、typed graph、backend plan、request session、KV cache、oracle 和 profiler 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `tiny_mlp_i8.txt`，核心命令或测试入口是 `lcqi_cli`。

Listing 1.1 第 1 章实践：端到端推理链路和项目总入口的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_cli / tiny_mlp_i8.txt
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：只跑一个 tiny MLP 推理请求。读者要先手写预期结果，再运行项目观察输出。动作是：把 `readfiles->tokenize->run_inference->logits->predicted_class` 串起来。如果输出里没有 `predicted_class` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：模型文件存在不等于推理链路可信；只有 CLI 输出、测试断言、输入合同和失败路径同时成立，才算完成入口闭环这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：最终大作业的总目录：每章报告都放入 `reports/volume3-practice/`，第一章提交项目地图、命令记录和一次成功/失败输入对照。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

代码走读：构建入口、项目边界与调用链总图

源码阅读覆盖范围

LCQI 的源码边界由三个层次组成。第一层是仓库构建入口，它决定 CMake 怎样把第三册的实验代码纳入构建和测试。第二层是 `labs/linux_cpu_inference`，它包含真正的库、CLI、benchmark、trace、ledger、serving probe 和测试目标。第三层是工具脚本，它们下载或复用外部模型资产，生成 smoke 和 benchmark 报告。

源码阅读任务

先读本章的 CMake 文件，再读 README。读者要能回答三个问题：一是 `lcqi` 静态库包含哪些实现文件；二是哪些可执行文件只依赖自研代码，哪些依赖可选外部库；三是真实 GPT-2 和 SmolLM2 smoke 为什么不进入默认 CTest。

根构建入口

第三册根目录的 CMake 文件只做项目级选择：是否构建第三册 labs，以及如何把 `labs/linux_cpu_inference` 接入。它应该保持薄层，避免把具体目标定义分散到多个地方。

Listing 1.2 cpu-volume-3/README.md

```

1 # 从 C++ 到 AI 计算：第三册
2
3 第三册用于承接 AI、AI 编译器、算子开发和本地大模型推理引擎。第二册只讨论硬件原
  ↳ 理、多核、高性能计算、并发、锁、无锁数据结构、并行算法和分布式计算，不再
  ↳ 直接进入 AI。
4
5 本册贯穿项目是一台能推理开源 7B 左右 decoder-only 大模型的本地引擎，并把 AI 编
  ↳ 译器作为引擎内部主线：模型图导入、typed tensor IR、算子融合、内存规划、
  ↳ CPU/GPU lowering、kernel selection 和 runtime dispatch 都要能落回编译原理
  ↳ 与底层机器账本。工程路线从 CPU 上可解释的 reference 和量化切片开始，逐步
  ↳ 扩展到 tokenizer、模型加载、Transformer 层、KV cache、CPU SIMD/多线程后
  ↳ 端、GPU 后端、正确性报告和性能对标。
6
7 正式书稿工程位于：
8
9 ```text
10 source/latex/
11 ```
12
13 构建命令：
14
15 ```bash
16 make -C source/latex check
17 make -C source/latex pdf
18 make -C source/latex epub
19 ```
20
21 生成文件：
22
23 ```text

```

```

24 source/latex/main.pdf
25 source/latex/main.epub
26 ```
27
28 当前保留的第一阶段切片：
29
30 ```text
31 labs/linux_cpu_inference/
32 ```
33
34 该目录目前包含最小 MLP/int8 权重教学切片、tiny reference decoder、GPT-2 ↵
    ↵ reference path、int8 scalar baseline、packed layout、AVX2 SIMD 路径和 ↵
    ↵ shape sweep benchmark，用来验证张量、线性层、量化权重、tokenizer、↵
    ↵ safetensors、GPT-2 forward、KV cache、oracle 和 benchmark 的基本边界。当↵
    ↵ 前 x86-64 AVX2 结果见：
35
36 ```text
37 results/lcqi-x86-avx2-benchmark-2026-06-25.csv
38 results/lcqi-x86-avx2-decode4096-2026-06-25.csv
39 ```
40
41 自研 GPT-2 smoke 入口如下：
42
43 ```bash
44 make cpu3-gpt2-smoke
45 make cpu3-gpt2-benchmark-compare
46 ```
47
48 第一条命令下载 `openai-community/gpt2` 的标准 safetensors 资产到用户缓存目录，↵
    ↵ 用 LCQI C++ reference path 生成 1 个 token，并把报告写入：
49
50 ```text
51 results/lcqi-gpt2-smoke.txt
52 ```
53
54 第二条命令用同一个 GPT-2 F32 模型对比 LCQI full-prefix 和 cached-KV 生成路径，↵
    ↵ 并在本机存在 llama.cpp/SmolLM2 缓存时附带成熟引擎参考，报告写入：
55
56 ```text
57 results/lcqi-gpt2-benchmark-compare.txt
58 ```
59
60 它仍然不是生产级 AI Infra。生产级门禁见：
61
62 ```text
63 reports/ai-infra-maturity-audit.md
64 ```
65
66 第三册后续必须把 lab 推进到“小型 CPU 推理 runtime + 手写高性能算子 + 严格 ↵
    ↵ benchmark 报告 + 真实系统对标”，否则只能证明系统学习，不能证明生产级能↵
    ↵ 力。

```

Listing 1.3 cpu-volume-3/CMakeLists.txt

```

1 cmake_minimum_required(VERSION 3.31)
2 project(CPUVolume3 LANGUAGES CXX)
3
4 set(CMAKE_CXX_STANDARD 20)
5 set(CMAKE_CXX_STANDARD_REQUIRED ON)
6 set(CMAKE_CXX_EXTENSIONS OFF)
7 set(CMAKE_EXPORT_COMPILE_COMMANDS ON)
8
9 option(CPU_VOLUME_3_BUILD_LABS "Build CPU Volume 3 labs" ON)
10
11 if(CPU_VOLUME_3_BUILD_LABS)
12     enable_testing()
13     add_subdirectory(labs/linux_cpu_inference)
14 endif()

```

实践与代码卷也提供一个薄 CMake 入口，它把同一份 LCQI 源码作为可测试对象接入。这样本卷不会复制实现，只复用第三册主工程。

Listing 1.4 cpu-volume-3-practice/CMakeLists.txt

```

1 cmake_minimum_required(VERSION 3.31)
2 project(CPUVolume3Practice LANGUAGES CXX)
3
4 set(CMAKE_CXX_STANDARD 20)
5 set(CMAKE_CXX_STANDARD_REQUIRED ON)
6 set(CMAKE_CXX_EXTENSIONS OFF)
7 set(CMAKE_EXPORT_COMPILE_COMMANDS ON)
8
9 enable_testing()
10 add_subdirectory(..cpu-volume-3/labs/linux_cpu_inference linux_cpu_inference)

```

本卷自身的 Makefile 负责构建 PDF/EPUB，并通过 `test` 目标回到第三册真实 LCQI 工程做 CMake/CTest 验证。实践与代码卷如果只排版、不测试，就不能宣称“完整源码可运行”。

Listing 1.5 cpu-volume-3-practice/Makefile

```

1 LATEX_DIR := source/latex
2
3 .PHONY: all check pdf epub test clean
4
5 all: pdf epub
6
7 check:
8     $(MAKE) -C $(LATEX_DIR) check
9
10 pdf:
11     $(MAKE) -C $(LATEX_DIR) pdf
12
13 epub:
14     $(MAKE) -C $(LATEX_DIR) epub
15

```

```

16 test:
17     cmake -S . -B build/lcqi-debug -DCMAKE_BUILD_TYPE=Debug
18     cmake --build build/lcqi-debug
19     ctest --test-dir build/lcqi-debug --output-on-failure
20
21 clean:
22     $(MAKE) -C $(LATEX_DIR) clean

```

Listing 1.6 cpu-volume-3-practice/source/latex/Makefile

```

1 LATEXMK ?= latexmk
2 ENGINE ?= -xelatex
3 MAIN ?= main.tex
4 EPUB_BUILDER ?= ../../../../cpu-volume-1/source/latex/scripts/build_epub.py
5
6 .PHONY: all pdf epub clean check
7
8 all: pdf epub
9
10 pdf:
11     $(LATEXMK) $(ENGINE) -interaction=nonstopmode -halt-on-error $(MAIN)
12
13 epub:
14     @BOOK_TITLE="从 C++ 到 AI 计算：第三册实践与代码卷" \
15     BOOK_IMPORT_TITLE="CPU Volume 3 Practice Code" \
16     BOOK_ID_SEED="cpu-volume-3-practice-code-weread-20260629" \
17     EPUB_ASCII_TITLES=1 \
18     BOOK_SUBTITLE="本地量化推理引擎、完整源码、KV Cache、后端优化与验收" \
19     BOOK_AUTHOR="CPU Performance Study" \
20     python3 $(EPUB_BUILDER) --book-dir . --output main.epub --no-cover-page --↵
    ↵ forbid-images --linearize-tables --wechat-compatible --no-embed-fonts
21
22 clean:
23     $(LATEXMK) -C
24
25 check:
26     @python3 scripts/check_inputs.py
27     @command -v xelatex >/dev/null 2>&1 || { echo "missing xelatex"; exit 1; }
28     @command -v latexmk >/dev/null 2>&1 || { echo "missing latexmk"; exit 1; }

```

LCQI 目标定义

`linux_cpu_inference/CMakeLists.txt` 是代码走读中的第一份核心文件。这里能看到 `lcqi` 库、`lcqi_cli`、`lcqi_bench`、`lcqi_trace`、`lcqi_gpt2`、`lcqi_ledger`、`lcqi_serving_probe`、可选 `lcqi_onednn_bench` 和 `lcqi_tests` 的完整关系。

Listing 1.7 linux_cpu_inference/CMakeLists.txt

```

1 find_package(Threads REQUIRED)
2
3 add_library(lcqi STATIC
4     src/f32_kernels.cpp

```

```

5     src/ggml_tensors.cpp
6     src/gguf.cpp
7     src/gpt2_reference.cpp
8     src/inference.cpp
9     src/int8_kernels.cpp
10    src/llama_gguf.cpp
11    src/model.cpp
12    src/q4_k.cpp
13    src/reference_decoder.cpp
14    src/safetensors.cpp
15    src/tokenizer.cpp
16 )
17
18 if(CMAKE_SYSTEM_PROCESSOR MATCHES "x86_64|AMD64|i[3-6]86")
19     if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
20         target_sources(lcqi PRIVATE src/f32_kernels_avx2.cpp)
21         target_sources(lcqi PRIVATE src/int8_kernels_avx2.cpp)
22         target_sources(lcqi PRIVATE src/q4_k_avx2.cpp)
23         set_source_files_properties(src/f32_kernels_avx2.cpp PROPERTIES ↵
↵ COMPILE_OPTIONS "-mavx2;-mfma")
24         set_source_files_properties(src/int8_kernels_avx2.cpp PROPERTIES ↵
↵ COMPILE_OPTIONS "-mavx2;-mfma")
25         set_source_files_properties(src/q4_k_avx2.cpp PROPERTIES ↵
↵ COMPILE_OPTIONS "-mavx2")
26         target_compile_definitions(lcqi PRIVATE LCQI_ENABLE_AVX2=1)
27     endif()
28 endif()
29
30 if(NOT CMAKE_SYSTEM_NAME STREQUAL "Linux")
31     message(WARNING "LCQI is written for the book's local Linux CPU target; ↵
↵ current system is ${CMAKE_SYSTEM_NAME}.")
32 endif()
33
34 target_include_directories(lcqi
35     PUBLIC
36     ${CMAKE_CURRENT_SOURCE_DIR}/include
37 )
38
39 target_compile_features(lcqi PUBLIC cxx_std_20)
40 target_link_libraries(lcqi PUBLIC Threads::Threads)
41
42 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
43     target_compile_options(lcqi PRIVATE -Wall -Wextra -Wpedantic)
44 endif()
45
46 add_executable(lcqi_cli
47     src/main.cpp
48 )
49
50 target_link_libraries(lcqi_cli PRIVATE lcqi)
51
52 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")

```

```
53     target_compile_options(lcqi_cli PRIVATE -Wall -Wextra -Wpedantic)
54 endif()
55
56 add_executable(lcqi_bench
57     src/bench.cpp
58 )
59
60 target_link_libraries(lcqi_bench PRIVATE lcqi)
61
62 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
63     target_compile_options(lcqi_bench PRIVATE -Wall -Wextra -Wpedantic)
64 endif()
65
66 add_executable(lcqi_gguf_q4_bench
67     src/gguf_q4_bench.cpp
68 )
69
70 target_link_libraries(lcqi_gguf_q4_bench PRIVATE lcqi)
71
72 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
73     target_compile_options(lcqi_gguf_q4_bench PRIVATE -Wall -Wextra -Wpedantic)
74 endif()
75
76 add_executable(lcqi_trace
77     src/trace.cpp
78 )
79
80 target_link_libraries(lcqi_trace PRIVATE lcqi)
81
82 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
83     target_compile_options(lcqi_trace PRIVATE -Wall -Wextra -Wpedantic)
84 endif()
85
86 add_executable(lcqi_gpt2
87     src/gpt2_cli.cpp
88 )
89
90 target_link_libraries(lcqi_gpt2 PRIVATE lcqi)
91
92 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
93     target_compile_options(lcqi_gpt2 PRIVATE -Wall -Wextra -Wpedantic)
94 endif()
95
96 add_executable(lcqi_llama_gguf
97     src/llama_cli.cpp
98 )
99
100 target_link_libraries(lcqi_llama_gguf PRIVATE lcqi)
101
102 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
103     target_compile_options(lcqi_llama_gguf PRIVATE -Wall -Wextra -Wpedantic)
104 endif()
```



```

105
106 add_executable(lcqi_ledger
107     src/ledger.cpp
108 )
109
110 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
111     target_compile_options(lcqi_ledger PRIVATE -Wall -Wextra -Wpedantic)
112 endif()
113
114 add_executable(lcqi_serving_probe
115     src/serving_probe.cpp
116 )
117
118 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
119     target_compile_options(lcqi_serving_probe PRIVATE -Wall -Wextra -Wpedantic)
120 endif()
121
122 find_package(dnnl QUIET)
123 if(dnnl_FOUND)
124     add_executable(lcqi_onednn_bench
125         src/onednn_bench.cpp
126     )
127     target_link_libraries(lcqi_onednn_bench PRIVATE DNNL::dnnl)
128     target_compile_features(lcqi_onednn_bench PRIVATE cxx_std_20)
129     if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
130         target_compile_options(lcqi_onednn_bench PRIVATE -Wall -Wextra ↵ ↵
131             ↵ Wpedantic)
132     endif()
133 endif()
134
135 add_executable(lcqi_tests
136     tests/lcqi_tests.cpp
137 )
138 target_link_libraries(lcqi_tests PRIVATE lcqi)
139
140 if(CMAKE_CXX_COMPILER_ID MATCHES "Clang|GNU")
141     target_compile_options(lcqi_tests PRIVATE -Wall -Wextra -Wpedantic)
142 endif()
143
144 add_test(NAME lcqi_tests COMMAND ${TARGET_FILE:lcqi_tests})
145 add_test(NAME lcqi_gpt2_tiny COMMAND ${TARGET_FILE:lcqi_gpt2})
146 set_tests_properties(lcqi_gpt2_tiny PROPERTIES
147     PASS_REGULAR_EXPRESSION "generated_ids 1 2 3 3 3"
148 )
149 add_test(NAME lcqi_trace COMMAND ${TARGET_FILE:lcqi_trace})
150 set_tests_properties(lcqi_trace PROPERTIES
151     PASS_REGULAR_EXPRESSION "tokenizer,0,-1,5,1,i32,contiguous,10,4,0;1;2;3;4"
152 )
153 add_test(NAME lcqi_ledger COMMAND ${TARGET_FILE:lcqi_ledger})
154 set_tests_properties(lcqi_ledger PROPERTIES
155     PASS_REGULAR_EXPRESSION "bandwidth_limit,int4_at_100_gbps,23.602,tokens/s"

```

```

156 )
157 add_test(NAME lcqi_serving_probe COMMAND $<TARGET_FILE:lcqi_serving_probe>)
158 set_tests_properties(lcqi_serving_probe PROPERTIES
159     PASS_REGULAR_EXPRESSION "request,req_too_long,0,memory_budget_exceeded"
160 )

```

项目说明与模型资产

README 是复现实验的入口。它写清楚当前能力、后续边界、构建命令、tiny 推理命令、benchmark 字段、reference trace 字段、GPT-2 smoke、7B ledger、serving SLO probe 和 SmolLM2 small smoke。读源码前先读 README，可以避免把 tiny fixture、GPT-2 reference path 和外部 llama.cpp smoke 混成同一个能力等级。

Listing 1.8 linux_cpu_inference/README.md

```

1 # Linux CPU Quantized Inference
2
3 这是第三册贯穿项目的第一阶段切片：Linux 本地 CPU 量化线性层和最小推理流程。第三
    ↳ 册的贯穿目标不是这个 MLP 本身，而是一台能推理开源 7B 左右 decoder-only 大
    ↳ 模型的本地引擎，并逐步覆盖 tokenizer、模型加载、Transformer 层、KV cache
    ↳ 、CPU/GPU 后端、正确性报告和性能对标。
4
5 目标平台是 x86-64 Linux 实验环境。完整性能报告应记录 CPU 型号、内核版本、编译器
    ↳ 版本、是否能使用 PMU、NUMA 和线程亲和能力。若运行环境缺少 perf、NUMA、线
    ↳ 程亲和性或硬件性能计数器权限，报告必须把相关结论降级为“未验证”，不能把源
    ↳ 码路径存在当作实测性能证据。后续 GPU 后端会作为独立 backend 接入，不改变
    ↳ 当前切片的语义合同。
6
7 当前版本支持：
8
9 - 教学文本模型格式 `LCQI_MODEL_V1`
10 - 最小 tokenizer 合同格式 `LCQI_TOKENIZER_V1`
11 - safetensors header manifest 解析：metadata、tensor name、dtype、shape、data
    ↳ offsets、offset 边界和 dtype/shape 字节数校验
12 - 两层 MLP 教学切片
13 - int8 权重加 per-layer scale
14 - float 输入和激活
15 - 量化线性层、ReLU、分类 argmax
16 - int8 linear scalar baseline、packed layout、AVX2 路径和 shape sweep benchmark
17 - GPT-2 F32 dot kernel: cached generation 热路径使用可替换 dot 边界，x86-64/GNU
    ↳ /Clang 构建接入 AVX2/FMA 实现，其他平台保留标量 fallback
18 - tiny reference decoder: RMSNorm、float linear、RoPE、GQA KV cache、decode
    ↳ attention、SwiGLU、lm head
19 - GPT-2 reference path: byte-level BPE tokenizer、`config.json`、F32/F16/BF16
    ↳ safetensors 张量读取、HuggingFace GPT-2 name mapping、LayerNorm、绝对位置
    ↳ 嵌入、packed QKV、causal attention、GELU、tied lm head、full-prefix
    ↳ greedy generation、KV cache greedy generation 和分阶段 benchmark
20 - reference trace CSV: 从 prompt/tokenizer 合同到 logits/sampler/token，逐
    ↳ checkpoint 输出 shape、stride、dtype、layout、checksum、max_abs、values
    ↳ 摘要
21 - 7B ledger CSV: 输出典型 7B shape 的 FLOPs、KV 容量、权重/KV 字节和 bandwidth-
    ↳ only tokens/s 上限

```

```

22 - serving SLO probe CSV: 输出 admission、batch state、KV 预算、queue wait、TTFT↵
    ↵、TPOT、取消回收和拒绝率
23 - 自研 GPT-2 冒烟: 加载 `openai-community/gpt2` 的 `config.json`、`vocab.json`↵
    ↵、`merges.txt`、`model.safetensors`, 在 LCQI C++ reference path 上生成 1 ↵
    ↵个 token, 并把文件 hash、prompt ids、generated ids 和文本输出写入报告
24 - 自研 GPT-2 benchmark 对比: 同一个 GPT-2 F32 safetensors 模型分别跑 full↵
    ↵prefix 和 cached-KV, 并在可用时附带 llama.cpp/SmolLM2 Q4_K_M 作为成熟引擎↵
    ↵参考
25 - 自研 GPT-2 优化 A/B: 从指定 baseline commit 建临时 worktree, 当前实现和 ↵
    ↵baseline 都通过 `books/cpu-volume-3-practice` 的 Release CMake 入口构建, ↵
    ↵再用同一个 GPT-2 F32 模型多轮测量 full-prefix 与 cached-KV
26 - 真实 small 开源模型冒烟: 通过 llama.cpp 加载 SmolLM2-135M-Instruct 的 GGUF 量↵
    ↵化文件, 在本地 CPU 上生成一段 assistant 文本, 并把模型 hash、命令、输出和↵
    ↵性能行写入报告
27 - 自研 SmolLM2/GGUF reference decode: 读取同一个 GGUF 内的 tokenizer tokens/↵
    ↵merges 和 `F32/Q8_0/Q5_0/Q6_K/Q4_K` 混合权重, 加载 30 层 SmolLM2, 执行 ↵
    ↵RMSNorm、normal RoPE、GQA KV cache、SwiGLU 和 tied lm head; 当前权重执行↵
    ↵方式是加载时解到 f32, 用作后续量化直算优化的 correctness baseline
28 - CLI 推理和简单 benchmark
29 - CTest 正确性测试
30
31 后续扩展方向:
32
33 - 真实 7B 级模型 metadata、tokenizer 和量化权重导入
34 - SentencePiece tokenizer、safetensors 分片加载和更多模型方言 name mapping
35 - GPT-2 reference path 的 Transformers/PyTorch golden logits 对齐报告
36 - KV cache 分页、内存规划和可选量化
37 - CPU packed weight、SIMD、NUMA 和线程亲和优化
38 - GPU backend adapter 和 device kernel
39 - 与现有框架的 prefill/decode/内存/误差对标报告
40
41 构建:
42
43 ```bash
44 cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -↵
    ↵DCMAKE_BUILD_TYPE=Debug
45 cmake --build books/cpu-volume-3/build/lcqi-debug
46 ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
47 ```
48
49 运行:
50
51 ```bash
52 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_cli \
53   books/cpu-volume-3/labs/linux_cpu_inference/models/tiny_mlp_i8.txt \
54   1.0 2.0 -1.0 0.5 1000
55 ```
56
57 kernel benchmark:
58
59 ```bash
60 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_bench 200

```

```

61  ```
62
63  输出 CSV 字段：
64
65  ```text
66  target_arch,input_size,output_size,output_block,repeat,backend,available,↵
    ↵ average_us,max_abs_diff,checksum
67  ```
68
69  当前 benchmark 按 backend 逐行输出：`scalar`、`packed_scalar`、`avx2`。x86-64 ↵
    ↵ 构建会编译 AVX2 路径；不支持 AVX2/FMA 的机器会把该 backend 标为 `↵
    ↵ available=0`，避免把源码存在误报成当前环境实测性能。
70
71  这还不是生产级高性能 kernel。当前 benchmark 建立 scalar baseline、packed layout↵
    ↵ 、AVX2 基线正确性和 shape sweep 口径。要达到完整 AI Infra 证据，还必须继↵
    ↵ 续补反汇编分析、AVX-512、perf counter、oneDNN/OpenBLAS/llama.cpp/ggml 对↵
    ↵ 照、sanitizer、fuzz 和 CI。
72
73  safetensors manifest gate:
74
75  `lcqi_tests` 会运行时生成一个最小 safetensors fixture，验证 header 长度、`↵
    ↵ __metadata__`、tensor name、dtype、shape、data offsets、payload 边界和 ↵
    ↵ dtype/shape 推导出的字节数。它还会生成 offset 超出 payload、dtype/shape ↵
    ↵ 字节数不匹配的坏文件，确认加载器会拒绝。这个 gate 只覆盖模型资产目录表，↵
    ↵ 不等于已经支持完整 LLaMA/Qwen 权重映射、分片合并或 mmap 执行；后续真实模↵
    ↵ 型加载必须在这个 manifest 上继续接 name mapping、shape verifier、quant ↵
    ↵ descriptor 和 first token trace。
76
77  reference decoder trace:
78
79  ```bash
80  books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_trace
81  ```
82
83  输出 CSV 字段：
84
85  ```text
86  name,token_position,layer_id,shape,stride,dtype,layout,checksum,max_abs,values
87  ```
88
89  这条 trace 固定 prompt fixture `tok1 tok2 tok3`，tokenizer 输出完整 `tokens↵
    ↵ =[0,1,2,3,4]`，其中 `0/4` 是 BOS/EOS；decoder trace 会剥离 BOS/EOS 后进入↵
    ↵ tiny decoder。这样报告能同时固定 tokenizer 合同和 decoder 输入合同，避免↵
    ↵ 把二者混成一个不可解释的 token 数组。随后 trace 把 embedding、RMSNorm、Q/↵
    ↵ K/V、RoPE、KV cache slot、attention、SwiGLU、layer output、final norm、↵
    ↵ logits、argmax sampler 和 generated token 串成可回归检查的硬链路。`↵
    ↵ lcqi_tests` 会检查 tokenizer contract hash、关键 checkpoint 的 shape、↵
    ↵ stride、dtype、layout 和 logits golden，CTest 也会直接运行 `lcqi_trace`。
90
91  当前 tokenizer fixture 的关键输出：
92
93  ```text

```

```

94 tokenizer,0,-1,5,1,i32,contiguous,10,4,0;1;2;3;4
95 tokenizer_contract,0,-1,scalar,scalar,u64,fnv1a,13452902845388333734,0,tiny-↵
    ↵ chat-template-v1
96 ```
97
98 GPT-2 自研 reference smoke:
99
100 ```bash
101 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_gpt2
102 make cpu3-gpt2-smoke
103 make cpu3-gpt2-benchmark-compare
104 make cpu3-gpt2-hotspot-profile
105 python3 books/cpu-volume-3/tools/run_gpt2_optimization_ab.py --rounds 3
106 python3 books/cpu-volume-3/tools/run_gpt2_hotspot_profile.py --token-counts ↵
    ↵ 4,16 --rounds 3
107 ```
108
109 第一条命令不需要下载模型，直接运行仓库内 tiny GPT-2 fixture，输出固定 prompt ↵
    ↵ ids、logits、predicted token 和 greedy generated ids。`lcqi_tests` 会覆盖↵
    ↵ tiny GPT-2 forward/generation、byte-level BPE 编码解码、F32/F16/BF16 ↵
    ↵ safetensors 读取、HuggingFace 风格 tiny GPT-2 checkpoint 加载和坏 shape ↵
    ↵ 拒绝。
110
111 第二条命令显式依赖网络，只下载 `openai-community/gpt2` 的 `config.json`、`vocab↵
    ↵ .json`、`merges.txt`、`model.safetensors` 到 `~/cache/lcqi-gpt2-smoke/↵
    ↵ openai-community--gpt2`，模型文件不进入 Git。脚本会校验四个文件的 bytes ↵
    ↵ 和 SHA256，构建 `lcqi_gpt2`，运行：
112
113 ```text
114 Hello, my name is
115 ```
116
117 当前验证报告写到：
118
119 ```text
120 books/cpu-volume-3/results/lcqi-gpt2-smoke.txt
121 ```
122
123 最近一次本机报告的关键行是：
124
125 ```text
126 prompt_ids 15496 11 616 1438 318
127 generated_ids 15496 11 616 1438 318 1757
128 generated_text Hello, my name is John
129 validation=PASS
130 ```
131
132 这说明 LCQI 自研 C++ reference path 已经能跑标准 GPT-2 safetensors 的 tokenizer↵
    ↵ 、权重导入、forward 和 greedy generation。默认真实模型路径现在使用 `↵
    ↵ cached_kv`，也可以用 `--engine full` 复现旧的 full-prefix 重算路径：
133
134 ```bash

```

```

135 books/cpu-volume-3/build/lcqi-release/labs/linux_cpu_inference/lcqi_gpt2 \
136 --benchmark --engine cached --threads 0 \
137 ~/.cache/lcqi-gpt2-smoke/openai-community--gpt2 \
138 "Hello, my name is" 4
139 ```
140
141 `--threads 0` 表示自动选择 cached decoder worker 数；`--threads 1` 强制串行；↵
    ↪ `--threads N` 固定 worker 数。当前 auto 上限是 16，避免在短 decode 上把线↵
    ↪ 程调度开销放大到超过收益。
142
143 `make cpu3-gpt2-benchmark-compare` 会生成：
144
145 ```text
146 books/cpu-volume-3/results/lcqi-gpt2-benchmark-compare.txt
147 ```
148
149 `run_gpt2_optimization_ab.py` 会生成：
150
151 ```text
152 books/cpu-volume-3/results/lcqi-gpt2-optimization-ab.txt
153 ```
154
155 这个 A/B 报告专门回答“当前 LCQI 代码相对指定 baseline 改快了吗”。脚本默认 ↵
    ↪ baseline 是 `4feb3f`，会临时创建 baseline worktree，分别构建 baseline 和↵
    ↪ 当前工作区的 Release `lcqi_gpt2`，再多轮运行同一条 prompt。上一轮工作区复↵
    ↪ 用优化的 2 轮 smoke 摘要保存在 `books/cpu-volume-3/reports/lcqi-gpt2-↵
    ↪ optimization-ab-summary.md`；本轮线程优化的正式 raw 报告保存在 `books/cpu-↵
    ↪ -volume-3/reports/lcqi-gpt2-threaded-manual-ab-caw.txt`，汇总保存在 `↵
    ↪ books/cpu-volume-3/reports/lcqi-gpt2-threaded-optimization-summary.md`。↵
    ↪ 以 `73f40c7` 作为 baseline，在 `caw` 上 5 轮中位数显示：4 token cached-KV↵
    ↪ 生成阶段从 `395.397 ms` 降到 `76.4141 ms`，16 token 从 `989.060 ms` 降到↵
    ↪ `185.479 ms`，生成吞吐分别提升到 `52.3464 token/s` 和 `86.2633 token/s`↵
    ↪ 。随后的 F32 dot kernel 优化以 `cb4d89f` 为 baseline，raw 报告保存在 `↵
    ↪ books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-ab-caw.txt`，汇总保存在 `↵
    ↪ books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-optimization-summary.md`。在↵
    ↪ 同一台 `caw`、同一 GPT-2 F32 模型、同样 `--threads 0` 自动 16 workers ↵
    ↪ 下，4 token 生成阶段中位数从 `74.4247 ms` 到 `62.3628 ms`，16 token 从 ↵
    ↪ `181.257 ms` 到 `157.230 ms`；decode token/s 分别提升到 `130.073` 和 ↵
    ↪ `127.405`。再往后，row-range kernel 和去 `std::function` 调度开销的 raw ↵
    ↪ 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-ab-caw.txt`↵
    ↪ ，汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-↵
    ↪ optimization-summary.md`。这一轮相对 `ab72ccb` 的端到端提升很小，16 token↵
    ↪ 生成中位数约 `153.354 ms` 到 `152.574 ms`；但直接比较 row-range 与 4-row↵
    ↪ AVX2 row-range 时，4 token 从 `62.0243 ms` 到 `60.3965 ms`，16 token 从 ↵
    ↪ `154.087 ms` 到 `151.222 ms`。本轮 prefill-skip 优化的 raw 报告保存在 `↵
    ↪ books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-ab-caw.txt`，汇总保存在↵
    ↪ `books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-optimization-summary.↵
    ↪ md`。它跳过 prompt 前缀 token 上不会被使用的 `lm_head` 预测：4 token 生成↵
    ↪ 中位数从 `60.5154 ms` 到 `52.1949 ms`，16 token 从 `153.574 ms` 到 ↵
    ↪ `147.424 ms`，生成文本和 token id 全部一致。端到端 GPT-2 数字受调度、温度↵
    ↪ 和共享机器状态影响很大，因此正式结论必须看多轮中位数、min/max 和生成文本↵
    ↪ 一致性，不能只挑一次最快结果。

```


156

157 ``run_gpt2_hotspot_profile.py`` 专门回答“现在热点到底在哪”。它通过 ``--profile-``
`hotspots`` 打开 `cached decode` 内部计时，字段包括 ``hotspot_lm_head_pct``、```
`hotspot_mlp_fc_pct``、``hotspot_mlp_projection_pct``、```
`hotspot_qkv_projection_pct`` 和 ``hotspot_attention_pct``。``caw`` 上的 3 轮
`Release profile` 摘要保存在 ``books/cpu-volume-3/reports/lcqi-gpt2-hotspot-``
`profile-summary.md``，原始报告保存在 ``books/cpu-volume-3/reports/lcqi-gpt2-``
`-hotspot-profile-caw.txt``。关键结论是：4 token 与 16 token 两个口径下，```
`lm_head`` 中位数约 ``30.2%``，MLP 两个线性投影合计约 ``46%``，QKV 投影约 ``17%``
```，`cached attention` 本体只有 ``0.06%`` 到 ``0.11%``。所以当前短上下文 GPT-2 的  
慢点不是 KV attention，而是标量 F32 matvec 和 ``50257 x 768`` vocab 扫描。

158

159 当前优化点集中在 `cached-KV generation`：``Gpt2CachedGreedyDecoder`` 把模型级 `shape``  
`validation`、KV cache、临时 workspace 和 worker pool 的生命周期提升到整段  
生成；``Gpt2ForwardWorkspace`` 复用 `hidden`、`normed`、`Q/K/V`、`packed QKV``、```  
`attention`、MLP、`scores` 和 `logits` 缓冲区；生成路径只需要 greedy token 时走  
`logits-free argmax`，避免把完整 `logits vector` 作为 API 结果反复分配；```  
`prompt prefill` 前缀通过 ``advance_without_prediction()`` 只更新 KV cache，```  
不跑 `final norm` 和 ``lm_head``，因为这些预测结果不会被用作输出；QKV、```  
`attention output projection`、MLP 两个 `projection` 和 `tied `lm_head`` 都在  
`cached session` 内按输出行并行；每个线程 chunk 再通过 ```  
`linear_f32_rows_unchecked`` 或 ``max_dot_f32_rows_unchecked`` 进入 F32 row-  
range kernel，x86-64 可用时使用 AVX2/FMA，其他平台走标量 fallback。KV  
cache 的 `checked public API` 仍保留，内部 `cached attention` 在一次边界校验  
后使用私有 `unchecked span` 扫描热路径。它不改变 GPT-2 数学语义，只改变会话  
生命周期、任务分片、无用工作剔除和热循环调度。

160

161 ``make cpu3-gpt2-benchmark-compare`` 仍用于和外部成熟引擎放在同一报告里看工程差  
距。同机 `llama.cpp/SmolLM2 Q4_K_M` 是成熟引擎参照，不是同模型质量排名：```  
LCQI 跑的是 GPT-2 F32 reference path，`llama.cpp` 跑的是 GGUF 量化 `small``  
`instruct` 模型。它揭示的是工程差距：LCQI 已经消除了“每步重算前缀”的主要算  
法错误，补上 `cached F32 row parallelism`，并把热路径接到 AVX2/FMA row-  
range kernel；但还缺 `packed/quantized weight`、GGUF 量化 `block`、`mmap/lazy``  
`load`、采样器、分页 KV cache、NUMA/亲和和成熟调度。

162

163 它仍然不是生产级推理引擎：当前 GPT-2 路径还没有把 `logits` 和 `Transformers/``  
`PyTorch` 逐数值对齐成外部 `golden` 报告，也没有 GGUF 自研导入、分页 KV cache  
```、量化权重执行和高性能 SIMD/packed kernel。

164

165 7B ledger:

166

167 ```bash

168 `books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_ledger \`

169 `> books/cpu-volume-3/results/lcqi-sevenb-ledger-2026-06-24.csv`

170 ```

171

172 输出 CSV 字段:

173

174 ```text

175 `section,item,value,unit,notes`

176 ```

177

```

178 这份账本固定 `L=32,H=4096,n_q=32,n_kv=8,head_dim=128,context=4096`, 报告 decode↵
    ↪ FLOPs、KV 容量、fp16/int8/int4 每 token 字节和不同有效带宽下的 tokens/s ↵
    ↪ 上限。CTest 会检查 `int4_at_100_gbps=23.602 tokens/s` 这一行, 防止账本公↵
    ↪ 式漂移。

179
180 serving SLO probe:
181
182 ```bash
183 books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_serving_probe↵
    ↪ \
184 > books/cpu-volume-3/results/lcqi-serving-slo-probe-2026-06-24.csv
185 ```
186
187 输出 CSV 字段:
188
189 ```text
190 row_type,request_id,accepted,rejection_reason,prompt_tokens,reserved_tokens,↵
    ↪ kv_reserved_mib,queue_wait_ms,prefill_ms,ttft_ms,decode_step_p95_ms,↵
    ↪ tpot_ms,completion_tokens,finish_reason,cancel_reclaim_ms,metric_name,↵
    ↪ metric_value
191 ```
192
193 probe 固定四个请求: 短请求、RAG 请求、取消请求和超长请求。它演示 admission 先按↵
    ↪ `prompt_tokens + max_new_tokens` 预留 KV, 取消请求释放预算, 超长请求返回↵
    ↪ `memory_budget_exceeded`, 并输出 `ttft_p95_ms`、`queue_wait_p95_ms`、↵
    ↪ `rejection_rate` 等服务化指标。

194
195 真实 small 开源模型 smoke:
196
197 ```bash
198 make cpu3-smolllm2-smoke
199 ```
200
201 这条命令显式依赖网络和外部构建, 因此不放进默认 CTest。脚本会把 llama.cpp 和模型↵
    ↪ 文件缓存到 `~/.cache/lcqi-smolllm2-small-smoke`, 模型不进入 Git。固定模型↵
    ↪ 是 `HuggingFaceTB/SmolLM2-135M-Instruct` 的 GGUF 量化版本, 来源仓库为 `↵
    ↪ bartowski/SmolLM2-135M-Instruct-GGUF`, 文件为 `SmolLM2-135M-Instruct-↵
    ↪ Q4_K_M.gguf`。脚本会校验文件大小 `105454432` 字节和 SHA256:

202
203 ```text
204 2e8040ceae7815abe0dcb3540b9995ead1fa0d2ca9e797d0a635ae4433c68c2d
205 ```
206
207 脚本构建固定 commit 的 `llama-simple-chat`, 用 `-ngl 0` 强制 CPU 路径, 输入短 ↵
    ↪ prompt, 检查 assistant response 非空, 并生成:

208
209 ```text
210 books/cpu-volume-3/results/lcqi-smolllm2-small-model-smoke.txt
211 ```
212
213 这份报告只能证明 llama.cpp 成熟外部引擎上的真实 GGUF small 模型已经完成加载、↵
    ↪ tokenizer/chat template、prefill、decode 和文本输出。它不能证明 LCQI 自研↵

```



```

    ↪ C++ 引擎已经达到 llama.cpp 的性能，也不能替代 tiny fixture 的逐层 golden↪
    ↪ trace。
214
215 LCQI 自研 SmolLM2/GGUF reference decode:
216
217 ```bash
218 books/cpu-volume-3/build/lcqi-release/labs/linux_cpu_inference/lcqi_llama_gguf ↪
    ↪ \
219 ~/.cache/lcqi-smolllm2-small-smoke/models/SmolLM2-135M-Instruct-Q4_K_M.gguf \
220 --prompt "Hello" --max-new 2 --benchmark --decode-text
221 ```
222
223 caw 上的最新报告写到:
224
225 ```text
226 books/cpu-volume-3/results/lcqi-smolllm2-gguf-reference-decode-caw.txt
227 ```
228
229 关键行是:
230
231 ```text
232 weight_execution f32_dequantized_reference
233 generated_text ... <|im_start|>assistant
234 Hello!
235 benchmark_prompt_tokens 31
236 benchmark_generated_tokens 2
237 benchmark_decode_tokens_per_second 22.1886
238 benchmark_quantized_weight_bytes 103668480
239 benchmark_f32_weight_bytes 538060032
240 benchmark_tensors_loaded 272
241 ```
242
243 这条路径证明 LCQI 自研代码已经能读取同一个 SmolLM2 GGUF、解析 tokenizer arrays↪
    ↪ 、解码混合量化权重并端到端生成文本。它仍然不是生产级高性能引擎：当前把 ↪
    ↪ 103.7 MB 量化权重解成约 538 MB f32 权重，再用通用 f32 matvec 执行；与 ↪
    ↪ llama.cpp 的差距主要来自低比特权重直算、packed kernel、线程调度、prefetch↪
    ↪ 、KV cache 和整图执行。

```

tiny MLP 模型文件是最短闭环的资产。它决定输入维度、hidden 维度、输出维度、权重量化值、scale 和 bias；测试中的 logits golden 依赖这份文件。

Listing 1.9 models/tiny_mlp_i8.txt

```

1 LCQI_MODEL_V1
2 input_size 4
3 hidden_size 3
4 output_size 2
5 hidden_scale 0.1
6 hidden_weights
7 5 -3 2 1
8 -4 6 1 -2
9 3 2 -5 4
10 hidden_bias

```

```
11 0.1 -0.2 0.0
12 output_scale 0.05
13 output_weights
14 4 -6 2
15 -3 5 8
16 output_bias
17 -0.5 0.4
```

tiny tokenizer fixture 固定 BOS、EOS、UNK、词表和模板 hash。tokenizer 合同如果漂移，后面的 decoder trace 和测试都会被污染。

Listing 1.10 models/tiny_tokenizer.txt

```
1 LCQI_TOKENIZER_V1
2 bos_id 0
3 eos_id 4
4 unk_id -1
5 chat_template_hash tiny-chat-template-v1
6 vocab_size 5
7 vocab
8 0 <bos>
9 1 tok1
10 2 tok2
11 3 tok3
12 4 <eos>
```

为什么要先讲调用链

实践与代码卷如果只把 *.cpp 和 *.hpp 全部贴出来，读者仍然不知道从哪里下手。LCQI 的代码同时包含 tiny MLP、int8 kernel、tokenizer、safetensors、reference decoder、GPT-2、benchmark、trace、ledger、serving probe 和外部模型脚本；这些文件不是平铺关系，而是几条调用链。先讲调用链，再给完整源码，读者才能把每个文件放到正确位置。

核心思想

读 LCQI 的顺序不是“哪个文件长先读哪个”，而是“入口、合同、资产、执行、证据”。入口说明目标怎样构建，合同说明类型怎样连接，资产说明模型从哪里来，执行说明 token 或张量怎样流动，证据说明测试和报告守住了哪些边界。

一、全工程入口：哪些文件决定能不能复现

| 文件 | 负责什么 | 读源码时要确认什么 |
|------------|---|--|
| 第三册 README | 第三册 AI 计算工程路线、当前 LCQI 能力、GPT-2 smoke 和结果报告位置。 | 能力声明必须能在 labs/linux_cpu_inference、tools 和 results 中找到证据。 |
| 第三册 CMake | 第三册根 CMake，负责把 linux_cpu_inference 接入。 | 根 CMake 应保持薄层；真正 target 定义必须在 lab 内部，避免入口文件承担实现细节。 |
| 本卷 CMake | 本卷复用同一份 LCQI lab。 | 本卷不复制源码，只把同一工程接入自己的测试构建，防止两份代码漂移。 |

| | | |
|------------|--|--|
| 本卷Makefile | 本卷的书籍构建入口。 | 它构建 PDF/EPUB，也提供 <code>test</code> 目标去构建第三册 LCQI；实践与代码卷不能只排版不验证。 |
| LCQICMake | <code>lcqi</code> 库、CLI、benchmark、trace、GPT-2、ledger、serving probe、CTest 目标。 | 读这里能看到哪些文件进入库，哪些是可执行程序，哪些依赖可选外部库。 |
| LCQIREADME | 本地构建、tiny 推理、benchmark、trace、GPT-2、SmolLM2 smoke 的命令。 | 先按 README 跑命令，再读源码；否则不知道某个函数是否真的能从命令行触达。 |

上表里的短标签对应下面这些真实路径：

- `books/cpu-volume-3/README.md`
- `books/cpu-volume-3/CMakeLists.txt`
- `books/cpu-volume-3-practice/CMakeLists.txt`
- `books/cpu-volume-3-practice/Makefile`
- `books/cpu-volume-3/labs/linux_cpu_inference/CMakeLists.txt`
- `books/cpu-volume-3/labs/linux_cpu_inference/README.md`

这些入口文件回答“代码在哪里”和“怎样运行”。读者进入本卷代码走读章时，先不要直接跳到 `gpt2_reference.cpp`。先确认 `lcqi` 静态库包含 `model.cpp`、`inference.cpp`、`int8_kernels.cpp`、`f32_kernels.cpp`、`gguf.cpp`、`ggml_tensors.cpp`、`q4_k.cpp`、`llama_gguf.cpp`、`reference_decoder.cpp`、`tokenizer.cpp`、`safetensors.cpp` 和 `gpt2_reference.cpp`；再确认 `lcqi_cli`、`lcqi_bench`、`lcqi_gguf_q4_bench`、`lcqi_llama_gguf`、`lcqi_trace`、`lcqi_gpt2`、`lcqi_ledger`、`lcqi_serving_probe` 和 `lcqi_tests` 分别从哪条路径进入。

二、Tiny MLP 调用链：最短闭环怎样跑通

Tiny MLP 是最小推理闭环，用来证明“模型资产、shape 检查、int8 linear、激活函数、logits、argmax、CLI 输出”可以完整连起来。它不是大模型能力本身，但它是后面所有复杂路径的可手算底座。

| 文件 | 关键类型或函数 | 这段代码的意思 |
|---|--|--|
| <code>models/tiny_mlp_i8.txt</code> | 输入维度、hidden 维度、输出维度、int8 权重、scale、bias。 | 这是模型资产。测试中的 <code>golden logits</code> 来自这里，资产改动必须带着测试改动。 |
| <code>include/lcqi/model.hpp</code> | <code>QuantizedLinearLayer</code> 、 <code>TinyMlpModel</code> 、 <code>load_model</code> 。 | 定义 <code>tiny MLP</code> 如何在内存中表示；只负责模型结构和加载入口，不做推理。 |
| <code>src/model.cpp</code> | <code>read_i8_vector</code> 、 <code>validate_layer</code> 、 <code>load_model</code> 。 | 解析文本模型，拒绝坏 key、坏维度、权重数量不匹配和非法 scale，把错误挡在执行前。 |
| <code>include/lcqi/inference.hpp</code> | <code>InferenceResult</code> 、 <code>linear_i8</code> 、 <code>relu_inplace</code> 、 <code>run_inference</code> 。 | 定义推理输出要包含 <code>hidden</code> 、 <code>logits</code> 和 <code>predicted class</code> ，方便测试定位错误层。 |
| <code>src/inference.cpp</code> | 输入 shape 检查、两层 int8 linear、ReLU、argmax。 | 最短执行路径：模型加输入变成 logits。这里的代码应保持清楚，不能混入 CLI 或文件解析。 |
| <code>src/main.cpp</code> | 解析命令行输入并打印结果。 | CLI 是薄层；它只接模型路径和输入，不复制 <code>run_inference</code> 逻辑。 |

这条链路按下面顺序读：`main()` 取得模型路径和输入向量，调用 `load_model()` 把 `tiny_mlp_i8.txt` 变成 `TinyMlpModel`，再调用 `run_inference()`。在 `run_inference()` 内部，

第一层 `linear_i8()` 把 float 输入和 int8 权重相乘，乘以 scale 后加 bias；`relu_inplace()` 把负 hidden 截断；第二层 `linear_i8()` 得到 logits；最后 `argmax()` 给出 predicted class。

读这个路径时要抓住边界条件。输入长度不等于 `input_size` 要立即拒绝；权重数量不是 `rows*columns` 要在加载时拒绝；logits 为空不能取 `argmax`。这些错误如果拖到 SIMD kernel 或 benchmark 里才暴露，定位成本会高很多。

三、Int8 Kernel 调用链：reference、packed、AVX2 和 benchmark 怎样对照

| 文件 | 关键代码 | 这段代码的意思 |
|--|---|---|
| <code>include/lcqi/int8_kernels.hpp</code> | <code>PackedLinearI8、KernelBenchmarkCase、benchmark_linear_i8_case。</code> | 定义 kernel 对标口径：shape、repeat、backend、平均耗时、checksum、max diff。 |
| <code>src/int8_kernels.cpp</code> | <code>pack_linear_i8、linear_i8_packed、linear_i8_packed_with_workspace、deterministic fixture。</code> | scalar 和 packed 路径是正确性基线，workspace 路径减少重复分配，fixture 让 benchmark 和测试可复现。 |
| <code>src/int8_kernels_avx2.cpp</code> | <code>linear_i8_packed_avx2_available、linear_i8_packed_avx2。</code> | 平台优化路径。必须先判断机器能力，再执行 AVX2/FMA；不可用时报告 unavailable，而不是假装测过。 |
| <code>include/lcqi/f32_kernels.hpp</code> 与 <code>src/f32_kernels.cpp</code> | <code>dot_f32、linear_f32_rows_unchecked、max_dot_f32_rows_unchecked。</code> | GPT-2 F32 hot path 的 kernel 边界。单行 dot 是基础，row-range linear 一次写一段输出行，row-range max 给 logits-free greedy <code>lm_head</code> 用。 |
| <code>src/f32_kernels_avx2.cpp</code> | <code>dot_f32_avx2_available、dot_f32_avx2_unchecked、4-row row-range kernel。</code> | x86 平台优化路径。它在一段输出行内部复用 input load，同时不改变 GPT-2 层结构、线程分片或 greedy tie-breaking。 |
| <code>include/lcqi/gguf.hpp</code> 与 <code>src/gguf.cpp</code> | <code>GgufManifest、GgufTensorInfo、load_gguf_manifest、read_gguf_tensor_bytes。</code> | 读取 GGUF header、metadata、tensor 表、alignment 和真实 tensor payload；它只负责格式边界，不做推理数学。 |
| <code>include/lcqi/ggml_tensors.hpp</code> 与 <code>src/ggml_tensors.cpp</code> | <code>dequantize_ggml_tensor、ggml_f16_to_f32。</code> | 把 GGUF 中的 <code>F32、Q8_0、Q5_0、Q6_K</code> 和 <code>Q4_K</code> 张量统一解到 f32。SmolLM2 <code>Q4_K_M</code> 不是单一 <code>Q4_K</code> 文件，完整加载必须先处理这些混合格式。 |
| <code>include/lcqi/q4_k.hpp</code> 与 <code>src/q4_k.cpp</code> | <code>dequantize_q4_k、quantize_q8_k_input、matvec_q4_k_q8。</code> | 实现 llama.cpp 同格式的 <code>Q4_K</code> block 解码和 <code>Q4_KxQ8_K</code> 单算子路径；单测用手写 block 固定 nibble 顺序、scale/min 解包和误差边界。 |

| | | |
|--|---|---|
| <code>include/lcqi/llama_gguf.hpp</code> 与 <code>src/llama_gguf.cpp</code> | <code>load_llama_gguf_reference_model</code> 、 <code>load_gpt2_tokenizer_from_gguf</code> 、 <code>llama_gguf_generate_greedy</code> 。 | 自研 SmolLM2/GGUF reference path。它从同一个 GGUF 读取 tokenizer tokens/merges、混合量化权重、RMSNorm、normal RoPE、GQA KV cache、SwiGLU 和 tied lm head；当前执行方式是加载时解到 f32，作为后续量化直算优化的正确性基线。 |
| <code>src/llama_cli.cpp</code> | <code>lcqi_llama_gguf</code> 。 | 命令行加载真实 SmolLM2 GGUF，支持 token id prompt 和文本 prompt，输出生成 token、文本、load/prefill/decode 时间、KV 字节、量化权重字节和解码后 f32 字节。 |
| <code>src/q4_k_avx2.cpp</code> | <code>q4_k_q8_avx2_available</code> 、 <code>matvec_q4_k_q8_avx2_unchecked</code> 。 | x86 AVX2 路径。它用 <code>maddubs</code> 把 32 个 unsigned nibble 与 32 个 signed int8 激活先归约成整数和，再套用 <code>Q4_K</code> 的 scale/min；报告必须同时给 scalar 与 auto 后端，防止把机器波动写成优化收益。 |
| <code>src/gguf_q4_bench.cpp</code> | <code>lcqi_gguf_q4_bench</code> 。 | 从真实 SmolLM2 GGUF 选择 <code>Q4_K</code> 矩阵，比较 f32 解码基线、 <code>Q4_Kxf32</code> 和 <code>Q4_KxQ8_K</code> ，输出速度、误差、checksum 和字节账本。 |
| <code>src/bench.cpp</code> | shape 列表、repeat 参数、back-end 输出。 | benchmark 程序只负责测 kernel 入口。性能结论必须和 <code>max_abs_diff</code> 、CPU、编译器、构建类型一起报告。 |

kernel 文件要按“可信基线到优化路径”的顺序读。先看 deterministic layer 和 input 怎么生成，保证每次运行输入相同；再看 `linear_i8_packed()` 怎样把 int8 权重、scale、bias 和 float 输入算成输出；再看 workspace 路径怎样复用临时缓冲；最后看 AVX2 路径怎样在支持平台上替换内层循环。Q4_K 路径也遵守同一顺序：先读 scalar 解码，再读 AVX2 `maddubs` 子块归约，最后看 benchmark 怎样强制 scalar 和 auto 后端并列输出。测试用 packed 与 scalar 的 `max_abs_diff` 对照，benchmark 再比较耗时。

四、Tokenizer、Safetensors 与 Reference Decoder：真实模型前的三道门

| 文件 | 关键类型或函数 | 这段代码的意思 |
|--|--|---|
| <code>models/tiny_tokenizer.txt</code>
<code>include/lcqi/tokenizer.hpp</code> 与 <code>src/tokenizer.cpp</code> | BOS、EOS、UNK、词表、模板 hash。
<code>TokenizerModel</code> 、 <code>load_tokenizer</code> 、 <code>encode_prompt</code> 、 <code>tokenizer_contract_hash</code> 。 | 固定 tokenizer 合同，避免 prompt 到 token id 的规则漂移。
把 prompt 切成稳定 token id；未知 token 是否允许，必须由 UNK 合同决定。 |
| <code>include/lcqi/safetensors.hpp</code> 与 <code>src/safetensors.cpp</code> | <code>SafeTensorManifest</code> 、 <code>SafeTensorFile</code> 、manifest parser、dtype 转换。 | 解析模型字节之前先验证 header、dtype、shape、offset 和 payload 边界，防止坏资产进入数学层。 |

| | | |
|---|--|--|
| <code>include/lcqi/reference_decoder.hpp</code> | embedding、RMSNorm、RoPE、KV cache、attention、SwiGLU、与 trace checkpoint。 | Tiny decoder-only reference path。它用 checkpoint 解释每一步张量状态，不只返回最终 token。 |
| <code>src/reference_decoder.cpp</code> | | |
| <code>src/trace.cpp</code> | trace CSV 输出。 | 把 reference decoder 的 checkpoint 打印成可复查表格，定位错误发生在哪个阶段。 |

这三道门分别解决三个问题。Tokenizer 解决“人写的 prompt 如何变成 token id”；Safetensors 解决“磁盘里的模型字节如何变成带 shape 的张量”；Reference Decoder 解决“token 和权重如何经过 decoder-only 层生成 logits”。任何一层含糊，后面的 GPT-2 路径都会变成看似能跑、实则无法定位的黑盒。

五、GPT-2 调用链：从 HuggingFace 资产到生成 token

GPT-2 路径是本卷代码走读里最长的一条链，因为它要处理真实模型格式。读 `gpt2_reference.cpp` 时按模块读，不要从第一行一路滚到最后。

| 文件 | 关键代码 | 这段代码的意思 |
|--|---|--|
| <code>include/lcqi/gpt2_reference.hpp</code> | 配置、模型、KV cache、cached decoder、tokenizer 与 forward/generate API。 | 公开 GPT-2 reference path 的合同：配置、权重、KV cache、tokenizer、full-prefix、cached generation，以及 prompt prefill 中只推进 cache、不做无用预测的入口。 |
| <code>src/gpt2_reference.cpp</code> | JSON parser、byte-level BPE、safetensors loader、LayerNorm、QKV、causal attention、GELU、lm head、KV cache。 | 正确性优先实现。它把 HuggingFace GPT-2 权重名、Conv1D 转置、tokenizer 和 forward 规则全部显式写出来。 |
| <code>src/gpt2_cli.cpp</code> | tiny mode、真实模型目录模式、generation、benchmark 输出。 | 命令行入口。无参数跑 tiny fixture；传模型目录时加载真实 GPT-2 资产；benchmark 对比 full-prefix 和 cached-KV。 |
| <code>tools/run_gpt2_smoke.py</code> | 下载标准 GPT-2 资产，调用 <code>lcqi_gpt2</code> 。 | 证明自研 safetensors、BPE、forward、KV cache 和 greedy generation 能跑真实开源模型。 |
| <code>tools/run_gpt2_benchmark_compare.py</code> | 运行 LCQI full-prefix、cached-KV，并可附带成熟引擎参考。 | 用同一模型口径比较算法差距和 kernel 差距，不把不同模型的速度混成一个结论。 |

GPT-2 代码按五段读。第一段是配置和权重加载：`load_gpt2_config()` 读 `config.json`，`load_gpt2_reference_model()` 从 safetensors 中取 embedding、每层 LayerNorm、QKV、MLP 和 final norm。HuggingFace GPT-2 的 Conv1D 权重形状和普通线性层方向不同，所以代码显式调用转置函数，不能跳过。

第二段是 tokenizer：`load_gpt2_tokenizer()` 读 `vocab.json` 和 `merges.txt`，`gpt2_encode()` 执行 byte-level BPE，`gpt2_decode()` 把 id 还原为文本。这里的重点是字节到 Unicode 映射、BPE merge rank 和 unknown token 边界。

第三段是 full-prefix forward：`run_gpt2_forward()` 每生成一个新 token 都用完整 token 前缀重新跑所有层。它慢，但容易验证。流程是 token embedding 加 position embedding，逐层执行 LayerNorm、packed QKV、causal attention、残差、MLP、残差，最后 final LayerNorm 和 tied lm head 得到 logits。

第四段是 cached-KV forward：`Gpt2KvCache` 保存每层每个 position 的 key/value，`run_gpt2_forward_cached()` 每次只处理最新 token，并让 attention 读取过去 cache。这里要特别

看 `validate_written()`：读取未写入 cache slot 必须报错，否则生成错误会很难定位。

第五段是 greedy generation: `gpt2_generate_greedy()` 用 full-prefix 路径, `gpt2_generate_greedy_cached()` 用 KV cache 路径。两个函数都要检查 prompt 非空、`max_new_tokens` 非负、不会超过 `max_positions`，并在生成 EOS 时停止。

六、报告、服务探针、外部对标和测试

| 文件 | 关键代码 | 这段代码的意思 |
|---|---|--|
| <code>src/ledger.cpp</code> | 7B 参数量、KV cache、bandwidth-only 上限。 | 输出的是工程账本，不是实测速度；它帮助读者估算权重和 KV 内存压力。 |
| <code>src/serving_probe.cpp</code> | 短请求、RAG 请求、取消请求、超长请求, TTFT/TPOT 字段。 | 把服务层边界变成字段：哪些请求接受，哪些拒绝，取消后资源是否回收。 |
| <code>src/onednn_bench.cpp</code> | oneDNN matmul benchmark。 | 可选外部库对标入口。没有 oneDNN 环境时不能声称已验证。 |
| <code>tools/run_smolLM2_small_smoke.py</code> | 下载 SmolLM2-135M-Instruct GGUF，构建固定 commit 的 llama.cpp smoke。 | 这是成熟外部引擎上的真实 small 模型 smoke, 用来建立外部参照, 不代表 LCQI 已具备同等能力。 |
| <code>tools/run_lcqi_benchmark.sh</code> | 收集 LCQI 本地 benchmark。 | 在固定机器上重复运行同一组 shape，形成可复查 CSV 或文本报告。 |
| <code>tests/lcqi_tests.cpp</code> | tiny MLP、tokenizer、safetensors、GPT-2 tiny、HF-style checkpoint、bad shape、decoder、trace、kernel。 | 这是本卷最重要的代码证据。新增功能如果没有进入这里，就不能写成已经守住的能力。 |

读测试文件时，不要只看最后是否返回 0。它里面的 fixture 本身就是源码讲解的一部分：测试会手写 safetensors header，构造 F16/BF16/F32 payload，生成 tiny GPT-2 权重，验证 bad shape 会被拒绝，比较 packed int8 kernel 和 scalar 路径。这些测试告诉读者“代码真正承诺了什么”。如果一个能力只出现在 README 或 CLI 输出里，却没有进入测试或 smoke 报告，就只能算实验入口，不能算稳定能力。

源码阅读任务

读完本章后，再去读相邻主题中的完整源码。每看到一个文件，都把它归到五类之一：入口、合同、资产解析、执行路径、证据。归不进去的文件，就说明工程边界需要重新解释或重构。

第2章实践：张量布局、地址流与第一个 MatVec

本章对应原书中的第三册“张量布局、地址流与第一个矩阵向量内核”。不要把它当作概念复述，本章要把 TensorView、dtype、shape、stride、row-major、地址公式、bias、ReLU、checksum 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `linear_i8`，核心命令或测试入口是 `lcqi_tests`。

Listing 2.1 第2章实践：张量布局、地址流与第一个 MatVec 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_tests / linear_i8
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：手算一个 2 行 4 列权重矩阵的地址偏移。读者要先手写预期结果，再运行项目观察输出。动作是：把 `weight[out,k]` 展开成 `base+(out*K+k)*sizeof(i8)`，再对应到 `linear_i8` 的循环。如果输出里没有 `logit` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：张量不是裸数组；每个 kernel 入口都要能解释 dtype、shape、stride、scale 和输出缓冲这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交一页地址流账本：输入向量、hidden 权重、输出 logits 各自的 shape、连续性、读写次数和错误边界。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 3 章实践：GEMM、packing 与 SIMD 证据

本章对应原书中的第三册“从矩阵向量乘走向 GEMM、packing 与 SIMD”。不要把它当作概念复述，本章要把 packed weight、scalar baseline、AVX2 backend、output block、shape sweep、反汇编和降级放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux_cpu_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `output_block`，核心命令或测试入口是 `lcqi_bench`。

Listing 3.1 第 3 章实践：GEMM、packing 与 SIMD 证据的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_bench / output_block
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：同一组输入同时跑 `scalar`、`packed_scalar` 和 `avx2`。读者要先手写预期结果，再运行项目观察输出。动作是：比较 CSV 中 `backend`、`available`、`average_us`、`max_abs_diff` 和 `checksum`。如果输出里没有 `max_abs_diff` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：SIMD 路径只有在结果和 baseline 对齐后才有资格谈速度；`available=0` 也是证据，不是失败这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux_cpu_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交一个 shape sweep 表，至少解释一个 SIMD 有收益、一个收益不明显、一个当前环境不可验证的场景。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第4章实践：int8 量化、累加器与重新量化

本章对应原书中的第三册“量化系统总览”和“int8 量化、累加器与重新量化”。不要把它当作概念复述，本章要把 int8 weight、float activation、scale、accumulator、requantize、oracle tolerance 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `TinyMlpModel`，核心命令或测试入口是 `lcqi_tests`。

Listing 4.1 第4章实践：int8 量化、累加器与重新量化的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_tests / TinyMlpModel
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：构造一行 int8 权重 `[2, -1, 3, 0]` 和输入 `[1, 2, -1, 0.5]`。读者要先手写预期结果，再运行项目观察输出。动作是：先手算 int32/float accumulator，再和 `linear_i8` 输出比较。如果输出里没有 `scale` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：量化不是把 float 强转成 int8；必须记录 scale 作用位置、累加宽度、误差阈值和 golden 输出这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交量化合同卡：每个 tensor 的 dtype、scale 粒度、accumulator 类型、误差门限和拒绝条件。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

代码走读：公共合同、Tiny MLP 与 Int8 Kernel

为什么先读头文件

头文件是系统对外承诺的最小合同。实现可以重写，kernel 可以替换，benchmark 可以补充平台路径，但头文件里暴露的数据结构、函数入口和返回字段决定了测试、CLI、工具脚本和章节讲解怎样调用这套工程。

核心思想

读 LCQI 时不要从最长的 `gpt2_reference.cpp` 开始。先读公共头文件，明确“模型如何表示”“推理返回什么”“trace 记录什么”“KV cache 怎样寻址”“tokenizer 的坏输入怎样处理”，再进入实现。这样读到长循环和 shape 检查时，知道它们在维护哪份合同。

Tiny MLP 与推理入口

`model.hpp` 定义 tiny MLP 的模型对象、权重、bias、scale 和加载入口。它的责任是把文本模型资产解释成 C++ 对象，不负责执行推理。

Listing 4.2 `include/lcqi/model.hpp`

```

1 #pragma once
2
3 #include <stdint>
4 #include <filesystem>
5 #include <string>
6 #include <vector>
7
8 namespace lcqi {
9
10 struct QuantizedLinearLayer {
11     std::int32_t input_size = 0;
12     std::int32_t output_size = 0;
13     float scale = 1.0F;
14     std::vector<std::int8_t> weights;
15     std::vector<float> bias;
16 };
17
18 struct TinyMlpModel {
19     std::int32_t input_size = 0;
20     std::int32_t hidden_size = 0;
21     std::int32_t output_size = 0;
22     QuantizedLinearLayer hidden;
23     QuantizedLinearLayer output;
24 };
25
26 TinyMlpModel load_model(const std::filesystem::path& path);
27
28 } // namespace lcqi

```

`inference.hpp` 定义推理结果和 `run_inference` 入口。返回值包含 hidden、logits 和 pre-

dicted class，测试可以用这些字段定位是加载错、kernel 错，还是 argmax 错。

Listing 4.3 include/lcqi/inference.hpp

```

1  #pragma once
2
3  #include <cstdint>
4  #include <span>
5  #include <vector>
6
7  #include <lcqi/model.hpp>
8
9  namespace lcqi {
10
11  struct InferenceResult {
12      std::vector<float> logits;
13      std::int32_t predicted_class = 0;
14  };
15
16  void linear_i8(const QuantizedLinearLayer& layer,
17               std::span<const float> input,
18               std::span<float> output);
19
20  void relu_inplace(std::span<float> values);
21
22  InferenceResult run_inference(const TinyMlpModel& model,
23                               std::span<const float> input);
24
25 } // namespace lcqi

```

Int8 Kernel、Tokenizer 与 Safetensors

int8 kernel 头文件定义 packed linear、deterministic fixture、scalar/workspace/AVX2 路径和误差比较。注意 AVX2 是运行时能力，不是源码存在就等于当前机器已验证。

Listing 4.4 include/lcqi/int8_kernels.hpp

```

1  #pragma once
2
3  #include <cstdint>
4  #include <span>
5  #include <vector>
6
7  #include <lcqi/model.hpp>
8
9  namespace lcqi {
10
11  struct PackedLinearI8 {
12      std::int32_t input_size = 0;
13      std::int32_t output_size = 0;
14      std::int32_t output_block_size = 0;
15      float scale = 1.0F;
16      std::vector<std::int8_t> packed_weights;

```

```

17     std::vector<float> bias;
18 };
19
20 struct KernelBenchmarkCase {
21     std::int32_t input_size = 0;
22     std::int32_t output_size = 0;
23     std::int32_t output_block_size = 0;
24     std::int32_t repeat = 0;
25 };
26
27 struct KernelBackendBenchmark {
28     const char* name = "";
29     bool available = false;
30     double average_us = 0.0;
31     float max_abs_diff = 0.0F;
32     float checksum = 0.0F;
33 };
34
35 struct KernelBenchmarkResult {
36     KernelBenchmarkCase benchmark_case;
37     std::vector<KernelBackendBenchmark> backends;
38 };
39
40 PackedLinearI8 pack_linear_i8(const QuantizedLinearLayer& layer,
41                               std::int32_t output_block_size);
42
43 void linear_i8_packed(const PackedLinearI8& layer,
44                      std::span<const float> input,
45                      std::span<float> output);
46
47 void linear_i8_packed_with_workspace(const PackedLinearI8& layer,
48                                     std::span<const float> input,
49                                     std::span<float> output,
50                                     std::span<float> accumulator_workspace);
51
52 bool linear_i8_packed_avx2_available() noexcept;
53
54 void linear_i8_packed_avx2(const PackedLinearI8& layer,
55                            std::span<const float> input,
56                            std::span<float> output);
57
58 QuantizedLinearLayer make_deterministic_i8_layer(std::int32_t input_size,
59                                                  std::int32_t output_size,
60                                                  float scale);
61
62 std::vector<float> make_deterministic_input(std::int32_t input_size);
63
64 float max_abs_diff(std::span<const float> lhs, std::span<const float> rhs);
65
66 KernelBenchmarkResult benchmark_linear_i8_case(const KernelBenchmarkCase& ↵
67     ↵ benchmark_case);

```



```

41         std::size_t row_begin,
42         std::size_t row_end,
43         float* output) noexcept;
44
45 void linear_f32_rows_avx2_unchecked(const float* weights,
46                                     const float* input,
47                                     const float* bias,
48                                     std::size_t input_size,
49                                     std::size_t row_begin,
50                                     std::size_t row_end,
51                                     float* output) noexcept;
52
53 [[nodiscard]] F32RowMax max_dot_f32_rows_scalar_unchecked(const float* weights,
54                                                            const float* input,
55                                                            std::size_t ↵
56                    ↵ input_size,
57                                                            std::size_t row_begin↵
58                    ↵ ,
59                                                            std::size_t row_end) ↵
60                    ↵ noexcept;
61
62 [[nodiscard]] F32RowMax max_dot_f32_rows_unchecked(const float* weights,
63                                                     const float* input,
64                                                     std::size_t input_size,
65                                                     std::size_t row_begin,
66                                                     std::size_t row_end) ↵
67                                                     ↵ noexcept;
68
69 [[nodiscard]] F32RowMax max_dot_f32_rows_avx2_unchecked(const float* weights,
70                                                           const float* input,
71                                                           std::size_t input_size,
72                                                           std::size_t row_begin,
73                                                           std::size_t row_end) ↵
74                                                           ↵ noexcept;
75
76 } // namespace lcqi

```

tokenizer 头文件固定 BOS/EOS/UNK、词表和 template hash。prompt 进入模型前，必须先变成稳定 token id；未知 token 行为也必须被写进合同。

Listing 4.6 include/lcqi/tokenizer.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <filesystem>
5 #include <string>
6 #include <string_view>
7 #include <vector>
8
9 namespace lcqi {
10
11 struct TokenEntry {

```



```

12     std::string text;
13     std::int32_t id = 0;
14 };
15
16 struct TokenizerModel {
17     std::int32_t bos_id = -1;
18     std::int32_t eos_id = -1;
19     std::int32_t unk_id = -1;
20     std::string chat_template_hash;
21     std::vector<TokenEntry> vocab;
22 };
23
24 [[nodiscard]] TokenizerModel load_tokenizer(const std::filesystem::path& path);
25
26 [[nodiscard]] std::vector<std::int32_t> encode_prompt(
27     const TokenizerModel& tokenizer,
28     std::string_view prompt);
29
30 [[nodiscard]] std::uint64_t tokenizer_contract_hash(const TokenizerModel& ↵
    ↵ tokenizer);
31
32 } // namespace lcqi

```

safetensors 头文件固定 dtype、shape、offset 和 payload 边界。它负责模型字节怎样被解释，不应该和数学推理逻辑混在一起。

Listing 4.7 include/lcqi/safetensors.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <filesystem>
5 #include <span>
6 #include <string>
7 #include <string_view>
8 #include <utility>
9 #include <vector>
10
11 namespace lcqi {
12
13 struct SafeTensorEntry {
14     std::string name;
15     std::string dtype;
16     std::vector<std::int64_t> shape;
17     std::uint64_t data_begin = 0;
18     std::uint64_t data_end = 0;
19
20     [[nodiscard]] std::uint64_t byte_size() const;
21 };
22
23 struct SafeTensorManifest {
24     std::uint64_t header_size = 0;
25     std::uint64_t data_start_offset = 0;

```

```

26     std::vector<std::pair<std::string, std::string>> metadata;
27     std::vector<SafeTensorEntry> tensors;
28
29     [[nodiscard]] const SafeTensorEntry* find_tensor(std::string_view name) ↵
    ↵ const;
30 };
31
32 struct SafeTensorFile {
33     std::filesystem::path path;
34     SafeTensorManifest manifest;
35     std::vector<std::uint8_t> bytes;
36 };
37
38 [[nodiscard]] SafeTensorManifest load_safetensors_manifest(
39     const std::filesystem::path& path);
40
41 [[nodiscard]] SafeTensorFile load_safetensors_file(const std::filesystem::path& ↵
    ↵ path);
42
43 [[nodiscard]] std::vector<float> read_safetensor_f32_tensor(
44     const SafeTensorFile& file,
45     std::string_view name);
46
47 [[nodiscard]] std::span<const std::uint8_t> read_safetensor_tensor_bytes(
48     const SafeTensorFile& file,
49     std::string_view name);
50
51 } // namespace lcqi

```

GGUF 与 Q4_K 合同

GGUF 头文件固定 llama.cpp 生态模型文件的读取边界：版本、metadata、alignment、tensor 名、shape、ggml type 和 tensor payload offset。它也保留 tokenizer 需要的字符串数组 metadata，例如 `tokenizer.ggml.tokens` 和 `tokenizer.ggml.merges`。它不负责模型图执行，只把“真实 GGUF 文件里有哪些 tensor、每个 tensor 在哪里、哪些 metadata 能支撑 tokenizer”变成可检查对象。

Listing 4.8 include/lcqi/gguf.hpp

```

1 #pragma once
2
3 #include <stdint>
4 #include <filesystem>
5 #include <span>
6 #include <string>
7 #include <string_view>
8 #include <vector>
9
10 namespace lcqi {
11
12     enum class GgmlType : std::int32_t {
13         f32 = 0,

```

```

14     f16 = 1,
15     q4_0 = 2,
16     q4_1 = 3,
17     q5_0 = 6,
18     q5_1 = 7,
19     q8_0 = 8,
20     q8_1 = 9,
21     q2_k = 10,
22     q3_k = 11,
23     q4_k = 12,
24     q5_k = 13,
25     q6_k = 14,
26     q8_k = 15,
27     i8 = 24,
28     i16 = 25,
29     i32 = 26,
30     i64 = 27,
31     f64 = 28,
32     bf16 = 30,
33     q1_0 = 41,
34 };
35
36 struct GgmlTypeLayout {
37     std::int64_t block_size = 1;
38     std::uint64_t type_size = 0;
39     const char* name = "";
40     bool quantized = false;
41 };
42
43 struct GgufMetadataEntry {
44     std::string key;
45     std::string type;
46     std::string value_preview;
47     std::vector<std::string> string_values;
48 };
49
50 struct GgufTensorInfo {
51     std::string name;
52     std::vector<std::int64_t> shape;
53     GgmlType type = GgmlType::f32;
54     std::uint64_t relative_offset = 0;
55     std::uint64_t absolute_offset = 0;
56     std::uint64_t byte_size = 0;
57
58     [[nodiscard]] std::uint64_t element_count() const;
59 };
60
61 struct GgufManifest {
62     std::uint32_t version = 0;
63     std::uint64_t tensor_count = 0;
64     std::uint64_t metadata_count = 0;
65     std::uint64_t alignment = 32;

```

```

66     std::uint64_t data_offset = 0;
67     std::vector<GgufMetadataEntry> metadata;
68     std::vector<GgufTensorInfo> tensors;
69
70     [[nodiscard]] const GgufTensorInfo* find_tensor(std::string_view name) ↵
    ↵ const;
71     [[nodiscard]] const GgufMetadataEntry* find_metadata(std::string_view key) ↵
    ↵ const;
72 };
73
74 [[nodiscard]] GgmlTypeLayout ggml_type_layout(GgmlType type);
75
76 [[nodiscard]] const char* ggml_type_name(GgmlType type);
77
78 [[nodiscard]] std::uint64_t ggml_tensor_byte_size(GgmlType type,
79                                                    std::span<const std::int64_t>↵
    ↵ shape);
80
81 [[nodiscard]] GgufManifest load_gguf_manifest(const std::filesystem::path& path↵
    ↵ );
82
83 [[nodiscard]] std::vector<std::uint8_t> read_gguf_tensor_bytes(
84     const std::filesystem::path& path,
85     const GgufTensorInfo& tensor);
86
87 } // namespace lcqi

```

GGML tensor 头文件是 GGUF 与模型数学之间的格式层。SmolLM2 的 `Q4_K_M` 文件内部并不是所有矩阵都是 `Q4_K`：远端 `caw` 扫描显示它包含 `F32`、`Q8_0`、`Q5_0`、`Q6_K` 和 `Q4_K`。因此完整模型加载不能只写一个 `Q4_K` 解码器，必须把这些格式统一变成后续执行层能消费的张量。

Listing 4.9 `include/lcqi/ggml_tensors.hpp`

```

1  #pragma once
2
3  #include <lcqi/gguf.hpp>
4
5  #include <cstdint>
6  #include <span>
7  #include <vector>
8
9  namespace lcqi {
10
11  constexpr std::int64_t LCQI_Q8_0_BLOCK_VALUES = 32;
12  constexpr std::int64_t LCQI_Q8_0_BLOCK_BYTES = 34;
13  constexpr std::int64_t LCQI_Q5_0_BLOCK_VALUES = 32;
14  constexpr std::int64_t LCQI_Q5_0_BLOCK_BYTES = 22;
15  constexpr std::int64_t LCQI_Q6_K_BLOCK_VALUES = 256;
16  constexpr std::int64_t LCQI_Q6_K_BLOCK_BYTES = 210;
17
18  [[nodiscard]] float ggml_f16_to_f32(std::uint16_t bits);
19
20  [[nodiscard]] std::vector<float> dequantize_ggml_tensor(

```

```

21     GgmlType type,
22     std::span<const std::uint8_t> bytes,
23     std::int64_t element_count);
24
25 void dequantize_ggml_tensor_to(GgmlType type,
26                               std::span<const std::uint8_t> bytes,
27                               std::span<float> output);
28
29 } // namespace lcqi

```

Q4_K 头文件固定本轮量化切片的数学合同。一个 Q4_K block 解码出 256 个权重, Q8_K input 是每 256 个激活临时量化一次。这里暴露的 `matvec_q4_k_q8` 是单算子入口, 不是完整 LLaMA-like decoder; 完整 decoder 还要接 RMSNorm、RoPE、attention、SwiGLU、KV cache 和 sampler。

Listing 4.10 `include/lcqi/q4_k.hpp`

```

1 #pragma once
2
3 #include <array>
4 #include <cstdint>
5 #include <span>
6 #include <vector>
7
8 namespace lcqi {
9
10 constexpr std::int64_t LCQI_QK_K_BLOCK_VALUES = 256;
11 constexpr std::int64_t LCQI_Q4_K_BLOCK_BYTES = 144;
12 constexpr std::int64_t LCQI_Q8_K_BLOCK_BYTES = 292;
13 constexpr std::int64_t LCQI_Q4_K_SUBBLOCK_VALUES = 32;
14 constexpr std::int64_t LCQI_Q4_K_SUBBLOCKS = 8;
15 constexpr std::int64_t LCQI_Q8_K_BSUM_GROUPS = 16;
16
17 struct Q8KBlock {
18     float d = 0.0F;
19     std::array<std::int8_t, LCQI_QK_K_BLOCK_VALUES> qs{};
20     std::array<std::int16_t, LCQI_Q8_K_BSUM_GROUPS> bsums{};
21 };
22
23 [[nodiscard]] std::vector<float> dequantize_q4_k(std::span<const std::uint8_t> ↵
    ↵ bytes,
24                                                  std::int64_t element_count);
25
26 void dequantize_q4_k_to(std::span<const std::uint8_t> bytes,
27                        std::span<float> output);
28
29 [[nodiscard]] std::vector<Q8KBlock> quantize_q8_k_input(std::span<const float> ↵
    ↵ input);
30
31 [[nodiscard]] bool q4_k_q8_avx2_available() noexcept;
32
33 [[nodiscard]] const char* q4_k_q8_active_backend() noexcept;
34
35 float dot_q4_k_f32(std::span<const std::uint8_t> q4_blocks,

```

```

36         std::span<const float> input);
37
38 float dot_q4_k_q8(std::span<const std::uint8_t> q4_blocks,
39                 std::span<const Q8KBlock> input);
40
41 void matvec_q4_k_f32(std::span<const std::uint8_t> q4_rows,
42                     std::int64_t row_count,
43                     std::int64_t column_count,
44                     std::span<const float> input,
45                     std::span<float> output);
46
47 void matvec_q4_k_q8(std::span<const std::uint8_t> q4_rows,
48                     std::int64_t row_count,
49                     std::int64_t column_count,
50                     std::span<const Q8KBlock> input,
51                     std::span<float> output);
52
53 } // namespace lcqi

```

LLaMA GGUF 头文件把真实 GGUF 文件接到 reference decoder 数学层。当前实现明确叫 **f32_dequantized_reference**：加载时把量化权重解到 f32，先建立端到端 correctness baseline，再逐步把热点替换为量化直算 micro-kernel。这个边界很重要，因为它防止把“能跑真实模型”误写成“已经拥有 llama.cpp 同等级的低比特执行引擎”。

Listing 4.11 include/lcqi/llama_gguf.hpp

```

1 #pragma once
2
3 #include <lcqi/gpt2_reference.hpp>
4 #include <lcqi/reference_decoder.hpp>
5
6 #include <cstdint>
7 #include <filesystem>
8 #include <span>
9 #include <string>
10 #include <vector>
11
12 namespace lcqi {
13
14 struct LlamaGgufModel {
15     ReferenceDecoderModel decoder;
16     std::string architecture;
17     std::string name;
18     std::int32_t bos_token_id = -1;
19     std::int32_t eos_token_id = -1;
20     std::int32_t unknown_token_id = -1;
21     bool lm_head_tied_to_embedding = true;
22 };
23
24 struct LlamaGgufLoadReport {
25     double manifest_ms = 0.0;
26     double weights_ms = 0.0;
27     std::uint64_t quantized_weight_bytes = 0;

```

```

28     std::uint64_t f32_weight_bytes = 0;
29     std::int64_t tensors_loaded = 0;
30 };
31
32 struct LlamaGgufLoadedModel {
33     LlamaGgufModel model;
34     LlamaGgufLoadReport report;
35 };
36
37 struct LlamaGgufGenerationResult {
38     std::vector<std::int32_t> prompt_ids;
39     std::vector<std::int32_t> generated_ids;
40     double load_ms = 0.0;
41     double prefill_ms = 0.0;
42     double decode_ms = 0.0;
43     double total_ms = 0.0;
44     std::size_t kv_cache_bytes = 0;
45     std::size_t prefill_steps = 0;
46     std::size_t decode_steps = 0;
47     std::int32_t predicted_first_token = -1;
48 };
49
50 [[nodiscard]] LlamaGgufLoadedModel load_llama_gguf_reference_model(
51     const std::filesystem::path& gguf_path);
52
53 [[nodiscard]] Gpt2Tokenizer load_gpt2_tokenizer_from_gguf(
54     const std::filesystem::path& gguf_path);
55
56 [[nodiscard]] std::vector<std::int32_t> llama_gguf_chat_prompt_ids(
57     const Gpt2Tokenizer& tokenizer,
58     std::string_view user_prompt,
59     std::int32_t bos_token_id);
60
61 [[nodiscard]] LlamaGgufGenerationResult llama_gguf_generate_greedy(
62     const LlamaGgufModel& model,
63     std::span<const std::int32_t> prompt_ids,
64     std::int32_t max_new_tokens);
65
66 } // namespace lcqi

```

Reference Decoder 与 GPT-2 合同

reference decoder 头文件定义张量、层配置、trace checkpoint、采样结果和 decode 函数。它是定位推理错误的主要合同，不只返回最终 token。

Listing 4.12 include/lcqi/reference_decoder.hpp

```

1 #pragma once
2
3 #include <cstdint>
4 #include <span>
5 #include <string>

```



```

6 #include <vector>
7
8 namespace lcqi {
9
10 struct LinearWeightsF32 {
11     std::int32_t input_size = 0;
12     std::int32_t output_size = 0;
13     std::vector<float> weights;
14     std::vector<float> bias;
15 };
16
17 struct DecoderConfig {
18     std::int32_t hidden_size = 0;
19     std::int32_t query_heads = 0;
20     std::int32_t kv_heads = 0;
21     std::int32_t head_dim = 0;
22     std::int32_t intermediate_size = 0;
23     std::int32_t vocab_size = 0;
24     std::int32_t max_context = 0;
25     float rms_epsilon = 1.0e-5F;
26     float rope_base = 10000.0F;
27 };
28
29 struct DecoderLayerWeightsF32 {
30     std::vector<float> rms_attention_weight;
31     LinearWeightsF32 wq;
32     LinearWeightsF32 wk;
33     LinearWeightsF32 wv;
34     LinearWeightsF32 wo;
35     std::vector<float> rms_mlp_weight;
36     LinearWeightsF32 w_gate;
37     LinearWeightsF32 w_up;
38     LinearWeightsF32 w_down;
39 };
40
41 struct ReferenceDecoderModel {
42     DecoderConfig config;
43     std::vector<float> token_embedding;
44     std::vector<DecoderLayerWeightsF32> layers;
45     std::vector<float> final_norm_weight;
46     LinearWeightsF32 lm_head;
47 };
48
49 struct ReferenceDecodeResult {
50     std::vector<float> logits;
51     std::int32_t predicted_token = 0;
52 };
53
54 struct ReferenceTensorCheckpoint {
55     std::string name;
56     std::int32_t token_position = 0;
57     std::int32_t layer_id = 0;

```

```

58     std::vector<std::int32_t> shape;
59     std::vector<std::int32_t> stride;
60     std::string dtype;
61     std::string layout;
62     float checksum = 0.0F;
63     float max_abs = 0.0F;
64     std::vector<float> values;
65 };
66
67 struct ReferenceDecodeTraceResult {
68     ReferenceDecodeResult result;
69     std::vector<ReferenceTensorCheckpoint> checkpoints;
70 };
71
72 class ReferenceKVCache {
73 public:
74     ReferenceKVCache(const DecoderConfig& config, std::int32_t layer_count);
75
76     void append(std::int32_t layer_id,
77                std::int32_t model_position,
78                std::span<const float> key,
79                std::span<const float> value);
80
81     [[nodiscard]] std::span<const float> key(std::int32_t layer_id,
82                                              std::int32_t model_position,
83                                              std::int32_t kv_head) const;
84
85     [[nodiscard]] std::span<const float> value(std::int32_t layer_id,
86                                              std::int32_t model_position,
87                                              std::int32_t kv_head) const;
88
89     [[nodiscard]] std::int32_t layer_count() const noexcept;
90     [[nodiscard]] std::int32_t filled_tokens() const noexcept;
91
92 private:
93     [[nodiscard]] std::size_t base_offset(std::int32_t layer_id,
94                                           std::int32_t model_position,
95                                           std::int32_t kv_head) const;
96     void validate_address(std::int32_t layer_id,
97                          std::int32_t model_position,
98                          std::int32_t kv_head) const;
99
100     DecoderConfig config_;
101     std::int32_t layer_count_ = 0;
102     std::int32_t filled_tokens_ = 0;
103     std::vector<float> keys_;
104     std::vector<float> values_;
105     std::vector<std::uint8_t> written_;
106 };
107
108 void rms_norm(std::span<const float> input,
109              std::span<const float> weight,

```

```

110         float epsilon,
111         std::span<float> output);
112
113 void linear_f32(const LinearWeightsF32& layer,
114               std::span<const float> input,
115               std::span<float> output);
116
117 void apply_rope(std::span<float> heads,
118               std::int32_t head_count,
119               std::int32_t head_dim,
120               std::int32_t model_position,
121               float rope_base);
122
123 void attention_decode(const DecoderConfig& config,
124                     const ReferenceKVCache& cache,
125                     std::int32_t layer_id,
126                     std::int32_t model_position,
127                     std::span<const float> query,
128                     std::span<float> output);
129
130 ReferenceDecodeResult run_reference_decode(const ReferenceDecoderModel& model,
131                                         std::span<const std::int32_t> ↵
132                                         ↵ token_ids);
133
134 ReferenceDecodeTraceResult run_reference_decode_with_trace(
135     const ReferenceDecoderModel& model,
136     std::span<const std::int32_t> token_ids);
137
138 ReferenceDecoderModel make_tiny_reference_decoder_model();
139 } // namespace lcqi

```

GPT-2 reference 头文件定义 GPT-2 配置、byte-level BPE tokenizer、HuggingFace safetensors 加载、forward、cached forward 和 greedy generation。它单独存在，是因为 GPT-2 使用 LayerNorm、绝对位置嵌入、packed QKV、GELU 和 tied lm head，不能直接塞进 LLaMA-like reference decoder。读 `Gpt2CachedGreedyDecoder` 时要区分三个入口：`step()` 推进一个 token 并返回 greedy 预测；`step_with_logits()` 保留完整 logits 给测试和对照；`advance_without_prediction()` 只推进 embedding、Transformer layers 和 KV cache，用在 prompt prefill 的前缀 token 上。第三个入口不是“少算一点但可能改变结果”，而是删掉不会被任何 caller 使用的预测结果。

Listing 4.13 include/lcqi/gpt2_reference.hpp

```

1 #pragma once
2
3 #include <stdint>
4 #include <filesystem>
5 #include <memory>
6 #include <span>
7 #include <string>
8 #include <string_view>
9 #include <unordered_map>
10 #include <vector>
11

```

```

12 namespace lcqi {
13
14 struct Gpt2Config {
15     std::int32_t hidden_size = 0;
16     std::int32_t head_count = 0;
17     std::int32_t layer_count = 0;
18     std::int32_t max_positions = 0;
19     std::int32_t vocab_size = 0;
20     std::int32_t intermediate_size = 0;
21     std::int32_t bos_token_id = -1;
22     std::int32_t eos_token_id = -1;
23     float layer_norm_epsilon = 1.0e-5F;
24     std::string activation_function = "gelu_new";
25 };
26
27 struct Gpt2LinearF32 {
28     std::int32_t input_size = 0;
29     std::int32_t output_size = 0;
30     std::vector<float> weights;
31     std::vector<float> bias;
32 };
33
34 struct Gpt2LayerWeightsF32 {
35     std::vector<float> ln_1_weight;
36     std::vector<float> ln_1_bias;
37     Gpt2LinearF32 c_attn;
38     Gpt2LinearF32 c_proj;
39     std::vector<float> ln_2_weight;
40     std::vector<float> ln_2_bias;
41     Gpt2LinearF32 c_fc;
42     Gpt2LinearF32 mlp_c_proj;
43 };
44
45 struct Gpt2ReferenceModel {
46     Gpt2Config config;
47     std::vector<float> token_embedding;
48     std::vector<float> position_embedding;
49     std::vector<Gpt2LayerWeightsF32> layers;
50     std::vector<float> final_ln_weight;
51     std::vector<float> final_ln_bias;
52     std::vector<float> lm_head_weight;
53     bool tie_lm_head_to_embedding = true;
54 };
55
56 struct Gpt2ForwardResult {
57     std::vector<float> logits;
58     std::int32_t predicted_token = 0;
59 };
60
61 struct Gpt2HotspotProfile {
62     std::int64_t decoder_steps = 0;
63     std::int64_t layer_steps = 0;

```

```

64     double total_step_ms = 0.0;
65     double embedding_ms = 0.0;
66     double layer_norm_ms = 0.0;
67     double final_norm_ms = 0.0;
68     double qkv_projection_ms = 0.0;
69     double kv_append_ms = 0.0;
70     double attention_ms = 0.0;
71     double attention_projection_ms = 0.0;
72     double residual_add_ms = 0.0;
73     double mlp_fc_ms = 0.0;
74     double gelu_ms = 0.0;
75     double mlp_projection_ms = 0.0;
76     double lm_head_ms = 0.0;
77     double logits_result_ms = 0.0;
78 };
79
80 struct Gpt2ExecutionOptions {
81     std::int32_t worker_count = 0;
82     std::int32_t parallel_min_rows = 512;
83 };
84
85 class Gpt2KvCache;
86
87 namespace detail {
88 class Gpt2ParallelWorkerPool;
89
90 void attend_cached_position(const Gpt2KvCache& cache,
91                             std::int32_t layer_id,
92                             std::int32_t model_position,
93                             std::span<const float> query,
94                             std::span<float> scores,
95                             std::span<float> output);
96 } // namespace detail
97
98 class Gpt2ForwardWorkspace {
99 public:
100     explicit Gpt2ForwardWorkspace(const Gpt2Config& config);
101
102     void reset_for_config(const Gpt2Config& config);
103
104     [[nodiscard]] std::span<float> hidden() noexcept;
105     [[nodiscard]] std::span<float> normed() noexcept;
106     [[nodiscard]] std::span<float> query() noexcept;
107     [[nodiscard]] std::span<float> key() noexcept;
108     [[nodiscard]] std::span<float> value() noexcept;
109     [[nodiscard]] std::span<float> attention() noexcept;
110     [[nodiscard]] std::span<float> projected() noexcept;
111     [[nodiscard]] std::span<float> qkv_packed() noexcept;
112     [[nodiscard]] std::span<float> mlp_fc() noexcept;
113     [[nodiscard]] std::span<float> mlp_out() noexcept;
114     [[nodiscard]] std::span<float> logits() noexcept;
115     [[nodiscard]] std::span<float> scores_prefix(std::int32_t count);

```

```

116
117 private:
118     Gpt2Config config_;
119     std::vector<float> hidden_;
120     std::vector<float> normed_;
121     std::vector<float> query_;
122     std::vector<float> key_;
123     std::vector<float> value_;
124     std::vector<float> attention_;
125     std::vector<float> projected_;
126     std::vector<float> qkv_packed_;
127     std::vector<float> mlp_fc_;
128     std::vector<float> mlp_out_;
129     std::vector<float> logits_;
130     std::vector<float> scores_;
131 };
132
133 class Gpt2KvCache {
134 public:
135     explicit Gpt2KvCache(const Gpt2Config& config);
136
137     void append(std::int32_t layer_id,
138                std::int32_t model_position,
139                std::span<const float> key,
140                std::span<const float> value);
141
142     [[nodiscard]] std::span<const float> key(std::int32_t layer_id,
143                                              std::int32_t model_position,
144                                              std::int32_t head) const;
145
146     [[nodiscard]] std::span<const float> value(std::int32_t layer_id,
147                                                std::int32_t model_position,
148                                                std::int32_t head) const;
149
150     [[nodiscard]] std::int32_t filled_tokens() const noexcept;
151     [[nodiscard]] std::size_t byte_size() const noexcept;
152
153 private:
154     friend void detail::attend_cached_position(const Gpt2KvCache& cache,
155                                               std::int32_t layer_id,
156                                               std::int32_t model_position,
157                                               std::span<const float> query,
158                                               std::span<float> scores,
159                                               std::span<float> output);
160
161     [[nodiscard]] std::size_t base_offset(std::int32_t layer_id,
162                                           std::int32_t model_position,
163                                           std::int32_t head) const;
164     [[nodiscard]] std::size_t base_offset_unchecked(std::int32_t layer_id,
165                                                     std::int32_t model_position↵
166
167     ↵ ,
168
169     std::int32_t head) const ↵

```

```

167     ↪ noexcept;
168     [[nodiscard]] std::span<const float> key_unchecked(std::int32_t layer_id,
169                                                         std::int32_t ↪
170                                                         ↪ model_position,
171                                                         std::int32_t head) const ↪
172     ↪ noexcept;
173     [[nodiscard]] std::span<const float> value_unchecked(std::int32_t layer_id,
174                                                         std::int32_t ↪
175                                                         ↪ model_position,
176                                                         std::int32_t head) ↪
177     ↪ const noexcept;
178     void validate_address(std::int32_t layer_id,
179                          std::int32_t model_position,
180                          std::int32_t head) const;
181     void validate_written(std::int32_t layer_id,
182                          std::int32_t model_position) const;
183
184     Gpt2Config config_;
185     std::int32_t filled_tokens_ = 0;
186     std::vector<float> keys_;
187     std::vector<float> values_;
188     std::vector<std::uint8_t> written_;
189 };
190
191 class Gpt2CachedGreedyDecoder {
192 public:
193     explicit Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& model);
194     Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& model,
195                             Gpt2HotspotProfile* hotspot_profile);
196     Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& model,
197                             Gpt2HotspotProfile* hotspot_profile,
198                             const Gpt2ExecutionOptions& execution_options);
199     ~Gpt2CachedGreedyDecoder();
200
201     Gpt2CachedGreedyDecoder(const Gpt2CachedGreedyDecoder&) = delete;
202     Gpt2CachedGreedyDecoder& operator=(const Gpt2CachedGreedyDecoder&) = delete ↪
203     ↪ ;
204     Gpt2CachedGreedyDecoder(Gpt2CachedGreedyDecoder&&) noexcept;
205     Gpt2CachedGreedyDecoder& operator=(Gpt2CachedGreedyDecoder&&) noexcept;
206
207     void advance_without_prediction(std::int32_t token_id);
208     [[nodiscard]] std::int32_t step(std::int32_t token_id);
209     [[nodiscard]] Gpt2ForwardResult step_with_logits(std::int32_t token_id);
210     [[nodiscard]] const Gpt2KvCache& cache() const noexcept;
211     [[nodiscard]] std::int32_t filled_tokens() const noexcept;
212     [[nodiscard]] std::size_t kv_cache_bytes() const noexcept;
213     [[nodiscard]] std::int32_t worker_count() const noexcept;
214
215 private:
216     const Gpt2ReferenceModel* model_;
217     Gpt2HotspotProfile* hotspot_profile_;
218     Gpt2ExecutionOptions execution_options_;

```



```

213     Gpt2KvCache cache_;
214     Gpt2ForwardWorkspace workspace_;
215     std::unique_ptr<detail::Gpt2ParallelWorkerPool> worker_pool_;
216 };
217
218 struct Gpt2Tokenizer {
219     std::int32_t bos_token_id = -1;
220     std::int32_t eos_token_id = -1;
221     std::unordered_map<std::string, std::int32_t> vocab;
222     std::vector<std::string> id_to_token;
223     std::unordered_map<std::string, std::int32_t> bpe_ranks;
224 };
225
226 [[nodiscard]] Gpt2Tokenizer load_gpt2_tokenizer(
227     const std::filesystem::path& vocab_json,
228     const std::filesystem::path& merges_txt,
229     std::int32_t bos_token_id,
230     std::int32_t eos_token_id);
231
232 [[nodiscard]] std::vector<std::int32_t> gpt2_encode(
233     const Gpt2Tokenizer& tokenizer,
234     std::string_view text);
235
236 [[nodiscard]] std::string gpt2_decode(
237     const Gpt2Tokenizer& tokenizer,
238     std::span<const std::int32_t> token_ids);
239
240 [[nodiscard]] Gpt2Config load_gpt2_config(const std::filesystem::path& ↵
    ↵ config_json);
241
242 [[nodiscard]] Gpt2ReferenceModel load_gpt2_reference_model(
243     const std::filesystem::path& config_json,
244     const std::filesystem::path& safetensors_path);
245
246 [[nodiscard]] Gpt2ReferenceModel load_gpt2_from_directory(
247     const std::filesystem::path& model_directory);
248
249 [[nodiscard]] Gpt2ForwardResult run_gpt2_forward(
250     const Gpt2ReferenceModel& model,
251     std::span<const std::int32_t> token_ids);
252
253 [[nodiscard]] Gpt2ForwardResult run_gpt2_forward_cached(
254     const Gpt2ReferenceModel& model,
255     Gpt2KvCache& cache,
256     std::int32_t token_id);
257
258 [[nodiscard]] std::vector<std::int32_t> gpt2_generate_greedy(
259     const Gpt2ReferenceModel& model,
260     std::span<const std::int32_t> prompt_token_ids,
261     std::int32_t max_new_tokens);
262
263 [[nodiscard]] std::vector<std::int32_t> gpt2_generate_greedy_cached(

```

```

264     const Gpt2ReferenceModel& model,
265     std::span<const std::int32_t> prompt_token_ids,
266     std::int32_t max_new_tokens);
267
268 [[nodiscard]] Gpt2ReferenceModel make_tiny_gpt2_reference_model();
269
270 } // namespace lcqi

```

最短闭环：从模型文件到 logits

tiny MLP 路径不是为了证明 LCQI 已经是完整大模型引擎，而是为了建立最短的、可手算的、可测试的推理闭环。读者要把模型加载、输入检查、int8 linear、ReLU、第二层 logits、argmax 和 CLI 输出连起来看。

模型加载与基础推理

`model.cpp` 负责解析 tiny 文本模型、权重、scale、bias 和 shape。坏格式、坏维度和坏数值都应该在这里被拒绝，不能拖到 kernel 内部才崩溃。

Listing 4.14 src/model.cpp

```

1  #include <lcqi/model.hpp>
2
3  #include <fstream>
4  #include <limits>
5  #include <stdexcept>
6  #include <string>
7
8  namespace lcqi {
9      namespace {
10
11         constexpr std::int32_t LCQI_INT8_MIN_VALUE = -128;
12         constexpr std::int32_t LCQI_INT8_MAX_VALUE = 127;
13
14         std::string read_key(std::istream& input) {
15             std::string key;
16             if (!(input >> key)) {
17                 throw std::runtime_error("unexpected end of model file");
18             }
19             return key;
20         }
21
22         void expect_key(std::istream& input, const std::string& expected) {
23             const std::string key = read_key(input);
24             if (key != expected) {
25                 throw std::runtime_error("expected key '" + expected + "', got '" + key +
26                 " + "'");
27             }
28         }
29
30         std::int32_t read_i32(std::istream& input, const std::string& key) {
31             expect_key(input, key);
32             std::int32_t value = 0;

```

```

32     if (!(input >> value)) {
33         throw std::runtime_error("invalid int32 value for key '" + key + "'");
34     }
35     return value;
36 }
37
38 float read_float(std::istream& input, const std::string& key) {
39     expect_key(input, key);
40     float value = 0.0F;
41     if (!(input >> value)) {
42         throw std::runtime_error("invalid float value for key '" + key + "'");
43     }
44     return value;
45 }
46
47 std::vector<std::int8_t> read_i8_vector(std::istream& input,
48                                         const std::string& key,
49                                         std::int32_t count) {
50     expect_key(input, key);
51     if (count < 0) {
52         throw std::runtime_error("negative vector size for key '" + key + "'");
53     }
54     std::vector<std::int8_t> values(static_cast<std::size_t>(count));
55     for (std::int32_t i = 0; i < count; ++i) {
56         std::int32_t raw = 0;
57         if (!(input >> raw)) {
58             throw std::runtime_error("invalid int8 payload for key '" + key + "↵
↵ '");
59         }
60         if (raw < LCQI_INT8_MIN_VALUE || raw > LCQI_INT8_MAX_VALUE) {
61             throw std::runtime_error("int8 payload out of range for key '" + ↵
↵ key + "'");
62         }
63         values[static_cast<std::size_t>(i)] = static_cast<std::int8_t>(raw);
64     }
65     return values;
66 }
67
68 std::vector<float> read_float_vector(std::istream& input,
69                                       const std::string& key,
70                                       std::int32_t count) {
71     expect_key(input, key);
72     if (count < 0) {
73         throw std::runtime_error("negative vector size for key '" + key + "'");
74     }
75     std::vector<float> values(static_cast<std::size_t>(count));
76     for (std::int32_t i = 0; i < count; ++i) {
77         if (!(input >> values[static_cast<std::size_t>(i)])) {
78             throw std::runtime_error("invalid float payload for key '" + key + ↵
↵ '");
79         }
80     }

```

```

81     return values;
82 }
83
84 void validate_layer(const QuantizedLinearLayer& layer, const std::string& name)↵
    ↵ {
85     if (layer.input_size <= 0 || layer.output_size <= 0) {
86         throw std::runtime_error(name + " layer has invalid shape");
87     }
88     const std::size_t expected_weights =
89         static_cast<std::size_t>(layer.input_size) *
90         static_cast<std::size_t>(layer.output_size);
91     if (layer.weights.size() != expected_weights) {
92         throw std::runtime_error(name + " layer weight size mismatch");
93     }
94     if (layer.bias.size() != static_cast<std::size_t>(layer.output_size)) {
95         throw std::runtime_error(name + " layer bias size mismatch");
96     }
97 }
98
99 std::int32_t checked_element_count(std::int32_t rows,
100                                   std::int32_t columns,
101                                   const std::string& name) {
102     if (rows <= 0 || columns <= 0) {
103         throw std::runtime_error(name + " layer has invalid shape");
104     }
105     const std::int64_t count =
106         static_cast<std::int64_t>(rows) * static_cast<std::int64_t>(columns);
107     if (count > static_cast<std::int64_t>(std::numeric_limits<std::int32_t>::↵
    ↵ max())) {
108         throw std::runtime_error(name + " layer element count is too large");
109     }
110     return static_cast<std::int32_t>(count);
111 }
112
113 } // namespace
114
115 TinyMlpModel load_model(const std::filesystem::path& path) {
116     std::ifstream input(path);
117     if (!input) {
118         throw std::runtime_error("cannot open model file: " + path.string());
119     }
120
121     expect_key(input, "LCQI_MODEL_V1");
122
123     TinyMlpModel model;
124     model.input_size = read_i32(input, "input_size");
125     model.hidden_size = read_i32(input, "hidden_size");
126     model.output_size = read_i32(input, "output_size");
127
128     const std::int32_t hidden_weight_count =
129         checked_element_count(model.input_size, model.hidden_size, "hidden");
130     const std::int32_t output_weight_count =

```

```

131     checked_element_count(model.hidden_size, model.output_size, "output");
132
133     model.hidden.input_size = model.input_size;
134     model.hidden.output_size = model.hidden_size;
135     model.hidden.scale = read_float(input, "hidden_scale");
136     model.hidden.weights = read_i8_vector(
137         input,
138         "hidden_weights",
139         hidden_weight_count);
140     model.hidden.bias = read_float_vector(input, "hidden_bias", model.↵
↵ hidden_size);
141
142     model.output.input_size = model.hidden_size;
143     model.output.output_size = model.output_size;
144     model.output.scale = read_float(input, "output_scale");
145     model.output.weights = read_i8_vector(
146         input,
147         "output_weights",
148         output_weight_count);
149     model.output.bias = read_float_vector(input, "output_bias", model.↵
↵ output_size);
150
151     validate_layer(model.hidden, "hidden");
152     validate_layer(model.output, "output");
153     return model;
154 }
155
156 } // namespace lcqi

```

`inference.cpp` 执行 int8 linear、ReLU、第二层 logits 和 predicted class。它是系统中最短的“模型资产到输出”路径。

Listing 4.15 src/inference.cpp

```

1 #include <lcqi/inference.hpp>
2
3 #include <algorithm>
4 #include <cmath>
5 #include <stdexcept>
6
7 namespace lcqi {
8 namespace {
9
10 void validate_input_size(std::span<const float> input, std::int32_t expected) {
11     if (input.size() != static_cast<std::size_t>(expected)) {
12         throw std::runtime_error("input size does not match model shape");
13     }
14 }
15
16 std::int32_t argmax(std::span<const float> values) {
17     if (values.empty()) {
18         throw std::runtime_error("cannot take argmax of empty logits");
19     }

```

```

20     std::int32_t best_index = 0;
21     float best_value = values[0];
22     for (std::int32_t i = 1; i < static_cast<std::int32_t>(values.size()); ++i) ↵
    ↵ {
23         const float candidate = values[static_cast<std::size_t>(i)];
24         if (candidate > best_value) {
25             best_value = candidate;
26             best_index = i;
27         }
28     }
29     return best_index;
30 }
31
32 } // namespace
33
34 void linear_i8(const QuantizedLinearLayer& layer,
35               std::span<const float> input,
36               std::span<float> output) {
37     if (input.size() != static_cast<std::size_t>(layer.input_size)) {
38         throw std::runtime_error("linear_i8 input size mismatch");
39     }
40     if (output.size() != static_cast<std::size_t>(layer.output_size)) {
41         throw std::runtime_error("linear_i8 output size mismatch");
42     }
43
44     for (std::int32_t out = 0; out < layer.output_size; ++out) {
45         const std::size_t row_base =
46             static_cast<std::size_t>(out) * static_cast<std::size_t>(layer. ↵
    ↵ input_size);
47         float acc = layer.bias[static_cast<std::size_t>(out)];
48         for (std::int32_t in = 0; in < layer.input_size; ++in) {
49             const std::int8_t weight =
50                 layer.weights[row_base + static_cast<std::size_t>(in)];
51             acc += static_cast<float>(weight) * layer.scale *
52                 input[static_cast<std::size_t>(in)];
53         }
54         output[static_cast<std::size_t>(out)] = acc;
55     }
56 }
57
58 void relu_inplace(std::span<float> values) {
59     for (float& value : values) {
60         value = std::max(0.0F, value);
61     }
62 }
63
64 InferenceResult run_inference(const TinyMlpModel& model,
65                               std::span<const float> input) {
66     validate_input_size(input, model.input_size);
67
68     std::vector<float> hidden(static_cast<std::size_t>(model.hidden_size), 0.0F ↵
    ↵ );

```

```

69     std::vector<float> logits(static_cast<std::size_t>(model.output_size), 0.0F ←
    ↪ );
70
71     linear_i8(model.hidden, input, hidden);
72     relu_inplace(hidden);
73     linear_i8(model.output, hidden, logits);
74
75     InferenceResult result;
76     result.predicted_class = argmax(logits);
77     result.logits = std::move(logits);
78     return result;
79 }
80
81 } // namespace lcqi

```

`main.cpp` 把模型路径和输入接到 `run_inference`，并打印结果。CLI 应保持薄层，不复制模型解析和推理逻辑。

Listing 4.16 `src/main.cpp`

```

1  #include <lcqi/inference.hpp>
2  #include <lcqi/model.hpp>
3
4  #include <chrono>
5  #include <cstdint>
6  #include <cstdlib>
7  #include <filesystem>
8  #include <iostream>
9  #include <span>
10 #include <stdexcept>
11 #include <string>
12 #include <vector>
13
14 namespace {
15
16 constexpr std::int32_t LCQI_DEFAULT_REPEAT = 1000;
17 constexpr int LCQI_EXPECTED_ARGC_MIN = 2;
18 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
19
20 std::vector<float> parse_input(int argc, char** argv, std::int32_t ←
    ↪ expected_size) {
21     if (argc < LCQI_EXPECTED_ARGC_MIN + expected_size) {
22         throw std::runtime_error("usage: lcqi_cli <model.txt> <input floats...> ←
    ↪ [repeat]");
23     }
24     std::vector<float> input(static_cast<std::size_t>(expected_size));
25     for (std::int32_t i = 0; i < expected_size; ++i) {
26         input[static_cast<std::size_t>(i)] =
27             std::stof(argv[LCQI_EXPECTED_ARGC_MIN + i]);
28     }
29     return input;
30 }
31

```



```

32 std::int32_t parse_repeat(int argc, char** argv, std::int32_t input_size) {
33     const int repeat_index = LCQI_EXPECTED_ARGC_MIN + input_size;
34     if (argc <= repeat_index) {
35         return LCQI_DEFAULT_REPEAT;
36     }
37     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
↵ repeat_index]));
38     if (repeat <= 0) {
39         throw std::runtime_error("repeat must be positive");
40     }
41     return repeat;
42 }
43
44 } // namespace
45
46 int main(int argc, char** argv) {
47     try {
48         if (argc < LCQI_EXPECTED_ARGC_MIN) {
49             throw std::runtime_error("usage: lcqi_cli <model.txt> <input floats↵
↵ ...> [repeat]");
50         }
51
52         const std::filesystem::path model_path = argv[1];
53         const lcqi::TinyMlpModel model = lcqi::load_model(model_path);
54         const std::vector<float> input = parse_input(argc, argv, model.↵
↵ input_size);
55         const std::int32_t repeat = parse_repeat(argc, argv, model.input_size);
56
57         const lcqi::InferenceResult result = lcqi::run_inference(model, input);
58         std::cout << "predicted_class " << result.predicted_class << "\n";
59         std::cout << "logits";
60         for (const float value : result.logits) {
61             std::cout << ' ' << value;
62         }
63         std::cout << "\n";
64
65         const auto begin = std::chrono::steady_clock::now();
66         std::int32_t checksum = 0;
67         for (std::int32_t i = 0; i < repeat; ++i) {
68             checksum += lcqi::run_inference(model, input).predicted_class;
69         }
70         const auto end = std::chrono::steady_clock::now();
71         const std::chrono::duration<double> elapsed = end - begin;
72         const double average_us =
73             elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double↵
↵ >(repeat);
74         std::cout << "repeat " << repeat << "\n";
75         std::cout << "average_us " << average_us << "\n";
76         std::cout << "checksum " << checksum << "\n";
77         return EXIT_SUCCESS;
78     } catch (const std::exception& error) {
79         std::cerr << "error: " << error.what() << "\n";

```

```

80         return EXIT_FAILURE;
81     }
82 }

```

Int8 Kernel 基础实现

`int8_kernels.cpp` 包含 scalar、packed、workspace、deterministic fixture 和误差比较。它是 AVX2 路径的 reference 对照，也是 benchmark 的正确性底座。

Listing 4.17 src/int8_kernels.cpp

```

1  #include <lcqi/int8_kernels.hpp>
2
3  #include <lcqi/inference.hpp>
4
5  #include <algorithm>
6  #include <chrono>
7  #include <cmath>
8  #include <stdexcept>
9  #include <string>
10
11 namespace lcqi {
12 namespace {
13
14 constexpr std::int32_t LCQI_I8_PATTERN_MODULUS = 23;
15 constexpr std::int32_t LCQI_I8_PATTERN_CENTER = 11;
16 constexpr std::int32_t LCQI_INPUT_PATTERN_MODULUS = 17;
17 constexpr std::int32_t LCQI_INPUT_PATTERN_CENTER = 8;
18 constexpr float LCQI_INPUT_PATTERN_SCALE = 0.125F;
19 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
20 constexpr std::int32_t LCQI_SIMD_OUTPUT_BLOCK = 8;
21 constexpr std::size_t LCQI_BENCHMARK_BACKEND_COUNT = 3;
22 constexpr const char* LCQI_BACKEND_SCALAR = "scalar";
23 constexpr const char* LCQI_BACKEND_PACKED_SCALAR = "packed_scalar";
24 constexpr const char* LCQI_BACKEND_AVX2 = "avx2";
25
26 void require_positive(std::int32_t value, const char* name) {
27     if (value <= 0) {
28         throw std::runtime_error(std::string(name) + " must be positive");
29     }
30 }
31
32 std::size_t checked_size(std::int32_t value, const char* name) {
33     if (value < 0) {
34         throw std::runtime_error(std::string(name) + " cannot be negative");
35     }
36     return static_cast<std::size_t>(value);
37 }
38
39 void validate_quantized_layer(const QuantizedLinearLayer& layer) {
40     require_positive(layer.input_size, "input_size");
41     require_positive(layer.output_size, "output_size");

```

```

42     const std::size_t expected_weights =
43         checked_size(layer.input_size, "input_size") *
44         checked_size(layer.output_size, "output_size");
45     if (layer.weights.size() != expected_weights) {
46         throw std::runtime_error("quantized layer weight size mismatch");
47     }
48     if (layer.bias.size() != checked_size(layer.output_size, "output_size")) {
49         throw std::runtime_error("quantized layer bias size mismatch");
50     }
51 }
52
53 std::int32_t round_up_to_block(std::int32_t value, std::int32_t block_size) {
54     require_positive(value, "value");
55     require_positive(block_size, "block_size");
56     return ((value + block_size - 1) / block_size) * block_size;
57 }
58
59 float checksum(std::span<const float> values) {
60     float sum = 0.0F;
61     for (const float value : values) {
62         sum += value;
63     }
64     return sum;
65 }
66
67 void add_unavailable_backend(KernelBenchmarkResult& result, const char* name) {
68     result.backends.push_back(KernelBackendBenchmark{
69         .name = name,
70         .available = false,
71         .average_us = 0.0,
72         .max_abs_diff = 0.0F,
73         .checksum = 0.0F,
74     });
75 }
76
77 template <typename Callable>
78 double measure_average_us(std::int32_t repeat, Callable&& callable) {
79     require_positive(repeat, "repeat");
80     const auto begin = std::chrono::steady_clock::now();
81     for (std::int32_t index = 0; index < repeat; ++index) {
82         callable();
83     }
84     const auto end = std::chrono::steady_clock::now();
85     const std::chrono::duration<double> elapsed = end - begin;
86     return elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double>
87     ↵ >(repeat);
88 }
89 } // namespace
90
91 PackedLinearI8 pack_linear_i8(const QuantizedLinearLayer& layer,
92     std::int32_t output_block_size) {

```

```

93 validate_quantized_layer(layer);
94 require_positive(output_block_size, "output_block_size");
95
96 PackedLinearI8 packed;
97 packed.input_size = layer.input_size;
98 packed.output_size = layer.output_size;
99 packed.output_block_size = output_block_size;
100 packed.scale = layer.scale;
101 packed.bias = layer.bias;
102
103 const std::int32_t padded_outputs =
104     round_up_to_block(layer.output_size, output_block_size);
105 packed.packed_weights.assign(
106     checked_size(padded_outputs, "padded_outputs") *
107     checked_size(layer.input_size, "input_size"),
108     0);
109
110 std::size_t packed_index = 0;
111 for (std::int32_t output_block = 0; output_block < padded_outputs;
112     output_block += output_block_size) {
113     for (std::int32_t input_index = 0; input_index < layer.input_size; ++input_index) {
114         for (std::int32_t offset = 0; offset < output_block_size; ++offset) {
115             const std::int32_t output_index = output_block + offset;
116             if (output_index < layer.output_size) {
117                 const std::size_t source =
118                     checked_size(output_index, "output_index") *
119                     checked_size(layer.input_size, "input_size") +
120                     checked_size(input_index, "input_index");
121                 packed.packed_weights[packed_index] = layer.weights[source];
122             }
123             ++packed_index;
124         }
125     }
126 }
127 return packed;
128 }
129
130 void linear_i8_packed(const PackedLinearI8& layer,
131     std::span<const float> input,
132     std::span<float> output) {
133     std::vector<float> accumulators(
134         checked_size(layer.output_block_size, "output_block_size"),
135         0.0F);
136     linear_i8_packed_with_workspace(layer, input, output, accumulators);
137 }
138
139 void linear_i8_packed_with_workspace(const PackedLinearI8& layer,
140     std::span<const float> input,
141     std::span<float> output,

```

```

142         std::span<float> accumulator_workspace) {
143     require_positive(layer.input_size, "packed input_size");
144     require_positive(layer.output_size, "packed output_size");
145     require_positive(layer.output_block_size, "packed output_block_size");
146     if (input.size() != checked_size(layer.input_size, "input_size")) {
147         throw std::runtime_error("linear_i8_packed input size mismatch");
148     }
149     if (output.size() != checked_size(layer.output_size, "output_size")) {
150         throw std::runtime_error("linear_i8_packed output size mismatch");
151     }
152     if (layer.bias.size() != checked_size(layer.output_size, "output_size")) {
153         throw std::runtime_error("linear_i8_packed bias size mismatch");
154     }
155
156     const std::int32_t padded_outputs =
157         round_up_to_block(layer.output_size, layer.output_block_size);
158     const std::size_t expected_packed_size =
159         checked_size(padded_outputs, "padded_outputs") *
160         checked_size(layer.input_size, "input_size");
161     if (layer.packed_weights.size() != expected_packed_size) {
162         throw std::runtime_error("linear_i8_packed packed weight size mismatch"
163     ↪ );
164     }
165     if (accumulator_workspace.size() != checked_size(layer.output_block_size, "
166     ↪ output_block_size")) {
167         throw std::runtime_error("linear_i8_packed accumulator workspace size
168     ↪ mismatch");
169     }
170
171     std::size_t packed_index = 0;
172     for (std::int32_t output_block = 0; output_block < padded_outputs;
173         output_block += layer.output_block_size) {
174         std::fill(accumulator_workspace.begin(), accumulator_workspace.end(),
175     ↪ 0.0F);
176         for (std::int32_t offset = 0; offset < layer.output_block_size; ++
177     ↪ offset) {
178             const std::int32_t output_index = output_block + offset;
179             if (output_index < layer.output_size) {
180                 accumulator_workspace[checked_size(offset, "offset")] =
181                 layer.bias[checked_size(output_index, "output_index")];
182             }
183         }
184
185         for (std::int32_t input_index = 0; input_index < layer.input_size; ++
186     ↪ input_index) {
187             const float input_value = input[checked_size(input_index, "
188     ↪ input_index")];
189             for (std::int32_t offset = 0; offset < layer.output_block_size; ++
190     ↪ offset) {
191                 const std::int32_t output_index = output_block + offset;
192                 const std::int8_t weight = layer.packed_weights[packed_index
193     ↪ ++];

```

```

185         if (output_index < layer.output_size) {
186             accumulator_workspace[checked_size(offset, "offset")] +=
187                 static_cast<float>(weight) * layer.scale * input_value;
188         }
189     }
190 }
191
192 for (std::int32_t offset = 0; offset < layer.output_block_size; ++offset) {
193     const std::int32_t output_index = output_block + offset;
194     if (output_index < layer.output_size) {
195         output[checked_size(output_index, "output_index")] =
196             accumulator_workspace[checked_size(offset, "offset")];
197     }
198 }
199 }
200 }
201
202 QuantizedLinearLayer make_deterministic_i8_layer(std::int32_t input_size,
203                                                  std::int32_t output_size,
204                                                  float scale) {
205     require_positive(input_size, "input_size");
206     require_positive(output_size, "output_size");
207     QuantizedLinearLayer layer;
208     layer.input_size = input_size;
209     layer.output_size = output_size;
210     layer.scale = scale;
211     const std::size_t weight_count =
212         checked_size(input_size, "input_size") * checked_size(output_size, "
213     output_size");
214     layer.weights.resize(weight_count);
215     layer.bias.resize(checked_size(output_size, "output_size"));
216     for (std::int32_t output_index = 0; output_index < output_size; ++output_index) {
217         layer.bias[checked_size(output_index, "output_index")] =
218             static_cast<float>((output_index % LCQI_INPUT_PATTERN_MODULUS) -
219                             LCQI_INPUT_PATTERN_CENTER) *
220             LCQI_INPUT_PATTERN_SCALE;
221         for (std::int32_t input_index = 0; input_index < input_size; ++input_index) {
222             const std::int32_t raw =
223                 ((output_index * input_size + input_index) % LCQI_I8_PATTERN_MODULUS) -
224                 LCQI_I8_PATTERN_CENTER;
225             layer.weights[checked_size(output_index, "output_index") *
226                           checked_size(input_size, "input_size") +
227                           checked_size(input_index, "input_index")] =
228                 static_cast<std::int8_t>(raw);
229         }
230     }
231     return layer;
232 }

```

```

232
233 std::vector<float> make_deterministic_input(std::int32_t input_size) {
234     require_positive(input_size, "input_size");
235     std::vector<float> input(checkered_size(input_size, "input_size"));
236     for (std::int32_t input_index = 0; input_index < input_size; ++input_index) ↵
    ↵ {
237         input[checked_size(input_index, "input_index")] =
238             static_cast<float>(((input_index % LCQI_INPUT_PATTERN_MODULUS) -
239                               LCQI_INPUT_PATTERN_CENTER) *
240                               LCQI_INPUT_PATTERN_SCALE);
241     }
242     return input;
243 }
244
245 float max_abs_diff(std::span<const float> lhs, std::span<const float> rhs) {
246     if (lhs.size() != rhs.size()) {
247         throw std::runtime_error("max_abs_diff size mismatch");
248     }
249     float diff = 0.0F;
250     for (std::size_t index = 0; index < lhs.size(); ++index) {
251         diff = std::max(diff, std::fabs(lhs[index] - rhs[index]));
252     }
253     return diff;
254 }
255
256 KernelBenchmarkResult benchmark_linear_i8_case(const KernelBenchmarkCase& ↵
    ↵ benchmark_case) {
257     require_positive(benchmark_case.input_size, "benchmark input_size");
258     require_positive(benchmark_case.output_size, "benchmark output_size");
259     require_positive(benchmark_case.output_block_size, "benchmark ↵
    ↵ output_block_size");
260     require_positive(benchmark_case.repeat, "benchmark repeat");
261
262     const QuantizedLinearLayer layer =
263         make_deterministic_i8_layer(benchmark_case.input_size,
264                                     benchmark_case.output_size,
265                                     0.015625F);
266     const PackedLinearI8 packed = pack_linear_i8(layer, benchmark_case.↵
    ↵ output_block_size);
267     const PackedLinearI8 packed_simd = pack_linear_i8(layer, ↵
    ↵ LCQI_SIMD_OUTPUT_BLOCK);
268     const std::vector<float> input = make_deterministic_input(benchmark_case.↵
    ↵ input_size);
269     std::vector<float> scalar_output(checked_size(benchmark_case.output_size, "↵
    ↵ output_size"),
270                                     0.0F);
271     std::vector<float> packed_output(scalar_output.size(), 0.0F);
272     std::vector<float> simd_output(scalar_output.size(), 0.0F);
273     std::vector<float> packed_workspace(
274         checked_size(benchmark_case.output_block_size, "output_block_size"),
275         0.0F);
276

```

```

277     linear_i8(layer, input, scalar_output);
278     linear_i8_packed_with_workspace(packed, input, packed_output, ↵
↵ packed_workspace);

279
280     KernelBenchmarkResult result;
281     result.benchmark_case = benchmark_case;
282     result.backends.reserve(LCQI_BENCHMARK_BACKEND_COUNT);
283     result.backends.push_back(KernelBackendBenchmark{
284         .name = LCQI_BACKEND_SCALAR,
285         .available = true,
286         .average_us = measure_average_us(benchmark_case.repeat,
287                                         [&]() { linear_i8(layer, input, ↵
↵ scalar_output); }),
288         .max_abs_diff = 0.0F,
289         .checksum = checksum(scalar_output),
290     });
291     result.backends.push_back(KernelBackendBenchmark{
292         .name = LCQI_BACKEND_PACKED_SCALAR,
293         .available = true,
294         .average_us = measure_average_us(
295             benchmark_case.repeat,
296             [&]() { linear_i8_packed_with_workspace(packed,
297                                                         input,
298                                                         packed_output,
299                                                         packed_workspace); }),
300         .max_abs_diff = max_abs_diff(scalar_output, packed_output),
301         .checksum = checksum(packed_output),
302     });
303     if (linear_i8_packed_avx2_available()) {
304         linear_i8_packed_avx2(packed_simd, input, simd_output);
305         result.backends.push_back(KernelBackendBenchmark{
306             .name = LCQI_BACKEND_AVX2,
307             .available = true,
308             .average_us = measure_average_us(
309                 benchmark_case.repeat,
310                 [&]() { linear_i8_packed_avx2(packed_simd, input, simd_output); ↵
↵ }),
311             .max_abs_diff = max_abs_diff(scalar_output, simd_output),
312             .checksum = checksum(simd_output),
313         });
314     } else {
315         add_unavailable_backend(result, LCQI_BACKEND_AVX2);
316     }
317     return result;
318 }
319
320 } // namespace lcqi
321
322 #if !defined(LCQI_ENABLE_AVX2)
323 namespace lcqi {
324
325 bool linear_i8_packed_avx2_available() noexcept {

```



```

326     return false;
327 }
328
329 void linear_i8_packed_avx2(const PackedLinearI8&, std::span<const float>, std::span<float>) {
330     throw std::runtime_error("AVX2 packed kernel is not enabled in this build");
331 }
332
333 } // namespace lcqi
334 #endif

```

`int8_kernels_avx2.cpp` 是平台优化路径。读者要先看可用性判断，再看向量化实现；如果机器不支持 AVX2/FMA，报告必须写 `unavailable`。

Listing 4.18 `src/int8_kernels_avx2.cpp`

```

1  #include <lcqi/int8_kernels.hpp>
2
3  #if defined(LCQI_ENABLE_AVX2)
4
5  #include <immintrin.h>
6
7  #include <algorithm>
8  #include <stdint>
9  #include <cstring>
10 #include <stdexcept>
11 #include <string>
12
13 namespace lcqi {
14 namespace {
15
16 constexpr std::int32_t LCQI_AVX2_OUTPUT_BLOCK = 8;
17
18 void require_positive(std::int32_t value, const char* name) {
19     if (value <= 0) {
20         throw std::runtime_error(std::string(name) + " must be positive");
21     }
22 }
23
24 std::size_t checked_size(std::int32_t value, const char* name) {
25     if (value < 0) {
26         throw std::runtime_error(std::string(name) + " cannot be negative");
27     }
28     return static_cast<std::size_t>(value);
29 }
30
31 std::int32_t round_up_to_block(std::int32_t value, std::int32_t block_size) {
32     require_positive(value, "value");
33     require_positive(block_size, "block_size");
34     return ((value + block_size - 1) / block_size) * block_size;
35 }
36

```

```

37 void validate_avx2_contract(const PackedLinearI8& layer,
38                             std::span<const float> input,
39                             std::span<float> output) {
40     require_positive(layer.input_size, "packed input_size");
41     require_positive(layer.output_size, "packed output_size");
42     if (layer.output_block_size != LCQI_AVX2_OUTPUT_BLOCK) {
43         throw std::runtime_error("AVX2 packed kernel requires output_block_size ←
↪ == 8");
44     }
45     if (input.size() != checked_size(layer.input_size, "input_size")) {
46         throw std::runtime_error("AVX2 packed kernel input size mismatch");
47     }
48     if (output.size() != checked_size(layer.output_size, "output_size")) {
49         throw std::runtime_error("AVX2 packed kernel output size mismatch");
50     }
51     if (layer.bias.size() != checked_size(layer.output_size, "output_size")) {
52         throw std::runtime_error("AVX2 packed kernel bias size mismatch");
53     }
54     const std::int32_t padded_outputs =
55         round_up_to_block(layer.output_size, layer.output_block_size);
56     const std::size_t expected_packed_size =
57         checked_size(padded_outputs, "padded_outputs") *
58         checked_size(layer.input_size, "input_size");
59     if (layer.packed_weights.size() != expected_packed_size) {
60         throw std::runtime_error("AVX2 packed kernel packed weight size ←
↪ mismatch");
61     }
62 }
63
64 } // namespace
65
66 bool linear_i8_packed_avx2_available() noexcept {
67     #if defined(__GNUC__) || defined(__clang__)
68         __builtin_cpu_init();
69         return __builtin_cpu_supports("avx2") && __builtin_cpu_supports("fma");
70     #else
71         return true;
72     #endif
73 }
74
75 void linear_i8_packed_avx2(const PackedLinearI8& layer,
76                             std::span<const float> input,
77                             std::span<float> output) {
78     if (!linear_i8_packed_avx2_available()) {
79         throw std::runtime_error("AVX2/FMA is not available on this CPU");
80     }
81     validate_avx2_contract(layer, input, output);
82
83     const std::int32_t padded_outputs =
84         round_up_to_block(layer.output_size, layer.output_block_size);
85     std::size_t packed_index = 0;
86     alignas(32) float block_output[LCQI_AVX2_OUTPUT_BLOCK] = {};

```

```

87
88     for (std::int32_t output_block = 0; output_block < padded_outputs;
89         output_block += LCQI_AVX2_OUTPUT_BLOCK) {
90         __m256 accumulator = _mm256_setzero_ps();
91
92         for (std::int32_t offset = 0; offset < LCQI_AVX2_OUTPUT_BLOCK; ++offset) {
93             const std::int32_t output_index = output_block + offset;
94             block_output[checked_size(offset, "offset")] =
95                 output_index < layer.output_size
96                 ? layer.bias[checked_size(output_index, "output_index")]
97                 : 0.0F;
98         }
99         accumulator = _mm256_load_ps(block_output);
100
101         for (std::int32_t input_index = 0; input_index < layer.input_size; ++input_index) {
102             std::uint64_t packed_bytes = 0;
103             std::memcpy(&packed_bytes,
104                 layer.packed_weights.data() + packed_index,
105                 sizeof(packed_bytes));
106             const __m128i packed_i8 =
107                 _mm_cvtsi64_si128(static_cast<long long>(packed_bytes));
108             packed_index += LCQI_AVX2_OUTPUT_BLOCK;
109             const __m256i weights_i32 = _mm256_cvtepi8_epi32(packed_i8);
110             const __m256 weights_f32 = _mm256_cvtepi32_ps(weights_i32);
111             const __m256 input_scaled =
112                 _mm256_set1_ps(input[checked_size(input_index, "input_index")])
113                 * layer.scale;
114             accumulator = _mm256_fmadd_ps(weights_f32, input_scaled,
115                 accumulator);
116             _mm256_store_ps(block_output, accumulator);
117             for (std::int32_t offset = 0; offset < LCQI_AVX2_OUTPUT_BLOCK; ++offset) {
118                 const std::int32_t output_index = output_block + offset;
119                 if (output_index < layer.output_size) {
120                     output[checked_size(output_index, "output_index")] =
121                         block_output[checked_size(offset, "offset")];
122                 }
123             }
124         }
125     }
126 }
127 } // namespace lcqi
128
129 #endif

```

第零部分

Transformer、KV Cache 与服务运行时

第 5 章实践：Transformer 切片与 KV Cache 状态

本章对应原书中的第三册“AI 推理原理、KV cache 与计算账本”和“KV cache 实现细节”。不要把它当作概念复述，本章要把 RMSNorm、Q/K/V、RoPE、GQA、KV slot、position、state edge 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `kv_cache`，核心命令或测试入口是 `lcqi_trace`。

Listing 5.1 第 5 章实践：Transformer 切片与 KV Cache 状态的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / kv_cache
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：固定 prompt `tok1tok2tok3`，观察第一个 decode step。读者要先手写预期结果，再运行项目观察输出。动作是：把 trace 中 tokenizer、q/k/v、`kv_cache_slot`、attention 和 logits 串成状态表。如果输出里没有 `token_position` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：KV cache 是请求状态，不是临时 activation；position、layer、head、layout 和生命周期必须进入 trace 这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 KV cache 状态表：每层每个 token 写入哪里、何时读取、何时禁止复用。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第6章实践：Reference Decoder、logits 与采样

本章对应原书中的第三册“Reference decoder 实现”和“推理算法：logits、softmax、采样”。不要把它当作概念复述，本章要把 reference decoder、logits golden、argmax sampler、tokenizer contract hash、BOS/EOS 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux_cpu_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 **argmax**，核心命令或测试入口是 **lcqi_trace**。

Listing 6.1 第6章实践：Reference Decoder、logits 与采样的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / argmax
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：让 trace 固定输出 tokenizer contract 和 logits golden。读者要先手写预期结果，再运行项目观察输出。动作是：比较 generated token、logits 数组和 tokenizer hash。如果输出里没有 **tokenizer_contract** 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：采样前必须先证明 logits 可复查；sampler 的随机性、温度和 top-k 只能建立在 reference 通过之后这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux_cpu_inference/README.md](#)、相关 **include/lcqi** 头文件和一个 **src** 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交采样报告：argmax、top-k、temperature 三种策略各自需要哪些额外字段才能复现。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 7 章实践：服务调度、SLO 与资源预算

本章对应原书中的第三册“业务服务实战：请求、调度、队列、取消与资源预算”。不要把它当作概念复述，本章要把 admission、KV budget、queue wait、TTFT、TPOT、cancel、reject、p95 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `ttft_ms`，核心命令或测试入口是 `lcqi_serving_probe`。

Listing 7.1 第 7 章实践：服务调度、SLO 与资源预算的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_serving_probe / ttft_ms
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：运行 serving probe 中短请求、RAG 请求、取消请求和超长请求。读者要先手写预期结果，再运行项目观察输出。动作是：解释 `memory_budget_exceeded`、cancel reclaim、`queue_wait_p95_ms` 和 `rejection_rate`。如果输出里没有 `rejection_rate` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：推理服务不是只看 tokens/s；是否接收请求、如何预留 KV、取消后是否回收，决定系统是否可运营这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 SLO 表：每类请求的 prompt/new token、预算、是否接受、TTFT、TPOT、拒绝原因和未验证边界。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 8 章实践：RAG、Agent 与状态图边界

本章对应原书中的第三册“上层 AI 应用编排：RAG、Agent、工具和状态图”。不要把它当作概念复述，本章要把 retrieval、tool call、prompt assembly、state graph、budget、trace id 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `req_rag`，核心命令或测试入口是 `lcqi_serving_probe`。

Listing 8.1 第 8 章实践：RAG、Agent 与状态图边界的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_serving_probe / req_rag
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：把 RAG 请求拆成 retrieve、rerank、prompt build、prefill、decode 五段。读者要先手写预期结果，再运行项目观察输出。动作是：为每段写输入输出和失败后能否重试。如果输出里没有 `prompt_tokens` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：上层编排不能把外部工具结果和模型状态混成字符串；每个边界都要有身份、预算和可重放记录这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交一个 RAG 状态图：节点、边、超时、重试、缓存键、prompt 版本和审计字段。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第零部分

模型导入、AI 编译器与内存计划

第9章实践：模型导入、manifest 与 Shape Verifier

本章对应原书中的第三册“模型导入、manifest 与 shape verifier”。不要把它当作概念复述，本章要把 safetensors header、metadata、tensor name、dtype、shape、`data_offsets`、payload boundary 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `load_safetensors_manifest`，核心命令或测试入口是 `lcqi_tests`。

Listing 9.1 第9章实践：模型导入、manifest 与 Shape Verifier 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<↵
↵ DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_tests / load_safetensors_manifest
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：加载测试临时 safetensors fixture。读者要先手写预期结果，再运行项目观察输出。动作是：故意构造 offset 越界和 dtype/shape 字节数不匹配。如果输出里没有 `data_offsets` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：模型导入的第一职责是早拒绝：名字、shape、dtype、offset 和 payload 都必须在 kernel 运行前被验证这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 manifest gate 报告：合法 fixture 和两个坏 fixture 的字段、错误原因和拒绝位置。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 10 章实践：真实模型加载闭环与首 Token Trace

本章对应原书中的第三册“真实模型加载闭环：Tokenizer、权重格式与首 token trace”。不要把它当作概念复述，本章要把 tokenizer、weight mapping、chat template、first token trace、checksum 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux_cpu_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

本章新增一个明确边界：tiny fixture 负责手算和逐层 oracle；LCQI GPT-2 reference path 负责证明自研代码已经能加载标准 GPT-2 safetensors、执行 byte-level BPE、跑 GPT-2 forward 并 greedy 生成；SmolLM2 small smoke 负责证明外部 llama.cpp 路径能加载 GGUF small 模型。三者不能互相冒充。tiny MLP 算对，不等于 GPT-2 完成；GPT-2 reference 能出一个 token，不等于高性能 KV cache runtime 完成；SmolLM2 能输出文本，也不等于 LCQI 自研 GGUF 后端完成。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里有三组核心对象。第一组是 [tiny_tokenizer.txt](#)、reference decoder 和 [lcqi_trace](#)，用于手算 token 合同和逐 checkpoint 追踪。第二组是 [gpt2_reference.hpp/.cpp](#)、[lcqi_gpt2](#) 和 [tools/run_gpt2_smoke.py](#)，用于自研 GPT-2 safetensors 路径。第三组是 [tools/run_smolllm2_small_smoke.py](#)，用于下载并校验 SmolLM2-135M-Instruct 的 GGUF 量化文件，再通过 llama.cpp 在 CPU 上实际生成 assistant 文本。

Listing 10.1 第 10 章实践：真实模型加载闭环与首 Token Trace 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↳ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / tiny_tokenizer.txt
books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_gpt2
make cpu3-gpt2-smoke
make cpu3-smolllm2-smoke
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：从 tokenizer fixture 开始，不跳到真实 7B。读者要先手写预期结果，再运行项目观察输出。动作是：解释 BOS/EOS、UNK、template hash 和 decoder 输入 tokens 的边界。如果输出里没有 `tokens=[0,1,2,3,4]` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：真实模型加载不是下载权重；必须先让 tokenizer 合同、权重 role、shape verifier 和首 token trace 形成闭环。这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、自研 GPT-2 reference smoke：先把标准 safetensors 跑通

GPT-2 不是 LLaMA。它使用 LayerNorm，不是 RMSNorm；使用 learned absolute position embedding，不是 RoPE；attention 里的 Q/K/V 在 HuggingFace 权重中是 packed Conv1D 权重；MLP 使用 GELU，不是 SwiGLU；`lm_head` 通常和 token embedding 共享权重。把 GPT-2 硬塞进原来的 LLaMA-like tiny decoder，会让概念混乱，所以 LCQI 新增了独立的 `gpt2_reference` 模块。

仓库内最小验收分两层。第一层不下载模型，运行 tiny GPT-2 fixture：

Listing 10.2 仓库内 tiny GPT-2 数学闭环

```
books/cpu-volume-3/build/lcqi-debug/labs/linux_cpu_inference/lcqi_gpt2
```

它固定 prompt ids 为 **123**，检查 logits、predicted token 和 greedy generated ids。对应测试还会临时写出一个 HuggingFace 风格 **model.safetensors**，再用 loader 读回来，覆盖 tensor name mapping、shape verifier、Conv1D 转置和 tied lm head。坏 shape 必须在 loader 阶段被拒绝。

第二层下载真实 GPT-2 资产：

Listing 10.3 LCQI 自研 GPT-2 smoke

```
make cpu3-gpt2-smoke
```

脚本下载 **openai-community/gpt2** 的 **config.json**、**vocab.json**、**merges.txt** 和 **model.safetensors** 到用户缓存目录，不提交模型文件。最近一次报告写到：

Listing 10.4 GPT-2 smoke 报告关键字段

```
books/cpu-volume-3/results/lcqi-gpt2-smoke.txt
prompt_ids 15496 11 616 1438 318
generated_ids 15496 11 616 1438 318 1757
generated_text Hello, my name is John
validation=PASS
```

这说明自研 LCQI C++ reference path 已经能跑标准 GPT-2 safetensors 的 tokenizer、权重导入、forward 和 greedy generation。边界也要写清：当前实现为了可读和可验证，会重跑完整前缀，没有 KV cache、没有 packed weight、没有多线程，也没有把 logits 与 PyTorch/Transformers 的逐数值 golden 报告纳入仓库。因此它是正确性闭环，不是性能结论。

四、真实 small 模型 smoke：不用 tiny，先让开源模型实际跑起来

本章的真实模型 smoke 固定为 **SmolLM2-135M-Instruct**，规模是 135M 参数，不是 tiny 随机夹具。脚本使用 GGUF 文件 **SmolLM2-135M-Instruct-Q4_K_M.gguf**，来源仓库是 **bartowski/SmolLM2-135M-Instruct-GGUF**，原模型组织是 **HuggingFaceTB**，license 字段为 **apache-2.0**。脚本不会把模型文件提交到仓库，而是缓存到用户目录；验收时检查文件大小 **105454432** 字节和 SHA256：

Listing 10.5 SmolLM2 small GGUF 文件身份

```
model_source_id=HuggingFaceTB/SmolLM2-135M-Instruct
model_gguf_repo_id=bartowski/SmolLM2-135M-Instruct-GGUF
model_file=SmolLM2-135M-Instruct-Q4_K_M.gguf
model_expected_bytes=105454432
model_expected_sha256=2↵
↵ e8040ceae7815abe0dcb3540b9995eaa1fa0d2ca9e797d0a635ae4433c68c2d
```

运行命令是：

Listing 10.6 真实 small 模型 CPU 冒烟

```
make cpu3-smolllm2-smoke
```

这条命令会做四步。第一，下载或复用 GGUF 文件，并校验 SHA256，防止“文件名对了但内容不对”。第二，拉取固定 commit 的 `llama.cpp` 并构建 `llama-simple-chat`，避免依赖本机临时安装。第三，用 `-ngl0` 强制 CPU 路径，输入一个短 prompt，确认程序能走过 tokenizer、chat template、prefill、decode 和 sampler。第四，把报告写到：

Listing 10.7 真实模型 smoke 报告路径

```
books/cpu-volume-3/results/lcqi-smolllm2-small-model-smoke.txt
```

报告里必须至少出现这些字段：`model_expected_sha256`、`llama_cpp_commit`、`command`、`exit_code=0`、`validation=PASS` 和 `assistant_response`。如果只有下载日志，没有 assistant response，不能算通过；如果 assistant response 内容答错了，也不能改写成“质量通过”。本章只验证真实模型资产可加载、可 tokenize、可 decode、可生成非空文本。

五、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：运行 GPT-2 smoke。报告不许只贴“生成了 John”，必须解释 `config.json`、`vocab.json`、`merges.txt`、`model.safetensors` 各自对应哪段代码，为什么 GPT-2 需要 byte-level BPE，为什么 HuggingFace Conv1D 权重要转置，为什么 `lm_head` 可以 tied 到 embedding。

任务四：运行真实 small 模型 smoke。报告不许只贴“命令执行成功”，必须解释脚本为什么校验模型 hash、为什么固定 `llama.cpp` commit、为什么使用 `-ngl0`，以及为什么 assistant response 非空只是 smoke，不是质量评测。

任务五：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。GPT-2 反例可以是：tensor name 缺失、shape 和 config 不一致、vocab 中缺少 BPE merge 后 token、`max_new_tokens` 超过 context。真实模型 smoke 的反例可以是：模型 SHA256 不匹配、网络不可用、`cmake` 缺失、assistant response 为空、命令超时。每个反例都要写清期望处理方式。

六、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。新增 GPT-2 smoke 后，最低验收还包括：真实 GPT-2 文件身份有 hash，prompt ids 可解释，generated ids 可解释，文本反解码可解释，并明确声明这不是高性能推理结论。新增 SmolLM2 smoke 后，最低验收还包括：模型身份有 hash，模型规模不是 tiny，CPU 路径实际运行，报告里有非空 assistant response，并明确声明这不是自研 LCQI 完整 GGUF 支持。

递进大作业：提交 GPT-2 reference path 的下一阶段迁移清单：KV cache、prefill/decode 分离、packed weight、线程并行、Transformers golden logits、更多模型方言 name mapping、分片 safetensors 和性能报告。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

代码走读: Reference Decoder、Tokenizer、Safetensors 与 GPT-2

从 tiny MLP 走向 decoder-only 模型

Reference decoder 和 GPT-2 reference path 是 LCQI 从教学闭环走向真实模型导入的关键层。这里的代码更长，是因为真实模型不只有矩阵乘法：它还需要 tokenizer、位置、归一化、QKV 拆分、attention mask、KV cache、激活函数、lm head、权重命名映射、dtype 转换和坏资产拒绝。

Tiny Reference Decoder

`reference_decoder.cpp` 实现 embedding、RMSNorm、Q/K/V、RoPE、KV cache、attention、SwiGLU、final norm、logits 和 sampler，并在关键位置生成 trace checkpoint。读者应按 checkpoint 顺序读，而不是从最终 logits 倒推。

Listing 10.8 `src/reference_decoder.cpp`

```

1 #include <lcqi/reference_decoder.hpp>
2
3 #include <algorithm>
4 #include <array>
5 #include <cmath>
6 #include <limits>
7 #include <stdexcept>
8 #include <string>
9
10 namespace lcqi {
11 namespace {
12
13 constexpr float LCQI_SOFTMAX_NEGATIVE_INFINITY = -std::numeric_limits<float>::infinity();
14 constexpr std::int32_t LCQI_ROPE_PAIR_WIDTH = 2;
15 constexpr const char* LCQI_TRACE_DTYPE_F32 = "f32";
16 constexpr const char* LCQI_TRACE_LAYOUT_CONTIGUOUS = "contiguous";
17 constexpr const char* LCQI_TRACE_LAYOUT_ROW_MAJOR = "row_major";
18 constexpr const char* LCQI_TRACE_LAYOUT_KV_CONTIGUOUS = "kv_contiguous";
19
20 void require_positive(std::int32_t value, const char* name) {
21     if (value <= 0) {
22         throw std::runtime_error(std::string(name) + " must be positive");
23     }
24 }
25
26 void validate_config(const DecoderConfig& config) {
27     require_positive(config.hidden_size, "hidden_size");
28     require_positive(config.query_heads, "query_heads");
29     require_positive(config.kv_heads, "kv_heads");
30     require_positive(config.head_dim, "head_dim");
31     require_positive(config.intermediate_size, "intermediate_size");
32     require_positive(config.vocab_size, "vocab_size");

```



```

33     require_positive(config.max_context, "max_context");
34     if (config.query_heads % config.kv_heads != 0) {
35         throw std::runtime_error("query_heads must be divisible by kv_heads");
36     }
37     if (config.hidden_size != config.query_heads * config.head_dim) {
38         throw std::runtime_error("hidden_size must equal query_heads * head_dim↵
↵ ");
39     }
40     if (config.head_dim % LCQI_ROPE_PAIR_WIDTH != 0) {
41         throw std::runtime_error("head_dim must be even for RoPE");
42     }
43     if (config.rope_base <= 1.0F) {
44         throw std::runtime_error("rope_base must be greater than 1");
45     }
46 }
47
48 std::size_t checked_size(std::int32_t value, const char* name) {
49     if (value < 0) {
50         throw std::runtime_error(std::string(name) + " cannot be negative");
51     }
52     return static_cast<std::size_t>(value);
53 }
54
55 std::int32_t kv_width(const DecoderConfig& config) {
56     return config.kv_heads * config.head_dim;
57 }
58
59 void validate_linear(const LinearWeightsF32& layer, const char* name) {
60     require_positive(layer.input_size, "linear input_size");
61     require_positive(layer.output_size, "linear output_size");
62     const std::size_t expected_weights =
63         checked_size(layer.input_size, name) * checked_size(layer.output_size, ↵
↵ name);
64     if (layer.weights.size() != expected_weights) {
65         throw std::runtime_error(std::string(name) + " weight size mismatch");
66     }
67     if (!layer.bias.empty() &&
68         layer.bias.size() != checked_size(layer.output_size, name)) {
69         throw std::runtime_error(std::string(name) + " bias size mismatch");
70     }
71 }
72
73 void validate_span_size(std::size_t actual, std::int32_t expected, const char* ↵
↵ name) {
74     if (actual != checked_size(expected, name)) {
75         throw std::runtime_error(std::string(name) + " size mismatch");
76     }
77 }
78
79 std::int32_t argmax(std::span<const float> values) {
80     if (values.empty()) {
81         throw std::runtime_error("cannot take argmax of empty values");

```



```

82     }
83     std::int32_t best_index = 0;
84     float best_value = values[0];
85     for (std::int32_t index = 1; index < static_cast<std::int32_t>(values.size()
↵    ↵ )); ++index) {
86         const float candidate = values[static_cast<std::size_t>(index)];
87         if (candidate > best_value) {
88             best_value = candidate;
89             best_index = index;
90         }
91     }
92     return best_index;
93 }
94
95 float silu(float value) {
96     return value / (1.0F + std::exp(-value));
97 }
98
99 std::span<const float> row_span(std::span<const float> values,
100                                std::int32_t row,
101                                std::int32_t row_width) {
102     const std::size_t begin = checked_size(row, "row") * checked_size(row_width,
↵    ↵ , "row_width");
103     return values.subspan(begin, checked_size(row_width, "row_width"));
104 }
105
106 void add_inplace(std::span<float> target, std::span<const float> source) {
107     if (target.size() != source.size()) {
108         throw std::runtime_error("residual add size mismatch");
109     }
110     for (std::size_t index = 0; index < target.size(); ++index) {
111         target[index] += source[index];
112     }
113 }
114
115 void swiglu_mlp(const DecoderLayerWeightsF32& layer,
116                std::span<const float> input,
117                std::span<float> output) {
118     std::vector<float> gate(checked_size(layer.w_gate.output_size, "gate"), 0.0F
↵    ↵ );
119     std::vector<float> up(checked_size(layer.w_up.output_size, "up"), 0.0F);
120     std::vector<float> hidden(gate.size(), 0.0F);
121     linear_f32(layer.w_gate, input, gate);
122     linear_f32(layer.w_up, input, up);
123     if (gate.size() != up.size()) {
124         throw std::runtime_error("SwiGLU gate/up size mismatch");
125     }
126     for (std::size_t index = 0; index < hidden.size(); ++index) {
127         hidden[index] = silu(gate[index]) * up[index];
128     }
129     linear_f32(layer.w_down, hidden, output);
130 }

```

```

131
132 float checksum(std::span<const float> values) {
133     float sum = 0.0F;
134     for (const float value : values) {
135         sum += value;
136     }
137     return sum;
138 }
139
140 float max_abs(std::span<const float> values) {
141     float result = 0.0F;
142     for (const float value : values) {
143         result = std::max(result, std::fabs(value));
144     }
145     return result;
146 }
147
148 std::vector<std::int32_t> contiguous_stride(std::span<const std::int32_t> shape ←
    ↪ ) {
149     std::vector<std::int32_t> stride(shape.size(), 1);
150     std::int32_t running = 1;
151     for (std::int32_t index = static_cast<std::int32_t>(shape.size()) - 1; ←
    ↪ index >= 0; --index) {
152         stride[checked_size(index, "stride index")] = running;
153         running *= shape[checked_size(index, "shape index")];
154     }
155     return stride;
156 }
157
158 void add_checkpoint(std::vector<ReferenceTensorCheckpoint>* checkpoints,
159                    const char* name,
160                    std::int32_t token_position,
161                    std::int32_t layer_id,
162                    std::span<const std::int32_t> shape,
163                    const char* layout,
164                    std::span<const float> values) {
165     if (checkpoints == nullptr) {
166         return;
167     }
168     ReferenceTensorCheckpoint checkpoint;
169     checkpoint.name = name;
170     checkpoint.token_position = token_position;
171     checkpoint.layer_id = layer_id;
172     checkpoint.shape.assign(shape.begin(), shape.end());
173     checkpoint.stride = contiguous_stride(shape);
174     checkpoint.dtype = LCQI_TRACE_DTYPE_F32;
175     checkpoint.layout = layout;
176     checkpoint.checksum = checksum(values);
177     checkpoint.max_abs = max_abs(values);
178     checkpoint.values.assign(values.begin(), values.end());
179     checkpoints->push_back(std::move(checkpoint));
180 }

```

```

181
182 void swiglu_mlp_with_trace(const DecoderLayerWeightsF32& layer,
183                             std::span<const float> input,
184                             std::span<float> output,
185                             std::int32_t token_position,
186                             std::int32_t layer_id,
187                             std::vector<ReferenceTensorCheckpoint>* checkpoints)↵
    ↵ {
188     std::vector<float> gate(check_size(layer.w_gate.output_size, "gate"), 0.0↵
    ↵ F);
189     std::vector<float> up(check_size(layer.w_up.output_size, "up"), 0.0F);
190     std::vector<float> hidden(gate.size(), 0.0F);
191     linear_f32(layer.w_gate, input, gate);
192     linear_f32(layer.w_up, input, up);
193     if (gate.size() != up.size()) {
194         throw std::runtime_error("SwiGLU gate/up size mismatch");
195     }
196     const std::array<std::int32_t, 1> intermediate_shape{layer.w_gate.↵
    ↵ output_size};
197     add_checkpoint(checkpoints,
198                     "mlp_gate",
199                     token_position,
200                     layer_id,
201                     intermediate_shape,
202                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
203                     gate);
204     add_checkpoint(checkpoints,
205                     "mlp_up",
206                     token_position,
207                     layer_id,
208                     intermediate_shape,
209                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
210                     up);
211     for (std::size_t index = 0; index < hidden.size(); ++index) {
212         hidden[index] = silu(gate[index]) * up[index];
213     }
214     add_checkpoint(checkpoints,
215                     "mlp_hidden",
216                     token_position,
217                     layer_id,
218                     intermediate_shape,
219                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
220                     hidden);
221     linear_f32(layer.w_down, hidden, output);
222 }
223
224 void forward_layer(const DecoderConfig& config,
225                   const DecoderLayerWeightsF32& layer,
226                   std::int32_t layer_id,
227                   std::int32_t model_position,
228                   ReferenceKVCache& cache,
229                   std::span<float> hidden,

```

```

230         std::vector<ReferenceTensorCheckpoint>* checkpoints = ↵
↵ nullptr) {
231     validate_span_size(hidden.size(), config.hidden_size, "hidden");
232
233     std::vector<float> normed(check_size(config.hidden_size, "hidden_size"), ↵
↵ 0.0F);
234     std::vector<float> query(check_size(config.hidden_size, "hidden_size"), ↵
↵ 0.0F);
235     std::vector<float> key(check_size(kv_width(config), "kv_width"), 0.0F);
236     std::vector<float> value(check_size(kv_width(config), "kv_width"), 0.0F);
237     std::vector<float> attention(check_size(config.hidden_size, "hidden_size"↵
↵ ), 0.0F);
238     std::vector<float> attention_projected(check_size(config.hidden_size, "↵
↵ hidden_size"), 0.0F);
239     std::vector<float> mlp(check_size(config.hidden_size, "hidden_size"), 0.0↵
↵ F);
240
241     const std::array<std::int32_t, 1> hidden_shape{config.hidden_size};
242     const std::array<std::int32_t, 2> query_shape{config.query_heads, config.↵
↵ head_dim};
243     const std::array<std::int32_t, 2> kv_shape{config.kv_heads, config.head_dim↵
↵ };
244     const std::array<std::int32_t, 4> kv_slot_shape{
245         1, 1, config.kv_heads, config.head_dim};
246
247     rms_norm(hidden, layer.rms_attention_weight, config.rms_epsilon, normed);
248     add_checkpoint(checkpoints,
249         "attn_norm",
250         model_position,
251         layer_id,
252         hidden_shape,
253         LCQI_TRACE_LAYOUT_CONTIGUOUS,
254         normed);
255     linear_f32(layer.wq, normed, query);
256     linear_f32(layer.wk, normed, key);
257     linear_f32(layer.wv, normed, value);
258     add_checkpoint(checkpoints,
259         "q_before_rope",
260         model_position,
261         layer_id,
262         query_shape,
263         LCQI_TRACE_LAYOUT_CONTIGUOUS,
264         query);
265     add_checkpoint(checkpoints,
266         "k_before_rope",
267         model_position,
268         layer_id,
269         kv_shape,
270         LCQI_TRACE_LAYOUT_CONTIGUOUS,
271         key);
272     add_checkpoint(checkpoints,
273         "v",

```

```

274         model_position,
275         layer_id,
276         kv_shape,
277         LCQI_TRACE_LAYOUT_CONTIGUOUS,
278         value);
279     apply_rope(query, config.query_heads, config.head_dim, model_position, ↵
↵ config.rope_base);
280     apply_rope(key, config.kv_heads, config.head_dim, model_position, config.↵
↵ rope_base);
281     add_checkpoint(checkpoints,
282         "q_after_rope",
283         model_position,
284         layer_id,
285         query_shape,
286         LCQI_TRACE_LAYOUT_CONTIGUOUS,
287         query);
288     add_checkpoint(checkpoints,
289         "k_after_rope",
290         model_position,
291         layer_id,
292         kv_shape,
293         LCQI_TRACE_LAYOUT_CONTIGUOUS,
294         key);
295
296     cache.append(layer_id, model_position, key, value);
297     add_checkpoint(checkpoints,
298         "kv_cache_key_slot",
299         model_position,
300         layer_id,
301         kv_slot_shape,
302         LCQI_TRACE_LAYOUT_KV_CONTIGUOUS,
303         key);
304     attention_decode(config, cache, layer_id, model_position, query, attention)↵
↵ ;
305     add_checkpoint(checkpoints,
306         "attention_out",
307         model_position,
308         layer_id,
309         hidden_shape,
310         LCQI_TRACE_LAYOUT_CONTIGUOUS,
311         attention);
312     linear_f32(layer.wo, attention, attention_projected);
313     add_inplace(hidden, attention_projected);
314     add_checkpoint(checkpoints,
315         "post_attention_residual",
316         model_position,
317         layer_id,
318         hidden_shape,
319         LCQI_TRACE_LAYOUT_CONTIGUOUS,
320         hidden);
321
322     rms_norm(hidden, layer.rms_mlp_weight, config.rms_epsilon, normed);

```

```

323     add_checkpoint(checkpoints,
324                     "mlp_norm",
325                     model_position,
326                     layer_id,
327                     hidden_shape,
328                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
329                     normed);
330     if (checkpoints == nullptr) {
331         swiglu_mlp(layer, normed, mlp);
332     } else {
333         swiglu_mlp_with_trace(layer, normed, mlp, model_position, layer_id, ↵
↵ checkpoints);
334     }
335     add_inplace(hidden, mlp);
336     add_checkpoint(checkpoints,
337                     "layer_out",
338                     model_position,
339                     layer_id,
340                     hidden_shape,
341                     LCQI_TRACE_LAYOUT_CONTIGUOUS,
342                     hidden);
343 }
344
345 LinearWeightsF32 make_linear(std::int32_t input_size,
346                               std::int32_t output_size,
347                               std::vector<float> weights,
348                               std::vector<float> bias = {}) {
349     LinearWeightsF32 layer;
350     layer.input_size = input_size;
351     layer.output_size = output_size;
352     layer.weights = std::move(weights);
353     layer.bias = std::move(bias);
354     validate_linear(layer, "tiny linear");
355     return layer;
356 }
357
358 std::vector<float> zeros(std::int32_t count) {
359     return std::vector<float>(checked_size(count, "zero count"), 0.0F);
360 }
361
362 } // namespace
363
364 ReferenceDecodeTraceResult run_reference_decode_with_trace(
365     const ReferenceDecoderModel& model,
366     std::span<const std::int32_t> token_ids) {
367     validate_config(model.config);
368     if (token_ids.empty()) {
369         throw std::runtime_error("reference decode needs at least one token");
370     }
371     if (token_ids.size() > checked_size(model.config.max_context, "max_context" ↵
↵ )) {
372         throw std::runtime_error("token_ids exceed max_context");

```

```

373     }
374     if (model.token_embedding.size() !=
375         checked_size(model.config.vocab_size, "vocab_size") *
376         checked_size(model.config.hidden_size, "hidden_size")) {
377         throw std::runtime_error("token embedding size mismatch");
378     }
379     if (model.final_norm_weight.size() != checked_size(model.config.hidden_size,
380 ↪ , "hidden_size")) {
381         throw std::runtime_error("final norm weight size mismatch");
382     }
383     validate_linear(model.lm_head, "lm_head");
384
385     ReferenceDecodeTraceResult trace;
386     ReferenceKVCache cache(model.config, static_cast<std::int32_t>(model.layers.
387 ↪ .size()));
388     std::vector<float> hidden(checked_size(model.config.hidden_size, "
389 ↪ hidden_size"), 0.0F);
390     const std::array<std::int32_t, 1> hidden_shape{model.config.hidden_size};
391     for (std::int32_t position = 0; position < static_cast<std::int32_t>(
392 ↪ token_ids.size());
393         ++position) {
394         const std::int32_t token_id = token_ids[checked_size(position, "
395 ↪ position")];
396         if (token_id < 0 || token_id >= model.config.vocab_size) {
397             throw std::runtime_error("token id out of vocabulary");
398         }
399         const std::span<const float> embedding =
400             row_span(model.token_embedding, token_id, model.config.hidden_size)
401 ↪ ;
402         std::copy(embedding.begin(), embedding.end(), hidden.begin());
403         add_checkpoint(&trace.checkpoints,
404             "embedding",
405             position,
406             -1,
407             hidden_shape,
408             LCQI_TRACE_LAYOUT_ROW_MAJOR,
409             hidden);
410         for (std::int32_t layer_id = 0; layer_id < static_cast<std::int32_t>(
411 ↪ model.layers.size());
412             ++layer_id) {
413             forward_layer(model.config,
414                 model.layers[checked_size(layer_id, "layer_id")],
415                 layer_id,
416                 position,
417                 cache,
418                 hidden,
419                 &trace.checkpoints);
420         }
421     }
422
423     std::vector<float> normed(checked_size(model.config.hidden_size, "
424 ↪ hidden_size"), 0.0F);

```



```

417     std::vector<float> logits(checkered_size(model.config.vocab_size, "vocab_size" ←
    ↪ "), 0.0F);
418     const std::array<std::int32_t, 1> logits_shape{model.config.vocab_size};
419     const std::int32_t last_position = static_cast<std::int32_t>(token_ids.size ←
    ↪ ()) - 1;
420     rms_norm(hidden, model.final_norm_weight, model.config.rms_epsilon, normed) ←
    ↪ ;
421     add_checkpoint(&trace.checkpoints,
422                   "final_norm",
423                   last_position,
424                   -1,
425                   hidden_shape,
426                   LCQI_TRACE_LAYOUT_CONTIGUOUS,
427                   normed);
428     linear_f32(model.lm_head, normed, logits);
429     add_checkpoint(&trace.checkpoints,
430                   "logits",
431                   last_position,
432                   -1,
433                   logits_shape,
434                   LCQI_TRACE_LAYOUT_CONTIGUOUS,
435                   logits);
436
437     trace.result.predicted_token = argmax(logits);
438     trace.result.logits = std::move(logits);
439     return trace;
440 }
441
442 ReferenceKVCache::ReferenceKVCache(const DecoderConfig& config, std::int32_t ←
    ↪ layer_count)
443 : config_(config),
444   layer_count_(layer_count) {
445     validate_config(this->config_);
446     require_positive(this->layer_count_, "layer_count");
447     const std::size_t element_count =
448         checked_size(this->layer_count_, "layer_count") *
449         checked_size(this->config_.max_context, "max_context") *
450         checked_size(this->config_.kv_heads, "kv_heads") *
451         checked_size(this->config_.head_dim, "head_dim");
452     this->keys_.assign(element_count, 0.0F);
453     this->values_.assign(element_count, 0.0F);
454     this->written_.assign(
455         checked_size(this->layer_count_, "layer_count") *
456         checked_size(this->config_.max_context, "max_context"),
457         0);
458 }
459
460 void ReferenceKVCache::append(std::int32_t layer_id,
461                               std::int32_t model_position,
462                               std::span<const float> key,
463                               std::span<const float> value) {
464     validate_span_size(key.size(), kv_width(this->config_), "KV key");

```

```

465     validate_span_size(value.size(), kv_width(this->config_), "KV value");
466     this->validate_address(layer_id, model_position, 0);
467     for (std::int32_t kv_head = 0; kv_head < this->config_.kv_heads; ++kv_head) {
468         const std::size_t target = this->base_offset(layer_id, model_position,
469         kv_head);
469         const std::size_t source =
470             checked_size(kv_head, "kv_head") * checked_size(this->config_.
471         head_dim, "head_dim");
471         std::copy_n(key.begin() + static_cast<std::ptrdiff_t>(source),
472             this->config_.head_dim,
473             this->keys_.begin() + static_cast<std::ptrdiff_t>(target));
474         std::copy_n(value.begin() + static_cast<std::ptrdiff_t>(source),
475             this->config_.head_dim,
476             this->values_.begin() + static_cast<std::ptrdiff_t>(target));
477     };
478     const std::size_t written_index =
479         checked_size(layer_id, "layer_id") * checked_size(this->config_.
480         max_context, "max_context") +
481         checked_size(model_position, "model_position");
481     this->written_[written_index] = 1;
482     this->filled_tokens_ = std::max(this->filled_tokens_, model_position + 1);
483 }
484
485 std::span<const float> ReferenceKVCache::key(std::int32_t layer_id,
486                                             std::int32_t model_position,
487                                             std::int32_t kv_head) const {
488     this->validate_address(layer_id, model_position, kv_head);
489     const std::size_t offset = this->base_offset(layer_id, model_position,
490     kv_head);
491     return std::span<const float>(this->keys_.data() + offset,
492     checked_size(this->config_.head_dim, "
493     head_dim"));
494 }
495
496 std::span<const float> ReferenceKVCache::value(std::int32_t layer_id,
497                                             std::int32_t model_position,
498                                             std::int32_t kv_head) const {
499     this->validate_address(layer_id, model_position, kv_head);
500     const std::size_t offset = this->base_offset(layer_id, model_position,
501     kv_head);
502     return std::span<const float>(this->values_.data() + offset,
503     checked_size(this->config_.head_dim, "
504     head_dim"));
505 }
506
507 std::int32_t ReferenceKVCache::layer_count() const noexcept {
508     return this->layer_count_;
509 }
510
511 std::int32_t ReferenceKVCache::filled_tokens() const noexcept {

```

```

508     return this->filled_tokens_;
509 }
510
511 std::size_t ReferenceKVCache::base_offset(std::int32_t layer_id,
512                                           std::int32_t model_position,
513                                           std::int32_t kv_head) const {
514     return (((checked_size(layer_id, "layer_id") *
515                    checked_size(this->config.max_context, "max_context")) +
516            checked_size(model_position, "model_position")) *
517            checked_size(this->config.kv_heads, "kv_heads") +
518            checked_size(kv_head, "kv_head")) *
519            checked_size(this->config.head_dim, "head_dim");
520 }
521
522 void ReferenceKVCache::validate_address(std::int32_t layer_id,
523                                         std::int32_t model_position,
524                                         std::int32_t kv_head) const {
525     if (layer_id < 0 || layer_id >= this->layer_count_) {
526         throw std::runtime_error("KV layer_id out of range");
527     }
528     if (model_position < 0 || model_position >= this->config.max_context) {
529         throw std::runtime_error("KV model_position out of range");
530     }
531     if (kv_head < 0 || kv_head >= this->config.kv_heads) {
532         throw std::runtime_error("KV head out of range");
533     }
534 }
535
536 void rms_norm(std::span<const float> input,
537              std::span<const float> weight,
538              float epsilon,
539              std::span<float> output) {
540     if (input.size() != weight.size() || input.size() != output.size()) {
541         throw std::runtime_error("RMSNorm size mismatch");
542     }
543     if (input.empty()) {
544         throw std::runtime_error("RMSNorm input is empty");
545     }
546     float sum_square = 0.0F;
547     for (const float value : input) {
548         sum_square += value * value;
549     }
550     const float mean_square = sum_square / static_cast<float>(input.size());
551     const float scale = 1.0F / std::sqrt(mean_square + epsilon);
552     for (std::size_t index = 0; index < input.size(); ++index) {
553         output[index] = input[index] * scale * weight[index];
554     }
555 }
556
557 void linear_f32(const LinearWeightsF32& layer,
558               std::span<const float> input,
559               std::span<float> output) {

```

```

560     validate_linear(layer, "linear_f32");
561     validate_span_size(input.size(), layer.input_size, "linear input");
562     validate_span_size(output.size(), layer.output_size, "linear output");
563     for (std::int32_t out = 0; out < layer.output_size; ++out) {
564         const std::size_t row_base =
565             checked_size(out, "out") * checked_size(layer.input_size, "↵
↵ input_size");
566         float accumulator = layer.bias.empty() ? 0.0F : layer.bias[checked_size↵
↵ (out, "out")];
567         for (std::int32_t in = 0; in < layer.input_size; ++in) {
568             accumulator += layer.weights[row_base + checked_size(in, "in")] *
569                 input[checked_size(in, "in")];
570         }
571         output[checked_size(out, "out")] = accumulator;
572     }
573 }
574
575 void apply_rope(std::span<float> heads,
576                std::int32_t head_count,
577                std::int32_t head_dim,
578                std::int32_t model_position,
579                float rope_base) {
580     require_positive(head_count, "head_count");
581     require_positive(head_dim, "head_dim");
582     validate_span_size(heads.size(), head_count * head_dim, "RoPE heads");
583     if (head_dim % LCQI_ROPE_PAIR_WIDTH != 0) {
584         throw std::runtime_error("RoPE head_dim must be even");
585     }
586     if (model_position < 0) {
587         throw std::runtime_error("RoPE model_position cannot be negative");
588     }
589     if (rope_base <= 1.0F) {
590         throw std::runtime_error("RoPE base must be greater than 1");
591     }
592     for (std::int32_t head = 0; head < head_count; ++head) {
593         for (std::int32_t offset = 0; offset < head_dim; offset += ↵
↵ LCQI_ROPE_PAIR_WIDTH) {
594             const float exponent = static_cast<float>(offset) / static_cast<↵
↵ float>(head_dim);
595             const float theta = 1.0F / std::pow(rope_base, exponent);
596             const float angle = static_cast<float>(model_position) * theta;
597             const float cosine = std::cos(angle);
598             const float sine = std::sin(angle);
599             const std::size_t first =
600                 checked_size(head, "head") * checked_size(head_dim, "head_dim")↵
↵ +
601                 checked_size(offset, "offset");
602             const float x0 = heads[first];
603             const float x1 = heads[first + 1];
604             heads[first] = x0 * cosine - x1 * sine;
605             heads[first + 1] = x0 * sine + x1 * cosine;
606         }

```

```

607     }
608 }
609
610 void attention_decode(const DecoderConfig& config,
611                     const ReferenceKVCache& cache,
612                     std::int32_t layer_id,
613                     std::int32_t model_position,
614                     std::span<const float> query,
615                     std::span<float> output) {
616     validate_config(config);
617     validate_span_size(query.size(), config.hidden_size, "attention query");
618     validate_span_size(output.size(), config.hidden_size, "attention output");
619     if (model_position < 0 || model_position >= cache.filled_tokens()) {
620         throw std::runtime_error("attention model_position has not been written
621     ↪ ");
622     }
623     std::fill(output.begin(), output.end(), 0.0F);
624
625     const std::int32_t group_size = config.query_heads / config.kv_heads;
626     const float score_scale = 1.0F / std::sqrt(static_cast<float>(config.kv_
627     ↪ head_dim));
628     std::vector<float> scores(check_size(model_position + 1, "score count"),
629                             LCQI_SOFTMAX_NEGATIVE_INFINITY);
630     std::vector<float> probabilities(scores.size(), 0.0F);
631
632     for (std::int32_t query_head = 0; query_head < config.query_heads; ++
633     ↪ query_head) {
634         const std::int32_t kv_head = query_head / group_size;
635         const std::size_t query_base =
636         ↪ checked_size(query_head, "query_head") * checked_size(config.kv_
637         ↪ head_dim, "head_dim");
638         float max_score = LCQI_SOFTMAX_NEGATIVE_INFINITY;
639         for (std::int32_t position = 0; position <= model_position; ++position)
640         ↪ {
641             const std::span<const float> key = cache.key(layer_id, position,
642             ↪ kv_head);
643             float dot = 0.0F;
644             for (std::int32_t dim = 0; dim < config.head_dim; ++dim) {
645                 dot += query[query_base + checked_size(dim, "dim")] *
646                 ↪ key[checked_size(dim, "dim")];
647             }
648             const float score = dot * score_scale;
649             scores[checked_size(position, "position")] = score;
650             max_score = std::max(max_score, score);
651         }
652
653         float probability_sum = 0.0F;
654         for (std::int32_t position = 0; position <= model_position; ++position)
655         ↪ {
656             const std::size_t index = checked_size(position, "position");
657             const float unnormalized = std::exp(scores[index] - max_score);
658             probabilities[index] = unnormalized;

```

```

652         probability_sum += unnormalized;
653     }
654     if (probability_sum <= 0.0F) {
655         throw std::runtime_error("attention probability sum is invalid");
656     }
657     for (std::int32_t position = 0; position <= model_position; ++position) ←
↪ {
658         const float probability =
659             probabilities[checked_size(position, "position")] / ←
↪ probability_sum;
660         const std::span<const float> value = cache.value(layer_id, position ←
↪ , kv_head);
661         for (std::int32_t dim = 0; dim < config.head_dim; ++dim) {
662             output[query_base + checked_size(dim, "dim")] +=
663                 probability * value[checked_size(dim, "dim")];
664         }
665     }
666 }
667 }
668
669 ReferenceDecodeResult run_reference_decode(const ReferenceDecoderModel& model,
670                                           std::span<const std::int32_t> ←
↪ token_ids) {
671     validate_config(model.config);
672     if (token_ids.empty()) {
673         throw std::runtime_error("reference decode needs at least one token");
674     }
675     if (token_ids.size() > checked_size(model.config.max_context, "max_context" ←
↪ )) {
676         throw std::runtime_error("token_ids exceed max_context");
677     }
678     if (model.token_embedding.size() !=
679         checked_size(model.config.vocab_size, "vocab_size") *
680         checked_size(model.config.hidden_size, "hidden_size")) {
681         throw std::runtime_error("token embedding size mismatch");
682     }
683     if (model.final_norm_weight.size() != checked_size(model.config.hidden_size ←
↪ , "hidden_size")) {
684         throw std::runtime_error("final norm weight size mismatch");
685     }
686     validate_linear(model.lm_head, "lm_head");
687
688     ReferenceKVCache cache(model.config, static_cast<std::int32_t>(model.layers ←
↪ .size()));
689     std::vector<float> hidden(checked_size(model.config.hidden_size, " ←
↪ hidden_size"), 0.0F);
690     for (std::int32_t position = 0; position < static_cast<std::int32_t>( ←
↪ token_ids.size());
691         ++position) {
692         const std::int32_t token_id = token_ids[checked_size(position, " ←
↪ position")];
693         if (token_id < 0 || token_id >= model.config.vocab_size) {

```

```

694         throw std::runtime_error("token id out of vocabulary");
695     }
696     const std::span<const float> embedding =
697         row_span(model.token_embedding, token_id, model.config.hidden_size)↵
↵ ;
698     std::copy(embedding.begin(), embedding.end(), hidden.begin());
699     for (std::int32_t layer_id = 0; layer_id < static_cast<std::int32_t>(↵
↵ model.layers.size());
700         ++layer_id) {
701         forward_layer(model.config,
702             model.layers[checked_size(layer_id, "layer_id")],
703             layer_id,
704             position,
705             cache,
706             hidden);
707     }
708 }
709
710 std::vector<float> normed(checked_size(model.config.hidden_size, "↵
↵ hidden_size"), 0.0F);
711 std::vector<float> logits(checked_size(model.config.vocab_size, "vocab_size↵
↵ "), 0.0F);
712 rms_norm(hidden, model.final_norm_weight, model.config.rms_epsilon, normed)↵
↵ ;
713 linear_f32(model.lm_head, normed, logits);
714
715 ReferenceDecodeResult result;
716 result.predicted_token = argmax(logits);
717 result.logits = std::move(logits);
718 return result;
719 }
720
721 ReferenceDecoderModel make_tiny_reference_decoder_model() {
722     ReferenceDecoderModel model;
723     model.config.hidden_size = 4;
724     model.config.query_heads = 2;
725     model.config.kv_heads = 1;
726     model.config.head_dim = 2;
727     model.config.intermediate_size = 6;
728     model.config.vocab_size = 5;
729     model.config.max_context = 8;
730     model.config.rms_epsilon = 1.0e-5F;
731     model.config.rope_base = 10000.0F;
732
733     model.token_embedding = {
734         0.20F, -0.10F, 0.05F, 0.30F,
735         -0.40F, 0.10F, 0.25F, -0.20F,
736         0.15F, 0.35F, -0.30F, 0.05F,
737         0.50F, -0.25F, 0.10F, -0.15F,
738         -0.05F, 0.20F, 0.40F, -0.35F,
739     };
740

```



```

741 DecoderLayerWeightsF32 layer;
742 layer.rms_attention_weight = {1.0F, 0.9F, 1.1F, 1.0F};
743 layer.wq = make_linear(4,
744                        4,
745                        {
746                            0.20F, -0.10F, 0.05F, 0.30F,
747                            -0.15F, 0.25F, 0.10F, -0.05F,
748                            0.05F, 0.10F, -0.20F, 0.15F,
749                            0.30F, -0.05F, 0.20F, 0.10F,
750                        });
751 layer.wk = make_linear(4,
752                        2,
753                        {
754                            0.10F, 0.20F, -0.10F, 0.05F,
755                            -0.05F, 0.15F, 0.25F, -0.20F,
756                        });
757 layer.wv = make_linear(4,
758                        2,
759                        {
760                            0.25F, -0.05F, 0.10F, 0.15F,
761                            -0.10F, 0.30F, 0.05F, 0.20F,
762                        });
763 layer.wo = make_linear(4,
764                        4,
765                        {
766                            0.20F, 0.10F, -0.05F, 0.15F,
767                            -0.10F, 0.25F, 0.20F, -0.05F,
768                            0.05F, -0.20F, 0.30F, 0.10F,
769                            0.15F, 0.05F, -0.10F, 0.25F,
770                        });
771 layer.rms_mlp_weight = {0.95F, 1.05F, 1.0F, 0.9F};
772 layer.w_gate = make_linear(4,
773                            6,
774                            {
775                                0.20F, -0.10F, 0.05F, 0.10F,
776                                -0.05F, 0.15F, 0.20F, -0.10F,
777                                0.10F, 0.05F, -0.15F, 0.25F,
778                                -0.20F, 0.10F, 0.15F, 0.05F,
779                                0.05F, -0.25F, 0.10F, 0.20F,
780                                0.15F, 0.05F, 0.05F, -0.15F,
781                            });
782 layer.w_up = make_linear(4,
783                          6,
784                          {
785                              0.10F, 0.20F, -0.05F, 0.05F,
786                              -0.15F, 0.05F, 0.25F, 0.10F,
787                              0.20F, -0.10F, 0.05F, 0.15F,
788                              0.05F, 0.10F, -0.20F, 0.25F,
789                              -0.05F, 0.15F, 0.10F, -0.10F,
790                              0.25F, -0.05F, 0.05F, 0.10F,
791                          });
792 layer.w_down = make_linear(6,

```

```

793         4,
794         {
795             0.10F, -0.05F, 0.15F, 0.20F, -0.10F, 0.05F,
796             -0.20F, 0.10F, 0.05F, -0.05F, 0.15F, 0.20F,
797             0.05F, 0.15F, -0.10F, 0.10F, 0.20F, -0.05F,
798             0.20F, 0.05F, 0.10F, -0.15F, -0.05F, 0.10F,
799         });
800     model.layers.push_back(std::move(layer));
801     model.final_norm_weight = {1.0F, 0.95F, 1.05F, 1.0F};
802     model.lm_head = make_linear(4,
803                                 5,
804                                 {
805                                     0.20F, -0.10F, 0.05F, 0.15F,
806                                     -0.05F, 0.25F, 0.10F, -0.20F,
807                                     0.15F, 0.05F, -0.25F, 0.10F,
808                                     -0.10F, 0.20F, 0.15F, 0.05F,
809                                     0.05F, -0.15F, 0.20F, 0.25F,
810                                 },
811                                 zeros(5));
812     return model;
813 }
814
815 } // namespace lcqi

```

Tokenizer 与 Safetensors 实现

`tokenizer.cpp` 把 prompt 变成 token id, 并固定 unknown token 行为。读者应检查 BOS/EOS 是否稳定插入, UNK 是否按合同处理。

Listing 10.9 src/tokenizer.cpp

```

1  #include <lcqi/tokenizer.hpp>
2
3  #include <cstdint>
4  #include <fstream>
5  #include <stdexcept>
6  #include <string>
7  #include <string_view>
8
9  namespace lcqi {
10 namespace {
11
12 constexpr std::uint64_t LCQI_FNV_OFFSET_BASIS = 1469598103934665603ULL;
13 constexpr std::uint64_t LCQI_FNV_PRIME = 1099511628211ULL;
14 constexpr std::int32_t LCQI_INVALID_TOKEN_ID = -1;
15
16 std::string read_key(std::istream& input) {
17     std::string key;
18     if (!(input >> key)) {
19         throw std::runtime_error("unexpected end of tokenizer file");
20     }
21     return key;

```

```

22 }
23
24 void expect_key(std::istream& input, const std::string& expected) {
25     const std::string key = read_key(input);
26     if (key != expected) {
27         throw std::runtime_error("expected tokenizer key '" + expected + "', ←
↪ got '" + key + "'");
28     }
29 }
30
31 std::int32_t read_i32(std::istream& input, const std::string& key) {
32     expect_key(input, key);
33     std::int32_t value = 0;
34     if (!(input >> value)) {
35         throw std::runtime_error("invalid tokenizer int32 value for key '" + ←
↪ key + "'");
36     }
37     return value;
38 }
39
40 std::string read_string(std::istream& input, const std::string& key) {
41     expect_key(input, key);
42     std::string value;
43     if (!(input >> value)) {
44         throw std::runtime_error("invalid tokenizer string value for key '" + ←
↪ key + "'");
45     }
46     return value;
47 }
48
49 std::int32_t lookup_token_id(const TokenizerModel& tokenizer, std::string_view ←
↪ text) {
50     for (const TokenEntry& entry : tokenizer.vocab) {
51         if (entry.text == text) {
52             return entry.id;
53         }
54     }
55     return LCQI_INVALID_TOKEN_ID;
56 }
57
58 void validate_tokenizer(const TokenizerModel& tokenizer) {
59     if (tokenizer.vocab.empty()) {
60         throw std::runtime_error("tokenizer vocab is empty");
61     }
62     for (std::size_t i = 0; i < tokenizer.vocab.size(); ++i) {
63         const TokenEntry& left = tokenizer.vocab[i];
64         if (left.id < 0) {
65             throw std::runtime_error("tokenizer token id must be non-negative")↪
↪ ;
66         }
67         for (std::size_t j = i + 1; j < tokenizer.vocab.size(); ++j) {
68             const TokenEntry& right = tokenizer.vocab[j];

```

```

69         if (left.id == right.id) {
70             throw std::runtime_error("duplicate tokenizer token id");
71         }
72         if (left.text == right.text) {
73             throw std::runtime_error("duplicate tokenizer token text");
74         }
75     }
76 }
77 if (tokenizer.bos_id != LCQI_INVALID_TOKEN_ID &&
78     lookup_token_id(tokenizer, "<bos>") != tokenizer.bos_id) {
79     throw std::runtime_error("tokenizer BOS id does not match vocab");
80 }
81 if (tokenizer.eos_id != LCQI_INVALID_TOKEN_ID &&
82     lookup_token_id(tokenizer, "<eos>") != tokenizer.eos_id) {
83     throw std::runtime_error("tokenizer EOS id does not match vocab");
84 }
85 }
86
87 std::uint64_t hash_byte(std::uint64_t hash, unsigned char value) {
88     hash ^= static_cast<std::uint64_t>(value);
89     hash *= LCQI_FNV_PRIME;
90     return hash;
91 }
92
93 std::uint64_t hash_string(std::uint64_t hash, std::string_view text) {
94     for (const unsigned char value : text) {
95         hash = hash_byte(hash, value);
96     }
97     return hash;
98 }
99
100 std::uint64_t hash_i32(std::uint64_t hash, std::int32_t value) {
101     const std::uint32_t bits = static_cast<std::uint32_t>(value);
102     for (std::int32_t shift = 0; shift < 32; shift += 8) {
103         hash = hash_byte(hash, static_cast<unsigned char>((bits >> shift) & 0x
104             ↪ xFFU));
105     }
106     return hash;
107 }
108 } // namespace
109
110 TokenizerModel load_tokenizer(const std::filesystem::path& path) {
111     std::ifstream input(path);
112     if (!input) {
113         throw std::runtime_error("cannot open tokenizer file: " + path.string()
114             ↪ );
115     }
116
117     expect_key(input, "LCQI_TOKENIZER_V1");
118
119     TokenizerModel tokenizer;

```

```

119     tokenizer.bos_id = read_i32(input, "bos_id");
120     tokenizer.eos_id = read_i32(input, "eos_id");
121     tokenizer.unk_id = read_i32(input, "unk_id");
122     tokenizer.chat_template_hash = read_string(input, "chat_template_hash");
123     const std::int32_t vocab_size = read_i32(input, "vocab_size");
124     if (vocab_size <= 0) {
125         throw std::runtime_error("tokenizer vocab_size must be positive");
126     }
127
128     tokenizer.vocab.reserve(static_cast<std::size_t>(vocab_size));
129     expect_key(input, "vocab");
130     for (std::int32_t i = 0; i < vocab_size; ++i) {
131         TokenEntry entry;
132         if (!(input >> entry.id >> entry.text)) {
133             throw std::runtime_error("invalid tokenizer vocab entry");
134         }
135         tokenizer.vocab.push_back(entry);
136     }
137
138     validate_tokenizer(tokenizer);
139     return tokenizer;
140 }
141
142 std::vector<std::int32_t> encode_prompt(const TokenizerModel& tokenizer,
143                                         std::string_view prompt) {
144     std::vector<std::int32_t> ids;
145     if (tokenizer.bos_id != LCQI_INVALID_TOKEN_ID) {
146         ids.push_back(tokenizer.bos_id);
147     }
148
149     std::size_t begin = 0;
150     while (begin < prompt.size()) {
151         while (begin < prompt.size() && prompt[begin] == ' ') {
152             ++begin;
153         }
154         if (begin >= prompt.size()) {
155             break;
156         }
157         std::size_t end = begin;
158         while (end < prompt.size() && prompt[end] != ' ') {
159             ++end;
160         }
161         const std::string_view token = prompt.substr(begin, end - begin);
162         const std::int32_t token_id = lookup_token_id(tokenizer, token);
163         if (token_id == LCQI_INVALID_TOKEN_ID) {
164             if (tokenizer.unk_id == LCQI_INVALID_TOKEN_ID) {
165                 throw std::runtime_error("tokenizer encountered unknown token");
166             }
167             ids.push_back(tokenizer.unk_id);
168         } else {
169             ids.push_back(token_id);

```

```

170     }
171     begin = end;
172 }
173
174 if (tokenizer.eos_id != LCQI_INVALID_TOKEN_ID) {
175     ids.push_back(tokenizer.eos_id);
176 }
177 return ids;
178 }
179
180 std::uint64_t tokenizer_contract_hash(const TokenizerModel& tokenizer) {
181     std::uint64_t hash = LCQI_FNV_OFFSET_BASIS;
182     hash = hash_string(hash, tokenizer.chat_template_hash);
183     hash = hash_i32(hash, tokenizer.bos_id);
184     hash = hash_i32(hash, tokenizer.eos_id);
185     hash = hash_i32(hash, tokenizer.unk_id);
186     for (const TokenEntry& entry : tokenizer.vocab) {
187         hash = hash_string(hash, entry.text);
188         hash = hash_i32(hash, entry.id);
189     }
190     return hash;
191 }
192
193 } // namespace lcqi

```

safetensors.cpp 解析 header、metadata、dtype、shape 和 data offsets。坏 dtype、坏 shape、坏 offset 和 payload 太短都应该被拒绝。

Listing 10.10 src/safetensors.cpp

```

1 #include <lcqi/safetensors.hpp>
2
3 #include <algorithm>
4 #include <cctype>
5 #include <cstdint>
6 #include <cmath>
7 #include <cstring>
8 #include <fstream>
9 #include <limits>
10 #include <stdexcept>
11 #include <string>
12 #include <string_view>
13 #include <vector>
14
15 namespace lcqi {
16 namespace {
17
18 constexpr std::size_t LCQI_SAFETENSORS_PREFIX_BYTES = 8;
19 constexpr std::uint64_t LCQI_SAFETENSORS_MAX_HEADER_BYTES = 16U * 1024U * 1024U↵
    ↵ ;
20 constexpr std::size_t LCQI_SAFETENSORS_OFFSET_COUNT = 2;
21 constexpr std::uint64_t LCQI_BYTE_BITS = 8;
22 constexpr std::uint64_t LCQI_DECIMAL_BASE = 10;

```

```

23 constexpr std::uint64_t LCQI_BYTE_MASK = 0xFFU;
24 constexpr unsigned char LCQI_JSON_CONTROL_CHAR_LIMIT = 0x20U;
25 constexpr std::uint64_t LCQI_DTYPE_BYTES_F64 = 8;
26 constexpr std::uint64_t LCQI_DTYPE_BYTES_F32 = 4;
27 constexpr std::uint64_t LCQI_DTYPE_BYTES_F16 = 2;
28 constexpr std::uint64_t LCQI_DTYPE_BYTES_I64 = 8;
29 constexpr std::uint64_t LCQI_DTYPE_BYTES_I32 = 4;
30 constexpr std::uint64_t LCQI_DTYPE_BYTES_I16 = 2;
31 constexpr std::uint64_t LCQI_DTYPE_BYTES_I8 = 1;
32 constexpr std::uint64_t LCQI_DTYPE_BYTES_U8 = 1;
33 constexpr std::uint32_t LCQI_F32_EXPONENT_MASK = 0x7F800000U;
34 constexpr std::uint32_t LCQI_F16_SIGN_MASK = 0x8000U;
35 constexpr std::uint32_t LCQI_F16_EXPONENT_MASK = 0x7C00U;
36 constexpr std::uint32_t LCQI_F16_MANTISSA_MASK = 0x03FFU;
37 constexpr std::uint32_t LCQI_F16_EXPONENT_BIAS = 15U;
38 constexpr std::uint32_t LCQI_F32_EXPONENT_BIAS = 127U;
39 constexpr std::uint32_t LCQI_F32_MANTISSA_BITS = 23U;
40 constexpr std::uint32_t LCQI_F16_MANTISSA_BITS = 10U;
41 constexpr std::uint32_t LCQI_F16_TO_F32_SIGN_SHIFT = 16U;
42
43 class HeaderCursor {
44 public:
45     explicit HeaderCursor(std::string_view text) : text_(text) {}
46
47     void expect(char expected) {
48         this->skip_ws();
49         if (this->eof() || this->text_[this->position_] != expected) {
50             throw std::runtime_error("invalid safetensors header syntax");
51         }
52         ++this->position_;
53     }
54
55     [[nodiscard]] bool consume(char expected) {
56         this->skip_ws();
57         if (!this->eof() && this->text_[this->position_] == expected) {
58             ++this->position_;
59             return true;
60         }
61         return false;
62     }
63
64     [[nodiscard]] std::string parse_string() {
65         this->skip_ws();
66         this->expect('\"');
67         std::string result;
68         while (!this->eof()) {
69             const char ch = this->text_[this->position_++];
70             if (ch == '\"') {
71                 return result;
72             }
73             if (static_cast<unsigned char>(ch) < LCQI_JSON_CONTROL_CHAR_LIMIT) ←
↪ {

```



```

74         throw std::runtime_error("control character in safetensors ↵
↵ string");
75     }
76     if (ch == '\\') {
77         result.push_back(this->parse_escape());
78     } else {
79         result.push_back(ch);
80     }
81 }
82 throw std::runtime_error("unterminated safetensors string");
83 }
84
85 [[nodiscard]] std::uint64_t parse_u64() {
86     this->skip_ws();
87     if (this->eof() ||
88 ↵ !std::isdigit(static_cast<unsigned char>(this->text_[this->↵
↵ position_]))) {
89         throw std::runtime_error("expected unsigned integer in safetensors ↵
↵ header");
90     }
91     std::uint64_t value = 0;
92     while (!this->eof() &&
93 ↵ std::isdigit(static_cast<unsigned char>(this->text_[this->↵
↵ position_]))) {
94         const std::uint64_t digit =
95             static_cast<std::uint64_t>(this->text_[this->position_] - '0');
96         if (value >
97 ↵ (std::numeric_limits<std::uint64_t>::max() - digit) / ↵
↵ LCQI_DECIMAL_BASE) {
98             throw std::runtime_error("safetensors integer overflow");
99         }
100         value = value * LCQI_DECIMAL_BASE + digit;
101         ++this->position_;
102     }
103     return value;
104 }
105
106 [[nodiscard]] std::vector<std::int64_t> parse_i64_array() {
107     std::vector<std::int64_t> values;
108     this->expect('[');
109     if (this->consume(' ')) {
110         return values;
111     }
112     while (true) {
113         const std::uint64_t raw = this->parse_u64();
114         if (raw > static_cast<std::uint64_t>(std::numeric_limits<std::↵
↵ int64_t>::max())) {
115             throw std::runtime_error("safetensors shape value is too large"↵
↵ );
116         }
117         values.push_back(static_cast<std::int64_t>(raw));
118         if (this->consume(' ')) {

```

```

119         return values;
120     }
121     this->expect(',');
122 }
123 }
124
125 [[nodiscard]] std::vector<std::uint64_t> parse_u64_array() {
126     std::vector<std::uint64_t> values;
127     this->expect('[');
128     if (this->consume(' ')) {
129         return values;
130     }
131     while (true) {
132         values.push_back(this->parse_u64());
133         if (this->consume(' ')) {
134             return values;
135         }
136         this->expect(',');
137     }
138 }
139
140 [[nodiscard]] bool eof_after_ws() {
141     this->skip_ws();
142     return this->eof();
143 }
144
145 private:
146     std::string_view text_;
147     std::size_t position_ = 0;
148
149     [[nodiscard]] bool eof() const {
150         return this->position_ >= this->text_.size();
151     }
152
153     void skip_ws() {
154         while (!this->eof() &&
155             std::isspace(static_cast<unsigned char>(this->text_[this->position_])) {
156             ++this->position_;
157         }
158     }
159
160     [[nodiscard]] char parse_escape() {
161         if (this->eof()) {
162             throw std::runtime_error("unterminated safetensors escape");
163         }
164         const char escaped = this->text_[this->position_++];
165         switch (escaped) {
166             case '"':
167             case '\\':
168             case '/':
169                 return escaped;

```

```

170         case 'b':
171             return '\\b';
172         case 'f':
173             return '\\f';
174         case 'n':
175             return '\\n';
176         case 'r':
177             return '\\r';
178         case 't':
179             return '\\t';
180         default:
181             throw std::runtime_error("unsupported safetensors string escape↵
↵ ");
182     }
183 }
184 };
185
186 std::uint64_t read_le_u64(std::span<const std::uint8_t> bytes) {
187     if (bytes.size() < LCQI_SAFETENSORS_PREFIX_BYTES) {
188         throw std::runtime_error("safetensors file is too small");
189     }
190     std::uint64_t value = 0;
191     for (std::size_t i = 0; i < LCQI_SAFETENSORS_PREFIX_BYTES; ++i) {
192         value |= static_cast<std::uint64_t>(bytes[i]) << (LCQI_BYTE_BITS * i);
193     }
194     return value;
195 }
196
197 std::string read_file_bytes(const std::filesystem::path& path) {
198     std::ifstream input(path, std::ios::binary);
199     if (!input) {
200         throw std::runtime_error("cannot open safetensors file: " + path.string↵
↵ ());
201     }
202     input.seekg(0, std::ios::end);
203     const std::streamoff size = input.tellg();
204     if (size < 0) {
205         throw std::runtime_error("cannot determine safetensors file size");
206     }
207     input.seekg(0, std::ios::beg);
208     std::string bytes(static_cast<std::size_t>(size), '\\0');
209     if (!bytes.empty() && !input.read(bytes.data(), static_cast<std::streamsize↵
↵ >(bytes.size())) {
210         throw std::runtime_error("cannot read safetensors file");
211     }
212     return bytes;
213 }
214
215 std::vector<std::uint8_t> read_file_u8(const std::filesystem::path& path) {
216     std::ifstream input(path, std::ios::binary);
217     if (!input) {
218         throw std::runtime_error("cannot open safetensors file: " + path.string↵

```

```

    ↪ ();
219 }
220 input.seekg(0, std::ios::end);
221 const std::streamoff size = input.tellg();
222 if (size < 0) {
223     throw std::runtime_error("cannot determine safetensors file size");
224 }
225 input.seekg(0, std::ios::beg);
226 std::vector<std::uint8_t> bytes(static_cast<std::size_t>(size), 0);
227 if (!bytes.empty() &&
228     !input.read(reinterpret_cast<char*>(bytes.data()),
229                 static_cast<std::streamsize>(bytes.size()))) {
230     throw std::runtime_error("cannot read safetensors file");
231 }
232 return bytes;
233 }
234
235 void parse_metadata_value(HeaderCursor& cursor,
236                           SafeTensorManifest& manifest,
237                           const std::string& key) {
238     if (key == "__metadata__") {
239         cursor.expect('{');
240         if (cursor.consume('}')) {
241             return;
242         }
243         while (true) {
244             const std::string metadata_key = cursor.parse_string();
245             cursor.expect(':');
246             const std::string metadata_value = cursor.parse_string();
247             manifest.metadata.emplace_back(metadata_key, metadata_value);
248             if (cursor.consume('}')) {
249                 return;
250             }
251             cursor.expect(',');
252         }
253     }
254     throw std::runtime_error("unexpected safetensors metadata value");
255 }
256
257 SafeTensorEntry parse_tensor_entry(HeaderCursor& cursor, const std::string& ↪
    ↪ name) {
258     SafeTensorEntry entry;
259     entry.name = name;
260     bool saw_dtype = false;
261     bool saw_shape = false;
262     bool saw_offsets = false;
263
264     cursor.expect('{');
265     if (cursor.consume('}')) {
266         throw std::runtime_error("empty safetensors tensor entry");
267     }
268     while (true) {

```

```

269     const std::string field = cursor.parse_string();
270     cursor.expect(':');
271     if (field == "dtype") {
272         entry.dtype = cursor.parse_string();
273         saw_dtype = true;
274     } else if (field == "shape") {
275         entry.shape = cursor.parse_i64_array();
276         saw_shape = true;
277     } else if (field == "data_offsets") {
278         const std::vector<std::uint64_t> offsets = cursor.parse_u64_array();
279         ↵ ;
280         if (offsets.size() != LCQI_SAFETENSORS_OFFSET_COUNT) {
281             throw std::runtime_error("safetensors data_offsets must contain ↵
282             ↵ two values");
283         }
284         entry.data_begin = offsets[0];
285         entry.data_end = offsets[1];
286         saw_offsets = true;
287     } else {
288         throw std::runtime_error("unsupported safetensors tensor field: " + ↵
289         ↵ field);
290     }
291     if (cursor.consume('}')) {
292         break;
293     }
294     cursor.expect(',');
295 }
296
297 if (!saw_dtype || !saw_shape || !saw_offsets) {
298     throw std::runtime_error("safetensors tensor entry is missing required ↵
299     ↵ fields");
300 }
301 if (entry.data_end < entry.data_begin) {
302     throw std::runtime_error("safetensors tensor offset range is invalid");
303 }
304 return entry;
305 }
306
307 void parse_header(std::string_view header, SafeTensorManifest& manifest) {
308     HeaderCursor cursor(header);
309     cursor.expect('{');
310     if (cursor.consume('}')) {
311         throw std::runtime_error("safetensors header is empty");
312     }
313
314     while (true) {
315         const std::string key = cursor.parse_string();
316         cursor.expect(':');
317         if (key == "__metadata__") {
318             parse_metadata_value(cursor, manifest, key);
319         } else {
320             manifest.tensors.push_back(parse_tensor_entry(cursor, key));

```

```

317     }
318     if (cursor.consume('}')) {
319         break;
320     }
321     cursor.expect(',');
322 }
323
324 if (!cursor.eof_after_ws()) {
325     throw std::runtime_error("trailing data after safetensors header");
326 }
327 }
328
329 std::uint64_t expected_tensor_bytes(const SafeTensorEntry& entry);
330
331 void validate_manifest(const SafeTensorManifest& manifest,
332                       std::uint64_t file_size) {
333     const std::uint64_t payload_size = file_size - manifest.data_start_offset;
334     for (std::size_t i = 0; i < manifest.tensors.size(); ++i) {
335         const SafeTensorEntry& left = manifest.tensors[i];
336         if (left.name.empty() || left.dtype.empty()) {
337             throw std::runtime_error("safetensors tensor has empty name or ↵
↵ dtype");
338         }
339         if (left.data_end > payload_size) {
340             throw std::runtime_error("safetensors tensor data_offsets exceed ↵
↵ payload size");
341         }
342         const std::uint64_t expected_bytes = expected_tensor_bytes(left);
343         if (left.byte_size() != expected_bytes) {
344             throw std::runtime_error("safetensors tensor byte size does not ↵
↵ match dtype and shape");
345         }
346         for (const std::int64_t dim : left.shape) {
347             if (dim < 0) {
348                 throw std::runtime_error("safetensors shape dimension is ↵
↵ negative");
349             }
350         }
351         for (std::size_t j = i + 1; j < manifest.tensors.size(); ++j) {
352             const SafeTensorEntry& right = manifest.tensors[j];
353             if (left.name == right.name) {
354                 throw std::runtime_error("duplicate safetensors tensor name");
355             }
356             const bool overlaps =
357                 left.data_begin < right.data_end && right.data_begin < left.↵
↵ data_end;
358             if (overlaps) {
359                 throw std::runtime_error("overlapping safetensors tensor data ↵
↵ range");
360             }
361         }
362     }

```

```

363 }
364
365 std::uint64_t dtype_byte_size(std::string_view dtype) {
366     if (dtype == "F64") {
367         return LCQI_DTYPE_BYTES_F64;
368     }
369     if (dtype == "F32") {
370         return LCQI_DTYPE_BYTES_F32;
371     }
372     if (dtype == "F16" || dtype == "BF16") {
373         return LCQI_DTYPE_BYTES_F16;
374     }
375     if (dtype == "I64" || dtype == "U64") {
376         return LCQI_DTYPE_BYTES_I64;
377     }
378     if (dtype == "I32" || dtype == "U32") {
379         return LCQI_DTYPE_BYTES_I32;
380     }
381     if (dtype == "I16" || dtype == "U16") {
382         return LCQI_DTYPE_BYTES_I16;
383     }
384     if (dtype == "I8") {
385         return LCQI_DTYPE_BYTES_I8;
386     }
387     if (dtype == "U8" || dtype == "B00L") {
388         return LCQI_DTYPE_BYTES_U8;
389     }
390     throw std::runtime_error("unsupported safetensors dtype: " + std::string(
391         dtype));
392 }
393
394 std::uint64_t expected_tensor_bytes(const SafeTensorEntry& entry) {
395     std::uint64_t element_count = 1;
396     for (const std::int64_t dim : entry.shape) {
397         if (dim < 0) {
398             throw std::runtime_error("safetensors shape dimension is negative");
399         }
400         const std::uint64_t extent = static_cast<std::uint64_t>(dim);
401         if (extent != 0 &&
402             element_count > std::numeric_limits<std::uint64_t>::max() / extent) {
403             throw std::runtime_error("safetensors shape element count overflow");
404         }
405         element_count *= extent;
406     }
407     const std::uint64_t dtype_bytes = dtype_byte_size(entry.dtype);
408     if (dtype_bytes != 0 &&
409         element_count > std::numeric_limits<std::uint64_t>::max() / dtype_bytes) {
410         throw std::runtime_error("safetensors tensor byte size overflow");
411     }
412 }

```



```

410     }
411     return element_count * dtype_bytes;
412 }
413
414 SafeTensorManifest parse_manifest_from_bytes(std::span<const std::uint8_t> ↵
    ↵ bytes) {
415     const std::uint64_t header_size = read_le_u64(bytes);
416     if (header_size == 0 || header_size > LCQI_SAFETENSORS_MAX_HEADER_BYTES) {
417         throw std::runtime_error("safetensors header size is invalid");
418     }
419     const std::uint64_t data_start_offset =
420         static_cast<std::uint64_t>(LCQI_SAFETENSORS_PREFIX_BYTES) + header_size↵
    ↵ ;
421     if (data_start_offset > bytes.size()) {
422         throw std::runtime_error("safetensors header extends beyond file size")↵
    ↵ ;
423     }
424
425     SafeTensorManifest manifest;
426     manifest.header_size = header_size;
427     manifest.data_start_offset = data_start_offset;
428     const char* header_begin =
429         reinterpret_cast<const char*>(bytes.data() + ↵
    ↵ LCQI_SAFETENSORS_PREFIX_BYTES);
430     const std::string_view header(header_begin, static_cast<std::size_t>(↵
    ↵ header_size));
431     parse_header(header, manifest);
432     validate_manifest(manifest, static_cast<std::uint64_t>(bytes.size()));
433     return manifest;
434 }
435
436 std::uint16_t read_le_u16(std::span<const std::uint8_t> bytes, std::size_t ↵
    ↵ offset) {
437     return static_cast<std::uint16_t>(
438         static_cast<std::uint16_t>(bytes[offset]) |
439         (static_cast<std::uint16_t>(bytes[offset + 1]) << LCQI_BYTE_BITS));
440 }
441
442 std::uint32_t read_le_u32(std::span<const std::uint8_t> bytes, std::size_t ↵
    ↵ offset) {
443     std::uint32_t value = 0;
444     for (std::size_t i = 0; i < LCQI_DTYPE_BYTES_F32; ++i) {
445         value |= static_cast<std::uint32_t>(bytes[offset + i]) << (↵
    ↵ LCQI_BYTE_BITS * i);
446     }
447     return value;
448 }
449
450 float bits_to_float(std::uint32_t bits) {
451     float value = 0.0F;
452     std::memcpy(&value, &bits, sizeof(value));
453     return value;

```

```

454 }
455
456 float f16_to_f32(std::uint16_t bits) {
457     const std::uint32_t sign = (static_cast<std::uint32_t>(bits) & ↵
    ↵ LCQI_F16_SIGN_MASK)
458         << LCQI_F16_TO_F32_SIGN_SHIFT;
459     const std::uint32_t exponent = static_cast<std::uint32_t>(bits) & ↵
    ↵ LCQI_F16_EXPONENT_MASK;
460     const std::uint32_t mantissa = static_cast<std::uint32_t>(bits) & ↵
    ↵ LCQI_F16_MANTISSA_MASK;
461
462     if (exponent == 0U) {
463         if (mantissa == 0U) {
464             return bits_to_float(sign);
465         }
466         float value = static_cast<float>(mantissa) /
467             static_cast<float>(1U << LCQI_F16_MANTISSA_BITS);
468         value = std::ldexp(value, 1 - static_cast<int>(LCQI_F16_EXPONENT_BIAS))↵
    ↵ ;
469         return (sign == 0U) ? value : -value;
470     }
471
472     if (exponent == LCQI_F16_EXPONENT_MASK) {
473         const std::uint32_t f32_mantissa =
474             mantissa << (LCQI_F32_MANTISSA_BITS - LCQI_F16_MANTISSA_BITS);
475         return bits_to_float(sign | LCQI_F32_EXPONENT_MASK | f32_mantissa);
476     }
477
478     const std::uint32_t f32_exponent =
479         ((exponent >> LCQI_F16_MANTISSA_BITS) - LCQI_F16_EXPONENT_BIAS +
480         LCQI_F32_EXPONENT_BIAS)
481         << LCQI_F32_MANTISSA_BITS;
482     const std::uint32_t f32_mantissa =
483         mantissa << (LCQI_F32_MANTISSA_BITS - LCQI_F16_MANTISSA_BITS);
484     return bits_to_float(sign | f32_exponent | f32_mantissa);
485 }
486
487 float bf16_to_f32(std::uint16_t bits) {
488     return bits_to_float(static_cast<std::uint32_t>(bits) << 16U);
489 }
490
491 std::span<const std::uint8_t> tensor_payload_span(const SafeTensorFile& file,
492     const SafeTensorEntry& entry)↵
    ↵ {
493     const std::uint64_t begin = file.manifest.data_start_offset + entry.↵
    ↵ data_begin;
494     const std::uint64_t end = file.manifest.data_start_offset + entry.data_end;
495     if (begin > end || end > file.bytes.size()) {
496         throw std::runtime_error("safetensors tensor byte range is invalid");
497     }
498     return std::span<const std::uint8_t>(
499         file.bytes.data() + static_cast<std::size_t>(begin),

```

```

500     static_cast<std::size_t>(end - begin));
501 }
502
503 } // namespace
504
505 std::uint64_t SafeTensorEntry::byte_size() const {
506     return this->data_end - this->data_begin;
507 }
508
509 const SafeTensorEntry* SafeTensorManifest::find_tensor(std::string_view name) ←
    ↪ const {
510     const auto found = std::find_if(
511         this->tensors.begin(),
512         this->tensors.end(),
513         [name](const SafeTensorEntry& entry) {
514             return entry.name == name;
515         });
516     if (found == this->tensors.end()) {
517         return nullptr;
518     }
519     return &(*found);
520 }
521
522 SafeTensorManifest load_safetensors_manifest(const std::filesystem::path& path) ←
    ↪ {
523     const std::string bytes = read_file_bytes(path);
524     return parse_manifest_from_bytes(
525         std::span<const std::uint8_t>(
526             reinterpret_cast<const std::uint8_t*>(bytes.data()),
527             bytes.size()));
528 }
529
530 SafeTensorFile load_safetensors_file(const std::filesystem::path& path) {
531     SafeTensorFile file;
532     file.path = path;
533     file.bytes = read_file_u8(path);
534     file.manifest = parse_manifest_from_bytes(file.bytes);
535     return file;
536 }
537
538 std::span<const std::uint8_t> read_safetensor_tensor_bytes(
539     const SafeTensorFile& file,
540     std::string_view name) {
541     const SafeTensorEntry* entry = file.manifest.find_tensor(name);
542     if (entry == nullptr) {
543         throw std::runtime_error("missing safetensors tensor: " + std::string(←
    ↪ name));
544     }
545     return tensor_payload_span(file, *entry);
546 }
547
548 std::vector<float> read_safetensor_f32_tensor(const SafeTensorFile& file,

```

```

549         std::string_view name) {
550     const SafeTensorEntry* entry = file.manifest.find_tensor(name);
551     if (entry == nullptr) {
552         throw std::runtime_error("missing safetensors tensor: " + std::string(↵
↵ name));
553     }
554     const std::span<const std::uint8_t> bytes = tensor_payload_span(file, *↵
↵ entry);
555     std::vector<float> values;
556     if (entry->dtype == "F32") {
557         values.resize(bytes.size() / LCQI_DTYPE_BYTES_F32);
558         for (std::size_t index = 0; index < values.size(); ++index) {
559             values[index] = bits_to_float(
560                 read_le_u32(bytes, index * LCQI_DTYPE_BYTES_F32));
561         }
562         return values;
563     }
564     if (entry->dtype == "F16") {
565         values.resize(bytes.size() / LCQI_DTYPE_BYTES_F16);
566         for (std::size_t index = 0; index < values.size(); ++index) {
567             values[index] = f16_to_f32(
568                 read_le_u16(bytes, index * LCQI_DTYPE_BYTES_F16));
569         }
570         return values;
571     }
572     if (entry->dtype == "BF16") {
573         values.resize(bytes.size() / LCQI_DTYPE_BYTES_F16);
574         for (std::size_t index = 0; index < values.size(); ++index) {
575             values[index] = bf16_to_f32(
576                 read_le_u16(bytes, index * LCQI_DTYPE_BYTES_F16));
577         }
578         return values;
579     }
580     throw std::runtime_error("safetensors tensor is not float-compatible: " + ↵
↵ entry->name);
581 }
582
583 } // namespace lcqi

```

GPT-2 Reference Path

`gpt2_reference.cpp` 包括 JSON 配置读取、byte-level BPE、safetensors F32/F16/BF16 张量读取、HuggingFace 权重名前缀解析、LayerNorm、packed QKV、causal attention、GELU、tied lm head、full-prefix greedy generation 和 cached-KV generation。它仍然是正确性优先的 reference path，但现在已经开始承接前三册的优化方法：先建立 baseline，再改 hot path，再用测试和 A/B benchmark 决定改动是否能进入书。

一次可审查的 cached-KV 优化

优化不能从“感觉慢”开始。LCQI 先用 `makecpu3-gpt2-benchmark-compare` 证明 full-prefix 每生成一个 token 都重算前缀，而 cached-KV 已经把前缀复用下来；随后再看 cached path 内部。旧实现每处理一个 token、每经过一层 Transformer，都会重新分配 `normed`、`query`、`key`、

`value`、`attention`、`projected`、`mlp_fc`、`mlp_out` 和 `attention scores`。同时，生成路径只需要下一个 greedy token，却仍然每步构造完整 `Gpt2ForwardResult.logits`。这些问题不是算法语义错误，但会在真实 GPT-2 的 decode loop 里形成额外分配、重复 shape validation、重复边界转换和不必要的结果写回。

本轮优化对应三册里的三条原则。第一册要求从热循环和二进制可观察行为出发，所以先量 `lm_head`、MLP、QKV、attention projection 和 attention 本体各占多少，再决定改哪里。第二册要求减少重复分配和内存扰动，所以新增 `Gpt2ForwardWorkspace`，把 hidden、QKV、MLP、scores 和 logits 缓冲区的生命周期提升到整段 cached generation。第三册要求按 decoder runtime 的状态组织代码，所以新增 `Gpt2CachedGreedyDecoder`，让 KV cache、workspace、worker pool 和模型验证跟随一次生成会话，而不是散落在每个 token 的临时对象里。

代码上有五处关键边界。第一，旧的 `run_gpt2_forward_cached` 仍然保留完整 logits API，测试和源码阅读可以继续用它和 full-prefix logits 做逐项对照。第二，CLI 的 `cached_kv` 生成路径改用 `Gpt2CachedGreedyDecoder::step`，只返回 greedy token；需要完整 logits 的测试使用 `step_with_logits`。第三，prompt prefill 的前缀 token 使用 `advance_without_prediction`，只更新 KV cache，不做 final norm 和 `lm_head`，因为这些中间预测会被后一个 prompt token 立即覆盖。第四，KV cache 对外仍保留 checked `key/value` API，内部 attention 在一次地址和 workspace 校验后使用私有 unchecked span 扫描热路径，避免把“不检查”的能力暴露给普通调用者。第五，线程池只属于 cached decoder 会话；`--threads0` 自动选择 worker 数，`--threads1` 强制串行，`--threadsN` 固定线程数，避免把全局可变线程服务塞进 reference model。

正确性证据在 `tests/lcqi_tests.cpp`。tiny GPT-2 同一个 prompt 同时覆盖 full-prefix logits、旧 cached logits、优化 decoder logits、强制两 worker 的并行 decoder logits 和 logits-free greedy token。它要求 predicted token、logits size、逐项 logit 误差、KV cache filled token 数和 greedy generated ids 全部一致。这样优化后的代码不是“跑得快但不知道还对不对”，而是先通过 tiny golden，再进入真实 GPT-2 benchmark。

性能证据不只看一次运行。新增 `tools/run_gpt2_optimization_ab.py` 会从指定 baseline commit 建临时 worktree，baseline 和当前工作区都通过 `books/cpu-volume-3-practice` 的 Release CMake 入口构建，再用同一份 GPT-2 F32 safetensors 模型跑多轮 full-prefix 与 cached-KV。后来远端 `caw` 不能直接复用本机 SSH remote 别名时，本轮采用等价的源码包 A/B：本机用 `gitarchive73f40c7` 导出 baseline，用当前工作区导出 current，二者在 `caw` 上分别 Release 构建并运行同一模型。raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-threaded-manual-ab-caw.txt`，汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-threaded-optimization-summary.md`。5 轮中位数显示，4 token cached-KV 生成阶段从 395.397ms 到 76.4141ms，16 token 从 989.060ms 到 185.479ms；生成吞吐分别到 52.3464token/s 和 86.2633token/s。

用户看到上一轮提升幅度后，正确反应不是继续猜，而是重新量热点。新增 `--profile-hotspots` 和 `tools/run_gpt2_hotspot_profile.py` 后，可以把 cached decode 的时间拆成 embedding、LayerNorm、QKV projection、KV append、attention、attention projection、MLP fc、GELU、MLP projection 和 lm head。远端 `caw` 使用 Intel Core Ultra 7 265KF、GCC 15.2.1、Release 构建、GPT-2 F32 safetensors，跑 `Hello, mynameis` 这个 prompt，优化前 3 轮中位数保存在 `books/cpu-volume-3/reports/lcqi-gpt2-hotspot-profile-summary.md`：生成 4 个 token 时，`lm_head` 约占 30.2423%，MLP 的 `c_fc` 和 `c_proj` 合计约 46.4228%，QKV projection 约 17.2127%，cached attention 本体只有 0.0558254%。这说明慢点不是 attention，而是大量互相独立的行点积。

这一轮优化据此把 `lm_head` greedy argmax 和 GPT-2 F32 `linear` 的输出行切给持久 worker pool。每一行仍然是同一个 dot product；不同线程只负责不同输出行。完整 logits 路径把 logits buffer 分片写入，logits-free greedy 路径每个分片先找本地 best token，再按分片顺序合并，用严格大于比较保留较小 token id 的 tie-breaking。这样并行改变调度，不改变数学定义。

还有一个细节说明为什么优化必须带着 profiler 走。第一次 auto 阈值只按“一百万次乘加”判断是否并行，结果 `768\times768` 的 attention output projection 仍走串行。中间 profile 立即显示 attention projection 从约 6% 反弹到约 25%，成为新瓶颈。最终阈值改成：输出行数至少 512，且输入列数至少 512 时也允许并行。最终 profile 保存在 `books/cpu-volume-3/reports/`

`lcqi-gpt2-threaded-hotspot-caw.txt`: 4 token 生成中位数 **75.9807ms**, 16 token **184.780ms**; `lm_head` 约 **20%**, MLP 两个投影合计约 **45%**, QKV 约 **19%**, attention projection 约 **12%**, attention 本体仍低于 **1%**。

用户继续问和 llama.cpp 的差距时, 下一步就不能再停留在“行并行”。LCQI 新增 `include/lcqi/f32_kernels.hpp`、`src/f32_kernels.cpp` 和 `src/f32_kernels_avx2.cpp`, 把 GPT-2 cached 热路径的单行点积抽成 `dot_f32_unchecked()`。checked `dot_f32()` 负责普通 span 长度检查, unchecked 入口给已经完成 shape validation 的 `linear`、logits 和 logits-free `lm_head` 使用; x86-64/GNU/Clang 构建会编译 AVX2/FMA 文件, 运行时再用 `dot_f32_avx2_available()` 判断是否能执行。这样 AVX intrinsics 不进入 GPT-2 层逻辑, 后续替换 AVX-512、int8 或 int4 block kernel 时, 边界仍然清楚。

`tests/lcqi_tests.cpp` 也随之增加 F32 dot kernel 对照: 同一组固定输入先跑 scalar, 再跑 dispatch; 如果当前 CPU 支持 AVX2/FMA, 再显式跑 AVX2 实现, 并要求误差落在 **1.0e-4** 内。测试还会传入长度不一致的 span, 确认 checked API 会拒绝。这样优化路径不是“编译过就算”, 而是有 scalar oracle、运行时能力判断和错误输入边界。

这轮 F32 dot 优化以 `cb4d89f` 为 baseline, 在 `caw` 上用同一 GPT-2 F32 模型、Release 构建和 `--threads0` 自动 16 workers 跑 5 轮。raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-dot-optimization-summary.md`。4 token 生成中位数从 **74.4247ms** 到 **62.3628ms**, 约 **1.19x**; 16 token 从 **181.257ms** 到 **157.230ms**, 约 **1.15x**。decode 中位数分别从 **26.1887ms** 到 **23.0640ms**, 从 **132.201ms** 到 **117.734ms**。这个提升比线程化小, 因为它只缩短每一行内部 dot, 而没有减少行数、层数、vocab 扫描或内存流量。

继续顺着热点走, 下一步不是在 GPT-2 层里散落更多 intrinsics, 而是把 kernel 边界从“单行 dot”扩成“row range”。新增的 `linear_f32_rows_unchecked()` 一次计算一段输出行, `max_dot_f32_rows_unchecked()` 一次扫描一段 vocab 行并返回局部最大值; `Gpt2ParallelWorkerPool::parallel_for_rows()` 改成模板入口, 去掉每个热路径任务里的 `std::function` 构造和间接调用。AVX2 文件里还增加 4-row row-range kernel: 同一组 input 向量 load 后, 同时喂给 4 个输出行累加器。这个写法已经接近真实 matvec kernel 的形状: 不是每一行各自重新走完整控制流, 而是在一个小 tile 里复用输入。

这轮 row-range 优化的 raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-f32-rowrange-optimization-summary.md`。相对 `ab72ccb` 这个 F32 dot baseline, 7 轮中位数显示 4 token 端到端生成从 **60.6291ms** 到 **61.1045ms**, 属于噪声内波动; 16 token 从 **153.354ms** 到 **152.574ms**, 约 **1.01x**。但把 row-range 版本和 4-row AVX2 row-range 版本直接对比, 4 token 从 **62.0243ms** 到 **60.3965ms**, 16 token 从 **154.087ms** 到 **151.222ms**。所以这段代码可以保留, 但结论要诚实: 它证明 tile 化 row-range 有正收益, 却不是数量级变化。

最终 hotspot 仍然说明差距在哪里。row-range 后, 16 token profile 中 `lm_head` 约 **22.3075%**, MLP `c_fc` 约 **22.4147%**, MLP projection 约 **22.448%**, QKV projection 约 **18.6294%**, attention projection 约 **11.1687%**, cached attention 本体只有 **0.653274%**。也就是说, 小 kernel 和调度优化已经把 reference path 又往前推了一点, 但主要时间仍在 F32 row-major matvec。真实 llama.cpp/ggml 的大优势来自 GGUF、block quantization、packed layout、cache-friendly matmul/matvec、mmap 和成熟线程调度, 不是只把裸 F32 行循环写得更精巧。

这个结论必须保守阅读。真实端到端 GPT-2 benchmark 受调度、频率、温度和共享机器状态影响明显, 单次结果可能反向波动。因此报告里必须同时看 `median/min/max`、生成文本是否一致、baseline 和 current 是否同一构建入口。更大的差距仍然不在这轮 F32 row-range 里: LCQI 还没有 GGUF 量化导入、packed/quantized lm head、量化 block 权重、分页 KV cache、NUMA/亲和和成熟调度。也就是说, 本轮把 reference path 从“行点积有 SIMD 内核边界”推进到“输出行段有可替换 tile kernel”, 但还没有把它变成 llama.cpp 级推理引擎。

继续看热点时, 有一类优化不在 kernel 内部, 而在调用链上。cached generation 的 prompt prefill 要把 prompt 的每个 token 写入 KV cache; 但如果 prompt 长度是 **N**, 前 **N-1** 个 prompt token 的 next-token prediction 根本不会被输出, 也不会作为后续输入。只有最后一个 prompt token 的 prediction 会成为第一个新生成 token。旧代码每个 prompt token 都跑 final norm 和 `lm_head`, 等

于把前缀 token 的预测算出来后马上丢掉。

因此新增 `Gpt2CachedGreedyDecoder::advance_without_prediction()`。它复用同一条 cached layer path, 照样执行 embedding、每层 QKV、attention、MLP 和 KV append, 只是在层结束后直接返回, 不进入 final norm 和 `lm_head`。CLI 和 `gpt2_generate_greedy_cached()` 的 prefill 改成: 前缀 token 调 `advance_without_prediction()`, 最后一个 prompt token 调 `step()`。这不是近似优化, 因为被跳过的预测值在旧流程中没有任何 caller 读取。

远端 `caw` 上 7 轮 A/B 证明这条优化有效, raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-prefill-skip-optimization-summary.md`。同一 GPT-2 F32 模型、同一 prompt、同一 `--threads0` 下, 4 token 生成中位数从 `60.5154ms` 到 `52.1949ms`, 约快 **13.75%**; 16 token 从 `153.574ms` 到 `147.424ms`, 约快 **4.00%**。收益在短生成里更明显, 因为 prompt prefill 占总时间比例更大; decode 段几乎不变, 因为真正生成阶段仍然必须预测下一个 token。所有轮次生成文本和 token id 都一致。

这一轮还做过一个没有进入主线的 packed F32 `lm_head` 实验: 把 `lm_head` 权重按 8 个输出一组重排, 让 AVX2 一次累计 8 个 token score。它能让 `lm_head_ms` 小幅下降, 但 `caw` 多轮 A/B 没有稳定端到端收益, 甚至会让部分 decode 口径略慢。这个结果本身也要写进工程判断: 不是每个看起来更接近 SIMD 的布局都值得进入教材主线; 没有稳定证据的优化只能留作实验记录, 不能伪装成成功案例。

继续追问“为什么提升还是不够大”时, 下一步不是再猜一个更复杂的内核, 而是把已经接受的 profile 再拆开看。row-range 之后, GPT-2 cached path 每个 token 会在 12 层里反复调用 QKV projection、attention projection、MLP `c_fc`、MLP projection; 再加上需要预测 token 时的 `lm_head`。这些矩阵向量乘每次本身不算巨型 GEMM, 却会频繁进入 `parallel_for_rows()`。也就是说, 热点不只有“每行点积慢”, 还有“每个小并行区唤醒 worker 的成本”。如果 worker pool 每次都拿 mutex、发 condition variable、再等完成信号, 那么短生成里会把调度成本支付几百次。

因此这一轮接受的优化没有改数学公式, 而是改 worker pool 的交接方式。旧版本用 `std::mutex`、`std::condition_variable` 和 `finished_` 条件变量发布任务; 新版本把任务元数据写入原子字段, 再用 `generation` 作为发布点。后台 worker 看到新的 generation 后按固定 worker index 取自己的行块, 做完后递减 `active_workers`; 主线程自己处理最后一个行块, 然后用 bounded spin/yield 等后台 worker 归零。代码里特意把这个池限制在 `Gpt2CachedGreedyDecoder` 会话内, 没有变成全局忙等线程服务; 当 `--threads1` 或模型规模不够并行时, 路径仍然直接串行执行。

这个设计有三个边界必须看懂。第一, 任务上下文仍然是栈上的 `ParallelTaskContext`, 所以所有任务元数据必须先写完, 再递增 `generation`; worker 对 `generation` 做 acquire load 后才允许读上下文。第二, 后台 worker 不再争抢全局 `next_worker_index`, 而是启动时固定自己的 worker index; 这样不会出现旧任务里空闲 worker 和下一轮任务元数据交叉的问题。第三, spin 不是无限空转: 短时间使用 x86 pause, 超过阈值后让出调度; 这适合当前 cached decode 的短小并行区, 但不应该被复制到长时间等待的 I/O 或服务端队列里。

`caw` 上 9 轮 A/B 证明这条优化是本轮真正站得住的提升。raw 报告保存在 `books/cpu-volume-3/reports/lcqi-gpt2-spinpool-ab-caw.txt`, 汇总保存在 `books/cpu-volume-3/reports/lcqi-gpt2-spinpool-optimization-summary.md`。相对旧 auto 16 worker baseline, 4 token 生成中位数从 `53.7887ms` 到 `37.5403ms`, 约快 **30.21%**; 16 token 从 `146.930ms` 到 `104.107ms`, 约快 **29.15%**; 64 token 从 `510.650ms` 到 `377.233ms`, 约快 **26.13%**。只和 auto8 condition-variable 版本比较, 也能看到 4/16/64 token 分别约 **25.67%**、**29.56%**、**20.44%** 的生成阶段下降; 固定 `--threads8` 下也有约 **18%** 到 **23%** 的下降。这说明收益不是只来自 worker 数从 16 改成 8, 而是高频并行区交接成本确实降下来了。

热点证据也吻合原因。auto 口径下, 4 token 的 QKV projection 中位数约快 **33.58%**, attention projection 约快 **63.30%**, MLP `c_fc` 约快 **30.36%**, MLP projection 约快 **29.08%**; 而 `lm_head` 只约快 **8.57%**。原因是 QKV 和 MLP 这些线性层在每层每 token 都调用小并行区, 调度成本占比高; `lm_head` 是一次更大的 vocab 扫描, 仍然主要受内存流量和点积内核影响。这个差异比一句“线程池更快”更重要, 因为它告诉读者优化到底打中了哪个成本。

本轮还试过 8-row AVX2 row-range 候选: 把现有一次处理 4 个输出行的 row-range kernel 扩

成一次处理 8 行。它没有进入主线，因为 `caw` 数据没有稳定收益：auto 口径下 4 token 从 **37.4937ms** 到 **37.7082ms**，16 token 从 **103.714ms** 到 **103.178ms**，64 token 从 **374.439ms** 到 **374.457ms**；固定 `--threads8` 下 64 token 还从 **371.807ms** 退到 **377.053ms**。这说明寄存器压力、load/store 形状和调度噪声抵消了多行复用 input 的设想，不能因为“8 行看起来比 4 行更充分”就合并。

至此可以重新回答和 llama.cpp 的差距。LCQI 现在已经有 cached KV、workspace 复用、prefill 跳过无用 prediction、F32 row-range AVX2、行并行和轻量 worker hand-off；这些能把 reference path 推进很多，但它仍然主要在 F32 row-major matvec 上花时间。llama.cpp/ggml 的优势不只是“也用了线程”，而是 GGUF 模型格式、block quantization、量化权重布局、成熟 matvec microkernel、mmap 加载、NUMA/亲和策略和大量 shape-specific 调参一起工作。LCQI 这轮优化的价值是把调度证据链讲清楚，并把失败候选挡在书外；下一阶段如果继续接近 llama.cpp，应优先做 GGUF/量化权重路径和量化 matvec，而不是继续在裸 F32 row-major 上堆小修小补。

Listing 10.11 src/f32_kernels.cpp

```
1 #include <lcqi/f32_kernels.hpp>
2
3 #include <limits>
4 #include <stdexcept>
5
6 namespace lcqi {
7 namespace {
8
9 constexpr std::size_t LCQI_F32_AVX2_MIN_DOT_SIZE = 32;
10 constexpr std::size_t LCQI_F32_AVX2_MIN_ROW_COUNT = 4;
11
12 } // namespace
13
14 float dot_f32(std::span<const float> lhs, std::span<const float> rhs) {
15     if (lhs.size() != rhs.size()) {
16         throw std::runtime_error("dot_f32 size mismatch");
17     }
18     return dot_f32_unchecked(lhs.data(), rhs.data(), lhs.size());
19 }
20
21 float dot_f32_scalar_unchecked(const float* lhs,
22                                const float* rhs,
23                                std::size_t size) noexcept {
24     float sum = 0.0F;
25     for (std::size_t index = 0; index < size; ++index) {
26         sum += lhs[index] * rhs[index];
27     }
28     return sum;
29 }
30
31 float dot_f32_unchecked(const float* lhs,
32                          const float* rhs,
33                          std::size_t size) noexcept {
34     static const bool AVX2_AVAILABLE = dot_f32_avx2_available();
35     if (AVX2_AVAILABLE && size >= LCQI_F32_AVX2_MIN_DOT_SIZE) {
36         return dot_f32_avx2_unchecked(lhs, rhs, size);
37     }
38     return dot_f32_scalar_unchecked(lhs, rhs, size);
39 }
```



```

40
41 void linear_f32_rows_scalar_unchecked(const float* weights,
42                                       const float* input,
43                                       const float* bias,
44                                       std::size_t input_size,
45                                       std::size_t row_begin,
46                                       std::size_t row_end,
47                                       float* output) noexcept {
48     for (std::size_t row = row_begin; row < row_end; ++row) {
49         const float* weight_row = weights + row * input_size;
50         output[row] = dot_f32_scalar_unchecked(weight_row, input, input_size) +
51                     (bias == nullptr ? 0.0F : bias[row]);
52     }
53 }
54
55 void linear_f32_rows_unchecked(const float* weights,
56                               const float* input,
57                               const float* bias,
58                               std::size_t input_size,
59                               std::size_t row_begin,
60                               std::size_t row_end,
61                               float* output) noexcept {
62     static const bool AVX2_AVAILABLE = dot_f32_avx2_available();
63     if (AVX2_AVAILABLE && input_size >= LCQI_F32_AVX2_MIN_DOT_SIZE &&
64         row_end - row_begin >= LCQI_F32_AVX2_MIN_ROW_COUNT) {
65         linear_f32_rows_avx2_unchecked(weights,
66                                         input,
67                                         bias,
68                                         input_size,
69                                         row_begin,
70                                         row_end,
71                                         output);
72         return;
73     }
74     linear_f32_rows_scalar_unchecked(weights, input, bias, input_size, ↵
75     ↵ row_begin, row_end, output);
76 }
77 F32RowMax max_dot_f32_rows_scalar_unchecked(const float* weights,
78                                              const float* input,
79                                              std::size_t input_size,
80                                              std::size_t row_begin,
81                                              std::size_t row_end) noexcept {
82     F32RowMax best;
83     best.row = row_begin;
84     best.value = -std::numeric_limits<float>::infinity();
85     for (std::size_t row = row_begin; row < row_end; ++row) {
86         const float* weight_row = weights + row * input_size;
87         const float value = dot_f32_scalar_unchecked(weight_row, input, ↵
88         ↵ input_size);
89         if (row == row_begin || value > best.value) {
90             best.row = row;

```

```

90         best.value = value;
91     }
92 }
93 return best;
94 }
95
96 F32RowMax max_dot_f32_rows_unchecked(const float* weights,
97                                     const float* input,
98                                     std::size_t input_size,
99                                     std::size_t row_begin,
100                                     std::size_t row_end) noexcept {
101     static const bool AVX2_AVAILABLE = dot_f32_avx2_available();
102     if (AVX2_AVAILABLE && input_size >= LCQI_F32_AVX2_MIN_DOT_SIZE &&
103         row_end - row_begin >= LCQI_F32_AVX2_MIN_ROW_COUNT) {
104         return max_dot_f32_rows_avx2_unchecked(weights,
105                                                input,
106                                                input_size,
107                                                row_begin,
108                                                row_end);
109     }
110     return max_dot_f32_rows_scalar_unchecked(weights, input, input_size, ↵
111 ↵ row_begin, row_end);
112 }
113 } // namespace lcqi
114
115 #if !defined(LCQI_ENABLE_AVX2)
116 namespace lcqi {
117
118 bool dot_f32_avx2_available() noexcept {
119     return false;
120 }
121
122 float dot_f32_avx2_unchecked(const float* lhs,
123                             const float* rhs,
124                             std::size_t size) noexcept {
125     return dot_f32_scalar_unchecked(lhs, rhs, size);
126 }
127
128 void linear_f32_rows_avx2_unchecked(const float* weights,
129                                    const float* input,
130                                    const float* bias,
131                                    std::size_t input_size,
132                                    std::size_t row_begin,
133                                    std::size_t row_end,
134                                    float* output) noexcept {
135     linear_f32_rows_scalar_unchecked(weights,
136                                     input,
137                                     bias,
138                                     input_size,
139                                     row_begin,
140                                     row_end,

```

```

141         output);
142     }
143
144     F32RowMax max_dot_f32_rows_avx2_unchecked(const float* weights,
145                                               const float* input,
146                                               std::size_t input_size,
147                                               std::size_t row_begin,
148                                               std::size_t row_end) noexcept {
149         return max_dot_f32_rows_scalar_unchecked(weights, input, input_size, ↵
            ↵ row_begin, row_end);
150     }
151
152 } // namespace lcqi
153 #endif

```

Listing 10.12 src/f32_kernels_avx2.cpp

```

1 #include <lcqi/f32_kernels.hpp>
2
3 #if defined(LCQI_ENABLE_AVX2)
4
5 #include <immintrin.h>
6
7 #include <array>
8 #include <cstdint>
9 #include <limits>
10
11 namespace lcqi {
12 namespace {
13
14 constexpr std::size_t LCQI_F32_AVX2_LANE_COUNT = 8;
15 constexpr std::size_t LCQI_F32_AVX2_UNROLL_COUNT = 4;
16 constexpr std::size_t LCQI_F32_AVX2_UNROLLED_FLOATS =
17     LCQI_F32_AVX2_LANE_COUNT * LCQI_F32_AVX2_UNROLL_COUNT;
18
19 [[nodiscard]] float horizontal_sum(__m256 values) noexcept {
20     const __m128 low = _mm256_castps256_ps128(values);
21     const __m128 high = _mm256_extractf128_ps(values, 1);
22     const __m128 pair_sum = _mm_add_ps(low, high);
23     const __m128 first_fold = _mm_hadd_ps(pair_sum, pair_sum);
24     const __m128 second_fold = _mm_hadd_ps(first_fold, first_fold);
25     return _mm_cvtss_f32(second_fold);
26 }
27
28 void compute_four_row_dot(const float* row0,
29                          const float* row1,
30                          const float* row2,
31                          const float* row3,
32                          const float* input,
33                          std::size_t input_size,
34                          float* sums) noexcept {
35     std::size_t index = 0;

```

```

36     __m256 accumulator0 = _mm256_setzero_ps();
37     __m256 accumulator1 = _mm256_setzero_ps();
38     __m256 accumulator2 = _mm256_setzero_ps();
39     __m256 accumulator3 = _mm256_setzero_ps();
40
41     for (; index + LCQI_F32_AVX2_LANE_COUNT <= input_size;
42          index += LCQI_F32_AVX2_LANE_COUNT) {
43         const __m256 input_values = _mm256_loadu_ps(input + index);
44         accumulator0 =
45             _mm256_fmadd_ps(_mm256_loadu_ps(row0 + index), input_values, ↵
↵ accumulator0);
46         accumulator1 =
47             _mm256_fmadd_ps(_mm256_loadu_ps(row1 + index), input_values, ↵
↵ accumulator1);
48         accumulator2 =
49             _mm256_fmadd_ps(_mm256_loadu_ps(row2 + index), input_values, ↵
↵ accumulator2);
50         accumulator3 =
51             _mm256_fmadd_ps(_mm256_loadu_ps(row3 + index), input_values, ↵
↵ accumulator3);
52     }
53
54     sums[0] = horizontal_sum(accumulator0);
55     sums[1] = horizontal_sum(accumulator1);
56     sums[2] = horizontal_sum(accumulator2);
57     sums[3] = horizontal_sum(accumulator3);
58     for (; index < input_size; ++index) {
59         const float input_value = input[index];
60         sums[0] += row0[index] * input_value;
61         sums[1] += row1[index] * input_value;
62         sums[2] += row2[index] * input_value;
63         sums[3] += row3[index] * input_value;
64     }
65 }
66
67 } // namespace
68
69 bool dot_f32_avx2_available() noexcept {
70     #if defined(__GNUC__) || defined(__clang__)
71         __builtin_cpu_init();
72         return __builtin_cpu_supports("avx2") && __builtin_cpu_supports("fma");
73     #else
74         return true;
75     #endif
76 }
77
78 float dot_f32_avx2_unchecked(const float* lhs,
79                              const float* rhs,
80                              std::size_t size) noexcept {
81     std::size_t index = 0;
82     __m256 accumulator0 = _mm256_setzero_ps();
83     __m256 accumulator1 = _mm256_setzero_ps();

```

```

84     __m256 accumulator2 = _mm256_setzero_ps();
85     __m256 accumulator3 = _mm256_setzero_ps();
86
87     for (; index + LCQI_F32_AVX2_UNROLLED_FLOATS <= size;
88          index += LCQI_F32_AVX2_UNROLLED_FLOATS) {
89         const __m256 lhs0 = _mm256_loadu_ps(lhs + index);
90         const __m256 rhs0 = _mm256_loadu_ps(rhs + index);
91         const __m256 lhs1 = _mm256_loadu_ps(lhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT);
92         const __m256 rhs1 = _mm256_loadu_ps(rhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT);
93         const __m256 lhs2 = _mm256_loadu_ps(lhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 2);
94         const __m256 rhs2 = _mm256_loadu_ps(rhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 2);
95         const __m256 lhs3 = _mm256_loadu_ps(lhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 3);
96         const __m256 rhs3 = _mm256_loadu_ps(rhs + index + ↵
↵ LCQI_F32_AVX2_LANE_COUNT * 3);
97
98         accumulator0 = _mm256_fmadd_ps(lhs0, rhs0, accumulator0);
99         accumulator1 = _mm256_fmadd_ps(lhs1, rhs1, accumulator1);
100        accumulator2 = _mm256_fmadd_ps(lhs2, rhs2, accumulator2);
101        accumulator3 = _mm256_fmadd_ps(lhs3, rhs3, accumulator3);
102    }
103
104    __m256 accumulator = _mm256_add_ps(
105        _mm256_add_ps(accumulator0, accumulator1),
106        _mm256_add_ps(accumulator2, accumulator3));
107    for (; index + LCQI_F32_AVX2_LANE_COUNT <= size;
108         index += LCQI_F32_AVX2_LANE_COUNT) {
109        const __m256 lhs_values = _mm256_loadu_ps(lhs + index);
110        const __m256 rhs_values = _mm256_loadu_ps(rhs + index);
111        accumulator = _mm256_fmadd_ps(lhs_values, rhs_values, accumulator);
112    }
113
114    float sum = horizontal_sum(accumulator);
115    for (; index < size; ++index) {
116        sum += lhs[index] * rhs[index];
117    }
118    return sum;
119 }
120
121 void linear_f32_rows_avx2_unchecked(const float* weights,
122                                     const float* input,
123                                     const float* bias,
124                                     std::size_t input_size,
125                                     std::size_t row_begin,
126                                     std::size_t row_end,
127                                     float* output) noexcept {
128     std::size_t row = row_begin;
129     for (; row + LCQI_F32_AVX2_UNROLL_COUNT <= row_end;

```

```

130     row += LCQI_F32_AVX2_UNROLL_COUNT) {
131     const float* row0 = weights + row * input_size;
132     const float* row1 = row0 + input_size;
133     const float* row2 = row1 + input_size;
134     const float* row3 = row2 + input_size;
135     std::array<float, LCQI_F32_AVX2_UNROLL_COUNT> sums{};
136     compute_four_row_dot(row0, row1, row2, row3, input, input_size, sums.↵
↵ data());
137     for (std::size_t offset = 0; offset < LCQI_F32_AVX2_UNROLL_COUNT; ++↵
↵ offset) {
138         output[row + offset] =
139             sums[offset] + (bias == nullptr ? 0.0F : bias[row + offset]);
140     }
141 }
142 for (; row < row_end; ++row) {
143     const float* weight_row = weights + row * input_size;
144     output[row] = dot_f32_avx2_unchecked(weight_row, input, input_size) +
145         (bias == nullptr ? 0.0F : bias[row]);
146 }
147 }
148
149 F32RowMax max_dot_f32_rows_avx2_unchecked(const float* weights,
150                                           const float* input,
151                                           std::size_t input_size,
152                                           std::size_t row_begin,
153                                           std::size_t row_end) noexcept {
154     F32RowMax best;
155     best.row = row_begin;
156     best.value = -std::numeric_limits<float>::infinity();
157     std::size_t row = row_begin;
158     for (; row + LCQI_F32_AVX2_UNROLL_COUNT <= row_end;
159         row += LCQI_F32_AVX2_UNROLL_COUNT) {
160         const float* row0 = weights + row * input_size;
161         const float* row1 = row0 + input_size;
162         const float* row2 = row1 + input_size;
163         const float* row3 = row2 + input_size;
164         std::array<float, LCQI_F32_AVX2_UNROLL_COUNT> sums{};
165         compute_four_row_dot(row0, row1, row2, row3, input, input_size, sums.↵
↵ data());
166         for (std::size_t offset = 0; offset < LCQI_F32_AVX2_UNROLL_COUNT; ++↵
↵ offset) {
167             if (row + offset == row_begin || sums[offset] > best.value) {
168                 best.row = row + offset;
169                 best.value = sums[offset];
170             }
171         }
172     }
173     for (; row < row_end; ++row) {
174         const float* weight_row = weights + row * input_size;
175         const float value = dot_f32_avx2_unchecked(weight_row, input, ↵
↵ input_size);
176         if (row == row_begin || value > best.value) {

```

```

177         best.row = row;
178         best.value = value;
179     }
180 }
181 return best;
182 }
183
184 } // namespace lcqi
185
186 #endif

```

Listing 10.13 src/gpt2_reference.cpp

```

1 #include <lcqi/gpt2_reference.hpp>
2
3 #include <lcqi/f32_kernels.hpp>
4 #include <lcqi/safetensors.hpp>
5
6 #include <algorithm>
7 #include <array>
8 #include <atomic>
9 #include <chrono>
10 #include <cmath>
11 #include <cstdint>
12 #include <fstream>
13 #include <limits>
14 #include <optional>
15 #include <sstream>
16 #include <stdexcept>
17 #include <string>
18 #include <string_view>
19 #include <thread>
20 #include <unordered_set>
21 #include <utility>
22
23 namespace lcqi {
24 namespace {
25
26 constexpr std::int32_t LCQI_GPT2_DEFAULT_BOS_EOS = 50256;
27 constexpr std::int32_t LCQI_GPT2_ATTENTION_PROJECTIONS = 3;
28 constexpr std::int32_t LCQI_GPT2_PAIR_TOKEN_COUNT = 2;
29 constexpr std::int32_t LCQI_GPT2_BYTE_COUNT = 256;
30 constexpr std::int32_t LCQI_GPT2_UNICODE_PRIVATE_BASE = 256;
31 constexpr std::int32_t LCQI_GPT2_ASCII_EXCLAMATION = 33;
32 constexpr std::int32_t LCQI_GPT2_ASCII_TILDE = 126;
33 constexpr std::int32_t LCQI_GPT2_LATIN_INVERTED_EXCLAMATION = 161;
34 constexpr std::int32_t LCQI_GPT2_LATIN_NOT_SIGN = 172;
35 constexpr std::int32_t LCQI_GPT2_LATIN_REGISTERED = 174;
36 constexpr std::int32_t LCQI_GPT2_LATIN_Y_DIAERESIS = 255;
37 constexpr std::int32_t LCQI_GPT2_UTF8_CONTINUATION_MASK = 0x3F;
38 constexpr std::int32_t LCQI_GPT2_UTF8_CONTINUATION_BITS = 0x80;
39 constexpr std::int32_t LCQI_GPT2_UTF8_TWO_BYTE_HEAD = 0xC0;

```



```

40 constexpr std::int32_t LCQI_GPT2_UTF8_THREE_BYTE_HEAD = 0xE0;
41 constexpr std::int32_t LCQI_GPT2_UTF8_FOUR_BYTE_HEAD = 0xF0;
42 constexpr std::int32_t LCQI_GPT2_UTF8_ONE_BYTE_LIMIT = 0x80;
43 constexpr std::int32_t LCQI_GPT2_UTF8_TWO_BYTE_LIMIT = 0x800;
44 constexpr std::int32_t LCQI_GPT2_UTF8_THREE_BYTE_LIMIT = 0x10000;
45 constexpr unsigned char LCQI_GPT2_ASCII_APOSTROPHE = 0x27U;
46 constexpr float LCQI_GPT2_GELU_SQRT_2_OVER_PI = 0.7978845608028654F;
47 constexpr float LCQI_GPT2_GELU_CUBIC_COEFF = 0.044715F;
48 constexpr float LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY =
49     -std::numeric_limits<float>::infinity();
50 constexpr std::int32_t LCQI_GPT2_MAX_AUTO_WORKERS = 8;
51 constexpr std::int32_t LCQI_GPT2_DEFAULT_PARALLEL_MIN_ROWS = 512;
52 constexpr std::int32_t LCQI_GPT2_PARALLEL_MIN_COLUMNS = 512;
53 constexpr std::int64_t LCQI_GPT2_PARALLEL_MIN_OPS = 1'000'000;
54 constexpr std::size_t LCQI_GPT2_PARALLEL_ROW_BLOCK_MULTIPLIER = 4;
55 constexpr std::size_t LCQI_GPT2_WORKER_SPIN_PAUSES_BEFORE_YIELD = 4096;
56
57 using Gpt2ProfileClock = std::chrono::steady_clock;
58
59 void pause_worker_spin(std::size_t& pause_count) noexcept {
60     #if defined(__GNUC__) && (defined(__x86_64__) || defined(__i386__))
61         __builtin_ia32_pause();
62     #endif
63     ++pause_count;
64     if (pause_count >= LCQI_GPT2_WORKER_SPIN_PAUSES_BEFORE_YIELD) {
65         pause_count = 0;
66         std::this_thread::yield();
67     }
68 }
69
70 } // namespace
71
72 namespace detail {
73
74 class Gpt2ParallelWorkerPool {
75 public:
76     explicit Gpt2ParallelWorkerPool(std::int32_t requested_workers)
77         : worker_count_(Gpt2ParallelWorkerPool::normalize_worker_count(↵
↵ requested_workers)) {
78         if (this->worker_count_ <= 1) {
79             return;
80         }
81         const std::size_t background_count =
82             static_cast<std::size_t>(this->worker_count_ - 1);
83         this->threads_.reserve(background_count);
84         for (std::size_t index = 0; index < background_count; ++index) {
85             this->threads_.emplace_back([this, index]() { this->worker_loop(↵
↵ index); });
86         }
87     }
88
89     ~Gpt2ParallelWorkerPool() {

```

```

90     this->stopping_.store(true, std::memory_order_release);
91     this->generation_.fetch_add(1, std::memory_order_release);
92     for (std::thread& thread : this->threads_) {
93         if (thread.joinable()) {
94             thread.join();
95         }
96     }
97 }
98
99 Gpt2ParallelWorkerPool(const Gpt2ParallelWorkerPool&) = delete;
100 Gpt2ParallelWorkerPool& operator=(const Gpt2ParallelWorkerPool&) = delete;
101
102 [[nodiscard]] std::int32_t worker_count() const noexcept {
103     return this->worker_count_;
104 }
105
106 template <typename Function>
107 void parallel_for_rows(std::size_t begin,
108                       std::size_t end,
109                       std::size_t grain,
110                       Function&& function) {
111     if (end <= begin) {
112         return;
113     }
114     const std::size_t row_count = end - begin;
115     if (this->worker_count_ <= 1 || row_count <= grain) {
116         function(begin, end);
117         return;
118     }
119
120     const std::size_t task_count =
121         std::min<std::size_t>(static_cast<std::size_t>(this->worker_count_)↵
↵ ,
122                               (row_count + grain - 1) / grain);
123     if (task_count <= 1) {
124         function(begin, end);
125         return;
126     }
127
128     auto task_context = ParallelTaskContext<Function>{function};
129     // Publish all task metadata before the generation bump; workers use ↵
↵ the
130     // generation acquire load as the hand-off point for this stack context↵
↵ .
131     this->task_begin_.store(begin, std::memory_order_release);
132     this->task_end_.store(end, std::memory_order_release);
133     this->task_count_.store(task_count, std::memory_order_release);
134     this->task_context_.store(&task_context, std::memory_order_release);
135     this->task_invoke_.store(&ParallelTaskContext<Function>::invoke,
136                             std::memory_order_release);
137     this->active_workers_.store(task_count - 1, std::memory_order_release);
138     this->generation_.fetch_add(1, std::memory_order_release);

```

```

139
140     const std::size_t main_worker = task_count - 1;
141     const auto [main_begin, main_end] = this->chunk_bounds(begin, end, ↵
↵ task_count, main_worker);
142     function(main_begin, main_end);
143
144     std::size_t pause_count = 0;
145     while (this->active_workers_.load(std::memory_order_acquire) != 0) {
146         pause_worker_spin(pause_count);
147     }
148     this->task_context_.store(nullptr, std::memory_order_release);
149     this->task_invoke_.store(nullptr, std::memory_order_release);
150 }
151
152 private:
153     template <typename Function>
154     struct ParallelTaskContext {
155         Function& function;
156
157         static void invoke(void* context, std::size_t begin, std::size_t end) {
158             auto* typed_context = static_cast<ParallelTaskContext*>(context);
159             typed_context->function(begin, end);
160         }
161     };
162
163     std::int32_t worker_count_ = 1;
164     std::vector<std::thread> threads_;
165     std::atomic<bool> stopping_ = false;
166     std::atomic<std::uint64_t> generation_ = 0;
167     std::atomic<std::size_t> task_begin_ = 0;
168     std::atomic<std::size_t> task_end_ = 0;
169     std::atomic<std::size_t> task_count_ = 0;
170     std::atomic<std::size_t> active_workers_ = 0;
171     std::atomic<void*> task_context_ = nullptr;
172     std::atomic<void (*) (void*, std::size_t, std::size_t)> task_invoke_ = ↵
↵ nullptr;
173
174     [[nodiscard]] static std::int32_t normalize_worker_count(std::int32_t ↵
↵ requested_workers) {
175         if (requested_workers > 0) {
176             return requested_workers;
177         }
178         const unsigned int hardware_workers = std::thread::hardware_concurrency↵
↵ ();
179         if (hardware_workers == 0) {
180             return 1;
181         }
182         const std::int32_t capped =
183             std::min<std::int32_t>(static_cast<std::int32_t>(hardware_workers),
184                                   LCQI_GPT2_MAX_AUTO_WORKERS);
185         return std::max(capped, 1);
186     }

```

```

187
188 [[nodiscard]] std::pair<std::size_t, std::size_t> chunk_bounds(
189     std::size_t begin,
190     std::size_t end,
191     std::size_t task_count,
192     std::size_t worker_index) const {
193     const std::size_t row_count = end - begin;
194     const std::size_t chunk_begin = begin + row_count * worker_index / ↵
↵ task_count;
195     const std::size_t chunk_end = begin + row_count * (worker_index + 1) / ↵
↵ task_count;
196     return {chunk_begin, chunk_end};
197 }
198
199 void worker_loop(std::size_t worker_index) {
200     std::uint64_t observed_generation = 0;
201     std::size_t pause_count = 0;
202     while (true) {
203         const std::uint64_t current_generation =
204             this->generation_.load(std::memory_order_acquire);
205         if (current_generation == observed_generation) {
206             if (this->stopping_.load(std::memory_order_acquire)) {
207                 return;
208             }
209             pause_worker_spin(pause_count);
210             continue;
211         }
212
213         observed_generation = current_generation;
214         pause_count = 0;
215         if (this->stopping_.load(std::memory_order_acquire)) {
216             return;
217         }
218
219         const std::size_t task_count = this->task_count_.load(std::↵
↵ memory_order_acquire);
220         if (this->generation_.load(std::memory_order_acquire) != ↵
↵ current_generation) {
221             continue;
222         }
223         const std::size_t background_task_count = task_count - 1;
224         if (worker_index >= background_task_count) {
225             continue;
226         }
227         void* task_context = this->task_context_.load(std::↵
↵ memory_order_acquire);
228         void (*task_invoke)(void*, std::size_t, std::size_t) =
229             this->task_invoke_.load(std::memory_order_acquire);
230         if (task_context == nullptr || task_invoke == nullptr) {
231             continue;
232         }
233         const std::size_t task_begin = this->task_begin_.load(std::↵

```

```

    ↪ memory_order_acquire);
234     const std::size_t task_end = this->task_end_.load(std::memory_order_acquire);
    ↪ memory_order_acquire);
235     if (this->generation_.load(std::memory_order_acquire) != current_generation) {
236         continue;
237     }
238     const auto [begin, end] =
239         this->chunk_bounds(task_begin,
240                             task_end,
241                             task_count,
242                             worker_index);
243     task_invoke(task_context, begin, end);
244
245     this->active_workers_.fetch_sub(1, std::memory_order_acq_rel);
246 }
247 }
248 };
249
250 } // namespace detail
251
252 namespace {
253
254 class JsonCursor {
255 public:
256     explicit JsonCursor(std::string_view text) : text_(text) {}
257
258     void expect(char expected) {
259         this->skip_ws();
260         if (this->eof() || this->text_[this->position_] != expected) {
261             throw std::runtime_error("invalid JSON syntax");
262         }
263         ++this->position_;
264     }
265
266     [[nodiscard]] bool consume(char expected) {
267         this->skip_ws();
268         if (!this->eof() && this->text_[this->position_] == expected) {
269             ++this->position_;
270             return true;
271         }
272         return false;
273     }
274
275     [[nodiscard]] std::string parse_string() {
276         this->skip_ws();
277         this->expect('"');
278         std::string result;
279         while (!this->eof()) {
280             const char ch = this->text_[this->position_++];
281             if (ch == '"') {
282                 return result;

```

```

283     }
284     if (ch == '\\') {
285         result += this->parse_escape();
286     } else {
287         result.push_back(ch);
288     }
289 }
290 throw std::runtime_error("unterminated JSON string");
291 }
292
293 [[nodiscard]] std::int64_t parse_i64() {
294     this->skip_ws();
295     bool negative = false;
296     if (this->consume('-')) {
297         negative = true;
298     }
299     if (this->eof() || this->text_[this->position_] < '0' ||
300         this->text_[this->position_] > '9') {
301         throw std::runtime_error("expected JSON integer");
302     }
303     std::int64_t value = 0;
304     while (!this->eof() && this->text_[this->position_] >= '0' &&
305         this->text_[this->position_] <= '9') {
306         const std::int64_t digit = this->text_[this->position_] - '0';
307         if (value >
308             (std::numeric_limits<std::int64_t>::max() - digit) / 10) {
309             throw std::runtime_error("JSON integer overflow");
310         }
311         value = value * 10 + digit;
312         ++this->position_;
313     }
314     return negative ? -value : value;
315 }
316
317 [[nodiscard]] double parse_number() {
318     this->skip_ws();
319     const std::size_t begin = this->position_;
320     if (!this->eof() && this->text_[this->position_] == '-') {
321         ++this->position_;
322     }
323     while (!this->eof() && this->text_[this->position_] >= '0' &&
324         this->text_[this->position_] <= '9') {
325         ++this->position_;
326     }
327     if (!this->eof() && this->text_[this->position_] == '.') {
328         ++this->position_;
329         while (!this->eof() && this->text_[this->position_] >= '0' &&
330             this->text_[this->position_] <= '9') {
331             ++this->position_;
332         }
333     }
334     if (!this->eof() &&

```

```

335         (this->text_[this->position_] == 'e' || this->text_[this->position_ ←
↪ ] == 'E')) {
336             ++this->position_;
337             if (!this->eof() &&
338                 (this->text_[this->position_] == '+' || this->text_[this-> ←
↪ position_] == '-')) {
339                 ++this->position_;
340             }
341             while (!this->eof() && this->text_[this->position_] >= '0' &&
342                 this->text_[this->position_] <= '9') {
343                 ++this->position_;
344             }
345         }
346         if (begin == this->position_) {
347             throw std::runtime_error("expected JSON number");
348         }
349         return std::stod(std::string(this->text_.substr(begin, this->position_ ←
↪ - begin)));
350     }
351
352     void skip_value() {
353         this->skip_ws();
354         if (this->eof()) {
355             throw std::runtime_error("unexpected end of JSON");
356         }
357         const char ch = this->text_[this->position_];
358         if (ch == '"') {
359             static_cast<void>(this->parse_string());
360             return;
361         }
362         if (ch == '{') {
363             this->expect('{');
364             if (this->consume(' ')) {
365                 return;
366             }
367             while (true) {
368                 static_cast<void>(this->parse_string());
369                 this->expect(':');
370                 this->skip_value();
371                 if (this->consume(' ')) {
372                     return;
373                 }
374                 this->expect(',');
375             }
376         }
377         if (ch == '[') {
378             this->expect('[');
379             if (this->consume(' ')) {
380                 return;
381             }
382             while (true) {
383                 this->skip_value();

```



```

384         if (this->consume(' ')) {
385             return;
386         }
387         this->expect(',');
388     }
389 }
390 if (ch == 't') {
391     this->expect_literal("true");
392     return;
393 }
394 if (ch == 'f') {
395     this->expect_literal("false");
396     return;
397 }
398 if (ch == 'n') {
399     this->expect_literal("null");
400     return;
401 }
402 static_cast<void>(this->parse_number());
403 }
404
405 [[nodiscard]] bool eof_after_ws() {
406     this->skip_ws();
407     return this->eof();
408 }
409
410 private:
411     std::string_view text_;
412     std::size_t position_ = 0;
413
414     [[nodiscard]] bool eof() const {
415         return this->position_ >= this->text_.size();
416     }
417
418     void skip_ws() {
419         while (!this->eof() &&
420             (this->text_[this->position_] == ' ' ||
421              this->text_[this->position_] == '\n' ||
422              this->text_[this->position_] == '\r' ||
423              this->text_[this->position_] == '\t')) {
424             ++this->position_;
425         }
426     }
427
428     void expect_literal(std::string_view literal) {
429         for (const char expected : literal) {
430             if (this->eof() || this->text_[this->position_] != expected) {
431                 throw std::runtime_error("invalid JSON literal");
432             }
433             ++this->position_;
434         }
435     }

```

```

436 [[nodiscard]] std::string parse_escape() {
437     if (this->eof()) {
438         throw std::runtime_error("unterminated JSON escape");
439     }
440     const char escaped = this->text_[this->position_++];
441     switch (escaped) {
442         case '"':
443         case '\\':
444         case '/':
445             return std::string(1, escaped);
446         case 'b':
447             return "\b";
448         case 'f':
449             return "\f";
450         case 'n':
451             return "\n";
452         case 'r':
453             return "\r";
454         case 't':
455             return "\t";
456         case 'u':
457             return this->parse_unicode_escape();
458         default:
459             throw std::runtime_error("unsupported JSON escape");
460     }
461 }
462 }
463
464 [[nodiscard]] std::string parse_unicode_escape() {
465     std::uint32_t value = 0;
466     for (std::int32_t i = 0; i < 4; ++i) {
467         if (this->eof()) {
468             throw std::runtime_error("unterminated JSON unicode escape");
469         }
470         const char ch = this->text_[this->position_++];
471         value <= 4U;
472         if (ch >= '0' && ch <= '9') {
473             value += static_cast<std::uint32_t>(ch - '0');
474         } else if (ch >= 'a' && ch <= 'f') {
475             value += static_cast<std::uint32_t>(10 + ch - 'a');
476         } else if (ch >= 'A' && ch <= 'F') {
477             value += static_cast<std::uint32_t>(10 + ch - 'A');
478         } else {
479             throw std::runtime_error("invalid JSON unicode escape");
480         }
481     }
482     std::string encoded;
483     if (value < LCQI_GPT2_UTF8_ONE_BYTE_LIMIT) {
484         encoded.push_back(static_cast<char>(value));
485     } else if (value < LCQI_GPT2_UTF8_TWO_BYTE_LIMIT) {
486         encoded.push_back(static_cast<char>(
487             LCQI_GPT2_UTF8_TWO_BYTE_HEAD | (value >> 6U)));

```

```

488         encoded.push_back(static_cast<char>(
489             LCQI_GPT2_UTF8_CONTINUATION_BITS |
490             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
491     } else {
492         encoded.push_back(static_cast<char>(
493             LCQI_GPT2_UTF8_THREE_BYTE_HEAD | (value >> 12U)));
494         encoded.push_back(static_cast<char>(
495             LCQI_GPT2_UTF8_CONTINUATION_BITS |
496             ((value >> 6U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
497         encoded.push_back(static_cast<char>(
498             LCQI_GPT2_UTF8_CONTINUATION_BITS |
499             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
500     }
501     return encoded;
502 }
503 };
504
505 std::string read_text_file(const std::filesystem::path& path) {
506     std::ifstream input(path);
507     if (!input) {
508         throw std::runtime_error("cannot open text file: " + path.string());
509     }
510     std::ostringstream buffer;
511     buffer << input.rdbuf();
512     return buffer.str();
513 }
514
515 std::size_t checked_size(std::int64_t value, const char* name) {
516     if (value < 0) {
517         throw std::runtime_error(std::string(name) + " cannot be negative");
518     }
519     return static_cast<std::size_t>(value);
520 }
521
522 std::string encode_codepoint(std::uint32_t value) {
523     std::string encoded;
524     if (value < LCQI_GPT2_UTF8_ONE_BYTE_LIMIT) {
525         encoded.push_back(static_cast<char>(value));
526     } else if (value < LCQI_GPT2_UTF8_TWO_BYTE_LIMIT) {
527         encoded.push_back(static_cast<char>(
528             LCQI_GPT2_UTF8_TWO_BYTE_HEAD | (value >> 6U)));
529         encoded.push_back(static_cast<char>(
530             LCQI_GPT2_UTF8_CONTINUATION_BITS |
531             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
532     } else if (value < LCQI_GPT2_UTF8_THREE_BYTE_LIMIT) {
533         encoded.push_back(static_cast<char>(
534             LCQI_GPT2_UTF8_THREE_BYTE_HEAD | (value >> 12U)));
535         encoded.push_back(static_cast<char>(
536             LCQI_GPT2_UTF8_CONTINUATION_BITS |
537             ((value >> 6U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
538         encoded.push_back(static_cast<char>(
539             LCQI_GPT2_UTF8_CONTINUATION_BITS |

```

```

540         (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
541     } else {
542         encoded.push_back(static_cast<char>(
543             LCQI_GPT2_UTF8_FOUR_BYTE_HEAD | (value >> 18U)));
544         encoded.push_back(static_cast<char>(
545             LCQI_GPT2_UTF8_CONTINUATION_BITS |
546             ((value >> 12U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
547         encoded.push_back(static_cast<char>(
548             LCQI_GPT2_UTF8_CONTINUATION_BITS |
549             ((value >> 6U) & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
550         encoded.push_back(static_cast<char>(
551             LCQI_GPT2_UTF8_CONTINUATION_BITS |
552             (value & LCQI_GPT2_UTF8_CONTINUATION_MASK)));
553     }
554     return encoded;
555 }
556
557 void require_positive(std::int32_t value, const char* name) {
558     if (value <= 0) {
559         throw std::runtime_error(std::string(name) + " must be positive");
560     }
561 }
562
563 void validate_config(const Gpt2Config& config) {
564     require_positive(config.hidden_size, "hidden_size");
565     require_positive(config.head_count, "head_count");
566     require_positive(config.layer_count, "layer_count");
567     require_positive(config.max_positions, "max_positions");
568     require_positive(config.vocab_size, "vocab_size");
569     require_positive(config.intermediate_size, "intermediate_size");
570     if (config.hidden_size % config.head_count != 0) {
571         throw std::runtime_error("hidden_size must be divisible by head_count")↵
572     };
573     if (config.layer_norm_epsilon <= 0.0F) {
574         throw std::runtime_error("layer_norm_epsilon must be positive");
575     }
576 }
577
578 std::int32_t head_dim(const Gpt2Config& config) {
579     return config.hidden_size / config.head_count;
580 }
581
582 std::size_t matrix_size(std::int32_t rows, std::int32_t columns) {
583     return checked_size(rows, "rows") * checked_size(columns, "columns");
584 }
585
586 Gpt2ProfileClock::time_point profile_start(const Gpt2HotspotProfile* profile) {
587     if (profile == nullptr) {
588         return Gpt2ProfileClock::time_point{};
589     }
590     return Gpt2ProfileClock::now();

```

```

591 }
592
593 void add_profile_ms(Gpt2HotspotProfile* profile,
594                   double Gpt2HotspotProfile::*field,
595                   Gpt2ProfileClock::time_point start) {
596     if (profile == nullptr) {
597         return;
598     }
599     (*profile).*field +=
600         std::chrono::duration<double, std::milli>(Gpt2ProfileClock::now() - ↵
↵ start).count());
601 }
602
603 void validate_vector_size(std::size_t actual, std::int32_t expected, const char↵
↵ * name) {
604     if (actual != checked_size(expected, name)) {
605         throw std::runtime_error(std::string(name) + " size mismatch");
606     }
607 }
608
609 void validate_linear(const Gpt2LinearF32& linear, const char* name) {
610     require_positive(linear.input_size, "linear input_size");
611     require_positive(linear.output_size, "linear output_size");
612     if (linear.weights.size() != matrix_size(linear.output_size, linear.↵
↵ input_size)) {
613         throw std::runtime_error(std::string(name) + " weight size mismatch");
614     }
615     if (!linear.bias.empty() &&
616         linear.bias.size() != checked_size(linear.output_size, "linear ↵
↵ output_size")) {
617         throw std::runtime_error(std::string(name) + " bias size mismatch");
618     }
619 }
620
621 void validate_model(const Gpt2ReferenceModel& model) {
622     validate_config(model.config);
623     if (model.layers.size() != checked_size(model.config.layer_count, "↵
↵ layer_count")) {
624         throw std::runtime_error("GPT-2 layer count mismatch");
625     }
626     if (model.token_embedding.size() !=
627         matrix_size(model.config.vocab_size, model.config.hidden_size)) {
628         throw std::runtime_error("GPT-2 token embedding size mismatch");
629     }
630     if (model.position_embedding.size() !=
631         matrix_size(model.config.max_positions, model.config.hidden_size)) {
632         throw std::runtime_error("GPT-2 position embedding size mismatch");
633     }
634     validate_vector_size(model.final_ln_weight.size(), model.config.hidden_size↵
↵ , "ln_f weight");
635     validate_vector_size(model.final_ln_bias.size(), model.config.hidden_size, ↵
↵ "ln_f bias");

```

```

636     if (model.tie_lm_head_to_embedding) {
637         if (!model.lm_head_weight.empty()) {
638             throw std::runtime_error("tied GPT-2 lm_head should not carry ↵
↵ separate weights");
639         }
640     } else if (model.lm_head_weight.size() !=
641         matrix_size(model.config.vocab_size, model.config.hidden_size)) ↵
↵ {
642         throw std::runtime_error("GPT-2 lm_head size mismatch");
643     }
644     for (const Gpt2LayerWeightsF32& layer : model.layers) {
645         validate_vector_size(layer.ln_1_weight.size(), model.config.hidden_size ↵
↵ , "ln_1 weight");
646         validate_vector_size(layer.ln_1_bias.size(), model.config.hidden_size, ↵
↵ "ln_1 bias");
647         validate_linear(layer.c_attn, "c_attn");
648         validate_linear(layer.c_proj, "c_proj");
649         validate_vector_size(layer.ln_2_weight.size(), model.config.hidden_size ↵
↵ , "ln_2 weight");
650         validate_vector_size(layer.ln_2_bias.size(), model.config.hidden_size, ↵
↵ "ln_2 bias");
651         validate_linear(layer.c_fc, "c_fc");
652         validate_linear(layer.mlp_c_proj, "mlp_c_proj");
653     }
654 }
655
656 std::span<const float> row_span(std::span<const float> values,
657     std::int32_t row,
658     std::int32_t row_width) {
659     return values.subspan(check_size(row, "row") * checked_size(row_width, "↵
↵ row_width"),
660         checked_size(row_width, "row_width"));
661 }
662
663 void add_inplace(std::span<float> target, std::span<const float> source) {
664     if (target.size() != source.size()) {
665         throw std::runtime_error("GPT-2 residual add size mismatch");
666     }
667     for (std::size_t index = 0; index < target.size(); ++index) {
668         target[index] += source[index];
669     }
670 }
671
672 void layer_norm(std::span<const float> input,
673     std::span<const float> weight,
674     std::span<const float> bias,
675     float epsilon,
676     std::span<float> output) {
677     if (input.empty() || input.size() != weight.size() || input.size() != bias.↵
↵ size() ||
678         input.size() != output.size()) {
679         throw std::runtime_error("GPT-2 LayerNorm size mismatch");

```

```

680     }
681     float mean = 0.0F;
682     for (const float value : input) {
683         mean += value;
684     }
685     mean /= static_cast<float>(input.size());
686
687     float variance = 0.0F;
688     for (const float value : input) {
689         const float centered = value - mean;
690         variance += centered * centered;
691     }
692     variance /= static_cast<float>(input.size());
693     const float scale = 1.0F / std::sqrt(variance + epsilon);
694
695     for (std::size_t index = 0; index < input.size(); ++index) {
696         output[index] = (input[index] - mean) * scale * weight[index] + bias[↵
↵ index];
697     }
698 }
699
700 void linear_f32(const Gpt2LinearF32& linear,
701                std::span<const float> input,
702                std::span<float> output) {
703     validate_linear(linear, "GPT-2 linear");
704     if (input.size() != checked_size(linear.input_size, "linear input") ||
705         output.size() != checked_size(linear.output_size, "linear output")) {
706         throw std::runtime_error("GPT-2 linear input/output size mismatch");
707     }
708     const std::size_t input_size = checked_size(linear.input_size, "linear ↵
↵ input");
709     const std::size_t output_size = checked_size(linear.output_size, "linear ↵
↵ output");
710     for (std::size_t out = 0; out < output_size; ++out) {
711         float sum = linear.bias.empty() ? 0.0F : linear.bias[out];
712         const std::size_t row_base = out * input_size;
713         for (std::size_t in = 0; in < input_size; ++in) {
714             sum += linear.weights[row_base + in] * input[in];
715         }
716         output[out] = sum;
717     }
718 }
719
720 void linear_f32_unchecked(const Gpt2LinearF32& linear,
721                           std::span<const float> input,
722                           std::span<float> output) {
723     const std::size_t input_size = static_cast<std::size_t>(linear.input_size);
724     const std::size_t output_size = static_cast<std::size_t>(linear.output_size↵
↵ );
725     const float* weights = linear.weights.data();
726     const float* bias = linear.bias.empty() ? nullptr : linear.bias.data();
727     const float* input_data = input.data();

```



```

728 linear_f32_rows_unchecked(weights, input_data, bias, input_size, 0, ↵
↵ output_size, output.data());
729 }
730
731 [[nodiscard]] bool should_parallelize_rows(const Gpt2ExecutionOptions& options,
732                                           std::size_t rows,
733                                           std::size_t columns,
734                                           const detail::Gpt2ParallelWorkerPool↵
↵ * worker_pool) {
735     if (worker_pool == nullptr || worker_pool->worker_count() <= 1) {
736         return false;
737     }
738     const std::int32_t min_rows =
739         options.parallel_min_rows > 0 ? options.parallel_min_rows
740         : LCQI_GPT2_DEFAULT_PARALLEL_MIN_ROWS;
741     if (rows < checked_size(min_rows, "parallel_min_rows")) {
742         return false;
743     }
744     if (options.worker_count > 1) {
745         return true;
746     }
747     return columns >= checked_size(LCQI_GPT2_PARALLEL_MIN_COLUMNS, "↵
↵ parallel_min_columns") ||
748         rows * columns >= static_cast<std::size_t>(↵
↵ LCQI_GPT2_PARALLEL_MIN_OPS);
749 }
750
751 [[nodiscard]] std::size_t parallel_row_grain(std::size_t rows,
752                                              const detail::↵
↵ Gpt2ParallelWorkerPool& worker_pool) {
753     const std::size_t workers = checked_size(worker_pool.worker_count(), "↵
↵ worker_count");
754     const std::size_t target_tasks = workers * ↵
↵ LCQI_GPT2_PARALLEL_ROW_BLOCK_MULTIPLIER;
755     return std::max<std::size_t>(1, (rows + target_tasks - 1) / target_tasks);
756 }
757
758 [[nodiscard]] bool model_has_parallel_work(const Gpt2ReferenceModel& model,
759                                             const Gpt2ExecutionOptions& options)↵
↵ {
760     const std::size_t hidden_size = checked_size(model.config.hidden_size, "↵
↵ hidden_size");
761     const std::size_t vocab_size = checked_size(model.config.vocab_size, "↵
↵ vocab_size");
762     const std::size_t intermediate_size =
763         checked_size(model.config.intermediate_size, "intermediate_size");
764     const std::size_t qkv_rows =
765         checked_size(model.config.hidden_size * LCQI_GPT2_ATTENTION_PROJECTIONS↵
↵ ,
766                     "qkv output rows");
767     const std::int32_t min_rows =
768         options.parallel_min_rows > 0 ? options.parallel_min_rows

```

```

769                                     : LCQI_GPT2_DEFAULT_PARALLEL_MIN_ROWS;
770     const auto large_enough = [min_rows](std::size_t rows, std::size_t columns)↵
↵ {
771         if (rows < checked_size(min_rows, "parallel_min_rows")) {
772             return false;
773         }
774         return columns >= checked_size(LCQI_GPT2_PARALLEL_MIN_COLUMNS,
775                                         "parallel_min_columns") ||
776                rows * columns >= static_cast<std::size_t>(↵
↵ LCQI_GPT2_PARALLEL_MIN_OPS);
777     };
778     if (options.worker_count > 1) {
779         return vocab_size >= checked_size(min_rows, "parallel_min_rows") ||
780                qkv_rows >= checked_size(min_rows, "parallel_min_rows") ||
781                intermediate_size >= checked_size(min_rows, "parallel_min_rows")↵
↵ ||
782                hidden_size >= checked_size(min_rows, "parallel_min_rows");
783     }
784     if (options.worker_count == 1) {
785         return false;
786     }
787     return large_enough(vocab_size, hidden_size) ||
788            large_enough(qkv_rows, hidden_size) ||
789            large_enough(hidden_size, hidden_size) ||
790            large_enough(intermediate_size, hidden_size) ||
791            large_enough(hidden_size, intermediate_size);
792 }
793
794 [[nodiscard]] std::unique_ptr<detail::Gpt2ParallelWorkerPool> make_worker_pool(
795     const Gpt2ReferenceModel& model,
796     const Gpt2ExecutionOptions& options) {
797     if (!model_has_parallel_work(model, options)) {
798         return nullptr;
799     }
800     return std::make_unique<detail::Gpt2ParallelWorkerPool>(options.↵
↵ worker_count);
801 }
802
803 void linear_f32_unchecked_parallel(const Gpt2LinearF32& linear,
804                                   std::span<const float> input,
805                                   std::span<float> output,
806                                   const Gpt2ExecutionOptions& options,
807                                   detail::Gpt2ParallelWorkerPool* worker_pool)↵
↵ {
808     const std::size_t input_size = static_cast<std::size_t>(linear.input_size);
809     const std::size_t output_size = static_cast<std::size_t>(linear.output_size)↵
↵ );
810     if (!should_parallelize_rows(options, output_size, input_size, worker_pool)↵
↵ ) {
811         linear_f32_unchecked(linear, input, output);
812         return;
813     }

```

[illegible]

```

864         0.0F);
865     linear_f32(layer.c_attn, hidden, packed);
866     const std::size_t width = checked_size(config.hidden_size, "hidden_size");
867     std::copy_n(packed.begin(), config.hidden_size, q.begin());
868     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width),
869                 config.hidden_size,
870                 k.begin());
871     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width * 2),
872                 config.hidden_size,
873                 v.begin());
874 }
875
876 void project_qkv_reuse_workspace(const Gpt2Config& config,
877                                 const Gpt2LayerWeightsF32& layer,
878                                 std::span<const float> hidden,
879                                 std::span<float> packed,
880                                 std::span<float> q,
881                                 std::span<float> k,
882                                 std::span<float> v,
883                                 const Gpt2ExecutionOptions& options,
884                                 detail::Gpt2ParallelWorkerPool* worker_pool) {
885     linear_f32_unchecked_parallel(layer.c_attn, hidden, packed, options, ↵
    ↵ worker_pool);
886     const std::size_t width = checked_size(config.hidden_size, "hidden_size");
887     std::copy_n(packed.begin(), config.hidden_size, q.begin());
888     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width),
889                 config.hidden_size,
890                 k.begin());
891     std::copy_n(packed.begin() + static_cast<std::ptrdiff_t>(width * 2),
892                 config.hidden_size,
893                 v.begin());
894 }
895
896 void attention_full_sequence(const Gpt2Config& config,
897                             std::span<const float> q_by_pos,
898                             std::span<const float> k_by_pos,
899                             std::span<const float> v_by_pos,
900                             std::int32_t sequence_length,
901                             std::span<float> output_by_pos) {
902     const std::int32_t local_head_dim = head_dim(config);
903     const float score_scale = 1.0F / std::sqrt(static_cast<float>(↵
    ↵ local_head_dim));
904     std::fill(output_by_pos.begin(), output_by_pos.end(), 0.0F);
905     std::vector<float> scores(checked_size(sequence_length, "sequence_length"),
906                             LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY);
907
908     for (std::int32_t position = 0; position < sequence_length; ++position) {
909         for (std::int32_t head = 0; head < config.head_count; ++head) {
910             float max_score = LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY;
911             for (std::int32_t past = 0; past <= position; ++past) {
912                 float dot = 0.0F;
913                 for (std::int32_t dim = 0; dim < local_head_dim; ++dim) {

```

```

914         const std::size_t q_index =
915             matrix_size(position, config.hidden_size) +
916             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
917         const std::size_t k_index =
918             matrix_size(past, config.hidden_size) +
919             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
920         dot += q_by_pos[q_index] * k_by_pos[k_index];
921     }
922     const float score = dot * score_scale;
923     scores[checked_size(past, "past")] = score;
924     max_score = std::max(max_score, score);
925 }
926
927 float denominator = 0.0F;
928 for (std::int32_t past = 0; past <= position; ++past) {
929     const std::size_t index = checked_size(past, "past");
930     scores[index] = std::exp(scores[index] - max_score);
931     denominator += scores[index];
932 }
933 if (denominator <= 0.0F) {
934     throw std::runtime_error("GPT-2 attention denominator is ↵
↵ invalid");
935 }
936 for (std::int32_t past = 0; past <= position; ++past) {
937     const float probability = scores[checked_size(past, "past")] / ↵
↵ denominator;
938     for (std::int32_t dim = 0; dim < local_head_dim; ++dim) {
939         const std::size_t output_index =
940             matrix_size(position, config.hidden_size) +
941             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
942         const std::size_t value_index =
943             matrix_size(past, config.hidden_size) +
944             matrix_size(head, local_head_dim) + checked_size(dim, "↵
↵ head dim");
945         output_by_pos[output_index] += probability * v_by_pos[↵
↵ value_index];
946     }
947 }
948 }
949 }
950 }
951
952 void forward_gpt2_layer(const Gpt2Config& config,
953     const Gpt2LayerWeightsF32& layer,
954     std::int32_t sequence_length,
955     std::vector<float>& hidden_by_pos) {
956     std::vector<float> normed(checked_size(config.hidden_size, "hidden_size"), ↵
↵ 0.0F);
957     std::vector<float> q_by_pos(matrix_size(sequence_length, config.hidden_size, ↵

```

```

↪ ), 0.0F);
958 std::vector<float> k_by_pos(q_by_pos.size(), 0.0F);
959 std::vector<float> v_by_pos(q_by_pos.size(), 0.0F);
960 std::vector<float> attention_by_pos(q_by_pos.size(), 0.0F);
961 std::vector<float> projected(checkered_size(config.hidden_size, "hidden_size"↪
↪ ), 0.0F);
962 std::vector<float> mlp_fc(checkered_size(config.intermediate_size, "↪
↪ intermediate_size"), 0.0F);
963 std::vector<float> mlp_out(checkered_size(config.hidden_size, "hidden_size"),↪
↪ 0.0F);
964
965 for (std::int32_t position = 0; position < sequence_length; ++position) {
966     const std::span<float> hidden = std::span<float>(
967         hidden_by_pos.data() + matrix_size(position, config.hidden_size),
968         checked_size(config.hidden_size, "hidden_size"));
969     layer_norm(hidden,
970         layer.ln_1_weight,
971         layer.ln_1_bias,
972         config.layer_norm_epsilon,
973         normed);
974     std::span<float> q = std::span<float>(
975         q_by_pos.data() + matrix_size(position, config.hidden_size),
976         checked_size(config.hidden_size, "hidden_size"));
977     std::span<float> k = std::span<float>(
978         k_by_pos.data() + matrix_size(position, config.hidden_size),
979         checked_size(config.hidden_size, "hidden_size"));
980     std::span<float> v = std::span<float>(
981         v_by_pos.data() + matrix_size(position, config.hidden_size),
982         checked_size(config.hidden_size, "hidden_size"));
983     project_qkv(config, layer, normed, q, k, v);
984 }
985
986 attention_full_sequence(config, q_by_pos, k_by_pos, v_by_pos, ↪
↪ sequence_length, attention_by_pos);
987
988 for (std::int32_t position = 0; position < sequence_length; ++position) {
989     const std::span<float> hidden = std::span<float>(
990         hidden_by_pos.data() + matrix_size(position, config.hidden_size),
991         checked_size(config.hidden_size, "hidden_size"));
992     const std::span<const float> attention = std::span<const float>(
993         attention_by_pos.data() + matrix_size(position, config.hidden_size)↪
↪ ,
994         checked_size(config.hidden_size, "hidden_size"));
995     linear_f32(layer.c_proj, attention, projected);
996     add_inplace(hidden, projected);
997
998     layer_norm(hidden,
999         layer.ln_2_weight,
1000         layer.ln_2_bias,
1001         config.layer_norm_epsilon,
1002         normed);
1003     linear_f32(layer.c_fc, normed, mlp_fc);

```

```

1004     for (float& value : mlp_fc) {
1005         value = gelu_new(value);
1006     }
1007     linear_f32(layer.mlp_c_proj, mlp_fc, mlp_out);
1008     add_inplace(hidden, mlp_out);
1009 }
1010 }
1011
1012 void forward_gpt2_layer_cached(const Gpt2Config& config,
1013                               const Gpt2LayerWeightsF32& layer,
1014                               std::int32_t layer_id,
1015                               std::int32_t model_position,
1016                               Gpt2KvCache& cache,
1017                               Gpt2ForwardWorkspace& workspace,
1018                               std::span<float> hidden,
1019                               Gpt2HotspotProfile* hotspot_profile,
1020                               const Gpt2ExecutionOptions& options,
1021                               detail::Gpt2ParallelWorkerPool* worker_pool) {
1022     if (hotspot_profile != nullptr) {
1023         ++hotspot_profile->layer_steps;
1024     }
1025     std::span<float> normed = workspace.normed();
1026     std::span<float> query = workspace.query();
1027     std::span<float> key = workspace.key();
1028     std::span<float> value = workspace.value();
1029     std::span<float> attention = workspace.attention();
1030     std::span<float> projected = workspace.projected();
1031     std::span<float> qkv_packed = workspace.qkv_packed();
1032     std::span<float> mlp_fc = workspace.mlp_fc();
1033     std::span<float> mlp_out = workspace.mlp_out();
1034     std::span<float> scores = workspace.scores_prefix(model_position + 1);
1035
1036     Gpt2ProfileClock::time_point start =
1037         profile_start(hotspot_profile);
1038     layer_norm(hidden,
1039               layer.ln_1_weight,
1040               layer.ln_1_bias,
1041               config.layer_norm_epsilon,
1042               normed);
1043     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::layer_norm_ms, start);
1044
1045     start = profile_start(hotspot_profile);
1046     project_qkv_reuse_workspace(config,
1047                                layer,
1048                                normed,
1049                                qkv_packed,
1050                                query,
1051                                key,
1052                                value,
1053                                options,
1054                                worker_pool);
1055     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::qkv_projection_ms, ←

```



```

↪ start);
1056
1057 start = profile_start(hotspot_profile);
1058 cache.append(layer_id, model_position, key, value);
1059 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::kv_append_ms, start);
1060
1061 start = profile_start(hotspot_profile);
1062 detail::attend_cached_position(cache,
1063                                layer_id,
1064                                model_position,
1065                                query,
1066                                scores,
1067                                attention);
1068 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::attention_ms, start);
1069
1070 start = profile_start(hotspot_profile);
1071 linear_f32_unchecked_parallel(layer.c_proj, attention, projected, options, ↪
↪ worker_pool);
1072 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::↪
↪ attention_projection_ms, start);
1073
1074 start = profile_start(hotspot_profile);
1075 add_inplace(hidden, projected);
1076 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::residual_add_ms, start↪
↪ );
1077
1078 start = profile_start(hotspot_profile);
1079 layer_norm(hidden,
1080            layer.ln_2_weight,
1081            layer.ln_2_bias,
1082            config.layer_norm_epsilon,
1083            normed);
1084 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::layer_norm_ms, start);
1085
1086 start = profile_start(hotspot_profile);
1087 linear_f32_unchecked_parallel(layer.c_fc, normed, mlp_fc, options, ↪
↪ worker_pool);
1088 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::mlp_fc_ms, start);
1089
1090 start = profile_start(hotspot_profile);
1091 for (float& value_ref : mlp_fc) {
1092     value_ref = gelu_new(value_ref);
1093 }
1094 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::gelu_ms, start);
1095
1096 start = profile_start(hotspot_profile);
1097 linear_f32_unchecked_parallel(layer.mlp_c_proj, mlp_fc, mlp_out, options, ↪
↪ worker_pool);
1098 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::mlp_projection_ms, ↪
↪ start);
1099
1100 start = profile_start(hotspot_profile);

```

```

1101     add_inplace(hidden, mlp_out);
1102     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::residual_add_ms, start↵
↵ );
1103 }
1104
1105 void compute_logits(const Gpt2ReferenceModel& model,
1106                    std::span<const float> hidden,
1107                    std::span<float> logits,
1108                    const Gpt2ExecutionOptions& options,
1109                    detail::Gpt2ParallelWorkerPool* worker_pool) {
1110     if (logits.size() != checked_size(model.config.vocab_size, "vocab_size")) {
1111         throw std::runtime_error("GPT-2 logits size mismatch");
1112     }
1113     const std::span<const float> weight =
1114         model.tie_lm_head_to_embedding ? std::span<const float>(model.↵
↵ token_embedding)
1115                                         : std::span<const float>(model.↵
↵ lm_head_weight);
1116     const std::size_t hidden_size = checked_size(model.config.hidden_size, "↵
↵ hidden_size");
1117     const std::size_t vocab_size = checked_size(model.config.vocab_size, "↵
↵ vocab_size");
1118     const float* weight_data = weight.data();
1119     const float* hidden_data = hidden.data();
1120     const auto rows_fn =
1121         [hidden_size, weight_data, hidden_data, logits_data = logits.data()](
1122             std::size_t begin,
1123             std::size_t end) {
1124             linear_f32_rows_unchecked(weight_data,
1125                                     hidden_data,
1126                                     nullptr,
1127                                     hidden_size,
1128                                     begin,
1129                                     end,
1130                                     logits_data);
1131         };
1132     if (should_parallelize_rows(options, vocab_size, hidden_size, worker_pool))↵
↵ {
1133         worker_pool->parallel_for_rows(0,
1134                                     vocab_size,
1135                                     parallel_row_grain(vocab_size, *↵
↵ worker_pool),
1136                                     rows_fn);
1137     } else {
1138         rows_fn(0, vocab_size);
1139     }
1140 }
1141
1142 void compute_logits(const Gpt2ReferenceModel& model,
1143                    std::span<const float> hidden,
1144                    std::span<float> logits) {
1145     Gpt2ExecutionOptions options;

```

```

1146     options.worker_count = 1;
1147     compute_logits(model, hidden, logits, options, nullptr);
1148 }
1149
1150 struct Gpt2TokenScore {
1151     std::int32_t token = 0;
1152     float value = -std::numeric_limits<float>::infinity();
1153 };
1154
1155 enum class Gpt2CachedStepMode {
1156     advance_only,
1157     predict_token,
1158     logits,
1159 };
1160
1161 [[nodiscard]] Gpt2TokenScore compute_predicted_token_range(const float* ↵
    ↵ weight_data,
1162                                                         const float* ↵
    ↵ hidden_data,
1163                                                         std::size_t ↵
    ↵ hidden_size,
1164                                                         std::size_t begin,
1165                                                         std::size_t end) {
1166     Gpt2TokenScore best;
1167     const F32RowMax row_max =
1168         max_dot_f32_rows_unchecked(weight_data, hidden_data, hidden_size, begin↵
    ↵ , end);
1169     best.token = static_cast<std::int32_t>(row_max.row);
1170     best.value = row_max.value;
1171     return best;
1172 }
1173
1174 std::int32_t compute_predicted_token(const Gpt2ReferenceModel& model,
1175                                     std::span<const float> hidden,
1176                                     const Gpt2ExecutionOptions& options,
1177                                     detail::Gpt2ParallelWorkerPool* ↵
    ↵ worker_pool) {
1178     const std::span<const float> weight =
1179         model.tie_lm_head_to_embedding ? std::span<const float>(model.↵
    ↵ token_embedding)
1180                                     : std::span<const float>(model.↵
    ↵ lm_head_weight);
1181     const std::size_t hidden_size = checked_size(model.config.hidden_size, "↵
    ↵ hidden_size");
1182     const std::size_t vocab_size = checked_size(model.config.vocab_size, "↵
    ↵ vocab_size");
1183     const float* weight_data = weight.data();
1184     const float* hidden_data = hidden.data();
1185     if (!should_parallelize_rows(options, vocab_size, hidden_size, worker_pool)↵
    ↵ ) {
1186         return compute_predicted_token_range(weight_data, hidden_data, ↵
    ↵ hidden_size, 0, vocab_size)

```

```

1187         .token;
1188     }
1189
1190     const std::size_t worker_count = checked_size(worker_pool->worker_count(), ↵
    ↵ "worker_count");
1191     const std::size_t chunk_count = std::min(worker_count, vocab_size);
1192     std::vector<Gpt2TokenScore> partials(chunk_count);
1193     const auto rows_fn =
1194         [chunk_count, hidden_size, vocab_size, weight_data, hidden_data, &↵
    ↵ partials](
1195         std::size_t begin,
1196         std::size_t end) {
1197         for (std::size_t chunk = begin; chunk < end; ++chunk) {
1198             const std::size_t token_begin = vocab_size * chunk / ↵
    ↵ chunk_count;
1199             const std::size_t token_end = vocab_size * (chunk + 1) / ↵
    ↵ chunk_count;
1200             partials[chunk] = compute_predicted_token_range(
1201                 weight_data,
1202                 hidden_data,
1203                 hidden_size,
1204                 token_begin,
1205                 token_end);
1206         }
1207     };
1208     worker_pool->parallel_for_rows(0, chunk_count, 1, rows_fn);
1209
1210     Gpt2TokenScore best = partials.front();
1211     for (std::size_t index = 1; index < partials.size(); ++index) {
1212         if (partials[index].value > best.value) {
1213             best = partials[index];
1214         }
1215     }
1216     return best.token;
1217 }
1218
1219 Gpt2ForwardResult run_gpt2_cached_step_unchecked(const Gpt2ReferenceModel& ↵
    ↵ model,
1220
    Gpt2KvCache& cache,
1221    Gpt2ForwardWorkspace& ↵
    ↵ workspace,
1222
    std::int32_t token_id,
1223    Gpt2CachedStepMode mode,
1224    Gpt2HotspotProfile* ↵
    ↵ hotspot_profile,
1225
    const Gpt2ExecutionOptions& ↵
    ↵ options,
1226
    detail::Gpt2ParallelWorkerPool↵
    ↵ * worker_pool) {
1227     const Gpt2ProfileClock::time_point step_start = profile_start(↵
    ↵ hotspot_profile);
1228     if (hotspot_profile != nullptr) {

```

```

1229     ++hotspot_profile->decoder_steps;
1230 }
1231 if (token_id < 0 || token_id >= model.config.vocab_size) {
1232     throw std::runtime_error("GPT-2 token id out of range");
1233 }
1234 const std::int32_t model_position = cache.filled_tokens();
1235 if (model_position >= model.config.max_positions) {
1236     throw std::runtime_error("GPT-2 cached forward would exceed ↵
↵ max_positions");
1237 }
1238
1239 std::span<float> hidden = workspace.hidden();
1240 const std::span<const float> token =
1241     row_span(model.token_embedding, token_id, model.config.hidden_size);
1242 const std::span<const float> position_embedding =
1243     row_span(model.position_embedding, model_position, model.config.↵
↵ hidden_size);
1244 Gpt2ProfileClock::time_point start = profile_start(hotspot_profile);
1245 for (std::int32_t dim = 0; dim < model.config.hidden_size; ++dim) {
1246     hidden[checked_size(dim, "dim")] =
1247         token[checked_size(dim, "dim")] + position_embedding[checked_size(↵
↵ dim, "dim")];
1248 }
1249 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::embedding_ms, start);
1250
1251 for (std::int32_t layer_id = 0;
1252     layer_id < static_cast<std::int32_t>(model.layers.size());
1253     ++layer_id) {
1254     forward_gpt2_layer_cached(model.config,
1255                             model.layers[checked_size(layer_id, "layer_id↵
↵ ")]),
1256                             layer_id,
1257                             model_position,
1258                             cache,
1259                             workspace,
1260                             hidden,
1261                             hotspot_profile,
1262                             options,
1263                             worker_pool);
1264 }
1265
1266 Gpt2ForwardResult result;
1267 if (mode == Gpt2CachedStepMode::advance_only) {
1268     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::total_step_ms, ↵
↵ step_start);
1269     return result;
1270 }
1271
1272 std::span<float> normed = workspace.normed();
1273 start = profile_start(hotspot_profile);
1274 layer_norm(hidden,
1275             model.final_ln_weight,

```

```

1276         model.final_ln_bias,
1277         model.config.layer_norm_epsilon,
1278         normed);
1279 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::final_norm_ms, start);
1280
1281 if (mode == Gpt2CachedStepMode::logits) {
1282     std::span<float> logits = workspace.logits();
1283     start = profile_start(hotspot_profile);
1284     compute_logits(model, normed, logits, options, worker_pool);
1285     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::lm_head_ms, start)↵
↵ ;
1286     start = profile_start(hotspot_profile);
1287     result.logits.assign(logits.begin(), logits.end());
1288     result.predicted_token = argmax(result.logits);
1289     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::logits_result_ms, ↵
↵ start);
1290 } else {
1291     start = profile_start(hotspot_profile);
1292     result.predicted_token = compute_predicted_token(model, normed, options↵
↵ , worker_pool);
1293     add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::lm_head_ms, start)↵
↵ ;
1294 }
1295 add_profile_ms(hotspot_profile, &Gpt2HotspotProfile::total_step_ms, ↵
↵ step_start);
1296 return result;
1297 }
1298
1299 std::vector<std::string> gpt2_bytes_to_unicode() {
1300     std::vector<std::int32_t> bytes;
1301     for (std::int32_t value = LCQI_GPT2_ASCII_EXCLAMATION;
1302          value <= LCQI_GPT2_ASCII_TILDE;
1303          ++value) {
1304         bytes.push_back(value);
1305     }
1306     for (std::int32_t value = LCQI_GPT2_LATIN_INVERTED_EXCLAMATION;
1307          value <= LCQI_GPT2_LATIN_NOT_SIGN;
1308          ++value) {
1309         bytes.push_back(value);
1310     }
1311     for (std::int32_t value = LCQI_GPT2_LATIN_REGISTERED;
1312          value <= LCQI_GPT2_LATIN_Y_DIAERESIS;
1313          ++value) {
1314         bytes.push_back(value);
1315     }
1316
1317     std::unordered_set<std::int32_t> byte_set(bytes.begin(), bytes.end());
1318     std::vector<std::int32_t> chars = bytes;
1319     std::int32_t private_offset = 0;
1320     for (std::int32_t byte = 0; byte < LCQI_GPT2_BYTE_COUNT; ++byte) {
1321         if (!byte_set.contains(byte)) {
1322             bytes.push_back(byte);

```

```

1323         chars.push_back(LCQI_GPT2_UNICODE_PRIVATE_BASE + private_offset);
1324         ++private_offset;
1325     }
1326 }
1327
1328 std::vector<std::string> mapping(LCQI_GPT2_BYTE_COUNT);
1329 for (std::size_t index = 0; index < bytes.size(); ++index) {
1330     mapping[checked_size(bytes[index], "byte unicode index")] =
1331         encode_codepoint(static_cast<std::uint32_t>(chars[index]));
1332 }
1333 return mapping;
1334 }
1335
1336 std::unordered_map<std::string, unsigned char> gpt2_unicode_to_bytes(
1337     const std::vector<std::string>& byte_encoder) {
1338     std::unordered_map<std::string, unsigned char> result;
1339     for (std::size_t byte = 0; byte < byte_encoder.size(); ++byte) {
1340         result.emplace(byte_encoder[byte], static_cast<unsigned char>(byte));
1341     }
1342     return result;
1343 }
1344
1345 std::string bytes_to_bpe_alphabet(std::string_view text) {
1346     static const std::vector<std::string> BYTE_ENCODER = gpt2_bytes_to_unicode(
1347         ↪ );
1348     std::string result;
1349     for (const unsigned char byte : text) {
1350         result += BYTE_ENCODER[byte];
1351     }
1352     return result;
1353 }
1354
1355 std::vector<std::string> utf8_symbols(std::string_view text) {
1356     std::vector<std::string> symbols;
1357     std::size_t offset = 0;
1358     while (offset < text.size()) {
1359         const unsigned char lead = static_cast<unsigned char>(text[offset]);
1360         std::size_t width = 1;
1361         if ((lead & 0xE0U) == LCQI_GPT2_UTF8_TWO_BYTE_HEAD) {
1362             width = 2;
1363         } else if ((lead & 0xF0U) == LCQI_GPT2_UTF8_THREE_BYTE_HEAD) {
1364             width = 3;
1365         } else if ((lead & 0xF8U) == LCQI_GPT2_UTF8_FOUR_BYTE_HEAD) {
1366             width = 4;
1367         }
1368         if (offset + width > text.size()) {
1369             throw std::runtime_error("invalid UTF-8/BPE symbol boundary");
1370         }
1371         symbols.emplace_back(text.substr(offset, width));
1372         offset += width;
1373     }
1374     return symbols;

```



```

1374 }
1375
1376 std::string pair_key(std::string_view left, std::string_view right) {
1377     std::string key(left);
1378     key.push_back('\n');
1379     key += right;
1380     return key;
1381 }
1382
1383 std::vector<std::string> bpe_apply(const Gpt2Tokenizer& tokenizer, std::↵
    ↵ string_view token) {
1384     std::vector<std::string> word = utf8_symbols(token);
1385     if (word.size() <= 1) {
1386         return word;
1387     }
1388
1389     while (true) {
1390         std::optional<std::int32_t> best_rank;
1391         std::string best_left;
1392         std::string best_right;
1393         for (std::size_t index = 0; index + 1 < word.size(); ++index) {
1394             const auto rank = tokenizer.bpe_ranks.find(pair_key(word[index], ↵
    ↵ word[index + 1]));
1395             if (rank != tokenizer.bpe_ranks.end() &&
1396                 (!best_rank.has_value() || rank->second < *best_rank)) {
1397                 best_rank = rank->second;
1398                 best_left = word[index];
1399                 best_right = word[index + 1];
1400             }
1401         }
1402         if (!best_rank.has_value()) {
1403             return word;
1404         }
1405
1406         std::vector<std::string> merged;
1407         std::size_t index = 0;
1408         while (index < word.size()) {
1409             if (index + 1 < word.size() && word[index] == best_left &&
1410                 word[index + 1] == best_right) {
1411                 merged.push_back(best_left + best_right);
1412                 index += LCQI_GPT2_PAIR_TOKEN_COUNT;
1413             } else {
1414                 merged.push_back(word[index]);
1415                 ++index;
1416             }
1417         }
1418         word = std::move(merged);
1419         if (word.size() == 1) {
1420             return word;
1421         }
1422     }
1423 }

```

```

1424
1425 bool is_ascii_letter(unsigned char value) {
1426     return (value >= 'A' && value <= 'Z') || (value >= 'a' && value <= 'z');
1427 }
1428
1429 bool is_ascii_digit(unsigned char value) {
1430     return value >= '0' && value <= '9';
1431 }
1432
1433 bool is_ascii_space(unsigned char value) {
1434     return value == ' ' || value == '\t' || value == '\n' ||
1435            value == '\r' || value == '\f' || value == '\v';
1436 }
1437
1438 bool is_gpt2_punctuation(unsigned char value) {
1439     return !is_ascii_letter(value) && !is_ascii_digit(value) &&
1440            !is_ascii_space(value);
1441 }
1442
1443 std::size_t consume_utf8_codepoint(std::string_view text, std::size_t offset) {
1444     const unsigned char lead = static_cast<unsigned char>(text[offset]);
1445     std::size_t width = 1;
1446     if ((lead & 0xE0U) == LCQI_GPT2_UTF8_TWO_BYTE_HEAD) {
1447         width = 2;
1448     } else if ((lead & 0xF0U) == LCQI_GPT2_UTF8_THREE_BYTE_HEAD) {
1449         width = 3;
1450     } else if ((lead & 0xF8U) == LCQI_GPT2_UTF8_FOUR_BYTE_HEAD) {
1451         width = 4;
1452     }
1453     if (offset + width > text.size()) {
1454         throw std::runtime_error("invalid UTF-8 text for GPT-2 tokenizer");
1455     }
1456     return offset + width;
1457 }
1458
1459 bool starts_with_contraction(std::string_view text,
1460                             std::size_t offset,
1461                             std::string_view contraction) {
1462     if (offset + contraction.size() > text.size()) {
1463         return false;
1464     }
1465     for (std::size_t index = 0; index < contraction.size(); ++index) {
1466         char left = text[offset + index];
1467         const char right = contraction[index];
1468         if (left >= 'A' && left <= 'Z') {
1469             left = static_cast<char>(left - 'A' + 'a');
1470         }
1471         if (left != right) {
1472             return false;
1473         }
1474     }
1475     return true;

```

```

1476 }
1477
1478 std::vector<std::string> gpt2_tokenize(std::string_view text) {
1479     std::vector<std::string> pieces;
1480     std::size_t offset = 0;
1481     const std::array<std::string_view, 7> contractions{
1482         "'s", "'t", "'re", "'ve", "'m", "'ll", "'d",
1483     };
1484     while (offset < text.size()) {
1485         bool emitted_contraction = false;
1486         for (const std::string_view contraction : contractions) {
1487             if (starts_with_contraction(text, offset, contraction)) {
1488                 pieces.emplace_back(text.substr(offset, contraction.size()));
1489                 offset += contraction.size();
1490                 emitted_contraction = true;
1491                 break;
1492             }
1493         }
1494         if (emitted_contraction) {
1495             continue;
1496         }
1497
1498         const std::size_t begin = offset;
1499         bool has_prefix_space = false;
1500         if (text[offset] == ' ' && offset + 1 < text.size() &&
1501             !is_ascii_space(static_cast<unsigned char>(text[offset + 1]))) {
1502             has_prefix_space = true;
1503             ++offset;
1504         }
1505
1506         if (offset >= text.size()) {
1507             pieces.emplace_back(text.substr(begin, offset - begin));
1508             continue;
1509         }
1510
1511         const unsigned char ch = static_cast<unsigned char>(text[offset]);
1512         if (is_ascii_letter(ch)) {
1513             while (offset < text.size() &&
1514                 is_ascii_letter(static_cast<unsigned char>(text[offset]))) {
1515                 ++offset;
1516             }
1517         } else if (is_ascii_digit(ch)) {
1518             while (offset < text.size() &&
1519                 is_ascii_digit(static_cast<unsigned char>(text[offset]))) {
1520                 ++offset;
1521             }
1522         } else if (is_gpt2_punctuation(ch) && ch != LCQI_GPT2_ASCII_APOSTROPHE) {
1523             while (offset < text.size() &&
1524                 is_gpt2_punctuation(static_cast<unsigned char>(text[offset]))) {
1525                 ++offset;
1526             }
1527         }
1528         pieces.emplace_back(text.substr(begin, offset - begin));
1529         offset = offset;
1530     }
1531     return pieces;
1532 }

```

```

↪ LCQI_GPT2_ASCII_APOSTROPHE) {
1526     offset = consume_utf8_codepoint(text, offset);
1527 }
1528 } else if (is_ascii_space(ch)) {
1529     while (offset < text.size() &&
1530           is_ascii_space(static_cast<unsigned char>(text[offset]))) {
1531         ++offset;
1532     }
1533 } else {
1534     offset = consume_utf8_codepoint(text, offset);
1535 }
1536
1537 if (has_prefix_space || offset > begin) {
1538     pieces.emplace_back(text.substr(begin, offset - begin));
1539 } else {
1540     ++offset;
1541 }
1542 }
1543 return pieces;
1544 }
1545
1546 std::int32_t parse_i32_field(JsonCursor& cursor) {
1547     const std::int64_t value = cursor.parse_i64();
1548     if (value < std::numeric_limits<std::int32_t>::min() ||
1549         value > std::numeric_limits<std::int32_t>::max()) {
1550         throw std::runtime_error("JSON int32 field out of range");
1551     }
1552     return static_cast<std::int32_t>(value);
1553 }
1554
1555 std::unordered_map<std::string, std::int32_t> parse_vocab_json(std::string_view↪
↪ text) {
1556     JsonCursor cursor(text);
1557     std::unordered_map<std::string, std::int32_t> vocab;
1558     cursor.expect('{');
1559     if (cursor.consume('}')) {
1560         return vocab;
1561     }
1562     while (true) {
1563         const std::string key = cursor.parse_string();
1564         cursor.expect(':');
1565         const std::int32_t id = parse_i32_field(cursor);
1566         vocab.emplace(key, id);
1567         if (cursor.consume('}')) {
1568             break;
1569         }
1570         cursor.expect(',');
1571     }
1572     if (!cursor.eof_after_ws()) {
1573         throw std::runtime_error("trailing data after vocab JSON");
1574     }
1575     return vocab;

```

```

1576 }
1577
1578 std::vector<std::string> build_id_to_token(
1579     const std::unordered_map<std::string, std::int32_t>& vocab) {
1580     std::int32_t max_id = -1;
1581     for (const auto& [token, id] : vocab) {
1582         if (id < 0) {
1583             throw std::runtime_error("GPT-2 vocab id cannot be negative");
1584         }
1585         max_id = std::max(max_id, id);
1586     }
1587     std::vector<std::string> result(check_size(max_id + 1, "max vocab id"));
1588     for (const auto& [token, id] : vocab) {
1589         std::string& slot = result[checked_size(id, "vocab id")];
1590         if (!slot.empty()) {
1591             throw std::runtime_error("duplicate GPT-2 vocab id");
1592         }
1593         slot = token;
1594     }
1595     return result;
1596 }
1597
1598 std::unordered_map<std::string, std::int32_t> parse_merges(std::string_view ↵
    ↵ text) {
1599     std::unordered_map<std::string, std::int32_t> ranks;
1600     std::istringstream input{std::string(text)};
1601     std::string line;
1602     std::int32_t rank = 0;
1603     while (std::getline(input, line)) {
1604         if (line.empty() || line[0] == '#') {
1605             continue;
1606         }
1607         std::istringstream line_stream(line);
1608         std::string left;
1609         std::string right;
1610         if (!(line_stream >> left >> right)) {
1611             throw std::runtime_error("invalid GPT-2 merges line");
1612         }
1613         ranks.emplace(pair_key(left, right), rank);
1614         ++rank;
1615     }
1616     return ranks;
1617 }
1618
1619 void transpose_hf_conv1d(std::vector<float>& values,
1620     std::int32_t input_size,
1621     std::int32_t output_size) {
1622     if (values.size() != matrix_size(input_size, output_size)) {
1623         throw std::runtime_error("GPT-2 Conv1D tensor size mismatch");
1624     }
1625     std::vector<float> transposed(matrix_size(output_size, input_size), 0.0F);
1626     for (std::int32_t input = 0; input < input_size; ++input) {

```

```

1627     for (std::int32_t output = 0; output < output_size; ++output) {
1628         transposed[matrix_size(output, input_size) + checked_size(input, "↵
↵ input")] =
1629             values[matrix_size(input, output_size) + checked_size(output, "↵
↵ output")];
1630     }
1631 }
1632 values = std::move(transposed);
1633 }
1634
1635 std::vector<float> require_tensor(const SafeTensorFile& file,
1636                                 std::string_view name,
1637                                 std::span<const std::int64_t> expected_shape)↵
↵ {
1638     const SafeTensorEntry* entry = file.manifest.find_tensor(name);
1639     if (entry == nullptr) {
1640         throw std::runtime_error("missing GPT-2 tensor: " + std::string(name));
1641     }
1642     if (entry->shape.size() != expected_shape.size()) {
1643         throw std::runtime_error("GPT-2 tensor rank mismatch: " + std::string(↵
↵ name));
1644     }
1645     for (std::size_t index = 0; index < expected_shape.size(); ++index) {
1646         if (entry->shape[index] != expected_shape[index]) {
1647             throw std::runtime_error("GPT-2 tensor shape mismatch: " + std::↵
↵ string(name));
1648         }
1649     }
1650     return read_safetensor_f32_tensor(file, name);
1651 }
1652
1653 std::vector<float> require_first_tensor(const SafeTensorFile& file,
1654                                         std::span<const std::string_view> names↵
↵ ,
1655                                         std::span<const std::int64_t> ↵
↵ expected_shape) {
1656     for (const std::string_view name : names) {
1657         if (file.manifest.find_tensor(name) != nullptr) {
1658             return require_tensor(file, name, expected_shape);
1659         }
1660     }
1661     throw std::runtime_error("missing GPT-2 tensor: " + std::string(names.front(↵
↵ ()))");
1662 }
1663
1664 Gpt2LinearF32 make_linear(std::int32_t input_size,
1665                           std::int32_t output_size,
1666                           std::vector<float> weights,
1667                           std::vector<float> bias = {}) {
1668     Gpt2LinearF32 linear;
1669     linear.input_size = input_size;
1670     linear.output_size = output_size;

```

```

1671     linear.weights = std::move(weights);
1672     linear.bias = std::move(bias);
1673     validate_linear(linear, "make GPT-2 linear");
1674     return linear;
1675 }
1676
1677 std::vector<float> zeros(std::int32_t count) {
1678     return std::vector<float>(checked_size(count, "zero count"), 0.0F);
1679 }
1680
1681 std::vector<float> ones(std::int32_t count) {
1682     return std::vector<float>(checked_size(count, "one count"), 1.0F);
1683 }
1684
1685 std::string tensor_name(std::int32_t layer, std::string_view suffix) {
1686     return "transformer.h." + std::to_string(layer) + "." + std::string(suffix) ↵
    ↵ ;
1687 }
1688
1689 std::string bare_tensor_name(std::int32_t layer, std::string_view suffix) {
1690     return "h." + std::to_string(layer) + "." + std::string(suffix);
1691 }
1692
1693 std::array<std::string_view, 2> embedding_names(std::string_view bare) {
1694     if (bare == "wte.weight") {
1695         return {"transformer.wte.weight", "wte.weight"};
1696     }
1697     if (bare == "wpe.weight") {
1698         return {"transformer.wpe.weight", "wpe.weight"};
1699     }
1700     if (bare == "ln_f.weight") {
1701         return {"transformer.ln_f.weight", "ln_f.weight"};
1702     }
1703     if (bare == "ln_f.bias") {
1704         return {"transformer.ln_f.bias", "ln_f.bias"};
1705     }
1706     throw std::runtime_error("unsupported GPT-2 embedding tensor alias");
1707 }
1708
1709 std::array<std::string, 2> layer_names(std::int32_t layer, std::string_view ↵
    ↵ suffix) {
1710     return {tensor_name(layer, suffix), bare_tensor_name(layer, suffix)};
1711 }
1712
1713 std::vector<float> require_layer_tensor(const SafeTensorFile& file,
1714                                         std::int32_t layer,
1715                                         std::string_view suffix,
1716                                         std::span<const std::int64_t> ↵
    ↵ expected_shape) {
1717     const std::array<std::string, 2> names = layer_names(layer, suffix);
1718     const std::array<std::string_view, 2> views{names[0], names[1]};
1719     return require_first_tensor(file, views, expected_shape);

```



```

1720 }
1721
1722 std::filesystem::path find_gpt2_weight_file(const std::filesystem::path& ↵
    ↵ model_directory) {
1723     const std::array<const char*, 2> candidates{
1724         "model.safetensors",
1725         "pytorch_model.safetensors",
1726     };
1727     for (const char* candidate : candidates) {
1728         const std::filesystem::path path = model_directory / candidate;
1729         if (std::filesystem::exists(path)) {
1730             return path;
1731         }
1732     }
1733     throw std::runtime_error("cannot find model.safetensors in GPT-2 directory" ↵
    ↵ );
1734 }
1735
1736 } // namespace
1737
1738 Gpt2Tokenizer load_gpt2_tokenizer(const std::filesystem::path& vocab_json,
1739                                   const std::filesystem::path& merges_txt,
1740                                   std::int32_t bos_token_id,
1741                                   std::int32_t eos_token_id) {
1742     Gpt2Tokenizer tokenizer;
1743     tokenizer.bos_token_id = bos_token_id;
1744     tokenizer.eos_token_id = eos_token_id;
1745     tokenizer.vocab = parse_vocab_json(read_text_file(vocab_json));
1746     tokenizer.id_to_token = build_id_to_token(tokenizer.vocab);
1747     tokenizer.bpe_ranks = parse_merges(read_text_file(merges_txt));
1748     if (tokenizer.vocab.empty() || tokenizer.bpe_ranks.empty()) {
1749         throw std::runtime_error("GPT-2 tokenizer assets are empty");
1750     }
1751     return tokenizer;
1752 }
1753
1754 std::vector<std::int32_t> gpt2_encode(const Gpt2Tokenizer& tokenizer,
1755                                       std::string_view text) {
1756     std::vector<std::int32_t> ids;
1757     for (const std::string& piece : gpt2_pretokenize(text)) {
1758         const std::string byte_piece = bytes_to_bpe_alphabet(piece);
1759         const std::vector<std::string> bpe_tokens = bpe_apply(tokenizer, ↵
    ↵ byte_piece);
1760         for (const std::string& token : bpe_tokens) {
1761             const auto found = tokenizer.vocab.find(token);
1762             if (found == tokenizer.vocab.end()) {
1763                 throw std::runtime_error("GPT-2 BPE token is missing from vocab" ↵
    ↵ );
1764             }
1765             ids.push_back(found->second);
1766         }
1767     }

```

```

1768     return ids;
1769 }
1770
1771 std::string gpt2_decode(const Gpt2Tokenizer& tokenizer,
1772                        std::span<const std::int32_t> token_ids) {
1773     static const std::vector<std::string> BYTE_ENCODER = gpt2_bytes_to_unicode(
1774         ↪ );
1775     static const std::unordered_map<std::string, unsigned char> BYTE_DECODER =
1776         ↪ gpt2_unicode_to_bytes(BYTE_ENCODER);
1777
1778     std::string token_text;
1779     for (const std::int32_t token_id : token_ids) {
1780         if (token_id < 0 || token_id >= static_cast<std::int32_t>(tokenizer.id_
1781             ↪ to_token.size())) {
1782             throw std::runtime_error("GPT-2 token id out of range");
1783         }
1784         token_text += tokenizer.id_to_token[checked_size(token_id, "token id")
1785             ↪ ];
1786     }
1787
1788     std::string result;
1789     const std::vector<std::string> symbols = utf8_symbols(token_text);
1790     for (const std::string& symbol : symbols) {
1791         const auto found = BYTE_DECODER.find(symbol);
1792         if (found == BYTE_DECODER.end()) {
1793             throw std::runtime_error("GPT-2 token cannot be decoded to byte");
1794         }
1795         result.push_back(static_cast<char>(found->second));
1796     }
1797     return result;
1798 }
1799
1800 Gpt2Config load_gpt2_config(const std::filesystem::path& config_json) {
1801     const std::string config_text = read_text_file(config_json);
1802     JsonCursor cursor(config_text);
1803     Gpt2Config config;
1804     config.bos_token_id = LCQI_GPT2_DEFAULT_BOS_EOS;
1805     config.eos_token_id = LCQI_GPT2_DEFAULT_BOS_EOS;
1806
1807     cursor.expect('{');
1808     if (cursor.consume('}')) {
1809         throw std::runtime_error("GPT-2 config is empty");
1810     }
1811     while (true) {
1812         const std::string key = cursor.parse_string();
1813         cursor.expect(':');
1814         if (key == "n_embd") {
1815             config.hidden_size = parse_i32_field(cursor);
1816         } else if (key == "n_head") {
1817             config.head_count = parse_i32_field(cursor);
1818         } else if (key == "n_layer") {
1819             config.layer_count = parse_i32_field(cursor);

```

```

1817     } else if (key == "n_positions" || key == "n_ctx") {
1818         const std::int32_t value = parse_i32_field(cursor);
1819         config.max_positions = std::max(config.max_positions, value);
1820     } else if (key == "vocab_size") {
1821         config.vocab_size = parse_i32_field(cursor);
1822     } else if (key == "n_inner") {
1823         config.intermediate_size = parse_i32_field(cursor);
1824     } else if (key == "layer_norm_epsilon") {
1825         config.layer_norm_epsilon = static_cast<float>(cursor.parse_number()
↪ ());
1826     } else if (key == "activation_function") {
1827         config.activation_function = cursor.parse_string();
1828     } else if (key == "bos_token_id") {
1829         config.bos_token_id = parse_i32_field(cursor);
1830     } else if (key == "eos_token_id") {
1831         config.eos_token_id = parse_i32_field(cursor);
1832     } else {
1833         cursor.skip_value();
1834     }
1835     if (cursor.consume('}')) {
1836         break;
1837     }
1838     cursor.expect(',');
1839 }
1840 if (!cursor.eof_after_ws()) {
1841     throw std::runtime_error("trailing data after GPT-2 config JSON");
1842 }
1843 if (config.intermediate_size == 0 && config.hidden_size > 0) {
1844     config.intermediate_size = config.hidden_size * 4;
1845 }
1846 validate_config(config);
1847 return config;
1848 }
1849
1850 Gpt2ReferenceModel load_gpt2_reference_model(const std::filesystem::path& ↪
↪ config_json,
1851                                             const std::filesystem::path& ↪
↪ safetensors_path) {
1852     Gpt2ReferenceModel model;
1853     model.config = load_gpt2_config(config_json);
1854     const SafeTensorFile file = load_safetensors_file(safetensors_path);
1855     const Gpt2Config& config = model.config;
1856     const std::array<std::int64_t, 2> wte_shape{config.vocab_size, config.↪
↪ hidden_size};
1857     const std::array<std::int64_t, 2> wpe_shape{config.max_positions, config.↪
↪ hidden_size};
1858     const std::array<std::int64_t, 1> hidden_shape{config.hidden_size};
1859
1860     const std::array<std::string_view, 2> wte_names = embedding_names("wte.↪
↪ weight");
1861     const std::array<std::string_view, 2> wpe_names = embedding_names("wpe.↪
↪ weight");

```

```

1862 model.token_embedding = require_first_tensor(file, wte_names, wte_shape);
1863 model.position_embedding = require_first_tensor(file, wpe_names, wpe_shape);
1864 model.layers.resize(check_size(config.layer_count, "layer_count"));
1865
1866 for (std::int32_t layer_id = 0; layer_id < config.layer_count; ++layer_id)
1867 {
1868     Gpt2LayerWeightsF32& layer = model.layers[check_size(layer_id, "
1869     layer_id")];
1870     layer.ln_1_weight = require_layer_tensor(file, layer_id, "ln_1.weight",
1871     hidden_shape);
1872     layer.ln_1_bias = require_layer_tensor(file, layer_id, "ln_1.bias",
1873     hidden_shape);
1874     layer.ln_2_weight = require_layer_tensor(file, layer_id, "ln_2.weight",
1875     hidden_shape);
1876     layer.ln_2_bias = require_layer_tensor(file, layer_id, "ln_2.bias",
1877     hidden_shape);
1878
1879     const std::array<std::int64_t, 2> c_attn_shape{
1880         config.hidden_size,
1881         config.hidden_size * LCQI_GPT2_ATTENTION_PROJECTIONS};
1882     std::vector<float> c_attn_weight =
1883         require_layer_tensor(file, layer_id, "attn.c_attn.weight",
1884         c_attn_shape);
1885     transpose_hf_conv1d(c_attn_weight,
1886         config.hidden_size,
1887         config.hidden_size *
1888         LCQI_GPT2_ATTENTION_PROJECTIONS);
1889     const std::array<std::int64_t, 1> c_attn_bias_shape{
1890         config.hidden_size * LCQI_GPT2_ATTENTION_PROJECTIONS};
1891     layer.c_attn = make_linear(config.hidden_size,
1892         config.hidden_size *
1893         LCQI_GPT2_ATTENTION_PROJECTIONS,
1894         std::move(c_attn_weight),
1895         require_layer_tensor(file,
1896             layer_id,
1897             "attn.c_attn.bias",
1898             c_attn_bias_shape));
1899
1900     const std::array<std::int64_t, 2> c_proj_shape{config.hidden_size,
1901     config.hidden_size};
1902     std::vector<float> c_proj_weight =
1903         require_layer_tensor(file, layer_id, "attn.c_proj.weight",
1904         c_proj_shape);
1905     transpose_hf_conv1d(c_proj_weight, config.hidden_size, config.
1906     hidden_size);
1907     layer.c_proj = make_linear(config.hidden_size,
1908         config.hidden_size,
1909         std::move(c_proj_weight),
1910         require_layer_tensor(file,
1911             layer_id,
1912             "attn.c_proj.bias",

```

```

1901                                     hidden_shape));
1902
1903     const std::array<std::int64_t, 2> c_fc_shape{
1904         config.hidden_size,
1905         config.intermediate_size};
1906     std::vector<float> c_fc_weight =
1907         require_layer_tensor(file, layer_id, "mlp.c_fc.weight", c_fc_shape)↵
1908 ↵ ;
1909     transpose_hf_conv1d(c_fc_weight, config.hidden_size, config.↵
1910 ↵ intermediate_size);
1911     const std::array<std::int64_t, 1> intermediate_shape{config.↵
1912 ↵ intermediate_size};
1913     layer.c_fc = make_linear(config.hidden_size,
1914                             config.intermediate_size,
1915                             std::move(c_fc_weight),
1916                             require_layer_tensor(file,
1917                                                     layer_id,
1918                                                     "mlp.c_fc.bias",
1919                                                     intermediate_shape));
1920
1921     const std::array<std::int64_t, 2> mlp_proj_shape{
1922         config.intermediate_size,
1923         config.hidden_size};
1924     std::vector<float> mlp_proj_weight =
1925         require_layer_tensor(file, layer_id, "mlp.c_proj.weight", ↵
1926 ↵ mlp_proj_shape);
1927     transpose_hf_conv1d(mlp_proj_weight, config.intermediate_size, config.↵
1928 ↵ hidden_size);
1929     layer.mlp_c_proj = make_linear(config.intermediate_size,
1930                                   config.hidden_size,
1931                                   std::move(mlp_proj_weight),
1932                                   require_layer_tensor(file,
1933                                                         layer_id,
1934                                                         "mlp.c_proj.bias",
1935                                                         hidden_shape));
1936 }
1937
1938 const std::array<std::string_view, 2> ln_weight_names = embedding_names("↵
1939 ↵ ln_f.weight");
1940 const std::array<std::string_view, 2> ln_bias_names = embedding_names("ln_f↵
1941 ↵ .bias");
1942 model.final_ln_weight = require_first_tensor(file, ln_weight_names, ↵
1943 ↵ hidden_shape);
1944 model.final_ln_bias = require_first_tensor(file, ln_bias_names, ↵
1945 ↵ hidden_shape);
1946 if (file.manifest.find_tensor("lm_head.weight") != nullptr) {
1947     model.tie_lm_head_to_embedding = false;
1948     model.lm_head_weight = require_tensor(file, "lm_head.weight", wte_shape↵
1949 ↵ );
1950 } else {
1951     model.tie_lm_head_to_embedding = true;
1952 }

```

```

1943     validate_model(model);
1944     return model;
1945 }
1946
1947 Gpt2ReferenceModel load_gpt2_from_directory(const std::filesystem::path& ↵
    ↵ model_directory) {
1948     return load_gpt2_reference_model(model_directory / "config.json",
1949                                     find_gpt2_weight_file(model_directory));
1950 }
1951
1952 Gpt2ForwardWorkspace::Gpt2ForwardWorkspace(const Gpt2Config& config) : config_(↵
    ↵ config) {
1953     this->reset_for_config(config);
1954 }
1955
1956 void Gpt2ForwardWorkspace::reset_for_config(const Gpt2Config& config) {
1957     validate_config(config);
1958     this->config_ = config;
1959     const std::size_t hidden_size = checked_size(this->config_.hidden_size, "↵
    ↵ hidden_size");
1960     const std::size_t intermediate_size =
1961         checked_size(this->config_.intermediate_size, "intermediate_size");
1962     this->hidden_.assign(hidden_size, 0.0F);
1963     this->normed_.assign(hidden_size, 0.0F);
1964     this->query_.assign(hidden_size, 0.0F);
1965     this->key_.assign(hidden_size, 0.0F);
1966     this->value_.assign(hidden_size, 0.0F);
1967     this->attention_.assign(hidden_size, 0.0F);
1968     this->projected_.assign(hidden_size, 0.0F);
1969     this->qkv_packed_.assign(
1970         matrix_size(LCQI_GPT2_ATTENTION_PROJECTIONS, this->config_.hidden_size)↵
    ↵ ,
1971         0.0F);
1972     this->mlp_fc_.assign(intermediate_size, 0.0F);
1973     this->mlp_out_.assign(hidden_size, 0.0F);
1974     this->logits_.assign(checked_size(this->config_.vocab_size, "vocab_size"), ↵
    ↵ 0.0F);
1975     this->scores_.assign(checked_size(this->config_.max_positions, "↵
    ↵ max_positions"), 0.0F);
1976 }
1977
1978 std::span<float> Gpt2ForwardWorkspace::hidden() noexcept {
1979     return this->hidden_;
1980 }
1981
1982 std::span<float> Gpt2ForwardWorkspace::normed() noexcept {
1983     return this->normed_;
1984 }
1985
1986 std::span<float> Gpt2ForwardWorkspace::query() noexcept {
1987     return this->query_;
1988 }

```

```

1989
1990 std::span<float> Gpt2ForwardWorkspace::key() noexcept {
1991     return this->key_;
1992 }
1993
1994 std::span<float> Gpt2ForwardWorkspace::value() noexcept {
1995     return this->value_;
1996 }
1997
1998 std::span<float> Gpt2ForwardWorkspace::attention() noexcept {
1999     return this->attention_;
2000 }
2001
2002 std::span<float> Gpt2ForwardWorkspace::projected() noexcept {
2003     return this->projected_;
2004 }
2005
2006 std::span<float> Gpt2ForwardWorkspace::qkv_packed() noexcept {
2007     return this->qkv_packed_;
2008 }
2009
2010 std::span<float> Gpt2ForwardWorkspace::mlp_fc() noexcept {
2011     return this->mlp_fc_;
2012 }
2013
2014 std::span<float> Gpt2ForwardWorkspace::mlp_out() noexcept {
2015     return this->mlp_out_;
2016 }
2017
2018 std::span<float> Gpt2ForwardWorkspace::logits() noexcept {
2019     return this->logits_;
2020 }
2021
2022 std::span<float> Gpt2ForwardWorkspace::scores_prefix(std::int32_t count) {
2023     if (count < 0 || count > this->config_.max_positions) {
2024         throw std::runtime_error("GPT-2 attention scores count out of range");
2025     }
2026     return std::span<float>(this->scores_.data(), checked_size(count, "scores ←
↪ count"));
2027 }
2028
2029 Gpt2KvCache::Gpt2KvCache(const Gpt2Config& config) : config_(config) {
2030     validate_config(this->config_);
2031     const std::size_t element_count =
2032         checked_size(this->config_.layer_count, "layer_count") *
2033         checked_size(this->config_.max_positions, "max_positions") *
2034         checked_size(this->config_.head_count, "head_count") *
2035         checked_size(head_dim(this->config_), "head_dim");
2036     this->keys_.assign(element_count, 0.0F);
2037     this->values_.assign(element_count, 0.0F);
2038     this->written_.assign(
2039         checked_size(this->config_.layer_count, "layer_count") *

```



```

2040         checked_size(this->config_.max_positions, "max_positions"),
2041         0);
2042     }
2043
2044     void Gpt2KvCache::append(std::int32_t layer_id,
2045                               std::int32_t model_position,
2046                               std::span<const float> key,
2047                               std::span<const float> value) {
2048         const std::size_t hidden_size = checked_size(this->config_.hidden_size, "←
        hidden_size");
2049         if (key.size() != hidden_size || value.size() != hidden_size) {
2050             throw std::runtime_error("GPT-2 KV key/value size mismatch");
2051         }
2052         this->validate_address(layer_id, model_position, 0);
2053         const std::int32_t local_head_dim = head_dim(this->config_);
2054         for (std::int32_t head = 0; head < this->config_.head_count; ++head) {
2055             const std::size_t source =
2056                 checked_size(head, "head") * checked_size(local_head_dim, "head_dim←
        ");
2057             const std::size_t target = this->base_offset(layer_id, model_position, ←
        head);
2058             std::copy_n(key.begin() + static_cast<std::ptrdiff_t>(source),
2059                         local_head_dim,
2060                         this->keys_.begin() + static_cast<std::ptrdiff_t>(target));
2061             std::copy_n(value.begin() + static_cast<std::ptrdiff_t>(source),
2062                         local_head_dim,
2063                         this->values_.begin() + static_cast<std::ptrdiff_t>(target)←
        );
2064         }
2065         const std::size_t written_index =
2066             checked_size(layer_id, "layer_id") *
2067             checked_size(this->config_.max_positions, "max_positions") +
2068             checked_size(model_position, "model_position");
2069         this->written_[written_index] = 1;
2070         this->filled_tokens_ = std::max(this->filled_tokens_, model_position + 1);
2071     }
2072
2073     std::span<const float> Gpt2KvCache::key(std::int32_t layer_id,
2074                                              std::int32_t model_position,
2075                                              std::int32_t head) const {
2076         this->validate_address(layer_id, model_position, head);
2077         this->validate_written(layer_id, model_position);
2078         return this->key_unchecked(layer_id, model_position, head);
2079     }
2080
2081     std::span<const float> Gpt2KvCache::value(std::int32_t layer_id,
2082                                               std::int32_t model_position,
2083                                               std::int32_t head) const {
2084         this->validate_address(layer_id, model_position, head);
2085         this->validate_written(layer_id, model_position);
2086         return this->value_unchecked(layer_id, model_position, head);
2087     }

```



```

2088
2089 void detail::attend_cached_position(const Gpt2KvCache& cache,
2090                                   std::int32_t layer_id,
2091                                   std::int32_t model_position,
2092                                   std::span<const float> query,
2093                                   std::span<float> scores,
2094                                   std::span<float> output) {
2095     cache.validate_address(layer_id, model_position, 0);
2096     cache.validate_written(layer_id, model_position);
2097     const std::int32_t local_head_dim = head_dim(cache.config_);
2098     const std::size_t hidden_size = checked_size(cache.config_.hidden_size, "↵
↵ hidden_size");
2099     const std::size_t local_head_dim_size = checked_size(local_head_dim, "↵
↵ head_dim");
2100     const std::size_t model_position_size =
2101         checked_size(model_position, "model_position");
2102     if (query.size() != hidden_size || output.size() != hidden_size ||
2103         scores.size() < model_position_size + 1) {
2104         throw std::runtime_error("GPT-2 cached attention workspace size ↵
↵ mismatch");
2105     }
2106
2107     const float score_scale = 1.0F / std::sqrt(static_cast<float>(↵
↵ local_head_dim));
2108     std::fill(output.begin(), output.end(), 0.0F);
2109     for (std::int32_t head = 0; head < cache.config_.head_count; ++head) {
2110         const std::size_t head_base =
2111             checked_size(head, "head") * local_head_dim_size;
2112         float max_score = LCQI_GPT2_SOFTMAX_NEGATIVE_INFINITY;
2113         for (std::size_t past = 0; past <= model_position_size; ++past) {
2114             const std::int32_t past_i32 = static_cast<std::int32_t>(past);
2115             const std::span<const float> key =
2116                 cache.key_unchecked(layer_id, past_i32, head);
2117             float dot = 0.0F;
2118             for (std::size_t dim = 0; dim < local_head_dim_size; ++dim) {
2119                 dot += query[head_base + dim] * key[dim];
2120             }
2121             const float score = dot * score_scale;
2122             scores[past] = score;
2123             max_score = std::max(max_score, score);
2124         }
2125
2126         float denominator = 0.0F;
2127         for (std::size_t past = 0; past <= model_position_size; ++past) {
2128             scores[past] = std::exp(scores[past] - max_score);
2129             denominator += scores[past];
2130         }
2131         if (denominator <= 0.0F) {
2132             throw std::runtime_error("GPT-2 cached attention denominator is ↵
↵ invalid");
2133         }
2134

```

```

2135     for (std::size_t past = 0; past <= model_position_size; ++past) {
2136         const std::int32_t past_i32 = static_cast<std::int32_t>(past);
2137         const float probability = scores[past] / denominator;
2138         const std::span<const float> value =
2139             cache.value_unchecked(layer_id, past_i32, head);
2140         for (std::size_t dim = 0; dim < local_head_dim_size; ++dim) {
2141             output[head_base + dim] += probability * value[dim];
2142         }
2143     }
2144 }
2145 }
2146
2147 std::span<const float> Gpt2KvCache::key_unchecked(std::int32_t layer_id,
2148                                                  std::int32_t model_position,
2149                                                  std::int32_t head) const {
2150     ↪ noexcept {
2151         const std::size_t offset = this->base_offset_unchecked(layer_id, ↪
2152             ↪ model_position, head);
2153         return std::span<const float>(
2154             this->keys_.data() + offset,
2155             static_cast<std::size_t>(head_dim(this->config_)));
2156     }
2157
2158 std::span<const float> Gpt2KvCache::value_unchecked(std::int32_t layer_id,
2159                                                    std::int32_t model_position, ↪
2160                                                    ↪ ,
2161                                                    std::int32_t head) const {
2162     ↪ noexcept {
2163         const std::size_t offset = this->base_offset_unchecked(layer_id, ↪
2164             ↪ model_position, head);
2165         return std::span<const float>(
2166             this->values_.data() + offset,
2167             static_cast<std::size_t>(head_dim(this->config_)));
2168     }
2169
2170 std::int32_t Gpt2KvCache::filled_tokens() const noexcept {
2171     return this->filled_tokens_;
2172 }
2173
2174 std::size_t Gpt2KvCache::byte_size() const noexcept {
2175     return (this->keys_.size() + this->values_.size()) * sizeof(float) +
2176         this->written_.size() * sizeof(std::uint8_t);
2177 }
2178
2179 std::size_t Gpt2KvCache::base_offset(std::int32_t layer_id,
2180                                       std::int32_t model_position,
2181                                       std::int32_t head) const {
2182     return (((checked_size(layer_id, "layer_id") *
2183         checked_size(this->config_.max_positions, "max_positions")) +
2184         checked_size(model_position, "model_position")) *
2185         checked_size(this->config_.head_count, "head_count") +
2186         checked_size(head, "head")) *

```

```

2182         checked_size(head_dim(this->config_), "head_dim");
2183     }
2184
2185     std::size_t Gpt2KvCache::base_offset_unchecked(std::int32_t layer_id,
2186                                                    std::int32_t model_position,
2187                                                    std::int32_t head) const ↵
2188     ↵ noexcept {
2189         return (((static_cast<std::size_t>(layer_id) *
2190                  static_cast<std::size_t>(this->config_.max_positions)) +
2191                  static_cast<std::size_t>(model_position)) *
2192                  static_cast<std::size_t>(this->config_.head_count) +
2193                  static_cast<std::size_t>(head)) *
2194                  static_cast<std::size_t>(head_dim(this->config_)));
2195     }
2196
2197     void Gpt2KvCache::validate_address(std::int32_t layer_id,
2198                                        std::int32_t model_position,
2199                                        std::int32_t head) const {
2200         if (layer_id < 0 || layer_id >= this->config_.layer_count) {
2201             throw std::runtime_error("GPT-2 KV layer_id out of range");
2202         }
2203         if (model_position < 0 || model_position >= this->config_.max_positions) {
2204             throw std::runtime_error("GPT-2 KV model_position out of range");
2205         }
2206         if (head < 0 || head >= this->config_.head_count) {
2207             throw std::runtime_error("GPT-2 KV head out of range");
2208         }
2209     }
2210
2211     void Gpt2KvCache::validate_written(std::int32_t layer_id,
2212                                        std::int32_t model_position) const {
2213         const std::size_t written_index =
2214             checked_size(layer_id, "layer_id") *
2215             checked_size(this->config_.max_positions, "max_positions") +
2216             checked_size(model_position, "model_position");
2217         if (written_index >= this->written_.size() || this->written_[written_index] ↵
2218             ↵ == 0) {
2219             throw std::runtime_error("GPT-2 KV slot has not been written");
2220         }
2221     }
2222
2223     Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(const Gpt2ReferenceModel& ↵
2224         ↵ model)
2225         : Gpt2CachedGreedyDecoder(model, nullptr) {}
2226
2227     Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(
2228         const Gpt2ReferenceModel& model,
2229         Gpt2HotspotProfile* hotspot_profile)
2230         : Gpt2CachedGreedyDecoder(model, hotspot_profile, Gpt2ExecutionOptions{}) ↵
2231         ↵ {}
2232
2233     Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(

```

```

2230     const Gpt2ReferenceModel& model,
2231     Gpt2HotspotProfile* hotspot_profile,
2232     const Gpt2ExecutionOptions& execution_options)
2233     : model_(&model),
2234       hotspot_profile_(hotspot_profile),
2235       execution_options_(execution_options),
2236       cache_(model.config),
2237       workspace_(model.config),
2238       worker_pool_(make_worker_pool(model, execution_options)) {
2239     validate_model(*this->model_);
2240 }
2241
2242 Gpt2CachedGreedyDecoder::~Gpt2CachedGreedyDecoder() = default;
2243
2244 Gpt2CachedGreedyDecoder::Gpt2CachedGreedyDecoder(Gpt2CachedGreedyDecoder&&) ←
    ↪ noexcept =
2245     default;
2246
2247 Gpt2CachedGreedyDecoder& Gpt2CachedGreedyDecoder::operator=(
2248     Gpt2CachedGreedyDecoder&&) noexcept = default;
2249
2250 void Gpt2CachedGreedyDecoder::advance_without_prediction(std::int32_t token_id) ←
    ↪ {
2251     static_cast<void>(run_gpt2_cached_step_unchecked(*this->model_,
2252                                                       this->cache_,
2253                                                       this->workspace_,
2254                                                       token_id,
2255                                                       Gpt2CachedStepMode::↪
    ↪ advance_only,
2256                                                       this->hotspot_profile_,
2257                                                       this->execution_options_,
2258                                                       this->worker_pool_.get()));
2259 }
2260
2261 std::int32_t Gpt2CachedGreedyDecoder::step(std::int32_t token_id) {
2262     return run_gpt2_cached_step_unchecked(*this->model_,
2263                                           this->cache_,
2264                                           this->workspace_,
2265                                           token_id,
2266                                           Gpt2CachedStepMode::predict_token,
2267                                           this->hotspot_profile_,
2268                                           this->execution_options_,
2269                                           this->worker_pool_.get())
2270         .predicted_token;
2271 }
2272
2273 Gpt2ForwardResult Gpt2CachedGreedyDecoder::step_with_logits(std::int32_t ↪
    ↪ token_id) {
2274     return run_gpt2_cached_step_unchecked(*this->model_,
2275                                           this->cache_,
2276                                           this->workspace_,
2277                                           token_id,

```

```

2278         Gpt2CachedStepMode::logits,
2279         this->hotspot_profile_,
2280         this->execution_options_,
2281         this->worker_pool_.get());
2282     }
2283
2284     const Gpt2KvCache& Gpt2CachedGreedyDecoder::cache() const noexcept {
2285         return this->cache_;
2286     }
2287
2288     std::int32_t Gpt2CachedGreedyDecoder::filled_tokens() const noexcept {
2289         return this->cache_.filled_tokens();
2290     }
2291
2292     std::size_t Gpt2CachedGreedyDecoder::kv_cache_bytes() const noexcept {
2293         return this->cache_.byte_size();
2294     }
2295
2296     std::int32_t Gpt2CachedGreedyDecoder::worker_count() const noexcept {
2297         return this->worker_pool_ == nullptr ? 1 : this->worker_pool_->worker_count();
2298     }
2299
2300     Gpt2ForwardResult run_gpt2_forward(const Gpt2ReferenceModel& model,
2301                                       std::span<const std::int32_t> token_ids) {
2302         validate_model(model);
2303         if (token_ids.empty()) {
2304             throw std::runtime_error("GPT-2 forward needs at least one token");
2305         }
2306         if (token_ids.size() > checked_size(model.config.max_positions, "max_positions")) {
2307             throw std::runtime_error("GPT-2 token_ids exceed max_positions");
2308         }
2309
2310         const std::int32_t sequence_length = static_cast<std::int32_t>(token_ids.size());
2311         std::vector<float> hidden_by_pos(matrix_size(sequence_length, model.config.hidden_size), 0.0F);
2312         for (std::int32_t position = 0; position < sequence_length; ++position) {
2313             const std::int32_t token_id = token_ids[checked_size(position, "position")];
2314             if (token_id < 0 || token_id >= model.config.vocab_size) {
2315                 throw std::runtime_error("GPT-2 token id out of range");
2316             }
2317             const std::span<const float> token =
2318                 row_span(model.token_embedding, token_id, model.config.hidden_size);
2319             const std::span<const float> position_embedding =
2320                 row_span(model.position_embedding, position, model.config.hidden_size);
2321             std::span<float> hidden = std::span<float>(
2322                 hidden_by_pos.data() + matrix_size(position, model.config.hidden_size),

```

```

    ↪ hidden_size),
2323         checked_size(model.config.hidden_size, "hidden_size"));
2324     for (std::int32_t dim = 0; dim < model.config.hidden_size; ++dim) {
2325         hidden[checked_size(dim, "dim")] =
2326             token[checked_size(dim, "dim")] + position_embedding[↪
    ↪ checked_size(dim, "dim")]);
2327     }
2328 }
2329
2330 for (const Gpt2LayerWeightsF32& layer : model.layers) {
2331     forward_gpt2_layer(model.config, layer, sequence_length, hidden_by_pos)↪
    ↪ ;
2332 }
2333
2334 std::vector<float> normed(checked_size(model.config.hidden_size, "↪
    ↪ hidden_size"), 0.0F);
2335 const std::span<const float> final_hidden = std::span<const float>(
2336     hidden_by_pos.data() + matrix_size(sequence_length - 1, model.config.↪
    ↪ hidden_size),
2337     checked_size(model.config.hidden_size, "hidden_size"));
2338 layer_norm(final_hidden,
2339     model.final_ln_weight,
2340     model.final_ln_bias,
2341     model.config.layer_norm_epsilon,
2342     normed);
2343
2344 Gpt2ForwardResult result;
2345 result.logits.assign(checked_size(model.config.vocab_size, "vocab_size"), ↪
    ↪ 0.0F);
2346 compute_logits(model, normed, result.logits);
2347 result.predicted_token = argmax(result.logits);
2348 return result;
2349 }
2350
2351 Gpt2ForwardResult run_gpt2_forward_cached(const Gpt2ReferenceModel& model,
2352     Gpt2KvCache& cache,
2353     std::int32_t token_id) {
2354     validate_model(model);
2355     Gpt2ForwardWorkspace workspace(model.config);
2356     Gpt2ExecutionOptions options;
2357     options.worker_count = 1;
2358     return run_gpt2_cached_step_unchecked(
2359         model,
2360         cache,
2361         workspace,
2362         token_id,
2363         Gpt2CachedStepMode::logits,
2364         nullptr,
2365         options,
2366         nullptr);
2367 }
2368

```

```

2369 std::vector<std::int32_t> gpt2_generate_greedy(const Gpt2ReferenceModel& model,
2370                                              std::span<const std::int32_t> ←
    prompt_token_ids,
2371                                              std::int32_t max_new_tokens) {
2372     if (max_new_tokens < 0) {
2373         throw std::runtime_error("max_new_tokens cannot be negative");
2374     }
2375     std::vector<std::int32_t> tokens(prompt_token_ids.begin(), prompt_token_ids ←
    .end());
2376     for (std::int32_t step = 0; step < max_new_tokens; ++step) {
2377         if (tokens.size() >= checked_size(model.config.max_positions, " ←
    max_positions")) {
2378             throw std::runtime_error("GPT-2 generation would exceed ←
    max_positions");
2379         }
2380         const Gpt2ForwardResult result = run_gpt2_forward(model, tokens);
2381         tokens.push_back(result.predicted_token);
2382         if (model.config.eos_token_id >= 0 && result.predicted_token == model. ←
    config.eos_token_id) {
2383             break;
2384         }
2385     }
2386     return tokens;
2387 }
2388
2389 std::vector<std::int32_t> gpt2_generate_greedy_cached(
2390     const Gpt2ReferenceModel& model,
2391     std::span<const std::int32_t> prompt_token_ids,
2392     std::int32_t max_new_tokens) {
2393     if (prompt_token_ids.empty()) {
2394         throw std::runtime_error("GPT-2 generation needs at least one prompt ←
    token");
2395     }
2396     if (max_new_tokens < 0) {
2397         throw std::runtime_error("max_new_tokens cannot be negative");
2398     }
2399     if (prompt_token_ids.size() > checked_size(model.config.max_positions, " ←
    max_positions")) {
2400         throw std::runtime_error("GPT-2 prompt exceeds max_positions");
2401     }
2402
2403     Gpt2CachedGreedyDecoder decoder(model);
2404     std::vector<std::int32_t> tokens(prompt_token_ids.begin(), prompt_token_ids ←
    .end());
2405     if (max_new_tokens == 0) {
2406         return tokens;
2407     }
2408     for (std::size_t index = 0; index + 1 < prompt_token_ids.size(); ++index) {
2409         decoder.advance_without_prediction(prompt_token_ids[index]);
2410     }
2411     std::int32_t predicted_token = decoder.step(prompt_token_ids.back());
2412     for (std::int32_t step = 0; step < max_new_tokens; ++step) {

```

```

2413     if (tokens.size() >= checked_size(model.config.max_positions, "↵
↵ max_positions")) {
2414         throw std::runtime_error("GPT-2 generation would exceed ↵
↵ max_positions");
2415     }
2416     tokens.push_back(predicted_token);
2417     if (model.config.eos_token_id >= 0 && predicted_token == model.config.↵
↵ eos_token_id) {
2418         break;
2419     }
2420     if (step + 1 < max_new_tokens) {
2421         predicted_token = decoder.step(predicted_token);
2422     }
2423 }
2424 return tokens;
2425 }
2426
2427 Gpt2ReferenceModel make_tiny_gpt2_reference_model() {
2428     Gpt2ReferenceModel model;
2429     model.config.hidden_size = 4;
2430     model.config.head_count = 2;
2431     model.config.layer_count = 1;
2432     model.config.max_positions = 8;
2433     model.config.vocab_size = 6;
2434     model.config.intermediate_size = 8;
2435     model.config.bos_token_id = 0;
2436     model.config.eos_token_id = 5;
2437     model.config.layer_norm_epsilon = 1.0e-5F;
2438     model.config.activation_function = "gelu_new";
2439
2440     model.token_embedding = {
2441         0.20F, -0.10F, 0.00F, 0.10F,
2442         -0.10F, 0.30F, 0.20F, -0.20F,
2443         0.05F, 0.10F, -0.30F, 0.25F,
2444         0.40F, -0.20F, 0.15F, 0.05F,
2445         -0.30F, 0.05F, 0.35F, -0.15F,
2446         0.10F, 0.20F, -0.10F, 0.30F,
2447     };
2448     model.position_embedding = {
2449         0.01F, 0.02F, 0.03F, 0.04F,
2450         -0.02F, 0.01F, 0.00F, 0.03F,
2451         0.03F, -0.01F, 0.02F, 0.00F,
2452         0.00F, 0.02F, -0.02F, 0.01F,
2453         0.02F, 0.00F, 0.01F, -0.01F,
2454         -0.01F, 0.03F, 0.00F, 0.02F,
2455         0.01F, -0.02F, 0.03F, 0.00F,
2456         0.00F, 0.01F, -0.01F, 0.02F,
2457     };
2458
2459     Gpt2LayerWeightsF32 layer;
2460     layer.ln_1_weight = ones(4);
2461     layer.ln_1_bias = zeros(4);

```



```

2462     layer.ln_2_weight = {0.9F, 1.1F, 1.0F, 0.95F};
2463     layer.ln_2_bias = {0.01F, -0.02F, 0.00F, 0.03F};
2464     layer.c_attn = make_linear(
2465         4,
2466         12,
2467         {
2468             0.10F, -0.20F, 0.05F, 0.15F,
2469             -0.05F, 0.12F, 0.20F, -0.10F,
2470             0.18F, 0.02F, -0.08F, 0.11F,
2471             -0.12F, 0.06F, 0.14F, 0.04F,
2472             0.07F, 0.16F, -0.09F, 0.03F,
2473             -0.15F, 0.05F, 0.13F, 0.10F,
2474             0.04F, -0.11F, 0.19F, -0.02F,
2475             0.09F, 0.08F, -0.06F, 0.17F,
2476             0.20F, -0.04F, 0.01F, 0.06F,
2477             -0.08F, 0.15F, 0.07F, -0.03F,
2478             0.05F, 0.10F, -0.14F, 0.12F,
2479             0.11F, -0.07F, 0.16F, 0.02F,
2480         },
2481         {
2482             0.01F, -0.02F, 0.00F, 0.03F,
2483             -0.01F, 0.02F, 0.01F, -0.02F,
2484             0.00F, 0.01F, -0.01F, 0.02F,
2485         });
2486     layer.c_proj = make_linear(
2487         4,
2488         4,
2489         {
2490             0.10F, 0.05F, -0.08F, 0.12F,
2491             -0.04F, 0.11F, 0.09F, -0.02F,
2492             0.07F, -0.10F, 0.13F, 0.05F,
2493             0.02F, 0.14F, -0.06F, 0.08F,
2494         },
2495         {0.01F, 0.00F, -0.01F, 0.02F});
2496     layer.c_fc = make_linear(
2497         4,
2498         8,
2499         {
2500             0.20F, -0.10F, 0.05F, 0.03F,
2501             -0.05F, 0.14F, 0.12F, -0.08F,
2502             0.09F, 0.07F, -0.11F, 0.16F,
2503             -0.12F, 0.05F, 0.18F, 0.02F,
2504             0.04F, -0.09F, 0.13F, 0.10F,
2505             0.11F, 0.03F, -0.07F, 0.15F,
2506             -0.02F, 0.17F, 0.06F, -0.04F,
2507             0.08F, -0.06F, 0.10F, 0.12F,
2508         },
2509         {0.01F, -0.02F, 0.00F, 0.03F, -0.01F, 0.02F, 0.01F, 0.00F});
2510     layer.mlp_c_proj = make_linear(
2511         8,
2512         4,
2513         {

```

```

2514         0.10F, -0.08F, 0.06F, 0.12F, -0.04F, 0.07F, 0.03F, 0.11F,
2515         -0.05F, 0.13F, 0.02F, -0.09F, 0.15F, 0.04F, -0.03F, 0.08F,
2516         0.07F, 0.01F, -0.12F, 0.10F, 0.06F, -0.05F, 0.14F, 0.02F,
2517         0.03F, 0.09F, 0.11F, -0.06F, 0.05F, 0.12F, -0.08F, 0.04F,
2518     },
2519     {0.00F, 0.01F, -0.01F, 0.02F});
2520     model.layers.push_back(std::move(layer));
2521     model.final_ln_weight = {1.0F, 0.95F, 1.05F, 1.0F};
2522     model.final_ln_bias = {0.0F, 0.01F, -0.01F, 0.02F};
2523     model.tie_lm_head_to_embedding = true;
2524     validate_model(model);
2525     return model;
2526 }
2527
2528 } // namespace lcqi

```

`gpt2_cli.cpp` 负责把模型目录、prompt、`max_new_tokens` 和 benchmark 选项接到 GPT-2 reference path。无参数时运行 tiny GPT-2 fixture；传入 HuggingFace 模型目录时加载真实资产。

Listing 10.14 src/gpt2_cli.cpp

```

1  #include <lcqi/gpt2_reference.hpp>
2
3  #include <chrono>
4  #include <cstdlib>
5  #include <exception>
6  #include <filesystem>
7  #include <iostream>
8  #include <span>
9  #include <stdexcept>
10 #include <string>
11 #include <string_view>
12 #include <vector>
13
14 namespace {
15
16 constexpr int LCQI_GPT2_TINY_ARGC = 1;
17 constexpr int LCQI_GPT2_MIN_REAL_ARGC = 3;
18 constexpr std::int32_t LCQI_GPT2_DEFAULT_MAX_NEW_TOKENS = 8;
19 constexpr std::int32_t LCQI_GPT2_TINY_PROMPT_0 = 1;
20 constexpr std::int32_t LCQI_GPT2_TINY_PROMPT_1 = 2;
21 constexpr std::int32_t LCQI_GPT2_TINY_PROMPT_2 = 3;
22
23 enum class Gpt2EngineMode {
24     full_prefix,
25     cached_kv,
26 };
27
28 struct Gpt2CliOptions {
29     Gpt2EngineMode engine = Gpt2EngineMode::cached_kv;
30     bool benchmark = false;
31     bool profile_hotspots = false;
32     std::filesystem::path model_dir;

```

```

33     std::string prompt;
34     std::int32_t max_new_tokens = LCQI_GPT2_DEFAULT_MAX_NEW_TOKENS;
35     std::int32_t worker_count = 0;
36 };
37
38 struct Gpt2BenchmarkTimings {
39     double load_ms = 0.0;
40     double tokenize_ms = 0.0;
41     double prefill_ms = 0.0;
42     double decode_ms = 0.0;
43     double generate_ms = 0.0;
44     double total_ms = 0.0;
45     std::size_t prefill_steps = 0;
46     std::size_t decode_steps = 0;
47     std::size_t kv_cache_bytes = 0;
48     std::int32_t worker_count = 1;
49 };
50
51 using Clock = std::chrono::steady_clock;
52
53 void print_ids(std::span<const std::int32_t> ids) {
54     for (std::size_t index = 0; index < ids.size(); ++index) {
55         if (index != 0) {
56             std::cout << ' ';
57         }
58         std::cout << ids[index];
59     }
60 }
61
62 [[nodiscard]] double elapsed_ms(Clock::time_point start, Clock::time_point end) ←
    ↪ {
63     return std::chrono::duration<double, std::milli>(end - start).count();
64 }
65
66 [[nodiscard]] std::size_t checked_size(std::int64_t value, const char* name) {
67     if (value < 0) {
68         throw std::runtime_error(std::string(name) + " cannot be negative");
69     }
70     return static_cast<std::size_t>(value);
71 }
72
73 [[nodiscard]] const char* engine_name(Gpt2EngineMode engine) {
74     switch (engine) {
75         case Gpt2EngineMode::full_prefix:
76             return "full_prefix";
77         case Gpt2EngineMode::cached_kv:
78             return "cached_kv";
79     }
80     return "unknown";
81 }
82
83 [[nodiscard]] Gpt2EngineMode parse_engine(std::string_view value) {

```

```

84     if (value == "full" || value == "full_prefix") {
85         return Gpt2EngineMode::full_prefix;
86     }
87     if (value == "cached" || value == "cached_kv") {
88         return Gpt2EngineMode::cached_kv;
89     }
90     throw std::runtime_error("unknown --engine value, expected cached or full");
91 }
92
93 [[nodiscard]] std::int32_t parse_non_negative_i32(const std::string& text,
94                                                  const char* name) {
95     const std::int32_t value = static_cast<std::int32_t>(std::stoi(text));
96     if (value < 0) {
97         throw std::runtime_error(std::string(name) + " cannot be negative");
98     }
99     return value;
100 }
101
102 [[nodiscard]] Gpt2CliOptions parse_options(int argc, char** argv) {
103     Gpt2CliOptions options;
104     std::vector<std::string> positional;
105     for (int index = 1; index < argc; ++index) {
106         const std::string arg = argv[index];
107         if (arg == "--benchmark") {
108             options.benchmark = true;
109         } else if (arg == "--profile-hotspots") {
110             options.profile_hotspots = true;
111             options.benchmark = true;
112         } else if (arg == "--engine") {
113             if (index + 1 >= argc) {
114                 throw std::runtime_error("--engine requires a value");
115             }
116             ++index;
117             options.engine = parse_engine(argv[index]);
118         } else if (arg.rfind("--engine=", 0) == 0) {
119             options.engine = parse_engine(std::string_view(arg).substr(std::string_view(
120                 "--engine=").size()));
121         } else if (arg == "--threads") {
122             if (index + 1 >= argc) {
123                 throw std::runtime_error("--threads requires a value");
124             }
125             ++index;
126             options.worker_count = parse_non_negative_i32(argv[index], "--
127             threads");
128         } else if (arg.rfind("--threads=", 0) == 0) {
129             options.worker_count =
130                 parse_non_negative_i32(arg.substr(std::string("--threads=").
131                 size()), "--threads");
132         } else if (arg == "--full") {
133             options.engine = Gpt2EngineMode::full_prefix;
134         } else if (arg == "--cached") {

```

```

132     options.engine = Gpt2EngineMode::cached_kv;
133 } else if (!arg.empty() && arg.front() == '-') {
134     throw std::runtime_error("unknown option: " + arg);
135 } else {
136     positional.push_back(arg);
137 }
138 }
139
140 if (positional.size() < 2 || positional.size() > 3) {
141     throw std::runtime_error(
142         "usage: lcqi_gpt2 [--engine cached|full] [--threads N] [--benchmark↵
↵ ] "
143         "model_dir prompt [max_new_tokens]");
144 }
145 options.model_dir = positional[0];
146 options.prompt = positional[1];
147 if (positional.size() == 3) {
148     options.max_new_tokens = parse_non_negative_i32(positional[2], "↵
↵ max_new_tokens");
149 }
150 return options;
151 }
152
153 [[nodiscard]] std::vector<std::int32_t> generate_full_prefix(
154     const lcqi::Gpt2ReferenceModel& model,
155     std::span<const std::int32_t> prompt_ids,
156     std::int32_t max_new_tokens,
157     Gpt2BenchmarkTimings& timings) {
158     const Clock::time_point generate_start = Clock::now();
159     std::vector<std::int32_t> tokens(prompt_ids.begin(), prompt_ids.end());
160     for (std::int32_t step = 0; step < max_new_tokens; ++step) {
161         if (tokens.size() >= checked_size(model.config.max_positions, "↵
↵ max_positions")) {
162             throw std::runtime_error("GPT-2 generation would exceed ↵
↵ max_positions");
163         }
164         const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(model, ↵
↵ tokens);
165         ++timings.decode_steps;
166         tokens.push_back(result.predicted_token);
167         if (model.config.eos_token_id >= 0 && result.predicted_token == model.↵
↵ config.eos_token_id) {
168             break;
169         }
170     }
171     const Clock::time_point generate_end = Clock::now();
172     timings.generate_ms = elapsed_ms(generate_start, generate_end);
173     timings.decode_ms = timings.generate_ms;
174     return tokens;
175 }
176
177 [[nodiscard]] std::vector<std::int32_t> generate_cached(

```

```

178     const lcqi::Gpt2ReferenceModel& model,
179     std::span<const std::int32_t> prompt_ids,
180     std::int32_t max_new_tokens,
181     Gpt2BenchmarkTimings& timings,
182     lcqi::Gpt2HotspotProfile* hotspot_profile,
183     std::int32_t worker_count) {
184     if (prompt_ids.empty()) {
185         throw std::runtime_error("GPT-2 generation needs at least one prompt ↵
↵ token");
186     }
187     lcqi::Gpt2ExecutionOptions execution_options;
188     execution_options.worker_count = worker_count;
189     lcqi::Gpt2CachedGreedyDecoder decoder(model, hotspot_profile, ↵
↵ execution_options);
190     timings.kv_cache_bytes = decoder.kv_cache_bytes();
191     timings.worker_count = decoder.worker_count();
192     std::vector<std::int32_t> tokens(prompt_ids.begin(), prompt_ids.end());
193     if (max_new_tokens == 0) {
194         return tokens;
195     }
196
197     const Clock::time_point generate_start = Clock::now();
198     const Clock::time_point prefill_start = Clock::now();
199     for (std::size_t index = 0; index + 1 < prompt_ids.size(); ++index) {
200         decoder.advance_without_prediction(prompt_ids[index]);
201         ++timings.prefill_steps;
202     }
203     std::int32_t predicted_token = decoder.step(prompt_ids.back());
204     ++timings.prefill_steps;
205     const Clock::time_point prefill_end = Clock::now();
206     timings.prefill_ms = elapsed_ms(prefill_start, prefill_end);
207
208     const Clock::time_point decode_start = Clock::now();
209     for (std::int32_t step = 0; step < max_new_tokens; ++step) {
210         if (tokens.size() >= checked_size(model.config.max_positions, "↵
↵ max_positions")) {
211             throw std::runtime_error("GPT-2 generation would exceed ↵
↵ max_positions");
212         }
213         tokens.push_back(predicted_token);
214         if (model.config.eos_token_id >= 0 && predicted_token == model.config.↵
↵ eos_token_id) {
215             break;
216         }
217         if (step + 1 < max_new_tokens) {
218             predicted_token = decoder.step(predicted_token);
219             ++timings.decode_steps;
220         }
221     }
222     const Clock::time_point decode_end = Clock::now();
223     const Clock::time_point generate_end = Clock::now();
224     timings.decode_ms = elapsed_ms(decode_start, decode_end);

```

```

225     timings.generate_ms = elapsed_ms(generate_start, generate_end);
226     return tokens;
227 }
228
229 void print_benchmark(const Gpt2BenchmarkTimings& timings,
230                     std::size_t prompt_tokens,
231                     std::size_t generated_tokens) {
232     const double new_tokens = static_cast<double>(generated_tokens);
233     const double generated_tokens_per_second =
234         timings.generate_ms > 0.0 ? new_tokens * 1000.0 / timings.generate_ms : ↵
        ↵ 0.0;
235     const double decode_tokens_per_second =
236         timings.decode_ms > 0.0
237         ? static_cast<double>(timings.decode_steps) * 1000.0 / timings.↵
        ↵ decode_ms
        : 0.0;
238
239     std::cout << "benchmark_load_ms " << timings.load_ms << "\n";
240     std::cout << "benchmark_tokenize_ms " << timings.tokenize_ms << "\n";
241     std::cout << "benchmark_prefill_ms " << timings.prefill_ms << "\n";
242     std::cout << "benchmark_decode_ms " << timings.decode_ms << "\n";
243     std::cout << "benchmark_generate_ms " << timings.generate_ms << "\n";
244     std::cout << "benchmark_total_ms " << timings.total_ms << "\n";
245     std::cout << "benchmark_prompt_tokens " << prompt_tokens << "\n";
246     std::cout << "benchmark_generated_tokens " << generated_tokens << "\n";
247     std::cout << "benchmark_prefill_steps " << timings.prefill_steps << "\n";
248     std::cout << "benchmark_decode_steps " << timings.decode_steps << "\n";
249     std::cout << "benchmark_worker_count " << timings.worker_count << "\n";
250     std::cout << "benchmark_generate_tokens_per_second "
251         << generated_tokens_per_second << "\n";
252     std::cout << "benchmark_decode_tokens_per_second "
253         << decode_tokens_per_second << "\n";
254     std::cout << "benchmark_kv_cache_bytes " << timings.kv_cache_bytes << "\n";
255 }
256
257 void print_hotspot_profile(const lcqi::Gpt2HotspotProfile& profile) {
258     const double total = profile.total_step_ms > 0.0 ? profile.total_step_ms : ↵
        ↵ 1.0;
259     const auto print_ms = [total](std::string_view name, double value) {
260         std::cout << "hotspot_" << name << "_ms " << value << "\n";
261         std::cout << "hotspot_" << name << "_pct " << (value * 100.0 / total) ↵
        ↵ << "\n";
262     };
263     std::cout << "hotspot_decoder_steps " << profile.decoder_steps << "\n";
264     std::cout << "hotspot_layer_steps " << profile.layer_steps << "\n";
265     print_ms("total_step", profile.total_step_ms);
266     print_ms("embedding", profile.embedding_ms);
267     print_ms("layer_norm", profile.layer_norm_ms);
268     print_ms("final_norm", profile.final_norm_ms);
269     print_ms("qkv_projection", profile.qkv_projection_ms);
270     print_ms("kv_append", profile.kv_append_ms);
271     print_ms("attention", profile.attention_ms);
272

```

```

273     print_ms("attention_projection", profile.attention_projection_ms);
274     print_ms("residual_add", profile.residual_add_ms);
275     print_ms("mlp_fc", profile.mlp_fc_ms);
276     print_ms("gelu", profile.gelu_ms);
277     print_ms("mlp_projection", profile.mlp_projection_ms);
278     print_ms("lm_head", profile.lm_head_ms);
279     print_ms("logits_result", profile.logits_result_ms);
280 }
281
282 int run_tiny_mode() {
283     const lcqi::Gpt2ReferenceModel model = lcqi::make_tiny_gpt2_reference_model(
284         ↵ );
285     const std::vector<std::int32_t> prompt{
286         LCQI_GPT2_TINY_PROMPT_0,
287         LCQI_GPT2_TINY_PROMPT_1,
288         LCQI_GPT2_TINY_PROMPT_2,
289     };
290     const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(model, prompt(
291         ↵ );
292     const std::vector<std::int32_t> generated =
293         lcqi::gpt2_generate_greedy(model, prompt, 2);
294     std::cout << "mode tiny\n";
295     std::cout << "prompt_ids ";
296     print_ids(prompt);
297     std::cout << "\n";
298     std::cout << "predicted_token " << result.predicted_token << "\n";
299     std::cout << "logits";
300     for (const float value : result.logits) {
301         std::cout << ' ' << value;
302     }
303     std::cout << "\n";
304     std::cout << "generated_ids ";
305     print_ids(generated);
306     std::cout << "\n";
307     return EXIT_SUCCESS;
308 }
309
310 int run_real_model_mode(int argc, char** argv) {
311     if (argc < LCQI_GPT2_MIN_REAL_ARGC) {
312         throw std::runtime_error(
313             ↵ "usage: lcqi_gpt2 [--engine cached|full] [--threads N] [--benchmark(
314             ↵ "model_dir prompt [max_new_tokens]");
315     }
316     const Clock::time_point total_start = Clock::now();
317     const Gpt2CliOptions options = parse_options(argc, argv);
318
319     Gpt2BenchmarkTimings timings;
320     const Clock::time_point load_start = Clock::now();
321     const lcqi::Gpt2ReferenceModel model = lcqi::load_gpt2_from_directory(
322         ↵ options.model_dir);
323     const lcqi::Gpt2Tokenizer tokenizer = lcqi::load_gpt2_tokenizer(

```



```

321     options.model_dir / "vocab.json",
322     options.model_dir / "merges.txt",
323     model.config.bos_token_id,
324     model.config.eos_token_id);
325     timings.load_ms = elapsed_ms(load_start, Clock::now());
326
327     const Clock::time_point tokenize_start = Clock::now();
328     const std::vector<std::int32_t> prompt_ids = lcqi::gpt2_encode(tokenizer, ←
    ↪ options.prompt);
329     timings.tokenize_ms = elapsed_ms(tokenize_start, Clock::now());
330
331     std::vector<std::int32_t> generated;
332     lcqi::Gpt2HotspotProfile hotspot_profile;
333     if (options.engine == Gpt2EngineMode::full_prefix) {
334         if (options.profile_hotspots) {
335             throw std::runtime_error("--profile-hotspots currently supports ←
    ↪ cached engine only");
336         }
337         generated = generate_full_prefix(model, prompt_ids, options.←
    ↪ max_new_tokens, timings);
338     } else {
339         generated = generate_cached(model,
340                                     prompt_ids,
341                                     options.max_new_tokens,
342                                     timings,
343                                     options.profile_hotspots ? &hotspot_profile←
    ↪ : nullptr,
344                                     options.worker_count);
345     }
346     timings.total_ms = elapsed_ms(total_start, Clock::now());
347
348     std::cout << "mode gpt2_directory\n";
349     std::cout << "engine " << engine_name(options.engine) << "\n";
350     std::cout << "model_dir " << options.model_dir.string() << "\n";
351     std::cout << "prompt_ids ";
352     print_ids(prompt_ids);
353     std::cout << "\n";
354     std::cout << "generated_ids ";
355     print_ids(generated);
356     std::cout << "\n";
357     std::cout << "generated_text " << lcqi::gpt2_decode(tokenizer, generated) ←
    ↪ << "\n";
358     if (options.benchmark) {
359         const std::size_t generated_tokens = generated.size() - prompt_ids.size←
    ↪ ();
360         print_benchmark(timings, prompt_ids.size(), generated_tokens);
361     }
362     if (options.profile_hotspots) {
363         print_hotspot_profile(hotspot_profile);
364     }
365     return EXIT_SUCCESS;
366 }

```

```
367
368 } // namespace
369
370 int main(int argc, char** argv) {
371     try {
372         if (argc == LCQI_GPT2_TINY_ARGC) {
373             return run_tiny_mode();
374         }
375         return run_real_model_mode(argc, argv);
376     } catch (const std::exception& error) {
377         std::cerr << "error: " << error.what() << "\n";
378         return EXIT_FAILURE;
379     }
380 }
```

第 11 章实践：运行时内存计划、liveness 与 Workspace

本章对应原书中的第三册“推理运行时与内存规划”。不要把它当作概念复述，本章要把 activation arena、workspace、liveness、state edge、debug dump、buffer reuse 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux_cpu_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `layout`，核心命令或测试入口是 `lcqi_trace`。

Listing 11.1 第 11 章实践：运行时内存计划、liveness 与 Workspace 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↪ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / layout
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：比较普通 activation 和 KV cache 的生命周期。读者要先手写预期结果，再运行项目观察输出。动作是：说明为什么 KV cache 不能被 activation planner 当作临时缓冲复用。如果输出里没有 `checksum` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：内存复用只在生命周期不重叠时成立；debug/oracle/materialization 会延长生命周期这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux_cpu_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 liveness 表：每个 checkpoint 的 `defined_at`、`last_used_at`、是否 state、是否允许复用。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 12 章实践：IR、Pass、Lowering 与 Executable Plan

本章对应原书中的第三册“AI 编译器细化：IR、pass、lowering 与 executable plan”。不要把它当作概念复述，本章要把 typed graph、pass、fusion、layout propagation、backend capability、fallback reason 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux_cpu_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `run_inference`，核心命令或测试入口是 `lcqi_cli`。

Listing 12.1 第 12 章实践：IR、Pass、Lowering 与 Executable Plan 的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_cli / run_inference
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：把 tiny MLP 写成三节点 IR：linear、relu、linear。读者要先手写预期结果，再运行项目观察输出。动作是：说明哪些 pass 可以融合，哪些必须保留为 oracle checkpoint。如果输出里没有 `backend` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：IR 的价值不是画图，而是让 shape、layout、量化、后端选择和 fallback 诊断有稳定承载物这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux_cpu_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 executable plan 草图：节点、输入输出 tensor、layout、kernel、fallback reason 和 profiler event。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第零部分

后端优化、工业专项与验收

第 13 章实践：CPU 后端、微内核与性能证据

本章对应原书中的第三册“CPU 后端优化细讲”和“CPU decode 实战”。不要把它当作概念复述，本章要把 scalar baseline、packed weight、AVX2、objdump、perf boundary、shape sweep 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `avx2`、`GGUF`、`Q4_K` 和 `Q8_K`，核心命令或测试入口是 `lcqi_bench` 与 `lcqi_gguf_q4_bench`。

Listing 13.1 第 13 章实践：CPU 后端、微内核与性能证据的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  < DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_bench / lcqi_gguf_q4_bench / avx2 / Q4_K
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：对同一 kernel 保存 benchmark CSV 和反汇编摘要。读者要先手写预期结果，再运行项目观察输出。动作是：解释速度差来自地址流、SIMD lane、packing、load 或 reduction 哪一项。如果输出里没有 `average_us` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

第二个特例必须对齐真实模型：`lcqi_gguf_q4_bench` 读取 llama.cpp smoke 使用的同一个 `SmolLM2-135M-Instruct-Q4_K_M.gguf`，从 GGUF tensor 表中选出真实 `Q4_K` 矩阵，比较三条路径：预先解成 f32 权重再 matvec、`Q4_K` 权重直接乘 f32 input、`Q4_K` 权重乘临时 `Q8_K` input。报告必须同时写 tensor 名称、shape、原始量化字节、解码后 f32 字节、重复轮次、最大误差和 checksum。

从这个特例推出的规律是：CPU 后端优化必须同时有 oracle、反汇编、shape sweep 和降级边界；只有耗时表不够这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA、GGUF 模型缓存或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。真实模型量化切片用下面命令生成报告：

Listing 13.2 SmolLM2 Q4_K 单算子证据入口

```
python3 books/cpu-volume-3/tools/run_smolm2_q4_k_benchmark.py --no-download --<
  < repeat 20
```

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系

统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 CPU kernel evidence：命令、机器、编译器、输入 shape、backend、checksum、反汇编摘要和未验证 PMU 边界。若使用真实 GGUF 模型，还要给出模型 repo、文件名、SHA-256、tensor 名、量化格式、同模型 llama.cpp 参考和“为什么不能把单 tensor us 直接等价于端到端 tokens/s”。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 14 章实践：GPU 后端边界、copy 与同步成本

本章对应原书中的第三册“GPU 硬件基础”和“GPU 后端优化细讲”。不要把它当作概念复述，本章要把 device buffer、stream、event、kernel launch、copy、sync、fallback 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux_cpu_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `StepTrace`，核心命令或测试入口是 `lcqi_trace`。

Listing 14.1 第 14 章实践：GPU 后端边界、copy 与同步成本的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_trace / StepTrace
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：当前项目没有 GPU kernel，也要写清 adapter 合同。读者要先手写预期结果，再运行项目观察输出。动作是：把 CPU trace 字段扩展成 GPU StepTrace 需要哪些字段。如果输出里没有 `copy_time_ns` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：GPU 快不快不能只看 kernel；copy、sync、sampler、fallback 和 batch wait 必须进入端到端 trace 这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux_cpu_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交 GPU adapter 设计：buffer owner、stream、event、H2D/D2H 字节、sync 点、fallback 和 oracle 对照。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 15 章实践：工业 LLM 专项能力对照

本章对应原书中的第三册“工业 LLM 专项：投机解码、MoE、LoRA、多 GPU 与源码对照”。不要把它当作概念复述，本章要把 speculative decoding、MoE routing、LoRA adapter、multi-GPU、source reading 放到同一个可运行项目里检查。贯穿代码仍然是 [books/cpu-volume-3/labs/linux_cpu_inference](#)；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `bandwidth_limit`，核心命令或测试入口是 `lcqi_ledger`。

Listing 15.1 第 15 章实践：工业 LLM 专项能力对照的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  <- DCMMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: lcqi_ledger / bandwidth_limit
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：选择一个专项，不写泛泛介绍。读者要先手写预期结果，再运行项目观察输出。动作是：把它接到现有 trace、memory、backend 或 serving 字段上。如果输出里没有 `tokens/s` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：工业专项不是功能清单；必须说明它购买什么能力、增加什么状态、破坏哪些假设这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

三、实践任务：把观察固定成报告字段

任务一：定位源码。至少阅读 [books/cpu-volume-3/labs/linux_cpu_inference/README.md](#)、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交专项设计卡：动机、状态新增、trace 字段、回归测试、性能账本和不能验证的部分。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

第 16 章实践：工程验收、回归门禁与最终项目

本章对应原书中的第三册“工程验收与回归体系”。不要把它当作概念复述，本章要把 reference、unit test、trace golden、benchmark gate、SLO gate、coverage、CI 放到同一个可运行项目里检查。贯穿代码仍然是 `books/cpu-volume-3/labs/linux_cpu_inference`；如果某个实验在当前机器上不能验证，报告必须写清降级原因，不能把源码存在当作实测证据。

一、对应原书问题：概念必须落到对象和命令

原书讲的是系统边界，本章实践要回答三个问题。第一，哪个代码对象承载这个概念；第二，哪个命令能产生可复查输出；第三，哪个坏输入或缺失字段能证明边界不是空话。这里的核心对象包括 `lcqi_tests`，核心命令或测试入口是 `ctest`。

Listing 16.1 第 16 章实践：工程验收、回归门禁与最终项目的最小命令入口

```
cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -<
  ↳ DCMAKE_BUILD_TYPE=Debug
cmake --build books/cpu-volume-3/build/lcqi-debug
ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure
# chapter focus: ctest / lcqi_tests
```

二、从特例推到规律：先抓一个能手算的切片

本章先看特例：把所有前面报告合成最终验收包。读者要先手写预期结果，再运行项目观察输出。动作是：运行 `ctest` 并说明每个 test 保护哪条合同。如果输出里没有 `100% tests passed` 或对应字段无法解释，就不要继续优化；先回到 reference、输入合同和 trace 字段。

从这个特例推出的规律是：实践卷最终目标是长期维护：结果正确、trace 稳定、性能可解释、服务指标可回归、失败边界不伪装这条规律要写进报告，不要只写“运行成功”。运行成功只说明某条路径没崩，解释清楚输入、输出、错误路径和不可验证边界，才说明学会了这个知识点。

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任务一：定位源码。至少阅读 `books/cpu-volume-3/labs/linux_cpu_inference/README.md`、相关 `include/lcqi` 头文件和一个 `src` 实现文件。报告要写出函数入口、输入对象、输出对象和错误处理方式。

任务二：运行命令。保存构建命令、测试命令、核心输出和环境字段。若命令依赖 AVX2、perf、GPU、NUMA 或外部库，必须记录当前机器是否支持；不支持时把结论降级为“接口已设计，性能未验证”。

任务三：构造反例。围绕本章知识点设计一个坏输入、缺失字段或不满足前提的场景，说明系统应拒绝、降级、fallback 还是继续执行。只给正常路径不合格。

四、验收和递进大作业

验收标准：报告能从一个具体输入推到工程规则；命令可以在仓库中复跑；输出字段能对应到源码；失败路径说得清；未验证边界不伪装成结论。

递进大作业：提交最终项目：README、命令记录、报告索引、测试结果、benchmark 表、trace 摘要、风险清单和下一阶段计划。这个作业会被后续章节继续引用，命名、目录和字段不要随意变化。

代码走读：工具、Smoke、Benchmark 与回归测试

工具不是附属品

LCQI 的工程证据不只在 C++ 源码里。smoke 脚本决定真实模型怎样下载、校验和运行；benchmark 程序决定性能口径；trace、ledger 和 serving probe 决定报告字段；CTest 决定哪些行为能长期回归。读者必须把这些文件当作工程合同的一部分。

Benchmark、Trace、Ledger 与 Serving Probe

`bench.cpp` 输出 int8 kernel benchmark。它服务性能观察，但必须和 max diff 对照一起看。

Listing 16.2 src/bench.cpp

```

1 #include <lcqi/int8_kernels.hpp>
2
3 #include <algorithm>
4 #include <cstdlib>
5 #include <iostream>
6 #include <stdexcept>
7 #include <string>
8 #include <vector>
9
10 namespace {
11
12 constexpr const char* LCQI_TARGET_ARCH_X86_64 = "x86_64";
13 constexpr const char* LCQI_TARGET_ARCH_I386 = "i386";
14 constexpr const char* LCQI_TARGET_ARCH_UNKNOWN = "unknown";
15 constexpr std::int32_t LCQI_DEFAULT_REPEAT = 200;
16 constexpr int LCQI_REPEAT_ARG_INDEX = 1;
17 constexpr std::int32_t LCQI_LARGE_CASE_REPEAT_DIVISOR = 4;
18 constexpr std::int32_t LCQI_MIN_LARGE_CASE_REPEAT = 1;
19 constexpr std::int32_t LCQI_DECODE_SHAPE_SIZE = 4096;
20
21 const char* target_arch() noexcept {
22     #if defined(__x86_64__) || defined(_M_X64)
23         return LCQI_TARGET_ARCH_X86_64;
24     #elif defined(__i386__) || defined(_M_IX86)
25         return LCQI_TARGET_ARCH_I386;
26     #else
27         return LCQI_TARGET_ARCH_UNKNOWN;
28     #endif
29 }
30
31 std::int32_t parse_repeat(int argc, char** argv) {
32     if (argc <= LCQI_REPEAT_ARG_INDEX) {
33         return LCQI_DEFAULT_REPEAT;
34     }
35     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
↵ LCQI_REPEAT_ARG_INDEX]));
36     if (repeat <= 0) {

```

```

37     throw std::runtime_error("repeat must be positive");
38 }
39 return repeat;
40 }
41
42 std::vector<lcqi::KernelBenchmarkCase> make_cases(std::int32_t repeat) {
43     return {
44         {128, 128, 16, repeat},
45         {256, 256, 16, repeat},
46         {512, 512, 16, repeat},
47         {513, 257, 16, repeat},
48         {1024, 4096, 16, std::max(LCQI_MIN_LARGE_CASE_REPEAT,
49                                     repeat / LCQI_LARGE_CASE_REPEAT_DIVISOR)},
50         {LCQI_DECODE_SHAPE_SIZE, LCQI_DECODE_SHAPE_SIZE, 16,
51          std::max(LCQI_MIN_LARGE_CASE_REPEAT, repeat / ↵
52 ↵ LCQI_LARGE_CASE_REPEAT_DIVISOR)}},
53     };
54 }
55 } // namespace
56
57 int main(int argc, char** argv) {
58     try {
59         const std::int32_t repeat = parse_repeat(argc, argv);
60         std::cout
61             << "target_arch,input_size,output_size,output_block,repeat,backend,↵
62 ↵ available,"
63             << "average_us,max_abs_diff,checksum\n";
64         for (const lcqi::KernelBenchmarkCase& benchmark_case : make_cases(↵
65 ↵ repeat)) {
66             const lcqi::KernelBenchmarkResult result =
67                 lcqi::benchmark_linear_i8_case(benchmark_case);
68             for (const lcqi::KernelBackendBenchmark& backend : result.backends)↵
69 ↵ {
70                 std::cout << target_arch() << ','
71                     << result.benchmark_case.input_size << ','
72                     << result.benchmark_case.output_size << ','
73                     << result.benchmark_case.output_block_size << ','
74                     << result.benchmark_case.repeat << ','
75                     << backend.name << ','
76                     << (backend.available ? 1 : 0) << ','
77                     << backend.average_us << ','
78                     << backend.max_abs_diff << ','
79                     << backend.checksum << '\n';
80             }
81         }
82         return EXIT_SUCCESS;
83     } catch (const std::exception& error) {
84         std::cerr << "error: " << error.what() << '\n';
85         return EXIT_FAILURE;
86     }
87 }

```

`trace.cpp` 输出 reference decoder checkpoint。它用于定位错误边界，不应该被简化成只打印最终 token。

Listing 16.3 `src/trace.cpp`

```

1  #include <lcqi/reference_decoder.hpp>
2  #include <lcqi/tokenizer.hpp>
3
4  #include <algorithm>
5  #include <array>
6  #include <cstdlib>
7  #include <exception>
8  #include <filesystem>
9  #include <iostream>
10 #include <span>
11 #include <sstream>
12 #include <string>
13 #include <string_view>
14
15 namespace {
16
17 constexpr std::array<std::int32_t, 3> LCQI_TRACE_TOKEN_IDS{1, 2, 3};
18 constexpr std::string_view LCQI_TRACE_PROMPT = "tok1 tok2 tok3";
19 constexpr std::int32_t LCQI_TRACE_METADATA_LAYER = -1;
20 constexpr std::int32_t LCQI_TRACE_BOS_ID = 0;
21 constexpr std::int32_t LCQI_TRACE_EOS_ID = 4;
22 constexpr std::size_t LCQI_TRACE_VALUE_LIMIT = 8;
23
24 std::filesystem::path tokenizer_fixture_path() {
25     return std::filesystem::path(__FILE__).parent_path().parent_path() /
26         "models" / "tiny_tokenizer.txt";
27 }
28
29 std::vector<std::int32_t> decoder_tokens_from_prompt(const lcqi::TokenizerModel&
    ↪ & tokenizer,
30                                                     std::string_view prompt) {
31     std::vector<std::int32_t> full_tokens = lcqi::encode_prompt(tokenizer,
    ↪ prompt);
32     if (full_tokens.size() != LCQI_TRACE_TOKEN_IDS.size() + 2 ||
33         full_tokens.front() != LCQI_TRACE_BOS_ID ||
34         full_tokens.back() != LCQI_TRACE_EOS_ID) {
35         throw std::runtime_error("unexpected tokenizer fixture ids");
36     }
37
38     std::vector<std::int32_t> decoder_tokens(
39         full_tokens.begin() + 1,
40         full_tokens.end() - 1);
41     if (!std::equal(decoder_tokens.begin(),
42                    decoder_tokens.end(),
43                    LCQI_TRACE_TOKEN_IDS.begin(),
44                    LCQI_TRACE_TOKEN_IDS.end())) {
45         throw std::runtime_error("unexpected decoder token ids");
46     }

```

```

47     return decoder_tokens;
48 }
49
50 std::string join_ints(std::span<const std::int32_t> values) {
51     std::ostringstream stream;
52     for (std::size_t index = 0; index < values.size(); ++index) {
53         if (index != 0) {
54             stream << 'x';
55         }
56         stream << values[index];
57     }
58     return stream.str();
59 }
60
61 std::string join_int_values(std::span<const std::int32_t> values) {
62     std::ostringstream stream;
63     for (std::size_t index = 0; index < values.size(); ++index) {
64         if (index != 0) {
65             stream << ';';
66         }
67         stream << values[index];
68     }
69     return stream.str();
70 }
71
72 std::int32_t checksum_ints(std::span<const std::int32_t> values) {
73     std::int32_t result = 0;
74     for (const std::int32_t value : values) {
75         result += value;
76     }
77     return result;
78 }
79
80 std::int32_t max_abs_ints(std::span<const std::int32_t> values) {
81     std::int32_t result = 0;
82     for (const std::int32_t value : values) {
83         result = std::max(result, std::abs(value));
84     }
85     return result;
86 }
87
88 std::string join_floats(std::span<const float> values) {
89     std::ostringstream stream;
90     const std::size_t count = std::min(values.size(), LCQI_TRACE_VALUE_LIMIT);
91     for (std::size_t index = 0; index < count; ++index) {
92         if (index != 0) {
93             stream << ';';
94         }
95         stream << values[index];
96     }
97     if (values.size() > LCQI_TRACE_VALUE_LIMIT) {
98         stream << "...";

```

```

99     }
100     return stream.str();
101 }
102
103 void print_trace_csv(const lcqi::ReferenceDecodeTraceResult& trace,
104                    const lcqi::TokenizerModel& tokenizer,
105                    std::span<const std::int32_t> full_tokens) {
106     std::cout << "name,token_position,layer_id,shape,stride,dtype,layout,↵
↵ checksum,max_abs,values\n";
107     std::cout << "prompt,0," << LCQI_TRACE_METADATA_LAYER
108         << ",scalar,scalar,utf8,text,0,0," << LCQI_TRACE_PROMPT << '\n';
109     std::cout << "tokenizer,0," << LCQI_TRACE_METADATA_LAYER
110         << ', ' << full_tokens.size() << ",1,i32,contiguous,"
111         << checksum_ints(full_tokens) << ', '
112         << max_abs_ints(full_tokens) << ', '
113         << join_int_values(full_tokens) << '\n';
114     std::cout << "tokenizer_contract,0," << LCQI_TRACE_METADATA_LAYER
115         << ",scalar,scalar,u64,fnv1a,"
116         << tokenizer_contract_hash(tokenizer) << ",0,"
117         << tokenizer.chat_template_hash << '\n';
118     for (const lcqi::ReferenceTensorCheckpoint& checkpoint : trace.checkpoints)↵
↵     {
119         std::cout << checkpoint.name << ', '
120             << checkpoint.token_position << ', '
121             << checkpoint.layer_id << ', '
122             << join_ints(checkpoint.shape) << ', '
123             << join_ints(checkpoint.stride) << ', '
124             << checkpoint.dtype << ', '
125             << checkpoint.layout << ', '
126             << checkpoint.checksum << ', '
127             << checkpoint.max_abs << ', '
128             << join_floats(checkpoint.values) << '\n';
129     }
130     std::cout << "sampler_argmax,"
131         << LCQI_TRACE_TOKEN_IDS.size() - 1 << ', '
132         << LCQI_TRACE_METADATA_LAYER
133         << ', ' << trace.result.logits.size() << ",1,f32,argmax,"
134         << "0,0,token=" << trace.result.predicted_token << '\n';
135     std::cout << "generated_token,"
136         << LCQI_TRACE_TOKEN_IDS.size() << ', '
137         << LCQI_TRACE_METADATA_LAYER
138         << ",scalar,scalar,i32,generated,"
139         << trace.result.predicted_token << ', '
140         << trace.result.predicted_token << ', '
141         << trace.result.predicted_token << '\n';
142 }
143
144 } // namespace
145
146 int main() {
147     try {
148         const lcqi::TokenizerModel tokenizer =

```



```

149         lcqi::load_tokenizer(tokenizer_fixture_path());
150         const std::vector<std::int32_t> full_tokens =
151             lcqi::encode_prompt(tokenizer, LCQI_TRACE_PROMPT);
152         const std::vector<std::int32_t> decoder_tokens =
153             decoder_tokens_from_prompt(tokenizer, LCQI_TRACE_PROMPT);
154         const lcqi::ReferenceDecoderModel model = lcqi::↵
↵ make_tiny_reference_decoder_model();
155         const lcqi::ReferenceDecodeTraceResult trace =
156             lcqi::run_reference_decode_with_trace(model, decoder_tokens);
157         print_trace_csv(trace, tokenizer, full_tokens);
158         return EXIT_SUCCESS;
159     } catch (const std::exception& error) {
160         std::cerr << "error: " << error.what() << '\n';
161         return EXIT_FAILURE;
162     }
163 }

```

ledger.cpp 输出 7B 规模账本。它是估算，不是实测吞吐；读者要区分 FLOPs、KV 容量、权重/KV 字节和 bandwidth-only tokens/s 上限。

Listing 16.4 src/ledger.cpp

```

1  #include <cstdlib>
2  #include <cstdint>
3  #include <exception>
4  #include <iomanip>
5  #include <iostream>
6  #include <string_view>
7
8  namespace {
9
10 constexpr double LCQI_BYTES_PER_GB = 1000.0 * 1000.0 * 1000.0;
11 constexpr double LCQI_BYTES_PER_MIB = 1024.0 * 1024.0;
12 constexpr double LCQI_FLOPS_PER_MAC = 2.0;
13 constexpr std::int32_t LCQI_SEVENB_LAYER_COUNT = 32;
14 constexpr std::int32_t LCQI_SEVENB_HIDDEN_SIZE = 4096;
15 constexpr std::int32_t LCQI_SEVENB_QUERY_HEADS = 32;
16 constexpr std::int32_t LCQI_SEVENB_KV_HEADS = 8;
17 constexpr std::int32_t LCQI_SEVENB_HEAD_DIM = 128;
18 constexpr std::int32_t LCQI_SEVENB_INTERMEDIATE_SIZE = 11008;
19 constexpr std::int32_t LCQI_SEVENB_CONTEXT_LENGTH = 4096;
20
21 struct DTypeLedger {
22     std::string_view name;
23     double weight_gb_per_token = 0.0;
24     double kv_element_bytes = 0.0;
25 };
26
27 constexpr DTypeLedger LCQI_DTYPE_LEDGERS[] = {
28     {"fp16", 14.0, 2.0},
29     {"int8", 7.0, 2.0},
30     {"int4", 3.7, 2.0},
31 };

```



```

32
33 constexpr double LCQI_BANDWIDTH_GB_PER_S[] = {
34     50.0,
35     100.0,
36     200.0,
37     600.0,
38     1000.0,
39     1600.0,
40 };
41
42 double linear_macs_per_layer() noexcept {
43     const double hidden = LCQI_SEVENB_HIDDEN_SIZE;
44     const double kv_width = LCQI_SEVENB_KV_HEADS * LCQI_SEVENB_HEAD_DIM;
45     const double qkv = hidden * (hidden + 2.0 * kv_width);
46     const double output_projection = hidden * hidden;
47     const double mlp = 3.0 * hidden * LCQI_SEVENB_INTERMEDIATE_SIZE;
48     return qkv + output_projection + mlp;
49 }
50
51 double attention_macs_per_layer() noexcept {
52     return 2.0 * LCQI_SEVENB_QUERY_HEADS * LCQI_SEVENB_CONTEXT_LENGTH *
53           LCQI_SEVENB_HEAD_DIM;
54 }
55
56 double kv_bytes_per_token_all_layers(double element_bytes) noexcept {
57     return static_cast<double>(LCQI_SEVENB_LAYER_COUNT) * 2.0 *
58           LCQI_SEVENB_KV_HEADS * LCQI_SEVENB_HEAD_DIM * element_bytes;
59 }
60
61 double kv_read_gb_per_decode_token(double element_bytes) noexcept {
62     const double bytes = static_cast<double>(LCQI_SEVENB_CONTEXT_LENGTH) *
63           kv_bytes_per_token_all_layers(element_bytes);
64     return bytes / LCQI_BYTES_PER_GB;
65 }
66
67 double kv_capacity_mib(double element_bytes, std::int32_t context_length) ↵
68     ↵ noexcept {
69     const double bytes = static_cast<double>(context_length) *
70           kv_bytes_per_token_all_layers(element_bytes);
71     return bytes / LCQI_BYTES_PER_MIB;
72 }
73
74 void print_model_rows() {
75     const double linear_macs = linear_macs_per_layer();
76     const double attention_macs = attention_macs_per_layer();
77     const double total_flops =
78         (linear_macs + attention_macs) * LCQI_SEVENB_LAYER_COUNT * ↵
79         ↵ LCQI_FLOPS_PER_MAC;
80     std::cout << "section,item,value,unit,notes\n";
81     std::cout << "model,layers," << LCQI_SEVENB_LAYER_COUNT << ",count,7B ↵
82         ↵ fixture\n";
83     std::cout << "model,hidden," << LCQI_SEVENB_HIDDEN_SIZE << ",elements,H\n";

```

```

81     std::cout << "model,query_heads," << LCQI_SEVENB_QUERY_HEADS << ",count,GQA" <<
    << query_heads << "\n";
82     std::cout << "model,kv_heads," << LCQI_SEVENB_KV_HEADS << ",count,GQA KV" <<
    << kv_heads << "\n";
83     std::cout << "model,head_dim," << LCQI_SEVENB_HEAD_DIM << ",elements,per" <<
    << head_dim << "\n";
84     std::cout << "compute,linear_macs_per_layer," << linear_macs << ",macs," <<
    << decode_token << "\n";
85     std::cout << "compute,attention_macs_per_layer," << attention_macs
86         << ",macs,context=4096\n";
87     std::cout << "compute,total_decode_flops_per_token," << total_flops
88         << ",flops,linear plus attention\n";
89 }
90
91 void print_kv_rows() {
92     for (const std::int32_t context : {1024, 2048, 4096, 8192}) {
93         std::cout << "kv,fp16_capacity_context_" << context << ', '
94             << kv_capacity_mib(2.0, context)
95             << ",MiB,single session\n";
96     }
97 }
98
99 void print_dtype_rows() {
100     for (const DTypeLedger& dtype : LCQI_DTYPE_LEDGERS) {
101         const double kv_read_gb = kv_read_gb_per_decode_token(dtype.
    << kv_element_bytes);
102         const double total_gb = dtype.weight_gb_per_token + kv_read_gb;
103         std::cout << "decode_bytes," << dtype.name << "_weight_gb,"
104             << dtype.weight_gb_per_token << ",GB/token,weight stream\n";
105         std::cout << "decode_bytes," << dtype.name << "_kv_read_gb,"
106             << kv_read_gb << ",GB/token,context=4096\n";
107         std::cout << "decode_bytes," << dtype.name << "_total_gb,"
108             << total_gb << ",GB/token,weight plus KV\n";
109         for (const double bandwidth : LCQI_BANDWIDTH_GB_PER_S) {
110             std::cout << "bandwidth_limit," << dtype.name << "_at_"
111                 << static_cast<std::int32_t>(bandwidth) << "_gbps,"
112                 << bandwidth / total_gb
113                 << ",tokens/s,bandwidth-only upper bound\n";
114         }
115     }
116 }
117
118 } // namespace
119
120 int main() {
121     try {
122         std::cout << std::fixed << std::setprecision(3);
123         print_model_rows();
124         print_kv_rows();
125         print_dtype_rows();
126         return EXIT_SUCCESS;
127     } catch (const std::exception& error) {

```

```

128         std::cerr << "error: " << error.what() << '\n';
129         return EXIT_FAILURE;
130     }
131 }

```

`serving_probe.cpp` 固定短请求、RAG 请求、取消请求和超长请求。它把 TTFT、TPOT、拒绝和取消回收变成可观察字段。

Listing 16.5 `src/serving_probe.cpp`

```

1  #include <algorithm>
2  #include <cstdlib>
3  #include <cstdint>
4  #include <exception>
5  #include <cmath>
6  #include <iomanip>
7  #include <iostream>
8  #include <string_view>
9  #include <vector>
10
11 namespace {
12
13 constexpr double LCQI_BYTES_PER_MIB = 1024.0 * 1024.0;
14 constexpr std::int32_t LCQI_PROBE_LAYER_COUNT = 32;
15 constexpr std::int32_t LCQI_PROBE_KV_HEADS = 8;
16 constexpr std::int32_t LCQI_PROBE_HEAD_DIM = 128;
17 constexpr double LCQI_PROBE_KV_ELEMENT_BYTES = 2.0;
18 constexpr double LCQI_PROBE_KV_BUDGET_MIB = 512.0;
19 constexpr double LCQI_PROBE_WORKSPACE_MIB = 64.0;
20 constexpr double LCQI_PROBE_ARENA_MIB = 32.0;
21 constexpr double LCQI_PROBE_TRACE_MIB = 4.0;
22 constexpr double LCQI_PROBE_QUEUE_WAIT_PER_REQUEST_MS = 8.0;
23 constexpr double LCQI_PROBE_PREFILL_MS_PER_TOKEN = 0.08;
24 constexpr double LCQI_PROBE_DECODE_STEP_MS = 14.0;
25 constexpr double LCQI_PROBE_CANCEL_RECLAIM_MS = 2.0;
26 constexpr std::size_t LCQI_PERCENTILE_MIN_INDEX = 1;
27
28 struct ProbeRequest {
29     std::string_view request_id;
30     std::int32_t prompt_tokens = 0;
31     std::int32_t max_new_tokens = 0;
32     bool cancel_after_prefill = false;
33 };
34
35 struct ProbeResult {
36     std::string_view request_id;
37     std::int32_t accepted = 0;
38     std::string_view rejection_reason;
39     std::int32_t prompt_tokens = 0;
40     std::int32_t reserved_tokens = 0;
41     double kv_reserved_mib = 0.0;
42     double queue_wait_ms = 0.0;
43     double prefill_ms = 0.0;

```

```

44     double ttft_ms = 0.0;
45     double decode_step_p95_ms = 0.0;
46     double tpot_ms = 0.0;
47     std::int32_t completion_tokens = 0;
48     std::string_view finish_reason;
49     double cancel_reclaim_ms = 0.0;
50 };
51
52 const std::vector<ProbeRequest> LCQI_PROBE_REQUESTS{
53     {"req_short", 32, 4, false},
54     {"req_rag", 512, 8, false},
55     {"req_cancel", 128, 16, true},
56     {"req_too_long", 4096, 512, false},
57 };
58
59 double kv_mib_for_tokens(std::int32_t tokens) noexcept {
60     const double bytes = static_cast<double>(tokens) * LCQI_PROBE_LAYER_COUNT *
        ↪ 2.0 *
61         LCQI_PROBE_KV_HEADS * LCQI_PROBE_HEAD_DIM *
62         LCQI_PROBE_KV_ELEMENT_BYTES;
63     return bytes / LCQI_BYTES_PER_MIB;
64 }
65
66 double percentile(std::vector<double> values, double fraction) {
67     if (values.empty()) {
68         return 0.0;
69     }
70     std::sort(values.begin(), values.end());
71     const std::size_t max_index = values.size() - LCQI_PERCENTILE_MIN_INDEX;
72     const std::size_t index =
73         std::min(max_index,
74             static_cast<std::size_t>(
75                 std::ceil(fraction * static_cast<double>(max_index))));
76     return values[index];
77 }
78
79 std::vector<ProbeResult> run_probe() {
80     std::vector<ProbeResult> results;
81     double reserved_kv_mib = 0.0;
82     std::int32_t accepted_before = 0;
83     for (const ProbeRequest& request : LCQI_PROBE_REQUESTS) {
84         ProbeResult result;
85         result.request_id = request.request_id;
86         result.prompt_tokens = request.prompt_tokens;
87         result.reserved_tokens = request.prompt_tokens + request.max_new_tokens
        ↪ ;
88         result.kv_reserved_mib = kv_mib_for_tokens(result.reserved_tokens);
89         const double static_session_mib =
90             result.kv_reserved_mib + LCQI_PROBE_WORKSPACE_MIB +
91             LCQI_PROBE_ARENA_MIB + LCQI_PROBE_TRACE_MIB;
92         if (reserved_kv_mib + result.kv_reserved_mib > LCQI_PROBE_KV_BUDGET_MIB
        ↪ ) {

```

```

93         result.accepted = 0;
94         result.rejection_reason = "memory_budget_exceeded";
95         result.finish_reason = "rejected";
96         results.push_back(result);
97         continue;
98     }
99
100     reserved_kv_mib += result.kv_reserved_mib;
101     result.accepted = 1;
102     result.rejection_reason = "none";
103     result.queue_wait_ms = static_cast<double>(accepted_before) *
104                             LCQI_PROBE_QUEUE_WAIT_PER_REQUEST_MS;
105     result.prefill_ms = static_cast<double>(request.prompt_tokens) *
106                             LCQI_PROBE_PREFILL_MS_PER_TOKEN;
107     result.ttft_ms = result.queue_wait_ms + result.prefill_ms + ↵
↵ LCQI_PROBE_DECODE_STEP_MS;
108     result.decode_step_p95_ms = LCQI_PROBE_DECODE_STEP_MS +
109                                 static_session_mib / 512.0;
110     result.tpot_ms = result.decode_step_p95_ms;
111     if (request.cancel_after_prefill) {
112         result.completion_tokens = 0;
113         result.finish_reason = "cancelled";
114         result.cancel_reclaim_ms = LCQI_PROBE_CANCEL_RECLAIM_MS;
115         reserved_kv_mib -= result.kv_reserved_mib;
116     } else {
117         result.completion_tokens = request.max_new_tokens;
118         result.finish_reason = "max_tokens";
119     }
120     ++accepted_before;
121     results.push_back(result);
122 }
123 return results;
124 }
125
126 void print_request_rows(const std::vector<ProbeResult>& results) {
127     std::cout << "row_type,request_id,accepted,rejection_reason,prompt_tokens,↵
↵ reserved_tokens,"
128                 "kv_reserved_mib,queue_wait_ms,prefill_ms,ttft_ms,↵
↵ decode_step_p95_ms,"
129                 "tpot_ms,completion_tokens,finish_reason,cancel_reclaim_ms,↵
↵ metric_name,"
130                 "metric_value\n";
131     for (const ProbeResult& result : results) {
132         std::cout << "request," << result.request_id << ',' <<
133                     << result.accepted << ',' <<
134                     << result.rejection_reason << ',' <<
135                     << result.prompt_tokens << ',' <<
136                     << result.reserved_tokens << ',' <<
137                     << result.kv_reserved_mib << ',' <<
138                     << result.queue_wait_ms << ',' <<
139                     << result.prefill_ms << ',' <<
140                     << result.ttft_ms << ',' <<

```

```

141         << result.decode_step_p95_ms << ', '
142         << result.tpot_ms << ', '
143         << result.completion_tokens << ', '
144         << result.finish_reason << ', '
145         << result.cancel_reclaim_ms << ",,\n";
146     }
147 }
148
149 void print_summary_row(const std::vector<ProbeResult>& results) {
150     std::vector<double> ttft_values;
151     std::vector<double> queue_values;
152     double accepted = 0.0;
153     double rejected = 0.0;
154     double cancelled = 0.0;
155     double reserved_peak_mib = 0.0;
156     double running_reserved_mib = 0.0;
157     for (const ProbeResult& result : results) {
158         if (result.accepted == 0) {
159             rejected += 1.0;
160             continue;
161         }
162         accepted += 1.0;
163         ttft_values.push_back(result.ttft_ms);
164         queue_values.push_back(result.queue_wait_ms);
165         running_reserved_mib += result.kv_reserved_mib;
166         reserved_peak_mib = std::max(reserved_peak_mib, running_reserved_mib);
167         if (result.finish_reason == "cancelled") {
168             cancelled += 1.0;
169             running_reserved_mib -= result.kv_reserved_mib;
170         }
171     }
172     const double total = accepted + rejected;
173     const double rejection_rate = total == 0.0 ? 0.0 : rejected / total;
174     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
175     << "accepted_count," << accepted << '\n';
176     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,"
177     << LCQI_PROBE_CANCEL_RECLAIM_MS << ",cancel_count," << cancelled <<
178     << '\n';
179     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
180     << "rejected_count," << rejected << '\n';
181     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
182     << "rejection_rate," << rejection_rate << '\n';
183     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
184     << "kv_reserved_peak_mib," << reserved_peak_mib << '\n';
185     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
186     << "queue_wait_p95_ms," << percentile(queue_values, 0.95) << '\n' <<
187     ;
188     std::cout << "summary,all,0,summary,0,0,0,0,0,0,0,0,summary,0,"
189     << "ttft_p95_ms," << percentile(ttft_values, 0.95) << '\n';
190 }
191 } // namespace

```

```

191
192 int main() {
193     try {
194         std::cout << std::fixed << std::setprecision(3);
195         const std::vector<ProbeResult> results = run_probe();
196         print_request_rows(results);
197         print_summary_row(results);
198         return EXIT_SUCCESS;
199     } catch (const std::exception& error) {
200         std::cerr << "error: " << error.what() << '\n';
201         return EXIT_FAILURE;
202     }
203 }

```

`onednn_bench.cpp` 是可选外部库对标入口。它只在环境安装 oneDNN 时构建，不能把不可用平台写成已验证性能。

Listing 16.6 `src/onednn_bench.cpp`

```

1 #include <dnnl.hpp>
2
3 #include <chrono>
4 #include <cstdint>
5 #include <cstdlib>
6 #include <iostream>
7 #include <numeric>
8 #include <stdexcept>
9 #include <string>
10 #include <unordered_map>
11 #include <vector>
12
13 namespace {
14
15 constexpr std::int64_t LCQI_ONEDNN_BATCH = 1;
16 constexpr std::int64_t LCQI_ONEDNN_K = 4096;
17 constexpr std::int64_t LCQI_ONEDNN_O = 4096;
18 constexpr std::int32_t LCQI_ONEDNN_DEFAULT_REPEAT = 20;
19 constexpr std::int32_t LCQI_ONEDNN_WARMUP = 3;
20 constexpr int LCQI_REPEAT_ARG_INDEX = 1;
21 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
22 constexpr float LCQI_INPUT_SCALE = 0.125F;
23 constexpr std::int32_t LCQI_INPUT_MODULUS = 17;
24 constexpr std::int32_t LCQI_INPUT_CENTER = 8;
25 constexpr std::int32_t LCQI_WEIGHT_MODULUS = 23;
26 constexpr std::int32_t LCQI_WEIGHT_CENTER = 11;
27 constexpr float LCQI_WEIGHT_SCALE = 0.03125F;
28
29 std::int32_t parse_repeat(int argc, char** argv) {
30     if (argc <= LCQI_REPEAT_ARG_INDEX) {
31         return LCQI_ONEDNN_DEFAULT_REPEAT;
32     }
33     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
↵ LCQI_REPEAT_ARG_INDEX]));

```



```

34     if (repeat <= 0) {
35         throw std::runtime_error("repeat must be positive");
36     }
37     return repeat;
38 }
39
40 std::vector<float> make_input() {
41     std::vector<float> input(static_cast<std::size_t>(LCQI_ONEDNN_K));
42     for (std::int64_t index = 0; index < LCQI_ONEDNN_K; ++index) {
43         input[static_cast<std::size_t>(index)] =
44             static_cast<float>((index % LCQI_INPUT_MODULUS) - LCQI_INPUT_CENTER) *
45             LCQI_INPUT_SCALE;
46     }
47     return input;
48 }
49
50 std::vector<float> make_weights_kn() {
51     std::vector<float> weights(static_cast<std::size_t>(LCQI_ONEDNN_K * LCQI_ONEDNN_0));
52     for (std::int64_t k = 0; k < LCQI_ONEDNN_K; ++k) {
53         for (std::int64_t out = 0; out < LCQI_ONEDNN_0; ++out) {
54             const std::int64_t row_major_out_k = out * LCQI_ONEDNN_K + k;
55             const float value =
56                 static_cast<float>((row_major_out_k % LCQI_WEIGHT_MODULUS) -
57                                     LCQI_WEIGHT_CENTER) *
58                 LCQI_WEIGHT_SCALE;
59             weights[static_cast<std::size_t>(k * LCQI_ONEDNN_0 + out)] = value;
60         }
61     }
62     return weights;
63 }
64
65 float checksum(const std::vector<float>& values) {
66     return std::accumulate(values.begin(), values.end(), 0.0F);
67 }
68
69 } // namespace
70
71 int main(int argc, char** argv) {
72     try {
73         const std::int32_t repeat = parse_repeat(argc, argv);
74         std::vector<float> input = make_input();
75         std::vector<float> weights = make_weights_kn();
76         std::vector<float> output(static_cast<std::size_t>(LCQI_ONEDNN_BATCH * LCQI_ONEDNN_0),
77                                0.0F);
78
79         dnnl::engine engine(dnnl::engine::kind::cpu, 0);
80         dnnl::stream stream(engine);
81
82         const dnnl::memory::dims source_dims = {LCQI_ONEDNN_BATCH, LCQI_ONEDNN_0};

```

```

↪ LCQI_ONEDNN_K};
83     const dnnl::memory::dims weight_dims = {LCQI_ONEDNN_K, LCQI_ONEDNN_0};
84     const dnnl::memory::dims output_dims = {LCQI_ONEDNN_BATCH, ↪
↪ LCQI_ONEDNN_0};

85
86     const dnnl::memory::desc source_desc(
87         source_dims, dnnl::memory::data_type::f32, dnnl::memory::format_tag↪
↪ ::ab);
88     const dnnl::memory::desc weight_desc(
89         weight_dims, dnnl::memory::data_type::f32, dnnl::memory::format_tag↪
↪ ::ab);
90     const dnnl::memory::desc output_desc(
91         output_dims, dnnl::memory::data_type::f32, dnnl::memory::format_tag↪
↪ ::ab);
92
93     dnnl::memory source_memory(source_desc, engine, input.data());
94     dnnl::memory weight_memory(weight_desc, engine, weights.data());
95     dnnl::memory output_memory(output_desc, engine, output.data());
96
97     dnnl::matmul::primitive_desc matmul_desc(engine, source_desc, ↪
↪ weight_desc, output_desc);
98     dnnl::matmul matmul(matmul_desc);
99     const std::unordered_map<int, dnnl::memory> arguments = {
100         {DNNL_ARG_SRC, source_memory},
101         {DNNL_ARG_WEIGHTS, weight_memory},
102         {DNNL_ARG_DST, output_memory},
103     };
104
105     for (std::int32_t index = 0; index < LCQI_ONEDNN_WARMUP; ++index) {
106         matmul.execute(stream, arguments);
107         stream.wait();
108     }
109
110     const auto begin = std::chrono::steady_clock::now();
111     for (std::int32_t index = 0; index < repeat; ++index) {
112         matmul.execute(stream, arguments);
113         stream.wait();
114     }
115     const auto end = std::chrono::steady_clock::now();
116     const std::chrono::duration<double> elapsed = end - begin;
117     const double average_us =
118         elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double>↪
↪ >(repeat);
119
120     std::cout << "library,input_size,output_size,dtype,repeat,average_us,↪
↪ checksum\n";
121     std::cout << "onednn," << LCQI_ONEDNN_K << ',' << LCQI_ONEDNN_0
122         << ",f32," << repeat << ',' << average_us << ','
123         << checksum(output) << '\n';
124     return EXIT_SUCCESS;
125 } catch (const std::exception& error) {
126     std::cerr << "error: " << error.what() << '\n';

```

```

127         return EXIT_FAILURE;
128     }
129 }

```

`gguf_q4_bench.cpp` 是真实 GGUF 量化 tensor 的单算子 benchmark。它读取同一个 SmolLM2 Q4_K_M GGUF, 选择真实 Q4_K 矩阵, 分别测 f32 解码基线、Q4_Kxf32 和 Q4_KxQ8_K。它输出的 `speedup` 只适用于这个单 tensor matvec, 不能写成完整模型 tokens/s。

Listing 16.7 `src/gguf_q4_bench.cpp`

```

1  #include <lcqi/gguf.hpp>
2  #include <lcqi/q4_k.hpp>
3
4  #include <algorithm>
5  #include <chrono>
6  #include <cmath>
7  #include <cstdlib>
8  #include <filesystem>
9  #include <iomanip>
10 #include <iostream>
11 #include <span>
12 #include <stdexcept>
13 #include <string>
14 #include <string_view>
15 #include <vector>
16
17 namespace {
18
19 constexpr int LCQI_ARG_MODEL = 1;
20 constexpr int LCQI_ARG_TENSOR = 2;
21 constexpr int LCQI_ARG_REPEAT = 3;
22 constexpr int LCQI_DEFAULT_REPEAT = 20;
23 constexpr std::int64_t LCQI_MAX_AUTO_ROWS = 4096;
24 constexpr double LCQI_SECONDS_TO_MICROSECONDS = 1000000.0;
25 constexpr float LCQI_INPUT_PATTERN_SCALE = 0.015625F;
26 constexpr int LCQI_INPUT_PATTERN_MODULUS = 31;
27 constexpr int LCQI_INPUT_PATTERN_CENTER = 15;
28
29 struct SelectedTensor {
30     const lcqi::GgufTensorInfo* tensor = nullptr;
31     std::int64_t rows = 0;
32     std::int64_t columns = 0;
33 };
34
35 std::int32_t parse_repeat(int argc, char** argv) {
36     if (argc <= LCQI_ARG_REPEAT) {
37         return LCQI_DEFAULT_REPEAT;
38     }
39     const std::int32_t repeat = static_cast<std::int32_t>(std::stoi(argv[↵
↵ LCQI_ARG_REPEAT]));
40     if (repeat <= 0) {
41         throw std::runtime_error("repeat must be positive");
42     }

```

```

43     return repeat;
44 }
45
46 bool is_q4_k_matrix(const lcqi::GgufTensorInfo& tensor) {
47     return tensor.type == lcqi::GgmlType::q4_k && tensor.shape.size() == 2 &&
48         tensor.shape[0] > 0 && tensor.shape[1] > 0 &&
49         tensor.shape[0] % lcqi::LCQI_QK_K_BLOCK_VALUES == 0;
50 }
51
52 bool preferred_tensor_name(std::string_view name) {
53     return name.find("ffn_up.weight") != std::string_view::npos ||
54         name.find("ffn_gate.weight") != std::string_view::npos ||
55         name.find("ffn_down.weight") != std::string_view::npos;
56 }
57
58 SelectedTensor select_tensor(const lcqi::GgufManifest& manifest, std::↵
    ↵ string_view requested) {
59     if (!requested.empty() && requested != "auto") {
60         const lcqi::GgufTensorInfo* tensor = manifest.find_tensor(requested);
61         if (tensor == nullptr) {
62             throw std::runtime_error("requested GGUF tensor not found");
63         }
64         if (!is_q4_k_matrix(*tensor)) {
65             throw std::runtime_error("requested GGUF tensor is not a rank-2 ↵
    ↵ Q4_K matrix");
66         }
67         return {tensor, tensor->shape[1], tensor->shape[0]};
68     }
69
70     const lcqi::GgufTensorInfo* best = nullptr;
71     for (const lcqi::GgufTensorInfo& tensor : manifest.tensors) {
72         if (!is_q4_k_matrix(tensor) || tensor.shape[1] > LCQI_MAX_AUTO_ROWS) {
73             continue;
74         }
75         if (best == nullptr) {
76             best = &tensor;
77             continue;
78         }
79         const bool tensor_preferred = preferred_tensor_name(tensor.name);
80         const bool best_preferred = preferred_tensor_name(best->name);
81         if ((tensor_preferred && !best_preferred) ||
82             (tensor_preferred == best_preferred &&
83              tensor.element_count() > best->element_count())) {
84             best = &tensor;
85         }
86     }
87     if (best == nullptr) {
88         throw std::runtime_error("no suitable rank-2 Q4_K tensor found in GGUF ↵
    ↵ model");
89     }
90     return {best, best->shape[1], best->shape[0]};
91 }

```

```

92
93 std::vector<float> make_input(std::int64_t columns) {
94     std::vector<float> input(static_cast<std::size_t>(columns), 0.0F);
95     for (std::int64_t index = 0; index < columns; ++index) {
96         input[static_cast<std::size_t>(index)] =
97             static_cast<float>((index % LCQI_INPUT_PATTERN_MODULUS) -
98                             LCQI_INPUT_PATTERN_CENTER) *
99             LCQI_INPUT_PATTERN_SCALE;
100     }
101     return input;
102 }
103
104 float max_abs_diff(std::span<const float> lhs, std::span<const float> rhs) {
105     if (lhs.size() != rhs.size()) {
106         throw std::runtime_error("max_abs_diff size mismatch");
107     }
108     float diff = 0.0F;
109     for (std::size_t index = 0; index < lhs.size(); ++index) {
110         diff = std::max(diff, std::fabs(lhs[index] - rhs[index]));
111     }
112     return diff;
113 }
114
115 float checksum(std::span<const float> values) {
116     float sum = 0.0F;
117     for (const float value : values) {
118         sum += value;
119     }
120     return sum;
121 }
122
123 void matvec_f32(std::span<const float> weights,
124                std::int64_t rows,
125                std::int64_t columns,
126                std::span<const float> input,
127                std::span<float> output) {
128     if (weights.size() != static_cast<std::size_t>(rows * columns) ||
129         input.size() != static_cast<std::size_t>(columns) ||
130         output.size() != static_cast<std::size_t>(rows)) {
131         throw std::runtime_error("F32 matvec shape mismatch");
132     }
133     for (std::int64_t row = 0; row < rows; ++row) {
134         const float* row_weights = weights.data() + row * columns;
135         float sum = 0.0F;
136         for (std::int64_t column = 0; column < columns; ++column) {
137             sum += row_weights[column] * input[static_cast<std::size_t>(column)];
138         }
139         output[static_cast<std::size_t>(row)] = sum;
140     }
141 }
142

```

```

143 template <typename Callable>
144 double measure_average_us(std::int32_t repeat, Callable&& callable) {
145     const auto begin = std::chrono::steady_clock::now();
146     for (std::int32_t index = 0; index < repeat; ++index) {
147         callable();
148     }
149     const auto end = std::chrono::steady_clock::now();
150     const std::chrono::duration<double> elapsed = end - begin;
151     return elapsed.count() * LCQI_SECONDS_TO_MICROSECONDS / static_cast<double>
    ↪ >(repeat);
152 }
153
154 void print_usage(const char* program) {
155     std::cerr << "usage: " << program << " <model.gguf> [tensor-name|auto] [
    ↪ repeat]\n";
156 }
157
158 } // namespace
159
160 int main(int argc, char** argv) {
161     try {
162         if (argc <= LCQI_ARG_MODEL) {
163             print_usage(argv[0]);
164             return EXIT_FAILURE;
165         }
166         const std::filesystem::path model_path = argv[LCQI_ARG_MODEL];
167         const std::string tensor_request =
168             argc > LCQI_ARG_TENSOR ? argv[LCQI_ARG_TENSOR] : "auto";
169         const std::int32_t repeat = parse_repeat(argc, argv);
170
171         const lcqi::GgufManifest manifest = lcqi::load_gguf_manifest(model_path
    ↪ );
172         const SelectedTensor selected = select_tensor(manifest, tensor_request)
    ↪ ;
173         const std::vector<std::uint8_t> tensor_bytes =
174             lcqi::read_gguf_tensor_bytes(model_path, *selected.tensor);
175         const std::vector<float> input = make_input(selected.columns);
176         const std::vector<lcqi::Q8KBlock> q8_input = lcqi::quantize_q8_k_input(
    ↪ input);
177         std::vector<float> q4_f32_output(static_cast<std::size_t>(selected.rows
    ↪ ), 0.0F);
178         std::vector<float> q4_q8_output(q4_f32_output.size(), 0.0F);
179         std::vector<float> f32_output(q4_f32_output.size(), 0.0F);
180
181         lcqi::matvec_q4_k_f32(tensor_bytes,
182                               selected.rows,
183                               selected.columns,
184                               input,
185                               q4_f32_output);
186         lcqi::matvec_q4_k_q8(tensor_bytes,
187                               selected.rows,
188                               selected.columns,

```

```

189         q8_input,
190         q4_q8_output);
191
192     const std::vector<float> f32_weights =
193         lcqi::dequantize_q4_k(tensor_bytes, selected.rows * selected.↵
↵ columns);
194     matvec_f32(f32_weights, selected.rows, selected.columns, input, ↵
↵ f32_output);
195
196     const double f32_us = measure_average_us(
197         repeat,
198         [&]() { matvec_f32(f32_weights, selected.rows, selected.columns, ↵
↵ input, f32_output); });
199     const double q4_f32_us = measure_average_us(
200         repeat,
201         [&]() {
202             lcqi::matvec_q4_k_f32(tensor_bytes,
203                                     selected.rows,
204                                     selected.columns,
205                                     input,
206                                     q4_f32_output);
207         });
208     const double q8_quantize_us = measure_average_us(
209         repeat,
210         [&]() {
211             const std::vector<lcqi::Q8KBlock> temporary =
212                 lcqi::quantize_q8_k_input(input);
213             static_cast<void>(temporary);
214         });
215     const double q4_q8_us = measure_average_us(
216         repeat,
217         [&]() {
218             lcqi::matvec_q4_k_q8(tensor_bytes,
219                                     selected.rows,
220                                     selected.columns,
221                                     q8_input,
222                                     q4_q8_output);
223         });
224
225     const double q4_q8_total_us = q8_quantize_us + q4_q8_us;
226     std::cout << std::fixed << std::setprecision(3);
227     std::cout << "lcqi_smol1m2_q4_k_benchmark\n";
228     std::cout << "model_path=" << model_path.string() << '\n';
229     std::cout << "gguf_version=" << manifest.version << '\n';
230     std::cout << "gguf_alignment=" << manifest.alignment << '\n';
231     std::cout << "gguf_tensors=" << manifest.tensors.size() << '\n';
232     std::cout << "tensor_name=" << selected.tensor->name << '\n';
233     std::cout << "tensor_type=" << lcqi::ggml_type_name(selected.tensor->↵
↵ type) << '\n';
234     std::cout << "tensor_shape_ne0_ne1=" << selected.columns << 'x' << ↵
↵ selected.rows << '\n';
235     std::cout << "tensor_bytes=" << selected.tensor->byte_size << '\n';

```



```

236     std::cout << "f32_dequantized_bytes=" << f32_weights.size() * sizeof(↵
↵ float) << '\n';
237     std::cout << "repeat=" << repeat << '\n';
238     std::cout << "f32_matvec_average_us=" << f32_us << '\n';
239     std::cout << "q4_f32_input_average_us=" << q4_f32_us << '\n';
240     std::cout << "q8_quantize_input_average_us=" << q8_quantize_us << '\n';
241     std::cout << "q4_q8_backend=" << lcqi::q4_k_q8_active_backend() << '\n'↵
↵ ;
242     std::cout << "q4_q8_matvec_average_us=" << q4_q8_us << '\n';
243     std::cout << "q4_q8_total_average_us=" << q4_q8_total_us << '\n';
244     std::cout << "speedup_q4_q8_total_vs_f32="
245         << (q4_q8_total_us > 0.0 ? f32_us / q4_q8_total_us : 0.0) << ↵
↵ '\n';
246     std::cout << "speedup_q4_f32_input_vs_f32="
247         << (q4_f32_us > 0.0 ? f32_us / q4_f32_us : 0.0) << '\n';
248     std::cout << "q4_f32_vs_f32_max_abs_diff="
249         << max_abs_diff(q4_f32_output, f32_output) << '\n';
250     std::cout << "q4_q8_vs_q4_f32_max_abs_diff="
251         << max_abs_diff(q4_q8_output, q4_f32_output) << '\n';
252     std::cout << "q4_q8_checksum=" << checksum(q4_q8_output) << '\n';
253     return EXIT_SUCCESS;
254 } catch (const std::exception& error) {
255     std::cerr << "error: " << error.what() << '\n';
256     return EXIT_FAILURE;
257 }
258 }

```

`ggml_tensors.cpp` 是真实 GGUF 混合量化进入 LCQI 的格式闸门。它按 GGML block 布局解码 `Q8_0`、`Q5_0`、`Q6_K` 和已有 `Q4_K`，也读取 `F32` 标量张量。测试用手造 block 固定每个格式的位布局，真实 SmolLM2 smoke 再验证同一个解码层能读完 272 个 tensor。

Listing 16.8 `src/ggml_tensors.cpp`

```

1  #include <lcqi/ggml_tensors.hpp>
2
3  #include <lcqi/q4_k.hpp>
4
5  #include <cmath>
6  #include <cstring>
7  #include <limits>
8  #include <stdexcept>
9  #include <string>
10
11 namespace lcqi {
12 namespace {
13
14 constexpr std::uint32_t LCQI_GGML_F32_EXPONENT_MASK = 0x7F800000U;
15 constexpr std::uint32_t LCQI_GGML_F16_SIGN_MASK = 0x8000U;
16 constexpr std::uint32_t LCQI_GGML_F16_EXPONENT_MASK = 0x7C00U;
17 constexpr std::uint32_t LCQI_GGML_F16_MANTISSA_MASK = 0x03FFU;
18 constexpr std::uint32_t LCQI_GGML_F16_EXPONENT_BIAS = 15U;
19 constexpr std::uint32_t LCQI_GGML_F32_EXPONENT_BIAS = 127U;
20 constexpr std::uint32_t LCQI_GGML_F32_MANTISSA_BITS = 23U;

```

```

21 constexpr std::uint32_t LCQI_GGML_F16_MANTISSA_BITS = 10U;
22 constexpr std::uint32_t LCQI_GGML_F16_TO_F32_SIGN_SHIFT = 16U;
23 constexpr std::uint32_t LCQI_GGML_BYTE_BITS = 8U;
24 constexpr std::uint8_t LCQI_GGML_LOW_NIBBLE_MASK = 0x0FU;
25 constexpr std::uint8_t LCQI_GGML_Q5_HIGH_BIT_MASK = 0x10U;
26 constexpr std::uint32_t LCQI_GGML_Q5_HIGH_BIT_SHIFT = 4U;
27 constexpr std::int32_t LCQI_GGML_Q5_SECOND_HALF_BIT_OFFSET = 16;
28 constexpr std::uint8_t LCQI_GGML_Q6_HIGH_TWO_BITS_MASK = 0x03U;
29 constexpr std::int32_t LCQI_GGML_Q5_ZERO_POINT = 16;
30 constexpr std::int32_t LCQI_GGML_Q6_ZERO_POINT = 32;
31 constexpr std::int32_t LCQI_GGML_Q6_HALF_BLOCK_VALUES = 128;
32 constexpr std::int32_t LCQI_GGML_Q6_QUARTER_BLOCK_VALUES = 64;
33 constexpr std::int32_t LCQI_GGML_Q6_SUBBLOCK_VALUES = 32;
34 constexpr std::int32_t LCQI_GGML_Q6_SCALE_STRIDE = 8;
35 constexpr std::int32_t LCQI_GGML_F32_BYTES = 4;
36
37 [[nodiscard]] std::size_t checked_size(std::int64_t value, const char* name) {
38     if (value < 0 ||
39         static_cast<std::uint64_t>(value) >
40         static_cast<std::uint64_t>(std::numeric_limits<std::size_t>::max())) ←
41     ) {
42         throw std::runtime_error(std::string(name) + " is outside size_t range" ←
43         );
44     }
45     return static_cast<std::size_t>(value);
46 }
47
48 void require_multiple(std::int64_t value, std::int64_t divisor, const char* ←
49     name) {
50     if (value <= 0 || value % divisor != 0) {
51         throw std::runtime_error(std::string(name) + " must be a positive ←
52         multiple of " +
53         std::to_string(divisor));
54     }
55 }
56
57 [[nodiscard]] std::uint16_t read_le_u16(const std::uint8_t* bytes) {
58     return static_cast<std::uint16_t>(
59         static_cast<std::uint16_t>(bytes[0]) |
60         (static_cast<std::uint16_t>(bytes[1]) << LCQI_GGML_BYTE_BITS));
61 }
62
63 [[nodiscard]] std::uint32_t read_le_u32(const std::uint8_t* bytes) {
64     std::uint32_t value = 0;
65     for (std::uint32_t index = 0; index < LCQI_GGML_F32_BYTES; ++index) {
66         value |= static_cast<std::uint32_t>(bytes[index]) <<
67             (LCQI_GGML_BYTE_BITS * index);
68     }
69     return value;
70 }
71
72 [[nodiscard]] float bits_to_float(std::uint32_t bits) {

```

```

69     float value = 0.0F;
70     std::memcpy(&value, &bits, sizeof(value));
71     return value;
72 }
73
74 void validate_quantized_bytes(std::span<const std::uint8_t> bytes,
75                               std::int64_t element_count,
76                               std::int64_t block_values,
77                               std::int64_t block_bytes,
78                               const char* name) {
79     require_multiple(element_count, block_values, name);
80     const std::int64_t expected_bytes = (element_count / block_values) * ↵
    ↵ block_bytes;
81     if (bytes.size() != checked_size(expected_bytes, name)) {
82         throw std::runtime_error(std::string(name) + " byte count does not ↵
    ↵ match element count");
83     }
84 }
85
86 void dequantize_f32_to(std::span<const std::uint8_t> bytes, std::span<float> ↵
    ↵ output) {
87     const std::size_t expected_bytes = output.size() * LCQI_GGML_F32_BYTES;
88     if (bytes.size() != expected_bytes) {
89         throw std::runtime_error("F32 byte count does not match element count")↵
    ↵ ;
90     }
91     for (std::size_t index = 0; index < output.size(); ++index) {
92         output[index] = bits_to_float(
93             read_le_u32(bytes.data() + index * LCQI_GGML_F32_BYTES));
94     }
95 }
96
97 void dequantize_q8_0_to(std::span<const std::uint8_t> bytes, std::span<float> ↵
    ↵ output) {
98     validate_quantized_bytes(bytes,
99                               static_cast<std::int64_t>(output.size()),
100                               LCQI_Q8_0_BLOCK_VALUES,
101                               LCQI_Q8_0_BLOCK_BYTES,
102                               "Q8_0");
103     const std::int64_t block_count =
104         static_cast<std::int64_t>(output.size()) / LCQI_Q8_0_BLOCK_VALUES;
105     for (std::int64_t block_index = 0; block_index < block_count; ++block_index↵
    ↵ ) {
106         const std::uint8_t* block =
107             bytes.data() + checked_size(block_index * LCQI_Q8_0_BLOCK_BYTES, "↵
    ↵ Q8_0 offset");
108         const float d = ggml_f16_to_f32(read_le_u16(block));
109         const std::uint8_t* qs = block + sizeof(std::uint16_t);
110         float* target =
111             output.data() + checked_size(block_index * LCQI_Q8_0_BLOCK_VALUES, ↵
    ↵ "Q8_0 target");
112         for (std::int32_t index = 0; index < LCQI_Q8_0_BLOCK_VALUES; ++index) {

```

```

113         target[index] = static_cast<float>(static_cast<std::int8_t>(qs[↵
↵ index])) * d;
114     }
115 }
116 }
117
118 void dequantize_q5_0_to(std::span<const std::uint8_t> bytes, std::span<float> ↵
↵ output) {
119     validate_quantized_bytes(bytes,
120                             static_cast<std::int64_t>(output.size()),
121                             LCQI_Q5_0_BLOCK_VALUES,
122                             LCQI_Q5_0_BLOCK_BYTES,
123                             "Q5_0");
124     const std::int64_t block_count =
125         static_cast<std::int64_t>(output.size()) / LCQI_Q5_0_BLOCK_VALUES;
126     for (std::int64_t block_index = 0; block_index < block_count; ++block_index↵
↵ ) {
127         const std::uint8_t* block =
128             bytes.data() + checked_size(block_index * LCQI_Q5_0_BLOCK_BYTES, "↵
↵ Q5_0 offset");
129         const float d = ggml_f16_to_f32(read_le_u16(block));
130         const std::uint32_t qh = read_le_u32(block + sizeof(std::uint16_t));
131         const std::uint8_t* qs = block + sizeof(std::uint16_t) + sizeof(std::↵
↵ uint32_t);
132         float* target =
133             output.data() + checked_size(block_index * LCQI_Q5_0_BLOCK_VALUES, ↵
↵ "Q5_0 target");
134         for (std::int32_t index = 0; index < LCQI_Q5_0_BLOCK_VALUES / 2; ++↵
↵ index) {
135             const std::uint8_t high_0 =
136                 static_cast<std::uint8_t>(((qh >> index) & 1U)
137                                         << LCQI_GGML_Q5_HIGH_BIT_SHIFT);
138             const std::uint8_t high_1 =
139                 static_cast<std::uint8_t>(((qh >> (index + ↵
↵ LCQI_GGML_Q5_SECOND_HALF_BIT_OFFSET)) &
140                                         1U)
141                                         << LCQI_GGML_Q5_HIGH_BIT_SHIFT);
142             const std::int32_t low =
143                 static_cast<std::int32_t>((qs[index] & ↵
↵ LCQI_GGML_LOW_NIBBLE_MASK) | high_0) -
144                 LCQI_GGML_Q5_ZERO_POINT;
145             const std::int32_t high =
146                 static_cast<std::int32_t>((qs[index] >> 4U) | high_1) -
147                 LCQI_GGML_Q5_ZERO_POINT;
148             target[index] = static_cast<float>(low) * d;
149             target[index + LCQI_Q5_0_BLOCK_VALUES / 2] = static_cast<float>(↵
↵ high) * d;
150         }
151     }
152 }
153
154 void dequantize_q6_k_to(std::span<const std::uint8_t> bytes, std::span<float> ↵

```

[illegible]

```

↪ )
195                                     << 4U)) -
196         LCQI_GGML_Q6_ZERO_POINT;
197         const std::int32_t q4 =
198             static_cast<std::int32_t>((ql[lane + ↪
↪ LCQI_GGML_Q6_SUBBLOCK_VALUES] >> 4U) |
199                                     (((qh[lane] >> 6U) &
200                                     LCQI_GGML_Q6_HIGH_TWO_BITS_MASK↪
↪ )
201                                     << 4U)) -
202         LCQI_GGML_Q6_ZERO_POINT;
203
204         y[lane] = d * static_cast<float>(scales[scale_index + 0]) *
205                     static_cast<float>(q1);
206         y[lane + LCQI_GGML_Q6_SUBBLOCK_VALUES] =
207             d * static_cast<float>(scales[scale_index + 2]) *
208                 static_cast<float>(q2);
209         y[lane + LCQI_GGML_Q6_QUARTER_BLOCK_VALUES] =
210             d * static_cast<float>(scales[scale_index + 4]) *
211                 static_cast<float>(q3);
212         y[lane + LCQI_GGML_Q6_QUARTER_BLOCK_VALUES +
213             LCQI_GGML_Q6_SUBBLOCK_VALUES] =
214             d * static_cast<float>(scales[scale_index + 6]) *
215                 static_cast<float>(q4);
216     }
217     y += LCQI_GGML_Q6_HALF_BLOCK_VALUES;
218     ql += LCQI_GGML_Q6_QUARTER_BLOCK_VALUES;
219     qh += LCQI_GGML_Q6_SUBBLOCK_VALUES;
220     scales += LCQI_GGML_Q6_SCALE_STRIDE;
221 }
222 }
223 }
224
225 } // namespace
226
227 float ggml_f16_to_f32(std::uint16_t bits) {
228     const std::uint32_t sign =
229         (static_cast<std::uint32_t>(bits) & LCQI_GGML_F16_SIGN_MASK)
230         << LCQI_GGML_F16_TO_F32_SIGN_SHIFT;
231     const std::uint32_t exponent =
232         static_cast<std::uint32_t>(bits) & LCQI_GGML_F16_EXPONENT_MASK;
233     const std::uint32_t mantissa =
234         static_cast<std::uint32_t>(bits) & LCQI_GGML_F16_MANTISSA_MASK;
235
236     if (exponent == 0U) {
237         if (mantissa == 0U) {
238             return bits_to_float(sign);
239         }
240         float value = static_cast<float>(mantissa) /
241                     static_cast<float>(1U << LCQI_GGML_F16_MANTISSA_BITS);
242         value = std::ldexp(value, 1 - static_cast<int>(↪
↪ LCQI_GGML_F16_EXPONENT_BIAS));

```

```

243     return sign == 0U ? value : -value;
244 }
245
246 if (exponent == LCQI_GGML_F16_EXPONENT_MASK) {
247     const std::uint32_t f32_mantissa =
248         mantissa << (LCQI_GGML_F32_MANTISSA_BITS -
↵ LCQI_GGML_F16_MANTISSA_BITS);
249     return bits_to_float(sign | LCQI_GGML_F32_EXPONENT_MASK | f32_mantissa)↵
↵ ;
250 }
251
252 const std::uint32_t f32_exponent =
253     ((exponent >> LCQI_GGML_F16_MANTISSA_BITS) -
↵ LCQI_GGML_F16_EXPONENT_BIAS +
254     LCQI_GGML_F32_EXPONENT_BIAS)
255     << LCQI_GGML_F32_MANTISSA_BITS;
256 const std::uint32_t f32_mantissa =
257     mantissa << (LCQI_GGML_F32_MANTISSA_BITS - LCQI_GGML_F16_MANTISSA_BITS)↵
↵ ;
258 return bits_to_float(sign | f32_exponent | f32_mantissa);
259 }
260
261 std::vector<float> dequantize_ggml_tensor(GgmlType type,
262                                         std::span<const std::uint8_t> bytes,
263                                         std::int64_t element_count) {
264     if (element_count < 0) {
265         throw std::runtime_error("GGML element_count cannot be negative");
266     }
267     std::vector<float> output(checked_size(element_count, "GGML element_count")↵
↵ , 0.0F);
268     dequantize_ggml_tensor_to(type, bytes, output);
269     return output;
270 }
271
272 void dequantize_ggml_tensor_to(GgmlType type,
273                               std::span<const std::uint8_t> bytes,
274                               std::span<float> output) {
275     switch (type) {
276     case GgmlType::f32:
277         dequantize_f32_to(bytes, output);
278         return;
279     case GgmlType::q8_0:
280         dequantize_q8_0_to(bytes, output);
281         return;
282     case GgmlType::q5_0:
283         dequantize_q5_0_to(bytes, output);
284         return;
285     case GgmlType::q6_k:
286         dequantize_q6_k_to(bytes, output);
287         return;
288     case GgmlType::q4_k:
289         dequantize_q4_k_to(bytes, output);

```



```

290         return;
291     default:
292         throw std::runtime_error(std::string("LCQI cannot dequantize GGML ↵
↵ type ") +
293                                   ggml_type_name(type));
294     }
295 }
296
297 } // namespace lcqi

```

`llama_gguf.cpp` 是本轮新增的自研 SmolLM2/GGUF reference path。它读取 GGUF meta-data 得到 hidden size、层数、GQA 头数、RoPE base、RMSNorm epsilon 和 tokenizer id；再读取 `token_embd.weight`、每层 attention/FFN 权重和 `output_norm.weight`。当前执行方式是加载时统一解到 f32，再走 RMSNorm、normal RoPE、GQA KV cache、SwiGLU 和 tied lm head。它的意义不是性能已经追上 llama.cpp，而是 LCQI 终于有了同一个真实 GGUF 文件上的自研端到端 correctness baseline。

Listing 16.9 src/llama_gguf.cpp

```

1  #include <lcqi/llama_gguf.hpp>
2
3  #include <lcqi/f32_kernels.hpp>
4  #include <lcqi/ggml_tensors.hpp>
5  #include <lcqi/gguf.hpp>
6
7  #include <algorithm>
8  #include <array>
9  #include <chrono>
10 #include <cmath>
11 #include <limits>
12 #include <sstream>
13 #include <stdexcept>
14 #include <string>
15 #include <unordered_map>
16 #include <utility>
17
18 namespace lcqi {
19 namespace {
20
21 constexpr std::int32_t LCQI_LLAMA_ROPE_PAIR_WIDTH = 2;
22 constexpr std::int32_t LCQI_LLAMA_DEFAULT_BOS_TOKEN = 1;
23 constexpr std::int32_t LCQI_LLAMA_DEFAULT_EOS_TOKEN = 2;
24 constexpr std::int32_t LCQI_LLAMA_DEFAULT_UNK_TOKEN = 0;
25 constexpr const char* LCQI_LLAMA_ARCHITECTURE_KEY = "general.architecture";
26 constexpr const char* LCQI_LLAMA_NAME_KEY = "general.name";
27 constexpr const char* LCQI_LLAMA_EMBEDDING_LENGTH_KEY = "llama.embedding_length↵
↵ ";
28 constexpr const char* LCQI_LLAMA_BLOCK_COUNT_KEY = "llama.block_count";
29 constexpr const char* LCQI_LLAMA_FEED_FORWARD_LENGTH_KEY = "llama.↵
↵ feed_forward_length";
30 constexpr const char* LCQI_LLAMA_HEAD_COUNT_KEY = "llama.attention.head_count";
31 constexpr const char* LCQI_LLAMA_KV_HEAD_COUNT_KEY = "llama.attention.↵
↵ head_count_kv";

```



```

32 constexpr const char* LCQI_LLAMA_ROPE_DIMENSION_KEY = "llama.ropes.↵
    ↵ dimension_count";
33 constexpr const char* LCQI_LLAMA_ROPE_BASE_KEY = "llama.ropes.freq_base";
34 constexpr const char* LCQI_LLAMA_CONTEXT_LENGTH_KEY = "llama.context_length";
35 constexpr const char* LCQI_LLAMA_VOCAB_SIZE_KEY = "llama.vocab_size";
36 constexpr const char* LCQI_LLAMA_RMS_EPSILON_KEY = "llama.attention.↵
    ↵ layer_norm_rms_epsilon";
37 constexpr const char* LCQI_LLAMA_TOKENIZER_TOKENS_KEY = "tokenizer.ggml.tokens"↵
    ↵ ;
38 constexpr const char* LCQI_LLAMA_TOKENIZER_MERGES_KEY = "tokenizer.ggml.merges"↵
    ↵ ;
39 constexpr const char* LCQI_LLAMA_BOS_TOKEN_KEY = "tokenizer.ggml.bos_token_id";
40 constexpr const char* LCQI_LLAMA_EOS_TOKEN_KEY = "tokenizer.ggml.eos_token_id";
41 constexpr const char* LCQI_LLAMA_UNKNOWN_TOKEN_KEY = "tokenizer.ggml.↵
    ↵ unknown_token_id";
42
43 using Clock = std::chrono::steady_clock;
44
45 [[nodiscard]] double elapsed_ms(Clock::time_point begin, Clock::time_point end)↵
    ↵ {
46     return std::chrono::duration<double, std::milli>(end - begin).count();
47 }
48
49 [[nodiscard]] std::size_t checked_size(std::int64_t value, const char* name) {
50     if (value < 0 ||
51         static_cast<std::uint64_t>(value) >
52         static_cast<std::uint64_t>(std::numeric_limits<std::size_t>::max()))↵
    ↵ ) {
53         throw std::runtime_error(std::string(name) + " is outside size_t range"↵
    ↵ );
54     }
55     return static_cast<std::size_t>(value);
56 }
57
58 void require_positive(std::int32_t value, const char* name) {
59     if (value <= 0) {
60         throw std::runtime_error(std::string(name) + " must be positive");
61     }
62 }
63
64 [[nodiscard]] const GgufMetadataEntry& require_metadata(const GgufManifest& ↵
    ↵ manifest,
65                                                         std::string_view key) {
66     const GgufMetadataEntry* entry = manifest.find_metadata(key);
67     if (entry == nullptr) {
68         throw std::runtime_error("missing GGUF metadata key: " + std::string(↵
    ↵ key));
69     }
70     return *entry;
71 }
72
73 [[nodiscard]] std::string metadata_string(const GgufManifest& manifest, std::↵

```

```

    ↪ string_view key) {
74     return require_metadata(manifest, key).value_preview;
75 }
76
77 [[nodiscard]] std::int32_t metadata_i32(const GgufManifest& manifest, std::↪
    ↪ string_view key) {
78     const std::int64_t value = std::stoll(require_metadata(manifest, key).↪
    ↪ value_preview);
79     if (value < std::numeric_limits<std::int32_t>::min() ||
80         value > std::numeric_limits<std::int32_t>::max()) {
81         throw std::runtime_error("GGUF metadata int32 out of range: " + std::↪
    ↪ string(key));
82     }
83     return static_cast<std::int32_t>(value);
84 }
85
86 [[nodiscard]] std::int32_t optional_metadata_i32(const GgufManifest& manifest,
87                                                  std::string_view key,
88                                                  std::int32_t fallback) {
89     const GgufMetadataEntry* entry = manifest.find_metadata(key);
90     if (entry == nullptr) {
91         return fallback;
92     }
93     const std::int64_t value = std::stoll(entry->value_preview);
94     if (value < std::numeric_limits<std::int32_t>::min() ||
95         value > std::numeric_limits<std::int32_t>::max()) {
96         throw std::runtime_error("GGUF metadata int32 out of range: " + std::↪
    ↪ string(key));
97     }
98     return static_cast<std::int32_t>(value);
99 }
100
101 [[nodiscard]] float metadata_f32(const GgufManifest& manifest, std::string_view↪
    ↪ key) {
102     return std::stof(require_metadata(manifest, key).value_preview);
103 }
104
105 void validate_shape(const GgufTensorInfo& tensor,
106                    std::span<const std::int64_t> expected_shape) {
107     if (tensor.shape.size() != expected_shape.size()) {
108         throw std::runtime_error("GGUF tensor rank mismatch: " + tensor.name);
109     }
110     for (std::size_t index = 0; index < expected_shape.size(); ++index) {
111         if (tensor.shape[index] != expected_shape[index]) {
112             throw std::runtime_error("GGUF tensor shape mismatch: " + tensor.↪
    ↪ name);
113         }
114     }
115 }
116
117 [[nodiscard]] std::vector<float> read_tensor_f32(const std::filesystem::path& ↪
    ↪ gguf_path,

```

```

118         const GgufManifest& manifest,
119         std::string_view name,
120         std::span<const std::int64_t> ↵
        ↵ expected_shape,
121         LlamaGgufLoadReport& report) {
122     const GgufTensorInfo* tensor = manifest.find_tensor(name);
123     if (tensor == nullptr) {
124         throw std::runtime_error("missing LLaMA GGUF tensor: " + std::string(↵
        ↵ name));
125     }
126     validate_shape(*tensor, expected_shape);
127     const std::vector<std::uint8_t> bytes = read_gguf_tensor_bytes(gguf_path, *↵
        ↵ tensor);
128     std::vector<float> values =
129         dequantize_ggml_tensor(tensor->type,
130                                bytes,
131                                static_cast<std::int64_t>(tensor->element_count↵
        ↵ ());
132     report.quantized_weight_bytes += tensor->byte_size;
133     report.f32_weight_bytes +=
134         static_cast<std::uint64_t>(values.size()) * static_cast<std::uint64_t>(↵
        ↵ sizeof(float));
135     ++report.tensors_loaded;
136     return values;
137 }
138
139 [[nodiscard]] LinearWeightsF32 make_linear_from_gguf(
140     const std::filesystem::path& gguf_path,
141     const GgufManifest& manifest,
142     std::string_view name,
143     std::int32_t input_size,
144     std::int32_t output_size,
145     LlamaGgufLoadReport& report) {
146     const std::array<std::int64_t, 2> shape{input_size, output_size};
147     LinearWeightsF32 linear;
148     linear.input_size = input_size;
149     linear.output_size = output_size;
150     linear.weights = read_tensor_f32(gguf_path, manifest, name, shape, report);
151     return linear;
152 }
153
154 [[nodiscard]] std::string layer_tensor_name(std::int32_t layer_id, std::↵
        ↵ string_view suffix) {
155     return "blk." + std::to_string(layer_id) + "." + std::string(suffix);
156 }
157
158 void validate_llama_config(const DecoderConfig& config, std::int32_t ↵
        ↵ layer_count) {
159     require_positive(layer_count, "layer_count");
160     require_positive(config.hidden_size, "hidden_size");
161     require_positive(config.query_heads, "query_heads");
162     require_positive(config.kv_heads, "kv_heads");

```

```

163     require_positive(config.head_dim, "head_dim");
164     require_positive(config.intermediate_size, "intermediate_size");
165     require_positive(config.vocab_size, "vocab_size");
166     require_positive(config.max_context, "max_context");
167     if (config.query_heads % config.kv_heads != 0) {
168         throw std::runtime_error("LLaMA query_heads must be divisible by ↵
↵ kv_heads");
169     }
170     if (config.hidden_size != config.query_heads * config.head_dim) {
171         throw std::runtime_error("LLaMA hidden_size must equal query_heads * ↵
↵ head_dim");
172     }
173     if (config.head_dim % LCQI_LLAMA_ROPE_PAIR_WIDTH != 0) {
174         throw std::runtime_error("LLaMA head_dim must be even for RoPE");
175     }
176 }
177
178 [[nodiscard]] std::span<const float> row_span(std::span<const float> values,
179                                               std::int32_t row,
180                                               std::int32_t row_width) {
181     return values.subspan(checked_size(row, "row") * checked_size(row_width, "↵
↵ row_width"),
182                           checked_size(row_width, "row_width"));
183 }
184
185 void add_inplace(std::span<float> target, std::span<const float> source) {
186     if (target.size() != source.size()) {
187         throw std::runtime_error("LLaMA residual add size mismatch");
188     }
189     for (std::size_t index = 0; index < target.size(); ++index) {
190         target[index] += source[index];
191     }
192 }
193
194 [[nodiscard]] float silu(float value) {
195     return value / (1.0F + std::exp(-value));
196 }
197
198 struct LlamaWorkspace {
199     std::vector<float> hidden;
200     std::vector<float> normed;
201     std::vector<float> query;
202     std::vector<float> key;
203     std::vector<float> value;
204     std::vector<float> attention;
205     std::vector<float> projected;
206     std::vector<float> gate;
207     std::vector<float> up;
208     std::vector<float> mlp_hidden;
209     std::vector<float> mlp_out;
210     std::vector<float> final_norm;
211

```

```

212     explicit LlamaWorkspace(const DecoderConfig& config)
213         : hidden(check_size(config.hidden_size, "hidden_size"), 0.0F),
214           normed(hidden.size(), 0.0F),
215           query(hidden.size(), 0.0F),
216           key(check_size(config.kv_heads * config.head_dim, "kv_width"), 0.0F) ←
    ↪ ),
217           value(key.size(), 0.0F),
218           attention(hidden.size(), 0.0F),
219           projected(hidden.size(), 0.0F),
220           gate(check_size(config.intermediate_size, "intermediate_size"), 0.0) ←
    ↪ F),
221           up(gate.size(), 0.0F),
222           mlp_hidden(gate.size(), 0.0F),
223           mlp_out(hidden.size(), 0.0F),
224           final_norm(hidden.size(), 0.0F) {}
225 };
226
227 void forward_llama_layer(const DecoderConfig& config,
228                         const DecoderLayerWeightsF32& layer,
229                         std::int32_t layer_id,
230                         std::int32_t model_position,
231                         ReferenceKVCache& cache,
232                         LlamaWorkspace& workspace) {
233     rms_norm(workspace.hidden,
234             layer.rms_attention_weight,
235             config.rms_epsilon,
236             workspace.normed);
237     linear_f32(layer.wq, workspace.normed, workspace.query);
238     linear_f32(layer.wk, workspace.normed, workspace.key);
239     linear_f32(layer.wv, workspace.normed, workspace.value);
240     apply_rope(workspace.query,
241             config.query_heads,
242             config.head_dim,
243             model_position,
244             config.rope_base);
245     apply_rope(workspace.key,
246             config.kv_heads,
247             config.head_dim,
248             model_position,
249             config.rope_base);
250     cache.append(layer_id, model_position, workspace.key, workspace.value);
251     attention_decode(config,
252             cache,
253             layer_id,
254             model_position,
255             workspace.query,
256             workspace.attention);
257     linear_f32(layer.wo, workspace.attention, workspace.projected);
258     add_inplace(workspace.hidden, workspace.projected);
259
260     rms_norm(workspace.hidden, layer.rms_mlp_weight, config.rms_epsilon, ←
    ↪ workspace.normed);

```

```

261 linear_f32(layer.w_gate, workspace.normed, workspace.gate);
262 linear_f32(layer.w_up, workspace.normed, workspace.up);
263 for (std::size_t index = 0; index < workspace.mlp_hidden.size(); ++index) {
264     workspace.mlp_hidden[index] = silu(workspace.gate[index]) * workspace.↵
↵ up[index];
265 }
266 linear_f32(layer.w_down, workspace.mlp_hidden, workspace.mlp_out);
267 add_inplace(workspace.hidden, workspace.mlp_out);
268 }
269
270 [[nodiscard]] std::int32_t predict_next_token(const LlamaGgufModel& model,
271                                             std::span<const float> ↵
↵ final_hidden) {
272     const DecoderConfig& config = model.decoder.config;
273     const std::span<const float> weights =
274         model.lm_head_tied_to_embedding
275         ? std::span<const float>(model.decoder.token_embedding)
276         : std::span<const float>(model.decoder.lm_head.weights);
277     const F32RowMax best = max_dot_f32_rows_unchecked(weights.data(),
278                                                         final_hidden.data(),
279                                                         checked_size(config.↵
↵ hidden_size, "hidden_size"),
280                                                         0,
281                                                         checked_size(config.↵
↵ vocab_size, "vocab_size"));
282     return static_cast<std::int32_t>(best.row);
283 }
284
285 [[nodiscard]] std::int32_t step_llama_decoder(const LlamaGgufModel& model,
286                                              const DecoderConfig& ↵
↵ active_config,
287                                              ReferenceKVCache& cache,
288                                              LlamaWorkspace& workspace,
289                                              std::int32_t token_id,
290                                              bool predict) {
291     const DecoderConfig& original_config = model.decoder.config;
292     if (token_id < 0 || token_id >= original_config.vocab_size) {
293         throw std::runtime_error("LLaMA token id out of vocabulary");
294     }
295     const std::int32_t model_position = cache.filled_tokens();
296     if (model_position >= active_config.max_context) {
297         throw std::runtime_error("LLaMA generation exceeded active context");
298     }
299
300     const std::span<const float> embedding =
301         row_span(model.decoder.token_embedding, token_id, original_config.↵
↵ hidden_size);
302     std::copy(embedding.begin(), embedding.end(), workspace.hidden.begin());
303     for (std::int32_t layer_id = 0;
304          layer_id < static_cast<std::int32_t>(model.decoder.layers.size());
305          ++layer_id) {
306         forward_llama_layer(active_config,

```

```

307         model.decoder.layers[checked_size(layer_id, "↵
↵ layer_id")]],
308         layer_id,
309         model_position,
310         cache,
311         workspace);
312     }
313     if (!predict) {
314         return -1;
315     }
316     rms_norm(workspace.hidden,
317             model.decoder.final_norm_weight,
318             active_config.rms_epsilon,
319             workspace.final_norm);
320     return predict_next_token(model, workspace.final_norm);
321 }
322
323 [[nodiscard]] std::size_t kv_cache_bytes(const DecoderConfig& config, std::↵
↵ int32_t layer_count) {
324     return checked_size(layer_count, "layer_count") *
325            checked_size(config.max_context, "max_context") *
326            checked_size(config.kv_heads, "kv_heads") *
327            checked_size(config.head_dim, "head_dim") * sizeof(float) * 2U;
328 }
329
330 [[nodiscard]] std::unordered_map<std::string, std::int32_t> ↵
↵ build_vocab_from_tokens(
331     std::span<const std::string> tokens) {
332     std::unordered_map<std::string, std::int32_t> vocab;
333     for (std::size_t index = 0; index < tokens.size(); ++index) {
334         vocab.emplace(tokens[index], static_cast<std::int32_t>(index));
335     }
336     return vocab;
337 }
338
339 [[nodiscard]] std::unordered_map<std::string, std::int32_t> build_merge_ranks(
340     std::span<const std::string> merges) {
341     std::unordered_map<std::string, std::int32_t> ranks;
342     for (std::size_t index = 0; index < merges.size(); ++index) {
343         std::stringstream line_stream(merges[index]);
344         std::string left;
345         std::string right;
346         if (!(line_stream >> left >> right)) {
347             throw std::runtime_error("invalid GGUF tokenizer merge: " + merges[↵
↵ index]);
348         }
349         ranks.emplace(left + "\n" + right, static_cast<std::int32_t>(index));
350     }
351     return ranks;
352 }
353
354 [[nodiscard]] bool starts_with(std::string_view text,

```

```

355         std::size_t offset,
356         std::string_view needle) {
357     return offset + needle.size() <= text.size() &&
358         text.substr(offset, needle.size()) == needle;
359 }
360
361 [[nodiscard]] std::vector<std::pair<std::string, std::int32_t>> special_tokens(
362     const Gpt2Tokenizer& tokenizer) {
363     std::vector<std::pair<std::string, std::int32_t>> result;
364     for (const auto& [token, id] : tokenizer.vocab) {
365         if (token.size() >= 4 && token.front() == '<' && token.find('|') != std::
↪ ::string::npos) {
366             result.emplace_back(token, id);
367         }
368     }
369     std::sort(result.begin(),
370             result.end(),
371             [](const auto& lhs, const auto& rhs) {
372                 return lhs.first.size() > rhs.first.size();
373             });
374     return result;
375 }
376
377 [[nodiscard]] std::vector<std::int32_t> encode_with_special_tokens(
378     const Gpt2Tokenizer& tokenizer,
379     std::string_view text) {
380     const std::vector<std::pair<std::string, std::int32_t>> specials = ↪
↪ special_tokens(tokenizer);
381     std::vector<std::int32_t> ids;
382     std::size_t offset = 0;
383     while (offset < text.size()) {
384         bool matched_special = false;
385         for (const auto& [token, id] : specials) {
386             if (starts_with(text, offset, token)) {
387                 ids.push_back(id);
388                 offset += token.size();
389                 matched_special = true;
390                 break;
391             }
392         }
393         if (matched_special) {
394             continue;
395         }
396
397         std::size_t next_special = text.size();
398         for (const auto& [token, id] : specials) {
399             (void)id;
400             const std::size_t found = text.find(token, offset);
401             if (found != std::string_view::npos) {
402                 next_special = std::min(next_special, found);
403             }
404         }

```



```

405     const std::vector<std::int32_t> chunk =
406         gpt2_encode(tokenizer, text.substr(offset, next_special - offset));
407     ids.insert(ids.end(), chunk.begin(), chunk.end());
408     offset = next_special;
409 }
410 return ids;
411 }
412
413 } // namespace
414
415 LlamaGgufLoadedModel load_llama_gguf_reference_model(const std::filesystem::↵
↵ path& gguf_path) {
416     LlamaGgufLoadedModel loaded;
417     const Clock::time_point manifest_begin = Clock::now();
418     const GgufManifest manifest = load_gguf_manifest(gguf_path);
419     loaded.report.manifest_ms = elapsed_ms(manifest_begin, Clock::now());
420
421     const Clock::time_point weights_begin = Clock::now();
422     loaded.model.architecture = metadata_string(manifest, ↵
↵ LCQI_LLAMA_ARCHITECTURE_KEY);
423     loaded.model.name = metadata_string(manifest, LCQI_LLAMA_NAME_KEY);
424     DecoderConfig& config = loaded.model.decoder.config;
425     config.hidden_size = metadata_i32(manifest, LCQI_LLAMA_EMBEDDING_LENGTH_KEY↵
↵ );
426     const std::int32_t layer_count = metadata_i32(manifest, ↵
↵ LCQI_LLAMA_BLOCK_COUNT_KEY);
427     config.intermediate_size = metadata_i32(manifest, ↵
↵ LCQI_LLAMA_FEED_FORWARD_LENGTH_KEY);
428     config.query_heads = metadata_i32(manifest, LCQI_LLAMA_HEAD_COUNT_KEY);
429     config.kv_heads = metadata_i32(manifest, LCQI_LLAMA_KV_HEAD_COUNT_KEY);
430     config.head_dim = config.hidden_size / config.query_heads;
431     const std::int32_t rope_dimension = metadata_i32(manifest, ↵
↵ LCQI_LLAMA_ROPE_DIMENSION_KEY);
432     if (rope_dimension != config.head_dim) {
433         throw std::runtime_error("LLaMA rope dimension does not match head_dim"↵
↵ );
434     }
435     config.rope_base = metadata_f32(manifest, LCQI_LLAMA_ROPE_BASE_KEY);
436     config.max_context = metadata_i32(manifest, LCQI_LLAMA_CONTEXT_LENGTH_KEY);
437     config.vocab_size = metadata_i32(manifest, LCQI_LLAMA_VOCAB_SIZE_KEY);
438     config.rms_epsilon = metadata_f32(manifest, LCQI_LLAMA_RMS_EPSILON_KEY);
439     loaded.model.bos_token_id =
440         optional_metadata_i32(manifest, LCQI_LLAMA_BOS_TOKEN_KEY, ↵
↵ LCQI_LLAMA_DEFAULT_BOS_TOKEN);
441     loaded.model.eos_token_id =
442         optional_metadata_i32(manifest, LCQI_LLAMA_EOS_TOKEN_KEY, ↵
↵ LCQI_LLAMA_DEFAULT_EOS_TOKEN);
443     loaded.model.unknown_token_id = optional_metadata_i32(
444         manifest,
445         LCQI_LLAMA_UNKNOWN_TOKEN_KEY,
446         LCQI_LLAMA_DEFAULT_UNK_TOKEN);
447     validate_llama_config(config, layer_count);

```



```

493         kv_width,
494         loaded.report);
495     layer.wo = make_linear_from_gguf(gguf_path,
496         manifest,
497         layer_tensor_name(layer_id, "←
↪ attn_output.weight"),
498         config.hidden_size,
499         config.hidden_size,
500         loaded.report);
501     layer.w_gate = make_linear_from_gguf(gguf_path,
502         manifest,
503         layer_tensor_name(layer_id, "←
↪ ffn_gate.weight"),
504         config.hidden_size,
505         config.intermediate_size,
506         loaded.report);
507     layer.w_up = make_linear_from_gguf(gguf_path,
508         manifest,
509         layer_tensor_name(layer_id, "ffn_up.←
↪ weight"),
510         config.hidden_size,
511         config.intermediate_size,
512         loaded.report);
513     layer.w_down = make_linear_from_gguf(gguf_path,
514         manifest,
515         layer_tensor_name(layer_id, "←
↪ ffn_down.weight"),
516         config.intermediate_size,
517         config.hidden_size,
518         loaded.report);
519 }
520
521 if (manifest.find_tensor("output.weight") != nullptr) {
522     loaded.model.lm_head_tied_to_embedding = false;
523     loaded.model.decoder.lm_head = make_linear_from_gguf(gguf_path,
524         manifest,
525         "output.weight",
526         config.hidden_size←
↪ ,
527         config.vocab_size,
528         loaded.report);
529 } else {
530     loaded.model.lm_head_tied_to_embedding = true;
531     loaded.model.decoder.lm_head.input_size = config.hidden_size;
532     loaded.model.decoder.lm_head.output_size = config.vocab_size;
533 }
534 loaded.report.weights_ms = elapsed_ms(weights_begin, Clock::now());
535 return loaded;
536 }
537
538 Gpt2Tokenizer load_gpt2_tokenizer_from_gguf(const std::filesystem::path& ←
↪ gguf_path) {

```

```

539 const GgufManifest manifest = load_gguf_manifest(gguf_path);
540 const GgufMetadataEntry& tokens_entry = require_metadata(manifest,
541                                     ↵
    ↵ LCQI_LLAMA_TOKENIZER_TOKENS_KEY);
542 const GgufMetadataEntry& merges_entry = require_metadata(manifest,
543                                     ↵
    ↵ LCQI_LLAMA_TOKENIZER_MERGES_KEY);
544 if (tokens_entry.string_values.empty() || merges_entry.string_values.empty()↵
    ↵ ()) {
545     throw std::runtime_error("GGUF tokenizer tokens or merges are empty");
546 }
547 Gpt2Tokenizer tokenizer;
548 tokenizer.bos_token_id =
549     optional_metadata_i32(manifest, LCQI_LLAMA_BOS_TOKEN_KEY, ↵
    ↵ LCQI_LLAMA_DEFAULT_BOS_TOKEN);
550 tokenizer.eos_token_id =
551     optional_metadata_i32(manifest, LCQI_LLAMA_EOS_TOKEN_KEY, ↵
    ↵ LCQI_LLAMA_DEFAULT_EOS_TOKEN);
552 tokenizer.id_to_token = tokens_entry.string_values;
553 tokenizer.vocab = build_vocab_from_tokens(tokenizer.id_to_token);
554 tokenizer.bpe_ranks = build_merge_ranks(merges_entry.string_values);
555 return tokenizer;
556 }
557
558 std::vector<std::int32_t> llama_gguf_chat_prompt_ids(const Gpt2Tokenizer& ↵
    ↵ tokenizer,
559                                     std::string_view ↵
    ↵ user_prompt,
560                                     std::int32_t bos_token_id)↵
    ↵ {
561     std::string prompt;
562     prompt += "<|im_start|>system\n";
563     prompt += "You are a helpful AI assistant named SmolLM, trained by Hugging ↵
    ↵ Face";
564     prompt += "<|im_end|>\n";
565     prompt += "<|im_start|>user\n";
566     prompt += user_prompt;
567     prompt += "<|im_end|>\n";
568     prompt += "<|im_start|>assistant\n";
569     std::vector<std::int32_t> ids = encode_with_special_tokens(tokenizer, ↵
    ↵ prompt);
570     if (ids.empty() && bos_token_id >= 0) {
571         ids.push_back(bos_token_id);
572     }
573     return ids;
574 }
575
576 LlamaGgufGenerationResult llama_gguf_generate_greedy(const LlamaGgufModel& ↵
    ↵ model,
577                                     std::span<const std::↵
    ↵ int32_t> prompt_ids,
578                                     std::int32_t ↵

```

```

↪ max_new_tokens) {
579     if (prompt_ids.empty()) {
580         throw std::runtime_error("LLaMA generation requires at least one prompt↪
↪ token");
581     }
582     if (max_new_tokens < 0) {
583         throw std::runtime_error("max_new_tokens cannot be negative");
584     }
585     const std::int32_t layer_count = static_cast<std::int32_t>(model.decoder.↪
↪ layers.size());
586     validate_llama_config(model.decoder.config, layer_count);
587     const std::int32_t active_context =
588         static_cast<std::int32_t>(prompt_ids.size()) + std::max(max_new_tokens,↪
↪ 1);
589     if (active_context > model.decoder.config.max_context) {
590         throw std::runtime_error("LLaMA prompt plus generation exceeds model ↪
↪ context");
591     }
592
593     DecoderConfig active_config = model.decoder.config;
594     active_config.max_context = active_context;
595     ReferenceKVCache cache(active_config, layer_count);
596     LlamaWorkspace workspace(active_config);
597
598     LlamaGgufGenerationResult result;
599     result.prompt_ids.assign(prompt_ids.begin(), prompt_ids.end());
600     result.generated_ids.assign(prompt_ids.begin(), prompt_ids.end());
601     result.kv_cache_bytes = kv_cache_bytes(active_config, layer_count);
602     if (max_new_tokens == 0) {
603         return result;
604     }
605
606     const Clock::time_point prefill_begin = Clock::now();
607     for (std::size_t index = 0; index + 1 < prompt_ids.size(); ++index) {
608         static_cast<void>(step_llama_decoder(model,
609                                             active_config,
610                                             cache,
611                                             workspace,
612                                             prompt_ids[index],
613                                             false));
614     }
615     std::int32_t predicted = step_llama_decoder(model,
616                                             active_config,
617                                             cache,
618                                             workspace,
619                                             prompt_ids.back(),
620                                             true);
621     result.predicted_first_token = predicted;
622     result.prefill_steps = prompt_ids.size();
623     result.prefill_ms = elapsed_ms(prefill_begin, Clock::now());
624
625     const Clock::time_point decode_begin = Clock::now();

```

```

626     for (std::int32_t step = 0; step < max_new_tokens; ++step) {
627         result.generated_ids.push_back(predicted);
628         if (predicted == model.eos_token_id) {
629             break;
630         }
631         if (step + 1 < max_new_tokens) {
632             predicted = step_llama_decoder(model,
633                                     active_config,
634                                     cache,
635                                     workspace,
636                                     predicted,
637                                     true);
638             ++result.decode_steps;
639         }
640     }
641     result.decode_ms = elapsed_ms(decode_begin, Clock::now());
642     result.total_ms = result.prefill_ms + result.decode_ms;
643     return result;
644 }
645
646 } // namespace lcqi

```

`llama_cli.cpp` 是 `lcqi_llama_gguf` 的命令行入口。它支持 `--ids` 直接给 token id，也支持 `--prompt` 从 GGUF tokenizer arrays 构造 chat prompt；`--benchmark` 输出 manifest、权重加载、prefill、decode、KV cache、量化权重字节和解码后 f32 字节。报告中 `decode_tokens_per_second` 只统计真正喂入新 token 后再次预测的 decode step，不能把首 token prefill 的预测误差算成绩 decode。

Listing 16.10 `src/llama_cli.cpp`

```

1  #include <lcqi/llama_gguf.hpp>
2
3  #include <chrono>
4  #include <cstdlib>
5  #include <exception>
6  #include <filesystem>
7  #include <iostream>
8  #include <sstream>
9  #include <stdexcept>
10 #include <string>
11 #include <string_view>
12 #include <vector>
13
14 namespace {
15
16 constexpr std::int32_t LCQI_LLAMA_DEFAULT_MAX_NEW_TOKENS = 1;
17 constexpr std::string_view LCQI_LLAMA_IDS_PREFIX = "--ids=";
18 constexpr std::string_view LCQI_LLAMA_PROMPT_PREFIX = "--prompt=";
19 constexpr std::string_view LCQI_LLAMA_MAX_NEW_PREFIX = "--max-new=";
20
21 using Clock = std::chrono::steady_clock;
22
23 struct LlamaCliOptions {

```

```

24     std::filesystem::path gguf_path;
25     std::string prompt;
26     std::vector<std::int32_t> ids;
27     std::int32_t max_new_tokens = LCQI_LLAMA_DEFAULT_MAX_NEW_TOKENS;
28     bool use_ids = false;
29     bool benchmark = false;
30     bool decode_text = false;
31 };
32
33 [[nodiscard]] double elapsed_ms(Clock::time_point begin, Clock::time_point end) ←
    ↪ {
34     return std::chrono::duration<double, std::milli>(end - begin).count();
35 }
36
37 void print_ids(std::string_view label, const std::vector<std::int32_t>& ids) {
38     std::cout << label;
39     for (const std::int32_t id : ids) {
40         std::cout << ' ' << id;
41     }
42     std::cout << "\n";
43 }
44
45 [[nodiscard]] std::int32_t parse_non_negative_i32(const std::string& text,
46                                                    const char* name) {
47     const int value = std::stoi(text);
48     if (value < 0) {
49         throw std::runtime_error(std::string(name) + " cannot be negative");
50     }
51     return static_cast<std::int32_t>(value);
52 }
53
54 [[nodiscard]] std::vector<std::int32_t> parse_ids(std::string_view text) {
55     std::vector<std::int32_t> ids;
56     std::string normalized(text);
57     for (char& ch : normalized) {
58         if (ch == ',') {
59             ch = ' ';
60         }
61     }
62     std::istringstream input(normalized);
63     std::int32_t id = 0;
64     while (input >> id) {
65         if (id < 0) {
66             throw std::runtime_error("token ids cannot be negative");
67         }
68         ids.push_back(id);
69     }
70     if (ids.empty()) {
71         throw std::runtime_error("--ids requires at least one token id");
72     }
73     return ids;
74 }

```

```

75
76 [[nodiscard]] bool consume_prefix(std::string_view arg,
77                                   std::string_view prefix,
78                                   std::string_view& value) {
79     if (arg.substr(0, prefix.size()) != prefix) {
80         return false;
81     }
82     value = arg.substr(prefix.size());
83     return true;
84 }
85
86 [[nodiscard]] LlamaCliOptions parse_options(int argc, char** argv) {
87     if (argc < 2) {
88         throw std::runtime_error(
89             "usage: lcqi_llama_gguf model.gguf [--prompt TEXT|--ids 1,2,3] "
90             "[--max-new N] [--benchmark] [--decode-text]");
91     }
92     LlamaCliOptions options;
93     options.gguf_path = argv[1];
94     for (int index = 2; index < argc; ++index) {
95         const std::string arg = argv[index];
96         std::string_view value;
97         if (arg == "--benchmark") {
98             options.benchmark = true;
99         } else if (arg == "--decode-text") {
100             options.decode_text = true;
101         } else if (arg == "--ids") {
102             if (index + 1 >= argc) {
103                 throw std::runtime_error("--ids requires a value");
104             }
105             ++index;
106             options.ids = parse_ids(argv[index]);
107             options.use_ids = true;
108         } else if (consume_prefix(arg, LCQI_LLAMA_IDS_PREFIX, value)) {
109             options.ids = parse_ids(value);
110             options.use_ids = true;
111         } else if (arg == "--prompt") {
112             if (index + 1 >= argc) {
113                 throw std::runtime_error("--prompt requires a value");
114             }
115             ++index;
116             options.prompt = argv[index];
117             options.use_ids = false;
118         } else if (consume_prefix(arg, LCQI_LLAMA_PROMPT_PREFIX, value)) {
119             options.prompt = std::string(value);
120             options.use_ids = false;
121         } else if (arg == "--max-new") {
122             if (index + 1 >= argc) {
123                 throw std::runtime_error("--max-new requires a value");
124             }
125             ++index;
126             options.max_new_tokens = parse_non_negative_i32(argv[index], "--max-

```



```

127     ↪ -new");
128     } else if (consume_prefix(arg, LCQI_LLAMA_MAX_NEW_PREFIX, value)) {
129         options.max_new_tokens =
130             parse_non_negative_i32(std::string(value), "--max-new");
131     } else {
132         throw std::runtime_error("unknown option: " + arg);
133     }
134     if (!options.use_ids && options.prompt.empty()) {
135         options.prompt = "Hello";
136     }
137     return options;
138 }
139
140 [[nodiscard]] std::vector<std::int32_t> make_prompt_ids(const LlamaCliOptions& ↪
141     ↪ options,
142                                     lcqi::Gpt2Tokenizer* ↪
143     ↪ tokenizer,
144                                     std::int32_t ↪
145     ↪ bos_token_id) {
146     if (options.use_ids) {
147         return options.ids;
148     }
149     return lcqi::llama_gguf_chat_prompt_ids(*tokenizer, options.prompt, ↪
150     ↪ bos_token_id);
151 }
152
153 void print_benchmark(const lcqi::LlamaGgufLoadedModel& loaded,
154                     const lcqi::LlamaGgufGenerationResult& result,
155                     std::size_t prompt_size,
156                     std::size_t generated_new_tokens) {
157     const double first_token_tokens_per_second =
158         result.prefill_ms > 0.0 ? 1000.0 / result.prefill_ms : 0.0;
159     const double decode_tokens_per_second =
160         result.decode_ms > 0.0
161         ? static_cast<double>(result.decode_steps) * 1000.0 / result.↪
162     ↪ decode_ms
163         : 0.0;
164     std::cout << "benchmark_manifest_ms " << loaded.report.manifest_ms << "\n";
165     std::cout << "benchmark_weight_load_ms " << loaded.report.weights_ms << "\n↪
166     ↪ ";
167     std::cout << "benchmark_load_ms " << result.load_ms << "\n";
168     std::cout << "benchmark_prefill_ms " << result.prefill_ms << "\n";
169     std::cout << "benchmark_decode_ms " << result.decode_ms << "\n";
170     std::cout << "benchmark_total_ms " << result.total_ms << "\n";
171     std::cout << "benchmark_prompt_tokens " << prompt_size << "\n";
172     std::cout << "benchmark_generated_tokens " << generated_new_tokens << "\n";
173     std::cout << "benchmark_prefill_steps " << result.prefill_steps << "\n";
174     std::cout << "benchmark_decode_steps " << result.decode_steps << "\n";
175     std::cout << "benchmark_first_token_tokens_per_second "
176         << first_token_tokens_per_second << "\n";
177     std::cout << "benchmark_decode_tokens_per_second " << ↪

```

```

172     decode_tokens_per_second << "\n";
173     std::cout << "benchmark_kv_cache_bytes " << result.kv_cache_bytes << "\n";
174     std::cout << "benchmark_quantized_weight_bytes "
175         << loaded.report.quantized_weight_bytes << "\n";
176     std::cout << "benchmark_f32_weight_bytes " << loaded.report.
177     f32_weight_bytes << "\n";
178     std::cout << "benchmark_tensors_loaded " << loaded.report.tensors_loaded <<
179     "\n";
180 }
181 } // namespace
182
183 int main(int argc, char** argv) {
184     try {
185         const Clock::time_point total_begin = Clock::now();
186         const LlamaCliOptions options = parse_options(argc, argv);
187         const Clock::time_point load_begin = Clock::now();
188         lcqi::LlamaGgufLoadedModel loaded =
189             lcqi::load_llama_gguf_reference_model(options.gguf_path);
190         const double load_ms = elapsed_ms(load_begin, Clock::now());
191
192         lcqi::Gpt2Tokenizer tokenizer;
193         if (!options.use_ids || options.decode_text) {
194             tokenizer = lcqi::load_gpt2_tokenizer_from_gguf(options.gguf_path);
195         }
196         const std::vector<std::int32_t> prompt_ids =
197             make_prompt_ids(options, &tokenizer, loaded.model.bos_token_id);
198         lcqi::LlamaGgufGenerationResult result =
199             lcqi::llama_gguf_generate_greedy(loaded.model,
200                 prompt_ids,
201                 options.max_new_tokens);
202
203         result.load_ms = load_ms;
204         result.total_ms = elapsed_ms(total_begin, Clock::now());
205
206         std::cout << "mode llama_gguf_reference\n";
207         std::cout << "weight_execution f32_dequantized_reference\n";
208         std::cout << "model_path " << options.gguf_path.string() << "\n";
209         std::cout << "architecture " << loaded.model.architecture << "\n";
210         std::cout << "model_name " << loaded.model.name << "\n";
211         std::cout << "layers " << loaded.model.decoder.layers.size() << "\n";
212         std::cout << "hidden_size " << loaded.model.decoder.config.hidden_size
213         << "\n";
214         std::cout << "query_heads " << loaded.model.decoder.config.query_heads
215         << "\n";
216         std::cout << "kv_heads " << loaded.model.decoder.config.kv_heads << "\n";
217         std::cout << "head_dim " << loaded.model.decoder.config.head_dim << "\n";
218         std::cout << "vocab_size " << loaded.model.decoder.config.vocab_size
219         << "\n";
220         std::cout << "tie_lm_head_to_embedding "
221             << (loaded.model.lm_head_tied_to_embedding ? 1 : 0) << "\n";

```

```

216     print_ids("prompt_ids", result.prompt_ids);
217     print_ids("generated_ids", result.generated_ids);
218     std::cout << "predicted_first_token " << result.predicted_first_token <<
    ↪ << "\n";
219     if (options.decode_text) {
220         std::cout << "generated_text "
221             << lcqi::gpt2_decode(tokenizer, result.generated_ids) <<
    ↪ "\n";
222     }
223     if (options.benchmark) {
224         const std::size_t generated_new_tokens =
225             result.generated_ids.size() >= result.prompt_ids.size()
226                 ? result.generated_ids.size() - result.prompt_ids.size()
227                 : 0;
228         print_benchmark(loader, result, result.prompt_ids.size(),
    ↪ generated_new_tokens);
229     }
230     return EXIT_SUCCESS;
231 } catch (const std::exception& error) {
232     std::cerr << "error: " << error.what() << "\n";
233     return EXIT_FAILURE;
234 }
235 }

```

`q4_k_avx2.cpp` 是本轮 Q4_K 优化入口。它不是完整 `llama.cpp` 后端，只把 Q4_KxQ8_K 的 32 元素子块点积换成 AVX2 `maddubs` 归约。读这份源码时要同时看 `fallback`：环境变量 `LCQI_Q4K_BACKEND=scalar` 可以强制走标量路径，报告脚本用它生成同机同模型的 A/B 证据。

Listing 16.11 `src/q4_k_avx2.cpp`

```

1  #include <lcqi/q4_k.hpp>
2
3  #if defined(LCQI_ENABLE_AVX2)
4
5  #include <immintrin.h>
6
7  #include <array>
8  #include <cmath>
9  #include <cstdint>
10 #include <cstdint>
11 #include <cstdlib>
12 #include <cstring>
13
14 namespace lcqi {
15 namespace {
16
17 constexpr std::uint32_t LCQI_Q4K_F32_EXPONENT_MASK = 0x7F800000U;
18 constexpr std::uint32_t LCQI_Q4K_F16_SIGN_MASK = 0x8000U;
19 constexpr std::uint32_t LCQI_Q4K_F16_EXPONENT_MASK = 0x7C00U;
20 constexpr std::uint32_t LCQI_Q4K_F16_MANTISSA_MASK = 0x03FFU;
21 constexpr std::uint32_t LCQI_Q4K_F16_EXPONENT_BIAS = 15U;
22 constexpr std::uint32_t LCQI_Q4K_F32_EXPONENT_BIAS = 127U;
23 constexpr std::uint32_t LCQI_Q4K_F32_MANTISSA_BITS = 23U;

```

```

24 constexpr std::uint32_t LCQI_Q4K_F16_MANTISSA_BITS = 10U;
25 constexpr std::uint32_t LCQI_Q4K_F16_TO_F32_SIGN_SHIFT = 16U;
26 constexpr std::uint32_t LCQI_Q4K_BYTE_BITS = 8U;
27 constexpr std::uint8_t LCQI_Q4K_LOW_NIBBLE_MASK = 0x0FU;
28 constexpr std::uint8_t LCQI_Q4K_SIX_BIT_MASK = 63U;
29 constexpr std::int64_t LCQI_Q4K_QS_OFFSET = 16;
30 constexpr std::int64_t LCQI_Q4K_QS_BYTES_PER_PAIR = 32;
31 constexpr std::int64_t LCQI_Q4K_SUBBLOCK_PAIR_COUNT = 4;
32 constexpr const char* LCQI_Q4K_BACKEND_ENV = "LCQI_Q4K_BACKEND";
33 constexpr const char* LCQI_Q4K_FORCE_SCALAR_ENV = "LCQI_Q4K_FORCE_SCALAR";
34 constexpr const char* LCQI_Q4K_BACKEND_SCALAR = "scalar";
35 constexpr const char* LCQI_Q4K_FORCE_TRUE = "1";
36
37 std::uint16_t read_le_u16(const std::uint8_t* bytes) noexcept {
38     return static_cast<std::uint16_t>(
39         static_cast<std::uint16_t>(bytes[0]) |
40         (static_cast<std::uint16_t>(bytes[1]) << LCQI_Q4K_BYTE_BITS));
41 }
42
43 float bits_to_float(std::uint32_t bits) noexcept {
44     float value = 0.0F;
45     std::memcpy(&value, &bits, sizeof(value));
46     return value;
47 }
48
49 float f16_to_f32(std::uint16_t bits) noexcept {
50     const std::uint32_t sign =
51         (static_cast<std::uint32_t>(bits) & LCQI_Q4K_F16_SIGN_MASK)
52         << LCQI_Q4K_F16_TO_F32_SIGN_SHIFT;
53     const std::uint32_t exponent =
54         static_cast<std::uint32_t>(bits) & LCQI_Q4K_F16_EXPONENT_MASK;
55     const std::uint32_t mantissa =
56         static_cast<std::uint32_t>(bits) & LCQI_Q4K_F16_MANTISSA_MASK;
57
58     if (exponent == 0U) {
59         if (mantissa == 0U) {
60             return bits_to_float(sign);
61         }
62         float value = static_cast<float>(mantissa) /
63             static_cast<float>(1U << LCQI_Q4K_F16_MANTISSA_BITS);
64         value = std::ldexp(value, 1 - static_cast<int>(<←
65             <← LCQI_Q4K_F16_EXPONENT_BIAS));
66         return sign == 0U ? value : -value;
67     }
68
69     if (exponent == LCQI_Q4K_F16_EXPONENT_MASK) {
70         const std::uint32_t f32_mantissa =
71             mantissa << (LCQI_Q4K_F32_MANTISSA_BITS - <←
72             <← LCQI_Q4K_F16_MANTISSA_BITS);
73         return bits_to_float(sign | LCQI_Q4K_F32_EXPONENT_MASK | f32_mantissa);
74     }
75 }

```

```

74     const std::uint32_t f32_exponent =
75         ((exponent >> LCQI_Q4K_F16_MANTISSA_BITS) - LCQI_Q4K_F16_EXPONENT_BIAS ←
    ↪ +
76         LCQI_Q4K_F32_EXPONENT_BIAS)
77         << LCQI_Q4K_F32_MANTISSA_BITS;
78     const std::uint32_t f32_mantissa =
79         mantissa << (LCQI_Q4K_F32_MANTISSA_BITS - LCQI_Q4K_F16_MANTISSA_BITS);
80     return bits_to_float(sign | f32_exponent | f32_mantissa);
81 }
82
83 void get_scale_min_k4(std::int32_t index,
84                     const std::uint8_t* packed,
85                     std::uint8_t& scale,
86                     std::uint8_t& min) noexcept {
87     if (index < LCQI_Q4K_SUBBLOCK_PAIR_COUNT) {
88         scale = packed[index] & LCQI_Q4K_SIX_BIT_MASK;
89         min = packed[index + LCQI_Q4K_SUBBLOCK_PAIR_COUNT] & ←
    ↪ LCQI_Q4K_SIX_BIT_MASK;
90         return;
91     }
92     scale = static_cast<std::uint8_t>((packed[index + ←
    ↪ LCQI_Q4K_SUBBLOCK_PAIR_COUNT] &
93         LCQI_Q4K_LOW_NIBBLE_MASK) |
94         ((packed[index - ←
    ↪ LCQI_Q4K_SUBBLOCK_PAIR_COUNT] >> 6U)
95         << 4U));
96     min = static_cast<std::uint8_t>((packed[index + ←
    ↪ LCQI_Q4K_SUBBLOCK_PAIR_COUNT] >> 4U) |
97         ((packed[index] >> 6U) << 4U));
98 }
99
100 std::int32_t horizontal_sum_i32(__m256i values) noexcept {
101     const __m128i low = _mm256_castsi256_si128(values);
102     const __m128i high = _mm256_extracti128_si256(values, 1);
103     __m128i sum = _mm_add_epi32(low, high);
104     sum = _mm_add_epi32(sum, _mm_shuffle_epi32(sum, _MM_SHUFFLE(1, 0, 3, 2)));
105     sum = _mm_add_epi32(sum, _mm_shuffle_epi32(sum, _MM_SHUFFLE(2, 3, 0, 1)));
106     return _mm_cvtsi128_si32(sum);
107 }
108
109 std::int32_t dot_32_u4_i8_avx2(__m256i q4_values, const std::int8_t* q8_values) ←
    ↪ noexcept {
110     const __m256i q8 = _mm256_loadu_si256(reinterpret_cast<const __m256i*>(←
    ↪ q8_values));
111     const __m256i pair_sums_i16 = _mm256_maddubs_epi16(q4_values, q8);
112     const __m256i ones = _mm256_set1_epi16(1);
113     const __m256i partial_i32 = _mm256_madd_epi16(pair_sums_i16, ones);
114     return horizontal_sum_i32(partial_i32);
115 }
116
117 float dot_block_q4_k_q8_avx2(const std::uint8_t* block, const Q8KBlock& input) ←
    ↪ noexcept {

```

```

118     const float d = f16_to_f32(read_le_u16(block));
119     const float dmin = f16_to_f32(read_le_u16(block + 2));
120     const std::uint8_t* scales = block + 4;
121     const std::uint8_t* qs = block + LCQI_Q4K_QS_OFFSET;
122     const __m256i nibble_mask = _mm256_set1_epi8(static_cast<char>(<
    ↪ LCQI_Q4K_LOW_NIBBLE_MASK));
123     std::int32_t weighted_sum = 0;
124     std::int32_t min_sum = 0;
125
126     for (std::int32_t pair = 0; pair < LCQI_Q4K_SUBBLOCK_PAIR_COUNT; ++pair) {
127         const __m256i packed =
128             _mm256_loadu_si256(reinterpret_cast<const __m256i*>(
129                 qs + pair * LCQI_Q4K_QS_BYTES_PER_PAIR));
130         const __m256i low_nibbles = _mm256_and_si256(packed, nibble_mask);
131         const __m256i high_nibbles =
132             _mm256_and_si256(_mm256_srli_epi16(packed, 4), nibble_mask);
133
134         const std::int32_t low_subblock = pair * 2;
135         const std::int32_t high_subblock = low_subblock + 1;
136         std::uint8_t low_scale = 0;
137         std::uint8_t low_min = 0;
138         std::uint8_t high_scale = 0;
139         std::uint8_t high_min = 0;
140         get_scale_min_k4(low_subblock, scales, low_scale, low_min);
141         get_scale_min_k4(high_subblock, scales, high_scale, high_min);
142
143         const std::int32_t low_dot =
144             dot_32_u4_i8_avx2(low_nibbles,
145                 input.qs.data() + low_subblock * <
    ↪ LCQI_Q4_K_SUBBLOCK_VALUES);
146         const std::int32_t high_dot =
147             dot_32_u4_i8_avx2(high_nibbles,
148                 input.qs.data() + high_subblock * <
    ↪ LCQI_Q4_K_SUBBLOCK_VALUES);
149         weighted_sum += static_cast<std::int32_t>(low_scale) * low_dot;
150         weighted_sum += static_cast<std::int32_t>(high_scale) * high_dot;
151
152         min_sum += static_cast<std::int32_t>(low_min) *
153             (static_cast<std::int32_t>(input.bsums[low_subblock * 2]) +
154                 static_cast<std::int32_t>(input.bsums[low_subblock * 2 + <
    ↪ 1]));
155         min_sum += static_cast<std::int32_t>(high_min) *
156             (static_cast<std::int32_t>(input.bsums[high_subblock * 2]) +
157                 static_cast<std::int32_t>(input.bsums[high_subblock * 2 + <
    ↪ 1]));
158     }
159
160     return (d * input.d) * static_cast<float>(weighted_sum) -
161         (dmin * input.d) * static_cast<float>(min_sum);
162 }
163
164 bool scalar_backend_forced() noexcept {

```

```

165     const char* backend = std::getenv(LCQI_Q4K_BACKEND_ENV);
166     if (backend != nullptr && std::strcmp(backend, LCQI_Q4K_BACKEND_SCALAR) == ↵
↵ 0) {
167         return true;
168     }
169     const char* force_scalar = std::getenv(LCQI_Q4K_FORCE_SCALAR_ENV);
170     return force_scalar != nullptr && std::strcmp(force_scalar, ↵
↵ LCQI_Q4K_FORCE_TRUE) == 0;
171 }
172
173 } // namespace
174
175 bool q4_k_q8_avx2_available() noexcept {
176     if (scalar_backend_forced()) {
177         return false;
178     }
179     #if defined(__GNUC__) || defined(__clang__)
180         __builtin_cpu_init();
181         return __builtin_cpu_supports("avx2");
182     #else
183         return true;
184     #endif
185 }
186
187 const char* q4_k_q8_active_backend() noexcept {
188     return q4_k_q8_avx2_available() ? "avx2_maddubs" : "scalar";
189 }
190
191 void matvec_q4_k_q8_avx2_unchecked(std::span<const std::uint8_t> q4_rows,
192                                     std::int64_t row_count,
193                                     std::int64_t column_count,
194                                     std::span<const Q8KBlock> input,
195                                     std::span<float> output) {
196     const std::int64_t row_blocks = column_count / LCQI_QK_K_BLOCK_VALUES;
197     const std::int64_t row_bytes = row_blocks * LCQI_Q4_K_BLOCK_BYTES;
198     for (std::int64_t row = 0; row < row_count; ++row) {
199         float sum = 0.0F;
200         const std::uint8_t* row_data = q4_rows.data() + row * row_bytes;
201         for (std::int64_t block = 0; block < row_blocks; ++block) {
202             sum += dot_block_q4_k_q8_avx2(
203                 row_data + block * LCQI_Q4_K_BLOCK_BYTES,
204                 input[static_cast<std::size_t>(block)]);
205         }
206         output[static_cast<std::size_t>(row)] = sum;
207     }
208 }
209
210 } // namespace lcqi
211
212 #endif

```


外部模型 Smoke 与对比脚本

`run_smollm2_small_smoke.py` 下载并校验 SmolLM2-135M-Instruct 的 GGUF 量化文件，构建固定 commit 的 `llama-simple-chat`，在 CPU 上运行一次真实开源 small 模型生成，并输出可复查报告。它是外部真实模型 smoke，不替代 LCQI tiny golden trace。

Listing 16.12 `tools/run_smollm2_small_smoke.py`

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import hashlib
7  import os
8  import re
9  import shlex
10 import shutil
11 import subprocess
12 import sys
13 from dataclasses import dataclass
14 from pathlib import Path
15
16
17 MODEL_SOURCE_ID = "HuggingFaceTB/SmolLM2-135M-Instruct"
18 MODEL_GGUF_REPO_ID = "bartowski/SmolLM2-135M-Instruct-GGUF"
19 MODEL_FILE_NAME = "SmolLM2-135M-Instruct-Q4_K_M.gguf"
20 MODEL_URL = (
21     "https://huggingface.co/bartowski/SmolLM2-135M-Instruct-GGUF"
22     f"/resolve/main/{MODEL_FILE_NAME}"
23 )
24 MODEL_EXPECTED_BYTES = 105454432
25 MODEL_EXPECTED_SHA256 = (
26     "2e8040ceae7815abe0dcb3540b9995eaa1fa0d2ca9e797d0a635ae4433c68c2d"
27 )
28 LLAMA_CPP_REPO_URL = "https://github.com/ggml-org/llama.cpp.git"
29 LLAMA_CPP_COMMIT = "b3fed31b99f9bd37725833674252bccb429bb183"
30 LLAMA_TARGET = "llama-simple-chat"
31 DEFAULT_PROMPT = "What is 2+3? Answer in one short sentence."
32 DEFAULT_CONTEXT_TOKENS = 512
33 DEFAULT_TIMEOUT_SECONDS = 180
34 DEFAULT_BUILD_JOBS = 2
35
36
37 @dataclass(frozen=True)
38 class CommandResult:
39     args: list[str]
40     returncode: int
41     stdout: str
42     stderr: str
43
44
45 def script_path() -> Path:

```



```

46     return Path(__file__).resolve()
47
48
49 def volume_dir() -> Path:
50     return script_path().parents[1]
51
52
53 def repo_dir() -> Path:
54     return script_path().parents[3]
55
56
57 def default_cache_dir() -> Path:
58     explicit_cache = os.environ.get("LCQI_SMOLLM2_CACHE_DIR")
59     if explicit_cache:
60         return Path(explicit_cache).expanduser()
61     xdg_cache = os.environ.get("XDG_CACHE_HOME")
62     cache_root = Path(xdg_cache).expanduser() if xdg_cache else Path.home() / "↵
↵ .cache"
63     return cache_root / "lcqi-smolllm2-small-smoke"
64
65
66 def default_report_path() -> Path:
67     return volume_dir() / "results" / "lcqi-smolllm2-small-model-smoke.txt"
68
69
70 def command_line(args: list[str]) -> str:
71     return " ".join(shlex.quote(arg) for arg in args)
72
73
74 def require_tool(name: str) -> str:
75     path = shutil.which(name)
76     if path is None:
77         raise RuntimeError(f"required tool not found in PATH: {name}")
78     return path
79
80
81 def run_command(
82     args: list[str],
83     *,
84     cwd: Path | None = None,
85     input_text: str | None = None,
86     timeout_seconds: int | None = None,
87 ) -> CommandResult:
88     try:
89         completed = subprocess.run(
90             args,
91             cwd=str(cwd) if cwd else None,
92             input=input_text,
93             text=True,
94             capture_output=True,
95             timeout=timeout_seconds,
96             check=False,

```

```

97         )
98     except subprocess.TimeoutExpired as exc:
99         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
100         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
101         return CommandResult(args=args, returncode=124, stdout=stdout, stderr=
↵ stderr)
102
103     return CommandResult(
104         args=args,
105         returncode=completed.returncode,
106         stdout=completed.stdout,
107         stderr=completed.stderr,
108     )
109
110
111 def ensure_success(result: CommandResult, action: str) -> None:
112     if result.returncode == 0:
113         return
114     raise RuntimeError(
115         f"{action} failed with exit code {result.returncode}\n"
116         f"command: {command_line(result.args)}\n"
117         f"stdout tail:\n{tail(result.stdout)}\n"
118         f"stderr tail:\n{tail(result.stderr)}"
119     )
120
121
122 def tail(text: str, max_chars: int = 4000) -> str:
123     if len(text) <= max_chars:
124         return text
125     return text[-max_chars:]
126
127
128 def sha256_file(path: Path) -> str:
129     digest = hashlib.sha256()
130     with path.open("rb") as handle:
131         for block in iter(lambda: handle.read(1024 * 1024), b''):
132             digest.update(block)
133     return digest.hexdigest()
134
135
136 def verify_model_file(path: Path) -> tuple[bool, str, int]:
137     if not path.exists():
138         return False, "", 0
139     actual_bytes = path.stat().st_size
140     actual_sha = sha256_file(path)
141     return (
142         actual_bytes == MODEL_EXPECTED_BYTES
143         and actual_sha == MODEL_EXPECTED_SHA256,
144         actual_sha,
145         actual_bytes,
146     )
147

```

```

148
149 def download_model(model_path: Path, *, force_download: bool, no_download: bool ↵
    ↵ ) -> None:
150     model_path.parent.mkdir(parents=True, exist_ok=True)
151
152     if force_download and model_path.exists():
153         model_path.unlink()
154
155     is_valid, actual_sha, actual_bytes = verify_model_file(model_path)
156     if is_valid:
157         print(f"[lcqi-smoke] model cache hit: {model_path}")
158         return
159
160     if model_path.exists():
161         print(
162             "[lcqi-smoke] cached model hash mismatch, redownloading: "
163             f"bytes={actual_bytes}, sha256={actual_sha}"
164         )
165         model_path.unlink()
166
167     if no_download:
168         raise RuntimeError(
169             f"model file is missing or invalid and --no-download was set: {↵
    ↵ model_path}"
170         )
171
172     require_tool("curl")
173     part_path = model_path.with_suffix(model_path.suffix + ".part")
174     if part_path.exists():
175         part_path.unlink()
176
177     print(f"[lcqi-smoke] downloading small GGUF model: {MODEL_GGUF_REPO_ID}/{↵
    ↵ MODEL_FILE_NAME}")
178     result = run_command(
179         [
180             "curl",
181             "-L",
182             "--fail",
183             "--retry",
184             "3",
185             "-o",
186             str(part_path),
187             MODEL_URL,
188         ]
189     )
190     ensure_success(result, "download SmolLM2 GGUF")
191
192     downloaded_ok, downloaded_sha, downloaded_bytes = verify_model_file(↵
    ↵ part_path)
193     if not downloaded_ok:
194         raise RuntimeError(
195             "downloaded model did not match expected identity: "

```

```

196         f"bytes={downloaded_bytes}, sha256={downloaded_sha}"
197     )
198     part_path.replace(model_path)
199     print(f"[lcqi-smoke] model verified: sha256={MODEL_EXPECTED_SHA256}")
200
201
202 def ensure_llama_simple_chat(cache_dir: Path, *, jobs: int, skip_build: bool) ←
    ↪ -> Path:
203     cached_binary = cache_dir / "llama.cpp" / "build" / "bin" / LLAMA_TARGET
204     source_dir = cache_dir / "llama.cpp"
205     build_dir = source_dir / "build"
206     if cached_binary.exists() and os.access(cached_binary, os.X_OK) and ←
    ↪ is_pinned_llama_checkout(source_dir):
207         print(f"[lcqi-smoke] llama.cpp binary cache hit: {cached_binary}")
208         return cached_binary
209
210     if skip_build:
211         raise RuntimeError(f"llama.cpp binary is missing and --skip-build was ←
    ↪ set: {cached_binary}")
212
213     require_tool("git")
214     require_tool("cmake")
215
216     cache_dir.mkdir(parents=True, exist_ok=True)
217
218     if not (source_dir / ".git").exists():
219         print(f"[lcqi-smoke] cloning llama.cpp into cache: {source_dir}")
220         result = run_command(
221             [
222                 "git",
223                 "clone",
224                 "--filter=blob:none",
225                 "--no-checkout",
226                 LLAMA_CPP_REPO_URL,
227                 str(source_dir),
228             ]
229         )
230         ensure_success(result, "clone llama.cpp")
231
232     print(f"[lcqi-smoke] checking out pinned llama.cpp commit: {←
    ↪ LLAMA_CPP_COMMIT}")
233     fetch = run_command(
234         ["git", "-C", str(source_dir), "fetch", "--depth", "1", "origin", ←
    ↪ LLAMA_CPP_COMMIT]
235     )
236     ensure_success(fetch, "fetch pinned llama.cpp commit")
237     checkout = run_command(["git", "-C", str(source_dir), "checkout", "--detach←
    ↪ ", LLAMA_CPP_COMMIT])
238     ensure_success(checkout, "checkout pinned llama.cpp commit")
239
240     print(f"[lcqi-smoke] configuring llama.cpp build: {build_dir}")
241     configure = run_command(

```

```

242     [
243         "cmake",
244         "-S",
245         str(source_dir),
246         "-B",
247         str(build_dir),
248         "-DCMAKE_BUILD_TYPE=Release",
249         "-DLLAMA_BUILD_TESTS=OFF",
250         "-DLLAMA_BUILD_EXAMPLES=ON",
251         "-DLLAMA_BUILD_SERVER=OFF",
252         "-DGGML_NATIVE=OFF",
253     ]
254 )
255 ensure_success(configure, "configure llama.cpp")
256
257 print(f"[lcqi-smoke] building {LLAMA_TARGET} with -j{jobs}")
258 build = run_command(
259     ["cmake", "--build", str(build_dir), "--target", LLAMA_TARGET, "-j", ←
↪ str(jobs)]
260 )
261 ensure_success(build, f"build {LLAMA_TARGET}")
262
263 if not cached_binary.exists():
264     raise RuntimeError(f"expected binary was not produced: {cached_binary}" ←
↪ )
265 return cached_binary
266
267
268 def is_pinned_llama_checkout(source_dir: Path) -> bool:
269     if not (source_dir / ".git").exists():
270         return False
271     result = run_command(["git", "-C", str(source_dir), "rev-parse", "HEAD"])
272     if result.returncode != 0:
273         return False
274     return result.stdout.strip() == LLAMA_CPP_COMMIT
275
276
277 def strip_ansi(text: str) -> str:
278     return re.sub(r"\x1b\[([0-?]*[ -/]*[@-~])", "", text)
279
280
281 def extract_response(stdout: str) -> str:
282     cleaned = strip_ansi(stdout).replace("\r", "")
283     chunks = [chunk.strip() for chunk in re.split(r"(:^|\n)> ?", cleaned) if ←
↪ chunk.strip()]
284     if not chunks:
285         return ""
286     return chunks[0]
287
288
289 def run_model_smoke(
290     llama_binary: Path,

```

```

291     model_path: Path,
292     *,
293     prompt: str,
294     context_tokens: int,
295     timeout_seconds: int,
296 ) -> tuple[CommandResult, str]:
297     command = [
298         str(llama_binary),
299         "-m",
300         str(model_path),
301         "-ngl",
302         "0",
303         "-c",
304         str(context_tokens),
305     ]
306     print("[lcqi-smoke] running real small model generation")
307     result = run_command(
308         command,
309         input_text=prompt.rstrip("\n") + "\n\n",
310         timeout_seconds=timeout_seconds,
311     )
312     ensure_success(result, "run small model smoke")
313
314     response = extract_response(result.stdout)
315     if not response:
316         raise RuntimeError(
317             "small model smoke exited successfully but produced no assistant ←
↵ text\n"
318             f"stdout:\n{result.stdout}\n"
319             f"stderr:\n{result.stderr}"
320         )
321     return result, response
322
323
324 def current_repo_commit() -> str:
325     result = run_command(["git", "-C", str(repo_dir()), "rev-parse", "--short", ←
↵ "HEAD"])
326     if result.returncode != 0:
327         return "unknown"
328     return result.stdout.strip()
329
330
331 def current_uname() -> str:
332     result = run_command(["uname", "-a"])
333     if result.returncode != 0:
334         return "unknown"
335     return result.stdout.strip()
336
337
338 def write_report(
339     report_path: Path,
340     *,

```

```

341     cache_dir: Path,
342     llama_binary: Path,
343     model_path: Path,
344     prompt: str,
345     result: CommandResult,
346     response: str,
347 ) -> None:
348     report_path.parent.mkdir(parents=True, exist_ok=True)
349     clean_stdout = strip_ansi(result.stdout).strip()
350     report = "\n".join(
351         [
352             "LCQI SmolLM2 Small Model Smoke",
353             "",
354             f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
355             f"repo_commit={current_repo_commit()}",
356             f"host={current_uname()}",
357             f"python={sys.version.split()[0]}",
358             f"cache_dir={cache_dir}",
359             "",
360             f"model_source_id={MODEL_SOURCE_ID}",
361             f"model_gguf_repo_id={MODEL_GGUF_REPO_ID}",
362             f"model_file={MODEL_FILE_NAME}",
363             f"model_url={MODEL_URL}",
364             f"model_expected_bytes={MODEL_EXPECTED_BYTES}",
365             f"model_expected_sha256={MODEL_EXPECTED_SHA256}",
366             f"model_path={model_path}",
367             "",
368             f"llama_cpp_repo={LLAMA_CPP_REPO_URL}",
369             f"llama_cpp_commit={LLAMA_CPP_COMMIT}",
370             f"llama_target={LLAMA_TARGET}",
371             f"llama_binary={llama_binary}",
372             "",
373             f"prompt={prompt}",
374             f"command={command_line(result.args)}",
375             f"exit_code={result.returncode}",
376             "",
377             "validation=PASS: model hash matched, llama-simple-chat exited 0, "
378             "assistant response was non-empty",
379             "",
380             "assistant_response:",
381             response,
382             "",
383             "stdout_clean_tail:",
384             tail(clean_stdout, 2000),
385             "",
386             "stderr_tail:",
387             tail(result.stderr.strip(), 2000),
388             "",
389         ]
390     )
391     report_path.write_text(report, encoding="utf-8")
392

```

```

393
394 def parse_args() -> argparse.Namespace:
395     parser = argparse.ArgumentParser(
396         description=(
397             "Run a real local small open-source model smoke for CPU Volume 3. "
398             "The script caches llama.cpp and the GGUF model outside the Git ↵
↵ repository."
399         )
400     )
401     parser.add_argument("--cache-dir", type=Path, default=default_cache_dir())
402     parser.add_argument("--model-path", type=Path, default=None)
403     parser.add_argument("--llama-bin", type=Path, default=None)
404     parser.add_argument("--report", type=Path, default=default_report_path())
405     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
406     parser.add_argument("--ctx", type=int, default=DEFAULT_CONTEXT_TOKENS)
407     parser.add_argument("--timeout", type=int, default=DEFAULT_TIMEOUT_SECONDS)
408     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
409     parser.add_argument("--force-download", action="store_true")
410     parser.add_argument("--no-download", action="store_true")
411     parser.add_argument("--skip-build", action="store_true")
412     return parser.parse_args()
413
414
415 def main() -> int:
416     args = parse_args()
417     cache_dir = args.cache_dir.expanduser().resolve()
418     model_path = (
419         args.model_path.expanduser().resolve()
420         if args.model_path
421         else cache_dir / "models" / MODEL_FILE_NAME
422     )
423     report_path = args.report.expanduser().resolve()
424
425     try:
426         download_model(
427             model_path,
428             force_download=args.force_download,
429             no_download=args.no_download,
430         )
431         if args.llama_bin:
432             llama_binary = args.llama_bin.expanduser().resolve()
433             if not llama_binary.exists() or not os.access(llama_binary, os.X_OK↵
↵ ):
434                 raise RuntimeError(f"--llama-bin is not executable: {↵
↵ llama_binary}")
435         else:
436             llama_binary = ensure_llama_simple_chat(
437                 cache_dir,
438                 jobs=args.jobs,
439                 skip_build=args.skip_build,
440             )
441

```



```

442     result, response = run_model_smoke(
443         llama_binary,
444         model_path,
445         prompt=args.prompt,
446         context_tokens=args.ctx,
447         timeout_seconds=args.timeout,
448     )
449     write_report(
450         report_path,
451         cache_dir=cache_dir,
452         llama_binary=llama_binary,
453         model_path=model_path,
454         prompt=args.prompt,
455         result=result,
456         response=response,
457     )
458     except RuntimeError as exc:
459         print(f"[lcqi-smoke] ERROR: {exc}", file=sys.stderr)
460         return 1
461
462     print(f"report={report_path}")
463     print(f"model_sha256={MODEL_EXPECTED_SHA256}")
464     print(f"assistant_response={response}")
465     return 0
466
467
468 if __name__ == "__main__":
469     raise SystemExit(main())

```

`run_smollm2_q4_k_benchmark.py` 复用同一个 SmolLM2 GGUF 缓存，构建 LCQI 的 `lcqi_gguf_q4_bench`，并可附带 llama.cpp `llama-bench` 端到端参考。它会先用 `LCQI_Q4K_BACKEND=scalar` 跑标量路径，再跑 `auto` 后端，因此报告里能直接看到 `avx2_maddubs` 相对 `scalar` 的提升。报告要同时说明模型 SHA-256、tensor 名、shape、量化字节、f32 解码字节、误差、checksum 和成熟引擎参考，避免把不同层级数字混在一起。

Listing 16.13 `tools/run_smollm2_q4_k_benchmark.py`

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import os
7  import re
8  import shlex
9  import subprocess
10 import sys
11 from dataclasses import dataclass
12 from pathlib import Path
13
14 import run_smollm2_small_smoke
15
16

```

```
17 DEFAULT_BUILD_JOBS = 2
18 DEFAULT_REPEAT = 20
19 DEFAULT_TIMEOUT_SECONDS = 300
20 BENCH_TARGET = "lcqi-gguf_q4_bench"
21
22
23 @dataclass(frozen=True)
24 class CommandResult:
25     name: str
26     args: list[str]
27     returncode: int
28     stdout: str
29     stderr: str
30     metrics: dict[str, str]
31
32
33 def volume_dir() -> Path:
34     return Path(__file__).resolve().parents[1]
35
36
37 def repo_dir() -> Path:
38     return Path(__file__).resolve().parents[3]
39
40
41 def default_build_dir() -> Path:
42     return volume_dir() / "build" / "lcqi-release"
43
44
45 def default_report_path() -> Path:
46     return volume_dir() / "results" / "lcqi-smollm2-q4-k-benchmark.txt"
47
48
49 def command_line(args: list[str]) -> str:
50     return " ".join(shlex.quote(arg) for arg in args)
51
52
53 def run_command(
54     name: str,
55     args: list[str],
56     *,
57     timeout_seconds: int,
58     cwd: Path | None = None,
59     env: dict[str, str] | None = None,
60 ) -> CommandResult:
61     command_env = os.environ.copy()
62     if env:
63         command_env.update(env)
64     try:
65         completed = subprocess.run(
66             args,
67             cwd=str(cwd) if cwd else str(repo_dir()),
68             env=command_env,
```

```

69         text=True,
70         capture_output=True,
71         timeout=timeout_seconds,
72         check=False,
73     )
74     except subprocess.TimeoutExpired as exc:
75         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
76         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
77         return CommandResult(name, args, 124, stdout, stderr, {})
78     return CommandResult(
79         name=name,
80         args=args,
81         returncode=completed.returncode,
82         stdout=completed.stdout,
83         stderr=completed.stderr,
84         metrics=parse_key_value_metrics(completed.stdout),
85     )
86
87
88 def ensure_success(result: CommandResult, action: str) -> None:
89     if result.returncode == 0:
90         return
91     raise RuntimeError(
92         f"{action} failed with exit code {result.returncode}\n"
93         f"command: {command_line(result.args)}\n"
94         f"stdout tail:\n{tail(result.stdout)}\n"
95         f"stderr tail:\n{tail(result.stderr)}"
96     )
97
98
99 def tail(text: str, max_chars: int = 4000) -> str:
100     return text if len(text) <= max_chars else text[-max_chars:]
101
102
103 def parse_key_value_metrics(stdout: str) -> dict[str, str]:
104     metrics: dict[str, str] = {}
105     for line in stdout.splitlines():
106         key, sep, value = line.partition("=")
107         if sep:
108             metrics[key.strip()] = value.strip()
109     return metrics
110
111
112 def parse_llama_bench_metrics(stdout: str) -> dict[str, str]:
113     metrics: dict[str, str] = {}
114     for line in stdout.splitlines():
115         stripped = line.strip()
116         if not stripped.startswith("|"):
117             continue
118         columns = [column.strip() for column in stripped.strip("|").split("|")]
119         if len(columns) != 7 or columns[0] == "model" or columns[0].startswith(↵
↵ "-"):

```

```

120         continue
121         metrics["llama_model"] = columns[0]
122         metrics["llama_size"] = columns[1]
123         metrics["llama_params"] = columns[2]
124         metrics["llama_backend"] = columns[3]
125         metrics["llama_threads"] = columns[4]
126         tokens_per_second = re.sub(r"\s+.*$", "", columns[6])
127         if columns[5].startswith("pp"):
128             metrics["llama_prefill_test"] = columns[5]
129             metrics["llama_prefill_tps"] = tokens_per_second
130         if columns[5].startswith("tg"):
131             metrics["llama_decode_test"] = columns[5]
132             metrics["llama_decode_tps"] = tokens_per_second
133     return metrics
134
135
136 def build_lcqi(build_dir: Path, jobs: int, timeout_seconds: int) -> Path:
137     configure = run_command(
138         "cmake_configure",
139         [
140             "cmake",
141             "-S",
142             str(volume_dir()),
143             "-B",
144             str(build_dir),
145             "-DCMAKE_BUILD_TYPE=Release",
146         ],
147         timeout_seconds=timeout_seconds,
148     )
149     ensure_success(configure, "configure LCQI")
150     build = run_command(
151         "cmake_build_lcqi_gguf_q4_bench",
152         [
153             "cmake",
154             "--build",
155             str(build_dir),
156             "--target",
157             BENCH_TARGET,
158             "-j",
159             str(jobs),
160         ],
161         timeout_seconds=timeout_seconds,
162     )
163     ensure_success(build, "build LCQI Q4_K benchmark")
164     binary = build_dir / "labs" / "linux_cpu_inference" / BENCH_TARGET
165     if not binary.exists() or not os.access(binary, os.X_OK):
166         raise RuntimeError(f"LCQI Q4_K benchmark binary not found: {binary}")
167     return binary
168
169
170 def llama_bench_binary(cache_dir: Path) -> Path:
171     return cache_dir / "llama.cpp" / "build" / "bin" / "llama-bench"

```

```

172
173
174 def model_path(cache_dir: Path) -> Path:
175     return cache_dir / "models" / run_smollm2_small_smoke.MODEL_FILE_NAME
176
177
178 def run_lcqi_benchmark(
179     binary: Path,
180     model: Path,
181     *,
182     name: str,
183     tensor: str,
184     repeat: int,
185     timeout_seconds: int,
186     env: dict[str, str] | None = None,
187 ) -> CommandResult:
188     return run_command(
189         name,
190         [str(binary), str(model), tensor, str(repeat)],
191         timeout_seconds=timeout_seconds,
192         env=env,
193     )
194
195
196 def run_llama_bench(cache_dir: Path, timeout_seconds: int) -> CommandResult | ←
    ↪ None:
197     binary = llama_bench_binary(cache_dir)
198     model = model_path(cache_dir)
199     if not binary.exists() or not os.access(binary, os.X_OK) or not model.↪
    ↪ exists():
200         return None
201     result = run_command(
202         "llama_cpp_smollm2_q4_k_m",
203         [
204             str(binary),
205             "-m",
206             str(model),
207             "-ngl",
208             "0",
209             "-p",
210             "5",
211             "-n",
212             "4",
213             "-r",
214             "1",
215         ],
216         timeout_seconds=timeout_seconds,
217     )
218     return CommandResult(
219         result.name,
220         result.args,
221         result.returncode,

```

```

222     result.stdout,
223     result.stderr,
224     parse_llama_bench_metrics(result.stdout),
225 )
226
227
228 def current_commit() -> str:
229     result = subprocess.run(
230         ["git", "-C", str(repo_dir()), "rev-parse", "--short", "HEAD"],
231         text=True,
232         capture_output=True,
233         check=False,
234     )
235     return result.stdout.strip() if result.returncode == 0 else "unknown"
236
237
238 def working_tree_dirty() -> str:
239     result = subprocess.run(
240         ["git", "-C", str(repo_dir()), "status", "--porcelain"],
241         text=True,
242         capture_output=True,
243         check=False,
244     )
245     if result.returncode != 0:
246         return "unknown"
247     return "true" if result.stdout.strip() else "false"
248
249
250 def write_report(path: Path, runs: list[CommandResult], *, model: Path) -> None:
251     ↪ :
252     path.parent.mkdir(parents=True, exist_ok=True)
253     model_matches, actual_sha256, actual_bytes = run_smolllm2_small_smoke.
254     ↪ verify_model_file(model)
255     lines = [
256         "LCQI SmolLM2 Q4_K Benchmark",
257         "",
258         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
259         f"repo_commit={current_commit()}",
260         f"working_tree_dirty={working_tree_dirty()}",
261         f"model={model}",
262         f"model_source={run_smolllm2_small_smoke.MODEL_GGUF_REPO_ID}",
263         f"model_file={run_smolllm2_small_smoke.MODEL_FILE_NAME}",
264         f"model_expected_bytes={run_smolllm2_small_smoke.MODEL_EXPECTED_BYTES}",
265         f"model_actual_bytes={actual_bytes}",
266         f"model_expected_sha256={run_smolllm2_small_smoke.MODEL_EXPECTED_SHA256}
267     ↪ ",
268         f"model_actual_sha256={actual_sha256}",
269         f"model_identity_match={str(model_matches).lower()}",
270         f"llama_cpp_commit={run_smolllm2_small_smoke.LLAMA_CPP_COMMIT}",
271         "",
272         "scope=LCQI reads the same SmolLM2 Q4_K_M GGUF used by llama.cpp and
273     ↪ runs a "

```

```

270     "single real Q4_K tensor matvec slice. llama.cpp numbers are end-to-end↵
↵ mature "
271     "engine context, not same-function microkernel parity.",
272     "",
273 ]
274 for run in runs:
275     lines.extend(
276         [
277             f"[{run.name}]",
278             f"returncode={run.returncode}",
279             f"command={command_line(run.args)}",
280         ]
281     )
282     for key in sorted(run.metrics):
283         lines.append(f"{key}={run.metrics[key]}")
284     lines.extend(
285         [
286             "stdout_tail:",
287             tail(run.stdout.strip(), 2400),
288             "stderr_tail:",
289             tail(run.stderr.strip(), 2400),
290             "",
291         ]
292     )
293     scalar = next((run for run in runs if run.name == "lcqi_smolllm2_q4_k_scalar↵
↵ "), None)
294     auto = next((run for run in runs if run.name == "lcqi_smolllm2_q4_k_auto"), ↵
↵ None)
295     if scalar is not None and auto is not None:
296         scalar_matvec = metric_as_float(scalar.metrics, "↵
↵ q4_q8_matvec_average_us")
297         auto_matvec = metric_as_float(auto.metrics, "q4_q8_matvec_average_us")
298         scalar_total = metric_as_float(scalar.metrics, "q4_q8_total_average_us"↵
↵ )
299         auto_total = metric_as_float(auto.metrics, "q4_q8_total_average_us")
300         if scalar_matvec is not None and auto_matvec is not None and ↵
↵ auto_matvec > 0.0:
301             lines.append("[lcqi_auto_vs_scalar]")
302             lines.append(f"q4_q8_matvec_speedup_auto_vs_scalar={scalar_matvec /↵
↵ auto_matvec:.3f}")
303             if scalar_total is not None and auto_total is not None and ↵
↵ auto_total > 0.0:
304                 lines.append(f"q4_q8_total_speedup_auto_vs_scalar={scalar_total↵
↵ / auto_total:.3f}")
305                 lines.append("")
306     while lines and lines[-1] == "":
307         lines.pop()
308     path.write_text("\n".join(lines) + "\n", encoding="utf-8")
309
310
311 def metric_as_float(metrics: dict[str, str], key: str) -> float | None:
312     value = metrics.get(key)

```

```

313     if value is None:
314         return None
315     try:
316         return float(value)
317     except ValueError:
318         return None
319
320
321 def parse_args() -> argparse.Namespace:
322     parser = argparse.ArgumentParser(
323         description="Benchmark LCQI Q4_K matvec on the same SmolLM2 GGUF used ↵
↵ by llama.cpp."
324     )
325     parser.add_argument("--cache-dir", type=Path, default=↵
↵ run_smolllm2_small_smoke.default_cache_dir())
326     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
327     parser.add_argument("--report", type=Path, default=default_report_path())
328     parser.add_argument("--tensor", default="auto")
329     parser.add_argument("--repeat", type=int, default=DEFAULT_REPEAT)
330     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
331     parser.add_argument("--timeout-seconds", type=int, default=↵
↵ DEFAULT_TIMEOUT_SECONDS)
332     parser.add_argument("--no-download", action="store_true")
333     parser.add_argument("--skip-llama", action="store_true")
334     return parser.parse_args()
335
336
337 def main() -> int:
338     args = parse_args()
339     if args.repeat <= 0:
340         print("[lcqi-smolllm2-q4] --repeat must be positive", file=sys.stderr)
341         return 1
342     try:
343         cache_dir = args.cache_dir.expanduser().resolve()
344         model = model_path(cache_dir)
345         run_smolllm2_small_smoke.download_model(
346             model,
347             force_download=False,
348             no_download=args.no_download,
349         )
350         binary = build_lcqi(args.build_dir, args.jobs, args.timeout_seconds)
351         runs = [
352             run_lcqi_benchmark(
353                 binary,
354                 model,
355                 name="lcqi_smolllm2_q4_k_scalar",
356                 tensor=args.tensor,
357                 repeat=args.repeat,
358                 timeout_seconds=args.timeout_seconds,
359                 env={"LCQI_Q4K_BACKEND": "scalar"},
360             ),
361             run_lcqi_benchmark(

```



```

362         binary,
363         model,
364         name="lcqi_smollm2_q4_k_auto",
365         tensor=args.tensor,
366         repeat=args.repeat,
367         timeout_seconds=args.timeout_seconds,
368     )
369 ]
370 if not args.skip_llama:
371     llama = run_llama_bench(cache_dir, args.timeout_seconds)
372     if llama is not None:
373         runs.append(llama)
374     write_report(args.report, runs, model=model)
375 except Exception as error:
376     print(f"[lcqi-smollm2-q4] error: {error}", file=sys.stderr)
377     return 1
378
379 for run in runs:
380     print(f"{run.name}: returncode={run.returncode}")
381     for key in sorted(run.metrics):
382         print(f"    {key}={run.metrics[key]}")
383 print(f"report={args.report}")
384 return 0 if all(run.returncode == 0 for run in runs) else 1
385
386
387 if __name__ == "__main__":
388     raise SystemExit(main())

```

`run_gpt2_smoke.py` 下载标准 GPT-2 的 `config.json`、`vocab.json`、`merges.txt` 和 `model.safetensors`，再调用 LCQI 自研 `lcqi_gpt2` reference path 生成 token。它证明自研 safetensors、BPE、GPT-2 forward、KV cache decode 和 greedy generation 已连成闭环。

Listing 16.14 tools/run_gpt2_smoke.py

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import hashlib
7  import os
8  import shlex
9  import subprocess
10 import sys
11 import urllib.request
12 from dataclasses import dataclass
13 from pathlib import Path
14
15
16 MODEL_REPO_ID = "openai-community/gpt2"
17 MODEL_REVISION = "main"
18 MODEL_FILE_IDENTITIES = {
19     "config.json": (

```

```

20         665,
21         "0daed7749b4f02b8f76240d5444551d7b08712dab4d0adb8239c56ba823bb7b4",
22     ),
23     "vocab.json": (
24         1042301,
25         "196139668be63f3b5d6574427317ae82f612a97c5d1cdf36ed2256dbf636783",
26     ),
27     "merges.txt": (
28         456318,
29         "1ce1664773c50f3e0cc8842619a93edc4624525b728b188a9e0be33b7726adc5",
30     ),
31     "model.safetensors": (
32         548105171,
33         "248dfc3911869ec493c76e65bf2fcf7f615828b0254c12b473182f0f81d3a707",
34     ),
35 }
36 DEFAULT_PROMPT = "Hello, my name is"
37 DEFAULT_MAX_NEW_TOKENS = 1
38 DEFAULT_TIMEOUT_SECONDS = 300
39 DEFAULT_BUILD_JOBS = 2
40
41
42 @dataclass(frozen=True)
43 class CommandResult:
44     args: list[str]
45     returncode: int
46     stdout: str
47     stderr: str
48
49
50 def script_path() -> Path:
51     return Path(__file__).resolve()
52
53
54 def volume_dir() -> Path:
55     return script_path().parents[1]
56
57
58 def repo_dir() -> Path:
59     return script_path().parents[3]
60
61
62 def default_cache_dir() -> Path:
63     explicit_cache = os.environ.get("LCQI_GPT2_CACHE_DIR")
64     if explicit_cache:
65         return Path(explicit_cache).expanduser()
66     xdg_cache = os.environ.get("XDG_CACHE_HOME")
67     cache_root = Path(xdg_cache).expanduser() if xdg_cache else Path.home() / "↵
↵ .cache"
68     return cache_root / "lcqi-gpt2-smoke" / MODEL_REPO_ID.replace("/", "--")
69
70

```

```

71 def default_build_dir() -> Path:
72     return volume_dir() / "build" / "lcqi-release"
73
74
75 def default_report_path() -> Path:
76     return volume_dir() / "results" / "lcqi-gpt2-smoke.txt"
77
78
79 def command_line(args: list[str]) -> str:
80     return " ".join(shlex.quote(arg) for arg in args)
81
82
83 def run_command(
84     args: list[str],
85     *,
86     cwd: Path | None = None,
87     timeout_seconds: int | None = None,
88 ) -> CommandResult:
89     try:
90         completed = subprocess.run(
91             args,
92             cwd=cwd if cwd else None,
93             text=True,
94             capture_output=True,
95             timeout=timeout_seconds,
96             check=False,
97         )
98     except subprocess.TimeoutExpired as exc:
99         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
100         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
101         return CommandResult(args=args, returncode=124, stdout=stdout, stderr=↵
↵ stderr)
102     return CommandResult(
103         args=args,
104         returncode=completed.returncode,
105         stdout=completed.stdout,
106         stderr=completed.stderr,
107     )
108
109
110 def ensure_success(result: CommandResult, action: str) -> None:
111     if result.returncode == 0:
112         return
113     raise RuntimeError(
114         f"{action} failed with exit code {result.returncode}\n"
115         f"command: {command_line(result.args)}\n"
116         f"stdout tail:\n{tail(result.stdout)}\n"
117         f"stderr tail:\n{tail(result.stderr)}"
118     )
119
120
121 def tail(text: str, max_chars: int = 4000) -> str:

```

```

122     if len(text) <= max_chars:
123         return text
124     return text[-max_chars:]
125
126
127 def sha256_file(path: Path) -> str:
128     digest = hashlib.sha256()
129     with path.open("rb") as handle:
130         for block in iter(lambda: handle.read(1024 * 1024), b''):
131             digest.update(block)
132     return digest.hexdigest()
133
134
135 def model_url(file_name: str) -> str:
136     return (
137         f"https://huggingface.co/{MODEL_REPO_ID}/resolve/"
138         f"{MODEL_REVISION}/{file_name}"
139     )
140
141
142 def download_file(url: str, target: Path, timeout_seconds: int) -> None:
143     part_path = target.with_suffix(target.suffix + ".part")
144     if part_path.exists():
145         part_path.unlink()
146     request = urllib.request.Request(url, headers={"User-Agent": "lcqi-gpt2-↵
↵ smoke"})
147     with urllib.request.urlopen(request, timeout=timeout_seconds) as response:
148         with part_path.open("wb") as output:
149             while True:
150                 block = response.read(1024 * 1024)
151                 if not block:
152                     break
153                 output.write(block)
154     part_path.replace(target)
155
156
157 def file_identity_matches(path: Path, expected_bytes: int, expected_sha256: str↵
↵ ) -> bool:
158     return (
159         path.exists()
160         and path.stat().st_size == expected_bytes
161         and sha256_file(path) == expected_sha256
162     )
163
164
165 def ensure_model_files(cache_dir: Path, *, no_download: bool, timeout_seconds: ↵
↵ int) -> None:
166     cache_dir.mkdir(parents=True, exist_ok=True)
167     missing = [
168         name
169         for name, (expected_bytes, expected_sha256) in MODEL_FILE_IDENTITYIES.↵
↵ items()

```

```

170         if not file_identity_matches(cache_dir / name, expected_bytes, ↵
↵ expected_sha256)
171     ]
172     if not missing:
173         print(f"[lcqi-gpt2] model cache hit: {cache_dir}")
174         return
175     if no_download:
176         raise RuntimeError(
177             f"missing GPT-2 model files and --no-download was set: {missing}"
178         )
179     for file_name in missing:
180         target = cache_dir / file_name
181         if target.exists():
182             target.unlink()
183         print(f"[lcqi-gpt2] downloading {MODEL_REPO_ID}/{file_name}")
184         download_file(model_url(file_name), target, timeout_seconds)
185         expected_bytes, expected_sha256 = MODEL_FILE_IDENTITIES[file_name]
186         if not file_identity_matches(target, expected_bytes, expected_sha256):
187             raise RuntimeError(
188                 f"downloaded GPT-2 file identity mismatch: {file_name}"
189             )
190
191
192 def build_lcqi_gpt2(build_dir: Path, jobs: int) -> Path:
193     configure = run_command(
194         [
195             "cmake",
196             "-S",
197             str(volume_dir()),
198             "-B",
199             str(build_dir),
200             "-DCMAKE_BUILD_TYPE=Release",
201         ],
202         timeout_seconds=DEFAULT_TIMEOUT_SECONDS,
203     )
204     ensure_success(configure, "configure LCQI")
205     build = run_command(
206         [
207             "cmake",
208             "--build",
209             str(build_dir),
210             "--target",
211             "lcqi_gpt2",
212             "-j",
213             str(jobs),
214         ],
215         timeout_seconds=DEFAULT_TIMEOUT_SECONDS,
216     )
217     ensure_success(build, "build lcqi_gpt2")
218     binary = build_dir / "labs" / "linux_cpu_inference" / "lcqi_gpt2"
219     if not binary.exists():
220         raise RuntimeError(f"lcqi_gpt2 binary not found: {binary}")

```

```

221     return binary
222
223
224 def write_report(
225     report_path: Path,
226     *,
227     cache_dir: Path,
228     binary: Path,
229     prompt: str,
230     max_new_tokens: int,
231     engine: str,
232     result: CommandResult,
233 ) -> None:
234     report_path.parent.mkdir(parents=True, exist_ok=True)
235     lines = [
236         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
237         f"model_repo_id={MODEL_REPO_ID}",
238         f"model_revision={MODEL_REVISION}",
239         f"model_cache_dir={cache_dir}",
240     ]
241     for file_name, (expected_bytes, expected_sha256) in MODEL_FILE_IDENTITYIES.↵
↵ items():
242         path = cache_dir / file_name
243         lines.append(f"file.{file_name}.bytes={path.stat().st_size}")
244         lines.append(f"file.{file_name}.sha256={sha256_file(path)}")
245         lines.append(f"file.{file_name}.expected_bytes={expected_bytes}")
246         lines.append(f"file.{file_name}.expected_sha256={expected_sha256}")
247     lines.extend(
248         [
249             f"binary={binary}",
250             f"prompt={prompt}",
251             f"max_new_tokens={max_new_tokens}",
252             f"engine={engine}",
253             f"command={command_line(result.args)}",
254             f"exit_code={result.returncode}",
255             "stdout_begin",
256             result.stdout.rstrip(),
257             "stdout_end",
258             "stderr_begin",
259             tail(result.stderr).rstrip(),
260             "stderr_end",
261         ]
262     )
263     passed = result.returncode == 0 and "generated_text " in result.stdout
264     lines.append(f"validation={'PASS' if passed else 'FAIL'}")
265     report_path.write_text("\n".join(lines) + "\n", encoding="utf-8")
266
267
268 def parse_args() -> argparse.Namespace:
269     parser = argparse.ArgumentParser(
270         description="Run LCQI self-owned GPT-2 smoke on a HuggingFace ↵
↵ safetensors directory."

```

```

271 )
272 parser.add_argument("--cache-dir", type=Path, default=default_cache_dir())
273 parser.add_argument("--build-dir", type=Path, default=default_build_dir())
274 parser.add_argument("--report", type=Path, default=default_report_path())
275 parser.add_argument("--prompt", default=DEFAULT_PROMPT)
276 parser.add_argument("--max-new-tokens", type=int, default=↵
↵ DEFAULT_MAX_NEW_TOKENS)
277 parser.add_argument("--engine", choices=("cached", "full"), default="cached↵
↵ ")
278 parser.add_argument("--benchmark", action="store_true")
279 parser.add_argument("--timeout-seconds", type=int, default=↵
↵ DEFAULT_TIMEOUT_SECONDS)
280 parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
281 parser.add_argument("--no-download", action="store_true")
282 return parser.parse_args()
283
284
285 def main() -> int:
286     args = parse_args()
287     try:
288         if args.max_new_tokens < 0:
289             raise RuntimeError("--max-new-tokens cannot be negative")
290         ensure_model_files(
291             args.cache_dir,
292             no_download=args.no_download,
293             timeout_seconds=args.timeout_seconds,
294         )
295         binary = build_lcqi_gpt2(args.build_dir, args.jobs)
296         command = [
297             str(binary),
298             "--engine",
299             args.engine,
300             str(args.cache_dir),
301             args.prompt,
302             str(args.max_new_tokens),
303         ]
304         if args.benchmark:
305             command.insert(1, "--benchmark")
306         result = run_command(command, cwd=repo_dir(), timeout_seconds=args.↵
↵ timeout_seconds)
307         write_report(
308             args.report,
309             cache_dir=args.cache_dir,
310             binary=binary,
311             prompt=args.prompt,
312             max_new_tokens=args.max_new_tokens,
313             engine=args.engine,
314             result=result,
315         )
316         print(f"[lcqi-gpt2] report: {args.report}")
317         print(tail(result.stdout, max_chars=2000))
318         return 0 if result.returncode == 0 else result.returncode

```

```

319     except Exception as error:
320         print(f"[lcqi-gpt2] error: {error}", file=sys.stderr)
321         return 1
322
323
324 if __name__ == "__main__":
325     raise SystemExit(main())

```

`run_gpt2_benchmark_compare.py` 用同一个 GPT-2 F32 safetensors 模型分别运行 LCQI full-prefix 与 cached-KV 路径，并在本机存在 llama.cpp/SmolLM2 缓存时附带成熟引擎参考。它的报告说明算法差距、kernel 差距和模型口径差异。

Listing 16.15 tools/run_gpt2_benchmark_compare.py

```

1  #!/usr/bin/env python3
2  from __future__ import annotations
3
4  import argparse
5  import datetime as dt
6  import os
7  import re
8  import shlex
9  import subprocess
10 import sys
11 from dataclasses import dataclass
12 from pathlib import Path
13
14 import run_gpt2_smoke
15
16
17 DEFAULT_PROMPT = "Hello, my name is"
18 DEFAULT_MAX_NEW_TOKENS = 4
19 DEFAULT_BUILD_JOBS = 2
20 DEFAULT_TIMEOUT_SECONDS = 300
21 DEFAULT_SMOLLM2_CACHE_DIR = Path.home() / ".cache" / "lcqi-smolllm2-small-smoke"
22 SMOLLM2_GGUF = "SmolLM2-135M-Instruct-Q4_K_M.gguf"
23
24
25 @dataclass(frozen=True)
26 class EngineRun:
27     name: str
28     command: list[str]
29     returncode: int
30     stdout: str
31     stderr: str
32     metrics: dict[str, str]
33
34
35 def volume_dir() -> Path:
36     return Path(__file__).resolve().parents[1]
37
38
39 def repo_dir() -> Path:

```



```

40     return Path(__file__).resolve().parents[3]
41
42
43 def default_report_path() -> Path:
44     return volume_dir() / "results" / "lcqi-gpt2-benchmark-compare.txt"
45
46
47 def default_build_dir() -> Path:
48     return volume_dir() / "build" / "lcqi-release"
49
50
51 def command_line(args: list[str]) -> str:
52     return " ".join(shlex.quote(arg) for arg in args)
53
54
55 def tail(text: str, max_chars: int = 4000) -> str:
56     return text if len(text) <= max_chars else text[-max_chars:]
57
58
59 def run_command(args: list[str], *, timeout_seconds: int) -> tuple[int, str, ←
    ↪ str]:
60     try:
61         completed = subprocess.run(
62             args,
63             cwd=repo_dir(),
64             text=True,
65             capture_output=True,
66             timeout=timeout_seconds,
67             check=False,
68         )
69     except subprocess.TimeoutExpired as exc:
70         stdout = exc.stdout if isinstance(exc.stdout, str) else ""
71         stderr = exc.stderr if isinstance(exc.stderr, str) else ""
72         return 124, stdout, stderr
73     return completed.returncode, completed.stdout, completed.stderr
74
75
76 def parse_lcqi_metrics(stdout: str) -> dict[str, str]:
77     metrics: dict[str, str] = {}
78     for line in stdout.splitlines():
79         if not line:
80             continue
81         key, _, value = line.partition(" ")
82         if key in {
83             "engine",
84             "generated_text",
85             "benchmark_load_ms",
86             "benchmark_tokenize_ms",
87             "benchmark_prefill_ms",
88             "benchmark_decode_ms",
89             "benchmark_generate_ms",
90             "benchmark_total_ms",

```

```

91         "benchmark_prompt_tokens",
92         "benchmark_generated_tokens",
93         "benchmark_prefill_steps",
94         "benchmark_decode_steps",
95         "benchmark_generate_tokens_per_second",
96         "benchmark_decode_tokens_per_second",
97         "benchmark_kv_cache_bytes",
98     }:
99         metrics[key] = value
100     return metrics
101
102
103 def run_lcqi(
104     binary: Path,
105     model_dir: Path,
106     *,
107     engine: str,
108     prompt: str,
109     max_new_tokens: int,
110     timeout_seconds: int,
111 ) -> EngineRun:
112     command = [
113         str(binary),
114         "--benchmark",
115         "--engine",
116         engine,
117         str(model_dir),
118         prompt,
119         str(max_new_tokens),
120     ]
121     returncode, stdout, stderr = run_command(command, timeout_seconds=↵
↵ timeout_seconds)
122     return EngineRun(
123         name=f"lcqi_{engine}",
124         command=command,
125         returncode=returncode,
126         stdout=stdout,
127         stderr=stderr,
128         metrics=parse_lcqi_metrics(stdout),
129     )
130
131
132 def llama_bench_binary(cache_dir: Path) -> Path:
133     return cache_dir / "llama.cpp" / "build" / "bin" / "llama-bench"
134
135
136 def smollm2_model_path(cache_dir: Path) -> Path:
137     return cache_dir / "models" / SMOLLM2_GGUF
138
139
140 def parse_llama_bench_metrics(stdout: str) -> dict[str, str]:
141     metrics: dict[str, str] = {}

```

```

142 lines = [line.strip() for line in stdout.splitlines() if line.strip()]
143 for line in lines:
144     if not line.startswith("|"):
145         continue
146     columns = [column.strip() for column in line.strip("|").split("|")]
147     if len(columns) != 7 or columns[0] == "model" or columns[0].startswith(↵
↵ "-"):
148         continue
149     metrics["llama_model"] = columns[0]
150     metrics["llama_size"] = columns[1]
151     metrics["llama_params"] = columns[2]
152     metrics["llama_backend"] = columns[3]
153     metrics["llama_threads"] = columns[4]
154     test_name = columns[5]
155     tokens_per_second = re.sub(r"\s+.*$", "", columns[6])
156     if test_name.startswith("pp"):
157         metrics["llama_prefill_test"] = test_name
158         metrics["llama_prefill_tps"] = tokens_per_second
159     elif test_name.startswith("tg"):
160         metrics["llama_decode_test"] = test_name
161         metrics["llama_decode_tps"] = tokens_per_second
162 return metrics
163
164
165 def run_llama_bench(cache_dir: Path, *, timeout_seconds: int) -> EngineRun | ↵
↵ None:
166     binary = llama_bench_binary(cache_dir)
167     model = smollm2_model_path(cache_dir)
168     if not binary.exists() or not os.access(binary, os.X_OK) or not model.↵
↵ exists():
169         return None
170     command = [
171         str(binary),
172         "-m",
173         str(model),
174         "-ngl",
175         "0",
176         "-p",
177         "5",
178         "-n",
179         "4",
180         "-r",
181         "1",
182     ]
183     returncode, stdout, stderr = run_command(command, timeout_seconds=↵
↵ timeout_seconds)
184     return EngineRun(
185         name="llama_cpp_smollm2_q4_k_m",
186         command=command,
187         returncode=returncode,
188         stdout=stdout,
189         stderr=stderr,

```

```

190     metrics=parse_llama_bench_metrics(stdout),
191 )
192
193
194 def write_report(path: Path, runs: list[EngineRun], *, prompt: str, ↵
    ↵ max_new_tokens: int) -> None:
195     path.parent.mkdir(parents=True, exist_ok=True)
196     lines = [
197         "LCQI GPT-2 Benchmark Compare",
198         "",
199         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
200         f"repo_commit={current_commit()}",
201         f"working_tree_dirty={working_tree_dirty()}",
202         f"python={sys.version.split()[0]}",
203         f"prompt={prompt}",
204         f"max_new_tokens={max_new_tokens}",
205         "",
206         "scope=LCQI full_prefix and cached_kv run the same GPT-2 F32 ↵
    ↵ safetensors model. "
207         "llama.cpp uses SmolLM2-135M-Instruct Q4_K_M as an external mature-↵
    ↵ engine reference, "
208         "so its numbers are engineering context, not same-model quality ranking↵
    ↵ .",
209         "",
210     ]
211     for run in runs:
212         lines.extend(
213             [
214                 f"[{run.name}]",
215                 f"returncode={run.returncode}",
216                 f"command={command_line(run.command)}",
217             ]
218         )
219         for key in sorted(run.metrics):
220             lines.append(f"{key}={run.metrics[key]}")
221         lines.extend(
222             [
223                 "stdout_tail:",
224                 tail(run.stdout.strip(), 2000),
225                 "stderr_tail:",
226                 tail(run.stderr.strip(), 2000),
227                 "",
228             ]
229         )
230     while lines and lines[-1] == "":
231         lines.pop()
232     path.write_text("\n".join(lines) + "\n", encoding="utf-8")
233
234
235 def current_commit() -> str:
236     result = subprocess.run(
237         ["git", "-C", str(repo_dir()), "rev-parse", "--short", "HEAD"],

```

```

238         text=True,
239         capture_output=True,
240         check=False,
241     )
242     return result.stdout.strip() if result.returncode == 0 else "unknown"
243
244
245 def working_tree_dirty() -> str:
246     result = subprocess.run(
247         ["git", "-C", str(repo_dir()), "status", "--porcelain"],
248         text=True,
249         capture_output=True,
250         check=False,
251     )
252     if result.returncode != 0:
253         return "unknown"
254     return "true" if result.stdout.strip() else "false"
255
256
257 def parse_args() -> argparse.Namespace:
258     parser = argparse.ArgumentParser(
259         description="Compare LCQI GPT-2 full-prefix and KV-cache paths, with ↵
↵ optional llama.cpp context."
260     )
261     parser.add_argument("--cache-dir", type=Path, default=run_gpt2_smoke.↵
↵ default_cache_dir())
262     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
263     parser.add_argument("--report", type=Path, default=default_report_path())
264     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
265     parser.add_argument("--max-new-tokens", type=int, default=↵
↵ DEFAULT_MAX_NEW_TOKENS)
266     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
267     parser.add_argument("--timeout-seconds", type=int, default=↵
↵ DEFAULT_TIMEOUT_SECONDS)
268     parser.add_argument("--no-download", action="store_true")
269     parser.add_argument("--skip-llama", action="store_true")
270     parser.add_argument("--smollm2-cache-dir", type=Path, default=↵
↵ DEFAULT_SMOLLM2_CACHE_DIR)
271     return parser.parse_args()
272
273
274 def main() -> int:
275     args = parse_args()
276     if args.max_new_tokens < 0:
277         print("[lcqi-compare] --max-new-tokens cannot be negative", file=sys.↵
↵ stderr)
278         return 1
279     try:
280         run_gpt2_smoke.ensure_model_files(
281             args.cache_dir,
282             no_download=args.no_download,
283             timeout_seconds=args.timeout_seconds,

```

```

284     )
285     binary = run_gpt2_smoke.build_lcqi_gpt2(args.build_dir, args.jobs)
286     runs = [
287         run_lcqi(
288             binary,
289             args.cache_dir,
290             engine="full",
291             prompt=args.prompt,
292             max_new_tokens=args.max_new_tokens,
293             timeout_seconds=args.timeout_seconds,
294         ),
295         run_lcqi(
296             binary,
297             args.cache_dir,
298             engine="cached",
299             prompt=args.prompt,
300             max_new_tokens=args.max_new_tokens,
301             timeout_seconds=args.timeout_seconds,
302         ),
303     ]
304     if not args.skip_llama:
305         llama_run = run_llama_bench(
306             args.smollm2_cache_dir.expanduser().resolve(),
307             timeout_seconds=args.timeout_seconds,
308         )
309         if llama_run is not None:
310             runs.append(llama_run)
311     write_report(args.report, runs, prompt=args.prompt, max_new_tokens=args.
↪ .max_new_tokens)
312     except Exception as error:
313         print(f"[lcqi-compare] error: {error}", file=sys.stderr)
314         return 1
315
316     for run in runs:
317         print(f"{run.name}: returncode={run.returncode}")
318         for key in sorted(run.metrics):
319             print(f"  {key}={run.metrics[key]}")
320     print(f"report={args.report}")
321     return 0 if all(run.returncode == 0 for run in runs) else 1
322
323
324 if __name__ == "__main__":
325     raise SystemExit(main())

```

`run_gpt2_optimization_ab.py` 专门比较 LCQI 自身改动前后。它从 baseline commit 建临时 worktree，baseline 和当前工作区都通过实践卷 CMake 入口构建 Release，再多轮运行同一个 GPT-2 F32 模型。报告中的 `median/min/max` 用来约束机器波动，`generated_text` 用来确认优化前后输出没有漂移，`benchmark_worker_count` 用来确认 cached decoder 到底是串行、自动线程数还是固定线程数。

Listing 16.16 tools/run_gpt2_optimization_ab.py

```
1 #!/usr/bin/env python3
```

```
2 from __future__ import annotations
3
4 import argparse
5 import datetime as dt
6 import shutil
7 import statistics
8 import subprocess
9 import sys
10 import tempfile
11 from dataclasses import dataclass
12 from pathlib import Path
13
14
15 DEFAULT_BASELINE_REF = "4febf3f"
16 DEFAULT_PROMPT = "Hello, my name is"
17 DEFAULT_MAX_NEW_TOKENS = 4
18 DEFAULT_ROUNDS = 3
19 DEFAULT_BUILD_JOBS = 2
20 DEFAULT_TIMEOUT_SECONDS = 300
21
22 METRIC_KEYS = [
23     "benchmark_prefill_ms",
24     "benchmark_decode_ms",
25     "benchmark_generate_ms",
26     "benchmark_worker_count",
27     "benchmark_generate_tokens_per_second",
28     "benchmark_decode_tokens_per_second",
29 ]
30
31
32 @dataclass(frozen=True)
33 class RunSample:
34     variant: str
35     engine: str
36     round_id: int
37     metrics: dict[str, float]
38     generated_text: str
39
40
41 def script_path() -> Path:
42     return Path(__file__).resolve()
43
44
45 def repo_dir() -> Path:
46     return script_path().parents[3]
47
48
49 def volume_dir() -> Path:
50     return script_path().parents[1]
51
52
53 def practice_dir(root: Path) -> Path:
```

```

54     return root / "books" / "cpu-volume-3-practice"
55
56
57 def default_report_path() -> Path:
58     return volume_dir() / "results" / "lcqi-gpt2-optimization-ab.txt"
59
60
61 def default_model_dir() -> Path:
62     return Path.home() / ".cache" / "lcqi-gpt2-smoke" / "openai-community--gpt2"
63     ↪ "
64
65 def run_command(
66     args: list[str],
67     *,
68     cwd: Path,
69     timeout_seconds: int,
70 ) -> subprocess.CompletedProcess[str]:
71     return subprocess.run(
72         args,
73         cwd=cwd,
74         text=True,
75         capture_output=True,
76         timeout=timeout_seconds,
77         check=False,
78     )
79
80
81 def ensure_success(
82     completed: subprocess.CompletedProcess[str],
83     action: str,
84 ) -> None:
85     if completed.returncode == 0:
86         return
87     command = (
88         " ".join(completed.args)
89         if isinstance(completed.args, list)
90         else str(completed.args)
91     )
92     raise RuntimeError(
93         f"{action} failed with exit code {completed.returncode}\n"
94         f"command: {command}\n"
95         f"stdout tail:\n{completed.stdout[-2000:]]}\n"
96         f"stderr tail:\n{completed.stderr[-2000:]]}"
97     )
98
99
100 def git_text(args: list[str]) -> str:
101     completed = run_command(["git", *args], cwd=repo_dir(), timeout_seconds=60)
102     ensure_success(completed, "git command")
103     return completed.stdout.strip()
104

```



```

105
106 def working_tree_dirty() -> bool:
107     completed = run_command(
108         ["git", "status", "--porcelain"],
109         cwd=repo_dir(),
110         timeout_seconds=60,
111     )
112     ensure_success(completed, "git status")
113     return bool(completed.stdout.strip())
114
115
116 def build_binary(root: Path, build_dir: Path, jobs: int, timeout_seconds: int) ←
    ↪ -> Path:
117     configure = run_command(
118         [
119             "cmake",
120             "-S",
121             str(practice_dir(root)),
122             "-B",
123             str(build_dir),
124             "-DCMAKE_BUILD_TYPE=Release",
125         ],
126         cwd=root,
127         timeout_seconds=timeout_seconds,
128     )
129     ensure_success(configure, f"configure {root}")
130     build = run_command(
131         [
132             "cmake",
133             "--build",
134             str(build_dir),
135             "--target",
136             "lcqi_gpt2",
137             "-j",
138             str(jobs),
139         ],
140         cwd=root,
141         timeout_seconds=timeout_seconds,
142     )
143     ensure_success(build, f"build {root}")
144     binary = build_dir / "linux_cpu_inference" / "lcqi_gpt2"
145     if not binary.exists():
146         raise RuntimeError(f"lcqi_gpt2 binary not found: {binary}")
147     return binary
148
149
150 def parse_metrics(stdout: str) -> tuple[dict[str, float], str]:
151     metrics: dict[str, float] = {}
152     generated_text = ""
153     for line in stdout.splitlines():
154         key, _, value = line.partition(" ")
155         if key in METRIC_KEYS:

```

```

156         metrics[key] = float(value)
157     elif key == "generated_text":
158         generated_text = value
159     metrics.setdefault("benchmark_worker_count", 1.0)
160     missing = sorted(set(METRIC_KEYS) - metrics.keys())
161     if missing:
162         raise RuntimeError(f"missing benchmark metrics: {missing}")
163     return metrics, generated_text
164
165
166 def run_sample(
167     binary: Path,
168     *,
169     variant: str,
170     engine: str,
171     round_id: int,
172     model_dir: Path,
173     prompt: str,
174     max_new_tokens: int,
175     threads: int,
176     timeout_seconds: int,
177 ) -> RunSample:
178     command = [
179         str(binary),
180         "--benchmark",
181         "--engine",
182         engine,
183     ]
184     if variant == "current":
185         command.extend(["--threads", str(threads)])
186     command.extend([
187         str(model_dir),
188         prompt,
189         str(max_new_tokens),
190     ])
191     completed = run_command(
192         command,
193         cwd=repo_dir(),
194         timeout_seconds=timeout_seconds,
195     )
196     ensure_success(completed, f"run {variant} {engine}")
197     metrics, generated_text = parse_metrics(completed.stdout)
198     return RunSample(
199         variant=variant,
200         engine=engine,
201         round_id=round_id,
202         metrics=metrics,
203         generated_text=generated_text,
204     )
205
206
207 def summarize(samples: list[RunSample]) -> list[str]:

```

```

208     lines: list[str] = []
209     variants = sorted({sample.variant for sample in samples})
210     engines = sorted({sample.engine for sample in samples})
211     for variant in variants:
212         for engine in engines:
213             selected = [
214                 sample for sample in samples
215                 if sample.variant == variant and sample.engine == engine
216             ]
217             if not selected:
218                 continue
219             lines.append(f"[{variant} {engine}]")
220             for key in METRIC_KEYS:
221                 values = [sample.metrics[key] for sample in selected]
222                 lines.append(
223                     f"{key}=median:{statistics.median(values):.6g},"
224                     f"min:{min(values):.6g},max:{max(values):.6g}"
225                 )
226             texts = sorted({sample.generated_text for sample in selected})
227             lines.append(f"generated_text={texts[0] if len(texts) == 1 else ↵
↵ texts}")
228             lines.append("")
229     return lines
230
231
232 def write_report(
233     path: Path,
234     samples: list[RunSample],
235     *,
236     baseline_ref: str,
237     baseline_commit: str,
238     current_commit: str,
239     prompt: str,
240     max_new_tokens: int,
241     rounds: int,
242     current_threads: int,
243 ) -> None:
244     path.parent.mkdir(parents=True, exist_ok=True)
245     lines = [
246         "LCQI GPT-2 Optimization A/B",
247         "",
248         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
249         f"baseline_ref={baseline_ref}",
250         f"baseline_commit={baseline_commit}",
251         f"current_commit={current_commit}",
252         f"current_working_tree_dirty={str(working_tree_dirty()).lower()}",
253         f"prompt={prompt}",
254         f"max_new_tokens={max_new_tokens}",
255         f"rounds={rounds}",
256         f"baseline_threads=1",
257         f"current_threads={current_threads}",
258         "",

```

```

259     "scope=Both variants are built through books/cpu-volume-3-practice with↵
↵ "
260     "CMAKE_BUILD_TYPE=Release and run on the same GPT-2 F32 safetensors ↵
↵ model. "
261     "The llama.cpp comparison remains in lcqi-gpt2-benchmark-compare.txt; ↵
↵ this "
262     "report isolates LCQI before/after code changes.",
263     "",
264     "[samples]",
265 ]
266 for sample in samples:
267     metric_text = ",".join(
268         f"{key}:{sample.metrics[key]:.6g}" for key in METRIC_KEYS
269     )
270     lines.append(
271         f"round={sample.round_id},variant={sample.variant},engine={sample.↵
↵ engine},"
272         f"{metric_text},generated_text={sample.generated_text}"
273     )
274     lines.append("")
275     lines.extend(summarize(samples))
276     while lines and lines[-1] == "":
277         lines.pop()
278     path.write_text("\n".join(lines) + "\n", encoding="utf-8")
279
280
281 def parse_args() -> argparse.Namespace:
282     parser = argparse.ArgumentParser(description="Run LCQI GPT-2 before/after A↵
↵ /B benchmark.")
283     parser.add_argument("--baseline-ref", default=DEFAULT_BASELINE_REF)
284     parser.add_argument("--model-dir", type=Path, default=default_model_dir())
285     parser.add_argument("--report", type=Path, default=default_report_path())
286     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
287     parser.add_argument("--max-new-tokens", type=int, default=↵
↵ DEFAULT_MAX_NEW_TOKENS)
288     parser.add_argument("--rounds", type=int, default=DEFAULT_ROUNDS)
289     parser.add_argument("--threads", type=int, default=0)
290     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
291     parser.add_argument("--timeout-seconds", type=int, default=↵
↵ DEFAULT_TIMEOUT_SECONDS)
292     return parser.parse_args()
293
294
295 def main() -> int:
296     args = parse_args()
297     if args.rounds <= 0:
298         print("[lcqi-ab] --rounds must be positive", file=sys.stderr)
299         return 1
300     if args.max_new_tokens < 0:
301         print("[lcqi-ab] --max-new-tokens cannot be negative", file=sys.stderr)
302         return 1
303     if args.threads < 0:

```



```

354         threads=1 if variant == "baseline" else args.↵
↵ threads,
355         timeout_seconds=args.timeout_seconds,
356     )
357     samples.append(sample)
358     print(
359         f"round={round_id} variant={variant} engine={engine}↵
↵ } "
360         f"generate_ms={sample.metrics['↵
↵ benchmark_generate_ms']:.3f} "
361         f"workers={sample.metrics['benchmark_worker_count↵
↵ ']:.0f} "
362         f"gen_tps={sample.metrics['↵
↵ benchmark_generate_tokens_per_second']:.5f}"
363     )
364     write_report(
365         args.report,
366         samples,
367         baseline_ref=args.baseline_ref,
368         baseline_commit=baseline_commit,
369         current_commit=current_commit,
370         prompt=args.prompt,
371         max_new_tokens=args.max_new_tokens,
372         rounds=args.rounds,
373         current_threads=args.threads,
374     )
375     finally:
376         subprocess.run(
377             ["git", "worktree", "remove", "--force", str(baseline_root)],
378             cwd=repo_dir(),
379             text=True,
380             capture_output=True,
381             check=False,
382         )
383     print(f"report={args.report}")
384     return 0
385
386
387 if __name__ == "__main__":
388     raise SystemExit(main())

```

`run_gpt2_hotspot_profile.py` 专门收集 cached decode 内部热点。它构建 Release `lcqi_gpt2`，用 `--profile-hotspots` 输出 `hotspot_lm_head_pct`、`hotspot_mlp_fc_pct`、`hotspot_mlp_projection_pct`、`hotspot_qkv_projection_pct`、`hotspot_attention_projection_pct`、`hotspot_attention_pct` 和 `benchmark_worker_count`，再按 token 数和轮次计算中位数。这个脚本回答“优化应该打哪里”，不是回答“某次提交快了多少”。本轮线程优化就是靠它发现第一次 auto 阈值漏掉了 attention output projection，随后把行数和列数阈值一起纳入并行判定。

Listing 16.17 tools/run_gpt2_hotspot_profile.py

```

1 #!/usr/bin/env python3
2 from __future__ import annotations

```

```

3
4 import argparse
5 import datetime as dt
6 import statistics
7 import subprocess
8 from dataclasses import dataclass
9 from pathlib import Path
10
11
12 DEFAULT_PROMPT = "Hello, my name is"
13 DEFAULT_TOKEN_COUNTS = "4,16"
14 DEFAULT_ROUNDS = 3
15 DEFAULT_BUILD_JOBS = 2
16 DEFAULT_TIMEOUT_SECONDS = 900
17
18 METRIC_PREFIXES = ("benchmark_", "hotspot_")
19
20
21 @dataclass(frozen=True)
22 class HotspotSample:
23     max_new_tokens: int
24     round_id: int
25     metrics: dict[str, float]
26     generated_text: str
27
28
29 def script_path() -> Path:
30     return Path(__file__).resolve()
31
32
33 def repo_dir() -> Path:
34     return script_path().parents[3]
35
36
37 def volume_dir() -> Path:
38     return script_path().parents[1]
39
40
41 def practice_dir() -> Path:
42     return repo_dir() / "books" / "cpu-volume-3-practice"
43
44
45 def default_build_dir() -> Path:
46     return volume_dir() / "build" / "lcqi-hotspot-profile"
47
48
49 def default_model_dir() -> Path:
50     return Path.home() / ".cache" / "lcqi-gpt2-smoke" / "openai-community--gpt2↵
    ↵ "
51
52
53 def default_report_path() -> Path:

```

```

54     return volume_dir() / "results" / "lcqi-gpt2-hotspot-profile.txt"
55
56
57 def run_command(
58     args: list[str],
59     *,
60     cwd: Path,
61     timeout_seconds: int,
62 ) -> subprocess.CompletedProcess[str]:
63     return subprocess.run(
64         args,
65         cwd=cwd,
66         text=True,
67         capture_output=True,
68         timeout=timeout_seconds,
69         check=False,
70     )
71
72
73 def ensure_success(completed: subprocess.CompletedProcess[str], action: str) -> None:
74     ↵ None:
75     if completed.returncode == 0:
76         return
77     command = (
78         " ".join(completed.args)
79         if isinstance(completed.args, list)
80         else str(completed.args)
81     )
82     raise RuntimeError(
83         f"{action} failed with exit code {completed.returncode}\n"
84         f"command: {command}\n"
85         f"stdout tail:\n{completed.stdout[-2000:]]}\n"
86         f"stderr tail:\n{completed.stderr[-2000:]]}"
87     )
88
89 def build_binary(build_dir: Path, jobs: int, timeout_seconds: int) -> Path:
90     configure = run_command(
91         [
92             "cmake",
93             "-S",
94             str(practice_dir()),
95             "-B",
96             str(build_dir),
97             "-DCMAKE_BUILD_TYPE=Release",
98         ],
99         cwd=repo_dir(),
100         timeout_seconds=timeout_seconds,
101     )
102     ensure_success(configure, "configure hotspot profile build")
103     build = run_command(
104         [

```



```

105         "cmake",
106         "--build",
107         str(build_dir),
108         "--target",
109         "lcqi_gpt2",
110         "-j",
111         str(jobs),
112     ],
113     cwd=repo_dir(),
114     timeout_seconds=timeout_seconds,
115 )
116 ensure_success(build, "build lcqi_gpt2")
117 binary = build_dir / "linux_cpu_inference" / "lcqi_gpt2"
118 if not binary.exists():
119     raise RuntimeError(f"lcqi_gpt2 binary not found: {binary}")
120 return binary
121
122
123 def parse_numeric_list(value: str) -> list[int]:
124     result: list[int] = []
125     for part in value.split(","):
126         stripped = part.strip()
127         if not stripped:
128             continue
129         parsed = int(stripped)
130         if parsed <= 0:
131             raise RuntimeError("--token-counts entries must be positive")
132         result.append(parsed)
133     if not result:
134         raise RuntimeError("--token-counts must contain at least one positive ↵
↵ integer")
135     return result
136
137
138 def parse_stdout(stdout: str) -> tuple[dict[str, float], str]:
139     metrics: dict[str, float] = {}
140     generated_text = ""
141     for line in stdout.splitlines():
142         key, _, value = line.partition(" ")
143         if key.startswith(METRIC_PREFIXES):
144             metrics[key] = float(value)
145         elif key == "generated_text":
146             generated_text = value
147     required = {
148         "benchmark_generate_ms",
149         "benchmark_worker_count",
150         "hotspot_total_step_ms",
151         "hotspot_lm_head_pct",
152         "hotspot_mlp_fc_pct",
153         "hotspot_mlp_projection_pct",
154         "hotspot_qkv_projection_pct",
155         "hotspot_attention_pct",

```

```

156     }
157     missing = sorted(required - metrics.keys())
158     if missing:
159         raise RuntimeError(f"missing hotspot metrics: {missing}")
160     if not generated_text:
161         raise RuntimeError("missing generated_text")
162     return metrics, generated_text
163
164
165 def run_sample(
166     binary: Path,
167     *,
168     model_dir: Path,
169     prompt: str,
170     max_new_tokens: int,
171     round_id: int,
172     threads: int,
173     timeout_seconds: int,
174 ) -> HotspotSample:
175     completed = run_command(
176         [
177             str(binary),
178             "--profile-hotspots",
179             "--engine",
180             "cached",
181             "--threads",
182             str(threads),
183             str(model_dir),
184             prompt,
185             str(max_new_tokens),
186         ],
187         cwd=repo_dir(),
188         timeout_seconds=timeout_seconds,
189     )
190     ensure_success(completed, f"run hotspot profile tokens={max_new_tokens}")
191     metrics, generated_text = parse_stdout(completed.stdout)
192     return HotspotSample(
193         max_new_tokens=max_new_tokens,
194         round_id=round_id,
195         metrics=metrics,
196         generated_text=generated_text,
197     )
198
199
200 def summarize(samples: list[HotspotSample]) -> list[str]:
201     lines: list[str] = []
202     metric_keys = [
203         "benchmark_load_ms",
204         "benchmark_generate_ms",
205         "benchmark_generate_tokens_per_second",
206         "benchmark_decode_tokens_per_second",
207         "hotspot_total_step_ms",

```

```

208     "benchmark_worker_count",
209     "hotspot_lm_head_pct",
210     "hotspot_mlp_fc_pct",
211     "hotspot_mlp_projection_pct",
212     "hotspot_qkv_projection_pct",
213     "hotspot_attention_projection_pct",
214     "hotspot_attention_pct",
215 ]
216 for token_count in sorted({sample.max_new_tokens for sample in samples}):
217     selected = [sample for sample in samples if sample.max_new_tokens == ←
↪ token_count]
218     lines.append(f"[summary max_new_tokens={token_count}]")
219     for key in metric_keys:
220         values = [sample.metrics[key] for sample in selected if key in ←
↪ sample.metrics]
221         if not values:
222             continue
223         lines.append(
224             f"{key}=median:{statistics.median(values):.6g},"
225             f"min:{min(values):.6g},max:{max(values):.6g}"
226         )
227         texts = sorted({sample.generated_text for sample in selected})
228         lines.append(f"generated_text={texts[0] if len(texts) == 1 else texts}"↪
↪ )
229         lines.append("")
230     return lines
231
232
233 def write_report(
234     path: Path,
235     samples: list[HotspotSample],
236     *,
237     model_dir: Path,
238     prompt: str,
239     token_counts: list[int],
240     rounds: int,
241     threads: int,
242 ) -> None:
243     path.parent.mkdir(parents=True, exist_ok=True)
244     lines = [
245         "LCQI GPT-2 Hotspot Profile",
246         "",
247         f"timestamp_utc={dt.datetime.now(dt.UTC).isoformat()}",
248         f"model_dir={model_dir}",
249         f"prompt={prompt}",
250         f"token_counts='{','.join(str(value) for value in token_counts)}'",
251         f"rounds={rounds}",
252         f"requested_threads={threads}",
253         "engine=cached",
254         "build=CMAKE_BUILD_TYPE=Release through books/cpu-volume-3-practice",
255         "",
256         "[samples]",

```

```

257 ]
258 for sample in samples:
259     metric_text = ",".join(
260         f"{key}:{sample.metrics[key]:.6g}"
261         for key in sorted(sample.metrics)
262         if key.startswith(METRIC_PREFIXES)
263     )
264     lines.append(
265         f"round={sample.round_id},max_new_tokens={sample.max_new_tokens},"
266         f"{metric_text},generated_text={sample.generated_text}"
267     )
268     lines.append("")
269     lines.extend(summarize(samples))
270 while lines and lines[-1] == "":
271     lines.pop()
272 path.write_text("\n".join(lines) + "\n", encoding="utf-8")
273
274 def parse_args() -> argparse.Namespace:
275     parser = argparse.ArgumentParser(description="Run LCQI GPT-2 cached hotspot↵
↵ profile.")
276     parser.add_argument("--model-dir", type=Path, default=default_model_dir())
277     parser.add_argument("--build-dir", type=Path, default=default_build_dir())
278     parser.add_argument("--report", type=Path, default=default_report_path())
279     parser.add_argument("--prompt", default=DEFAULT_PROMPT)
280     parser.add_argument("--token-counts", default=DEFAULT_TOKEN_COUNTS)
281     parser.add_argument("--rounds", type=int, default=DEFAULT_ROUNDS)
282     parser.add_argument("--threads", type=int, default=0)
283     parser.add_argument("--jobs", type=int, default=DEFAULT_BUILD_JOBS)
284     parser.add_argument("--timeout-seconds", type=int, default=↵
↵ DEFAULT_TIMEOUT_SECONDS)
285     return parser.parse_args()
286
287
288
289 def main() -> int:
290     args = parse_args()
291     if args.rounds <= 0:
292         print("[lcqi-hotspot] --rounds must be positive")
293         return 1
294     if args.threads < 0:
295         print("[lcqi-hotspot] --threads cannot be negative")
296         return 1
297     if not args.model_dir.exists():
298         print(f"[lcqi-hotspot] model directory not found: {args.model_dir}")
299         return 1
300     token_counts = parse_numeric_list(args.token_counts)
301     binary = build_binary(args.build_dir, args.jobs, args.timeout_seconds)
302     samples: list[HotspotSample] = []
303     for round_id in range(1, args.rounds + 1):
304         for token_count in token_counts:
305             sample = run_sample(
306                 binary,

```

```

307         model_dir=args.model_dir,
308         prompt=args.prompt,
309         max_new_tokens=token_count,
310         round_id=round_id,
311         threads=args.threads,
312         timeout_seconds=args.timeout_seconds,
313     )
314     samples.append(sample)
315     print(
316         f"round={round_id} max_new_tokens={token_count} "
317         f"generate_ms={sample.metrics['benchmark_generate_ms']:.3f} "
318         f"workers={sample.metrics['benchmark_worker_count']:.0f} "
319         f"lm_head_pct={sample.metrics['hotspot_lm_head_pct']:.3f} "
320         f"mlp_pct={sample.metrics['hotspot_mlp_fc_pct']} + sample.↵
↵ metrics['hotspot_mlp_projection_pct']:.3f"
321     )
322     write_report(
323         args.report,
324         samples,
325         model_dir=args.model_dir,
326         prompt=args.prompt,
327         token_counts=token_counts,
328         rounds=args.rounds,
329         threads=args.threads,
330     )
331     print(f"report={args.report}")
332     return 0
333
334
335 if __name__ == "__main__":
336     raise SystemExit(main())

```

`run_lcqi_benchmark.sh` 是本地 benchmark 收集脚本。它适合在固定机器上反复运行，把同一组 shape 和 backend 的结果落到报告文件。

Listing 16.18 tools/run_lcqi_benchmark.sh

```

1  #!/usr/bin/env bash
2  set -euo pipefail
3
4  SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
5  VOLUME_DIR="$(cd "${SCRIPT_DIR}/.." && pwd)"
6  REPO_DIR="$(cd "${VOLUME_DIR}/../.." && pwd)"
7  BUILD_DIR="${VOLUME_DIR}/build/lcqi-release"
8  RESULT_DIR="${VOLUME_DIR}/results"
9  REPEAT="${1:-200}"
10 STAMP="$(date -u +%Y%m%dT%H%M%S)"
11 CSV_PATH="${RESULT_DIR}/lcqi-bench-${STAMP}.csv"
12 META_PATH="${RESULT_DIR}/lcqi-bench-${STAMP}.meta.txt"
13 PERF_PATH="${RESULT_DIR}/lcqi-bench-${STAMP}.perf.txt"
14
15 mkdir -p "${RESULT_DIR}"
16

```

```

17 cmake -S "${VOLUME_DIR}" -B "${BUILD_DIR}" -DCMAKE_BUILD_TYPE=Release
18 cmake --build "${BUILD_DIR}"
19 ctest --test-dir "${BUILD_DIR}" --output-on-failure
20
21 {
22     echo "timestamp_utc=${STAMP}"
23     echo "repeat=${REPEAT}"
24     echo "commit=$(git -C "${REPO_DIR}" rev-parse --short HEAD)"
25     echo "uname=$(uname -a)"
26     echo "compiler=$((${CXX:-c++} --version | sed -n '1p')"
27     echo "cmake=$(cmake --version | sed -n '1p')"
28     echo "bench=${BUILD_DIR}/labs/linux_cpu_inference/lcqi_bench"
29 } > "${META_PATH}"
30
31 "${BUILD_DIR}/labs/linux_cpu_inference/lcqi_bench" "${REPEAT}" > "${CSV_PATH}"
32
33 if command -v perf >/dev/null 2>&1; then
34     perf stat \
35         -e cycles,instructions,cache-misses,branches,branch-misses \
36         "${BUILD_DIR}/labs/linux_cpu_inference/lcqi_bench" "${REPEAT}" \
37         > /dev/null 2> "${PERF_PATH}" || true
38 else
39     echo "perf_unavailable=1" > "${PERF_PATH}"
40 fi
41
42 echo "csv=${CSV_PATH}"
43 echo "meta=${META_PATH}"
44 echo "perf=${PERF_PATH}"

```

完整测试

`lcqi_tests.cpp` 把 safetensors、tokenizer、tiny MLP、int8 kernel、reference decoder、GPT-2 tiny path、trace、ledger 和 serving probe 串成验收网。GPT-2 优化不是只看速度：测试会把 full-prefix logits、旧 cached logits、优化 decoder logits、强制两 worker 的并行 decoder logits 和 logits-free greedy token 放在同一个 tiny prompt 下对齐，确认 predicted token、logits、KV filled token 数和 generated ids 不漂移。新增优化或 shape 时，必须先扩展这里的 golden 和边界样例。

Listing 16.19 tests/lcqi_tests.cpp

```

1 #include <lcqi/gpt2_reference.hpp>
2 #include <lcqi/f32_kernels.hpp>
3 #include <lcqi/ggml_tensors.hpp>
4 #include <lcqi/gguf.hpp>
5 #include <lcqi/inference.hpp>
6 #include <lcqi/int8_kernels.hpp>
7 #include <lcqi/llama_gguf.hpp>
8 #include <lcqi/model.hpp>
9 #include <lcqi/q4_k.hpp>
10 #include <lcqi/reference_decoder.hpp>
11 #include <lcqi/safetensors.hpp>
12 #include <lcqi/tokenizer.hpp>
13

```

```

14 #include <array>
15 #include <cmath>
16 #include <cstring>
17 #include <cstdlib>
18 #include <filesystem>
19 #include <fstream>
20 #include <iostream>
21 #include <span>
22 #include <sstream>
23 #include <stdexcept>
24 #include <string>
25 #include <utility>
26 #include <vector>
27
28 namespace {
29
30 constexpr float LCQI_TEST_EPSILON = 1.0e-5F;
31 constexpr std::int32_t LCQI_TEST_INPUT_SIZE = 4;
32 constexpr std::int32_t LCQI_TEST_HIDDEN_SIZE = 3;
33 constexpr std::int32_t LCQI_TEST_OUTPUT_SIZE = 2;
34 constexpr std::int32_t LCQI_TEST_PREDICTED_CLASS = 1;
35 constexpr float LCQI_TEST_LOGIT_0 = -0.48F;
36 constexpr float LCQI_TEST_LOGIT_1 = 1.06F;
37 constexpr float LCQI_TEST_HIDDEN_0 = 0.0F;
38 constexpr float LCQI_TEST_HIDDEN_1 = 0.4F;
39 constexpr float LCQI_TEST_HIDDEN_2 = 1.4F;
40 constexpr float LCQI_RMS_EXPECTED_0 = 0.848528F;
41 constexpr float LCQI_RMS_EXPECTED_1 = 0.565685F;
42 constexpr float LCQI_ATTENTION_EXPECTED_0 = 2.0F;
43 constexpr float LCQI_ATTENTION_EXPECTED_1 = 3.0F;
44 constexpr float LCQI_KERNEL_MAX_DIFF = 1.0e-5F;
45 constexpr float LCQI_F32_KERNEL_MAX_DIFF = 1.0e-4F;
46 constexpr std::int32_t LCQI_SIMD_TEST_BLOCK = 8;
47 constexpr std::size_t LCQI_F32_DOT_TEST_SIZE = 257;
48 constexpr std::size_t LCQI_F32_ROW_TEST_INPUT_SIZE = 65;
49 constexpr std::size_t LCQI_F32_ROW_TEST_OUTPUT_SIZE = 17;
50 constexpr std::int32_t LCQI_REFERENCE_DECODER_PREDICTED_TOKEN = 0;
51 constexpr std::size_t LCQI_REFERENCE_DECODER_VOCAB_SIZE = 5;
52 constexpr std::array<float, LCQI_REFERENCE_DECODER_VOCAB_SIZE> ↵
    ↵ LCQI_REFERENCE_DECODER_LOGITS{
53     0.352516979F,
54     -0.126884326F,
55     0.0797837451F,
56     -0.29797405F,
57     0.127030417F,
58 };
59 constexpr std::int32_t LCQI_REFERENCE_TRACE_TOKEN_COUNT = 3;
60 constexpr std::int32_t LCQI_REFERENCE_TRACE_HIDDEN_SIZE = 4;
61 constexpr std::int32_t LCQI_REFERENCE_TRACE_FIRST_TOKEN = 0;
62 constexpr std::uint64_t LCQI_TOKENIZER_HASH_EXPECTED = 13452902845388333734ULL;
63 constexpr std::int32_t LCQI_SAFETENSORS_PREFIX_BYTES = 8;
64 constexpr std::int32_t LCQI_TEST_BYTE_BITS = 8;

```

```

65 constexpr std::uint64_t LCQI_SAFETENSORS_VALID_PAYLOAD_BYTES = 48;
66 constexpr std::uint64_t LCQI_SAFETENSORS_EMBED_BEGIN = 0;
67 constexpr std::uint64_t LCQI_SAFETENSORS_EMBED_END = 24;
68 constexpr std::uint64_t LCQI_SAFETENSORS_QPROJ_BEGIN = 24;
69 constexpr std::uint64_t LCQI_SAFETENSORS_QPROJ_END = 48;
70 constexpr unsigned int LCQI_BYTE_MASK = 0xFFU;
71 constexpr std::int32_t LCQI_GPT2_PREDICTED_TOKEN = 3;
72 constexpr std::size_t LCQI_GPT2_VOCAB_SIZE = 6;
73 constexpr std::array<float, LCQI_GPT2_VOCAB_SIZE> LCQI_GPT2_LOGITS{
74     0.407358F,
75     -0.504049F,
76     -0.107834F,
77     0.840337F,
78     -0.450123F,
79     -0.156763F,
80 };
81 constexpr std::int32_t LCQI_GPT2_GENERATED_LENGTH = 5;
82 constexpr std::int32_t LCQI_GPT2_TOKENIZER_HELLO_ID = 7;
83 constexpr std::uint16_t LCQI_TEST_F16_ONE = 0x3C00U;
84 constexpr std::uint16_t LCQI_TEST_F16_HALF = 0x3800U;
85 constexpr std::uint16_t LCQI_TEST_F16_NEGATIVE_TWO = 0xC000U;
86 constexpr std::uint16_t LCQI_TEST_BF16_ONE_POINT_FIVE = 0x3FC0U;
87 constexpr std::uint16_t LCQI_TEST_BF16_NEGATIVE_HALF = 0xBF00U;
88 constexpr std::uint32_t LCQI_GGUF_TEST_VERSION = 3;
89 constexpr std::uint32_t LCQI_GGUF_TEST_ALIGNMENT = 32;
90 constexpr std::int64_t LCQI_GGUF_TEST_TENSOR_COUNT = 1;
91 constexpr std::int64_t LCQI_GGUF_TEST_METADATA_COUNT = 3;
92 constexpr std::int32_t LCQI_GGUF_TYPE_UINT32 = 4;
93 constexpr std::int32_t LCQI_GGUF_TYPE_FLOAT32 = 6;
94 constexpr std::int32_t LCQI_GGUF_TYPE_STRING = 8;
95 constexpr std::int32_t LCQI_GGUF_TYPE_ARRAY = 9;
96 constexpr std::int32_t LCQI_GGUF_TENSOR_TYPE_Q4_K = 12;
97 constexpr std::uint64_t LCQI_GGUF_TEST_TENSOR_OFFSET = 0;
98 constexpr float LCQI_Q4K_TEST_BLOCK_SCALE = 0.5F;
99 constexpr float LCQI_Q4K_TEST_BLOCK_MIN = 0.25F;
100 constexpr float LCQI_Q4K_Q8_DIFF_LIMIT = 0.08F;
101 constexpr std::int32_t LCQI_GGML_TEST_Q5_HALF_VALUES = 16;
102 constexpr std::int32_t LCQI_GGML_TEST_Q5_ZERO_POINT = 16;
103 constexpr std::int32_t LCQI_GGML_TEST_Q6_LANE = 5;
104 constexpr std::int32_t LCQI_LLAMA_TEST_HIDDEN = 4;
105 constexpr std::int32_t LCQI_LLAMA_TEST_HEADS = 2;
106 constexpr std::int32_t LCQI_LLAMA_TEST_KV_HEADS = 1;
107 constexpr std::int32_t LCQI_LLAMA_TEST_HEAD_DIM = 2;
108 constexpr std::int32_t LCQI_LLAMA_TEST_INTERMEDIATE = 4;
109 constexpr std::int32_t LCQI_LLAMA_TEST_VOCAB = 4;
110 constexpr std::int32_t LCQI_LLAMA_TEST_CONTEXT = 8;
111 constexpr std::int32_t LCQI_LLAMA_TEST_LAYER_COUNT = 1;
112 constexpr std::int32_t LCQI_LLAMA_TEST_EXPECTED_TOKEN = 2;
113 constexpr float LCQI_LLAMA_TEST_SMALL_ATTENTION_VALUE = 0.1F;
114 constexpr float LCQI_LLAMA_TEST_SMALL_OUTPUT_SCALE = 0.1F;
115
116 struct KernelShapeCase {

```



```

117     std::int32_t input_size = 0;
118     std::int32_t output_size = 0;
119     std::int32_t output_block_size = 0;
120 };
121
122 struct RawTensorFixture {
123     std::string name;
124     std::string dtype;
125     std::vector<std::int64_t> shape;
126     std::vector<std::uint8_t> bytes;
127 };
128
129 struct FloatTensorFixture {
130     std::string name;
131     std::vector<std::int64_t> shape;
132     std::vector<float> values;
133 };
134
135 struct GgufTensorFixture {
136     std::string name;
137     std::vector<std::int64_t> shape;
138     lcqi::GgmlType type = lcqi::GgmlType::f32;
139     std::vector<std::uint8_t> bytes;
140     std::uint64_t relative_offset = 0;
141 };
142
143 constexpr std::array<KernelShapeCase, 4> LCQI_KERNEL_SHAPE_CASES{{
144     {17, 19, 8},
145     {33, 7, 8},
146     {64, 32, 16},
147     {65, 31, 16},
148 }};
149
150 void require(bool condition, const char* message) {
151     if (!condition) {
152         throw std::runtime_error(message);
153     }
154 }
155
156 void require_close(float actual, float expected, const char* message) {
157     if (std::fabs(actual - expected) > LCQI_TEST_EPSILON) {
158         throw std::runtime_error(message);
159     }
160 }
161
162 std::filesystem::path test_model_path() {
163     return std::filesystem::path(__FILE__).parent_path().parent_path() /
164         "models" / "tiny_mlp_i8.txt";
165 }
166
167 std::filesystem::path test_tokenizer_path() {
168     return std::filesystem::path(__FILE__).parent_path().parent_path() /

```

```

169         "models" / "tiny_tokenizer.txt";
170     }
171
172     std::filesystem::path temp_file_path(const std::string& name) {
173         return std::filesystem::temp_directory_path() / name;
174     }
175
176     void write_le_u64(std::ofstream& output, std::uint64_t value) {
177         for (std::int32_t i = 0; i < LCQI_SAFETENSORS_PREFIX_BYTES; ++i) {
178             const char byte = static_cast<char>(
179                 (value >> (LCQI_TEST_BYTE_BITS * i)) & LCQI_BYTE_MASK);
180             output.write(&byte, 1);
181         }
182     }
183
184     void append_le_u16(std::vector<std::uint8_t>& output, std::uint16_t value) {
185         output.push_back(static_cast<std::uint8_t>(value & LCQI_BYTE_MASK));
186         output.push_back(static_cast<std::uint8_t>((value >> LCQI_TEST_BYTE_BITS) &
187             ↪ LCQI_BYTE_MASK));
188     }
189
190     void append_le_u32(std::vector<std::uint8_t>& output, std::uint32_t value) {
191         for (std::int32_t i = 0; i < 4; ++i) {
192             output.push_back(static_cast<std::uint8_t>(
193                 (value >> (LCQI_TEST_BYTE_BITS * i)) & LCQI_BYTE_MASK));
194         }
195     }
196
197     void append_le_i32(std::vector<std::uint8_t>& output, std::int32_t value) {
198         std::uint32_t bits = 0;
199         std::memcpy(&bits, &value, sizeof(bits));
200         append_le_u32(output, bits);
201     }
202
203     void append_le_u64(std::vector<std::uint8_t>& output, std::uint64_t value) {
204         for (std::int32_t i = 0; i < 8; ++i) {
205             output.push_back(static_cast<std::uint8_t>(
206                 (value >> (LCQI_TEST_BYTE_BITS * i)) & LCQI_BYTE_MASK));
207         }
208     }
209
210     void append_le_i64(std::vector<std::uint8_t>& output, std::int64_t value) {
211         std::uint64_t bits = 0;
212         std::memcpy(&bits, &value, sizeof(bits));
213         append_le_u64(output, bits);
214     }
215
216     void append_gguf_string(std::vector<std::uint8_t>& output, const std::string& ↪
217         ↪ text) {
218         append_le_u64(output, static_cast<std::uint64_t>(text.size()));
219         output.insert(output.end(), text.begin(), text.end());
220     }

```

```

219
220 void append_le_f32(std::vector<std::uint8_t>& output, float value) {
221     std::uint32_t bits = 0;
222     std::memcpy(&bits, &value, sizeof(bits));
223     for (std::int32_t i = 0; i < 4; ++i) {
224         output.push_back(static_cast<std::uint8_t>(
225             (bits >> (LCQI_TEST_BYTE_BITS * i)) & LCQI_BYTE_MASK));
226     }
227 }
228
229 void write_safetensors_fixture(const std::filesystem::path& path,
230                                const std::string& header,
231                                std::uint64_t payload_bytes) {
232     std::ofstream output(path, std::ios::binary);
233     if (!output) {
234         throw std::runtime_error("cannot create safetensors test fixture");
235     }
236     write_le_u64(output, static_cast<std::uint64_t>(header.size()));
237     output.write(header.data(), static_cast<std::streamsize>(header.size()));
238     for (std::uint64_t i = 0; i < payload_bytes; ++i) {
239         const char byte = static_cast<char>(i & 0xFFU);
240         output.write(&byte, 1);
241     }
242 }
243
244 std::string json_shape(std::span<const std::int64_t> shape) {
245     std::ostringstream stream;
246     stream << '[';
247     for (std::size_t index = 0; index < shape.size(); ++index) {
248         if (index != 0) {
249             stream << ',';
250         }
251         stream << shape[index];
252     }
253     stream << ']';
254     return stream.str();
255 }
256
257 void write_safetensors_raw_fixture(const std::filesystem::path& path,
258                                    std::span<const RawTensorFixture> tensors) {
259     std::ostringstream header;
260     header << "{\"__metadata__\":{\"format\":\"lcqi-test\"}";
261     std::uint64_t offset = 0;
262     for (const RawTensorFixture& tensor : tensors) {
263         header << ",\"" << tensor.name << "\":{\"dtype\":\"" << tensor.dtype
264             << "\",\"shape\":\"" << json_shape(tensor.shape)
265             << "\",\"data_offsets\":[\"" << offset << ','
266             << offset + tensor.bytes.size() << "]}\"";
267         offset += tensor.bytes.size();
268     }
269     header << '}'

```

```

271     std::ofstream output(path, std::ios::binary);
272     if (!output) {
273         throw std::runtime_error("cannot create safetensors raw test fixture");
274     }
275     const std::string header_text = header.str();
276     write_le_u64(output, static_cast<std::uint64_t>(header_text.size()));
277     output.write(header_text.data(), static_cast<std::streamsize>(header_text.↵
↵ size()));
278     for (const RawTensorFixture& tensor : tensors) {
279         output.write(reinterpret_cast<const char*>(tensor.bytes.data()),
280                     static_cast<std::streamsize>(tensor.bytes.size()));
281     }
282 }
283
284 std::vector<std::uint8_t> f32_payload(std::span<const float> values) {
285     std::vector<std::uint8_t> bytes;
286     bytes.reserve(values.size() * 4);
287     for (const float value : values) {
288         append_le_f32(bytes, value);
289     }
290     return bytes;
291 }
292
293 std::vector<float> transpose_linear_to_hf_conv1d(const lcqi::Gpt2LinearF32& ↵
↵ linear) {
294     std::vector<float> transposed(
295         static_cast<std::size_t>(linear.input_size) *
296         static_cast<std::size_t>(linear.output_size),
297         0.0F);
298     for (std::int32_t out = 0; out < linear.output_size; ++out) {
299         for (std::int32_t in = 0; in < linear.input_size; ++in) {
300             transposed[static_cast<std::size_t>(in) *
301                       static_cast<std::size_t>(linear.output_size) +
302                       static_cast<std::size_t>(out)] =
303                 linear.weights[static_cast<std::size_t>(out) *
304                               static_cast<std::size_t>(linear.input_size) ↵
↵ +
305                               static_cast<std::size_t>(in)];
306         }
307     }
308     return transposed;
309 }
310
311 void add_f32_tensor(std::vector<RawTensorFixture>& tensors,
312                    std::string name,
313                    std::vector<std::int64_t> shape,
314                    std::span<const float> values) {
315     tensors.push_back(RawTensorFixture{
316         std::move(name),
317         "F32",
318         std::move(shape),
319         f32_payload(values),

```

```

320     });
321 }
322
323 void write_text_file(const std::filesystem::path& path, const std::string& text ←
    ↪ ) {
324     std::ofstream output(path);
325     if (!output) {
326         throw std::runtime_error("cannot create text fixture");
327     }
328     output << text;
329 }
330
331 std::vector<std::uint8_t> make_q4_k_test_block() {
332     std::vector<std::uint8_t> block(
333         static_cast<std::size_t>(lcqi::LCQI_Q4_K_BLOCK_BYTES),
334         0);
335     block[0] = 0x00U;
336     block[1] = 0x38U;
337     block[2] = 0x00U;
338     block[3] = 0x34U;
339
340     std::array<std::uint8_t, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS ←
    ↪ )> scales{
341         1, 2, 3, 4, 5, 6, 7, 8,
342     };
343     std::array<std::uint8_t, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS ←
    ↪ )> mins{
344         1, 1, 2, 2, 3, 3, 4, 4,
345     };
346     for (std::size_t index = 0; index < 4; ++index) {
347         block[4 + index] = static_cast<std::uint8_t>(
348             (scales[index] & 0x3FU) | ((scales[index + 4] >> 4U) << 6U));
349         block[8 + index] = static_cast<std::uint8_t>(
350             (mins[index] & 0x3FU) | ((mins[index + 4] >> 4U) << 6U));
351         block[12 + index] = static_cast<std::uint8_t>(
352             (scales[index + 4] & 0x0FU) | ((mins[index + 4] & 0x0FU) << 4U));
353     }
354
355     for (std::int32_t pair = 0; pair < 4; ++pair) {
356         const std::int32_t low_subblock = pair * 2;
357         const std::int32_t high_subblock = low_subblock + 1;
358         for (std::int32_t index = 0; index < lcqi::LCQI_Q4_K_SUBBLOCK_VALUES; ←
    ↪ ++index) {
359             const std::uint8_t low = static_cast<std::uint8_t>(
360                 (low_subblock + index) & 0x0F);
361             const std::uint8_t high = static_cast<std::uint8_t>(
362                 (high_subblock + index + 1) & 0x0F);
363             block[static_cast<std::size_t>(16 + pair * 32 + index)] =
364                 static_cast<std::uint8_t>(low | (high << 4U));
365         }
366     }
367     return block;

```

```

368 }
369
370 std::vector<float> expected_q4_k_test_values() {
371     const std::array<float, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS)>
↵ > scales{
372         1, 2, 3, 4, 5, 6, 7, 8,
373     };
374     const std::array<float, static_cast<std::size_t>(lcqi::LCQI_Q4_K_SUBBLOCKS)>
↵ > mins{
375         1, 1, 2, 2, 3, 3, 4, 4,
376     };
377     std::vector<float> values(
378         static_cast<std::size_t>(lcqi::LCQI_QK_K_BLOCK_VALUES),
379         0.0F);
380     for (std::int32_t subblock = 0; subblock < lcqi::LCQI_Q4_K_SUBBLOCKS; ++
↵ subblock) {
381         for (std::int32_t index = 0; index < lcqi::LCQI_Q4_K_SUBBLOCK_VALUES;
↵ ++index) {
382             const std::int32_t quantized =
383                 (subblock % 2 == 0)
384                 ? ((subblock + index) & 0x0F)
385                 : ((subblock + index + 1) & 0x0F);
386             values[static_cast<std::size_t>(
387                 subblock * lcqi::LCQI_Q4_K_SUBBLOCK_VALUES + index)] =
388                 LCQI_Q4K_TEST_BLOCK_SCALE * scales[static_cast<std::size_t>(
↵ subblock)] *
389                 static_cast<float>(quantized) -
390                 LCQI_Q4K_TEST_BLOCK_MIN * mins[static_cast<std::size_t>(
↵ subblock)];
391         }
392     }
393     return values;
394 }
395
396 std::vector<std::uint8_t> make_q8_0_test_block() {
397     std::vector<std::uint8_t> block(
398         static_cast<std::size_t>(lcqi::LCQI_Q8_0_BLOCK_BYTES),
399         0);
400     block[0] = static_cast<std::uint8_t>(LCQI_TEST_F16_HALF & LCQI_BYTE_MASK);
401     block[1] = static_cast<std::uint8_t>((LCQI_TEST_F16_HALF >>
↵ LCQI_TEST_BYTE_BITS) &
402                                         LCQI_BYTE_MASK);
403     for (std::int32_t index = 0; index < lcqi::LCQI_Q8_0_BLOCK_VALUES; ++index)
↵ {
404         block[static_cast<std::size_t>(2 + index)] =
405             static_cast<std::uint8_t>(static_cast<std::int8_t>(index - 16));
406     }
407     return block;
408 }
409
410 std::vector<std::uint8_t> make_q5_0_test_block() {
411     std::vector<std::uint8_t> block(

```

```

412     static_cast<std::size_t>(lcqi::LCQI_Q5_0_BLOCK_BYTES),
413     0);
414     block[0] = static_cast<std::uint8_t>(LCQI_TEST_F16_HALF & LCQI_BYTE_MASK);
415     block[1] = static_cast<std::uint8_t>((LCQI_TEST_F16_HALF >> ↵
↵ LCQI_TEST_BYTE_BITS) &
416                                     LCQI_BYTE_MASK);
417
418     std::uint32_t qh = 0;
419     for (std::int32_t index = 0; index < lcqi::LCQI_Q5_0_BLOCK_VALUES / 2; ++↵
↵ index) {
420         const std::int32_t first_quantized = index - ↵
↵ LCQI_GGML_TEST_Q5_ZERO_POINT;
421         const std::int32_t second_quantized = index;
422         const std::uint8_t first_encoded =
423             static_cast<std::uint8_t>(first_quantized + ↵
↵ LCQI_GGML_TEST_Q5_ZERO_POINT);
424         const std::uint8_t second_encoded =
425             static_cast<std::uint8_t>(second_quantized + ↵
↵ LCQI_GGML_TEST_Q5_ZERO_POINT);
426         if ((first_encoded & 0x10U) != 0U) {
427             qh |= (1U << index);
428         }
429         if ((second_encoded & 0x10U) != 0U) {
430             qh |= (1U << (index + LCQI_GGML_TEST_Q5_HALF_VALUES));
431         }
432         const std::uint8_t low = static_cast<std::uint8_t>(first_encoded & 0↵
↵ x0FU);
433         const std::uint8_t high = static_cast<std::uint8_t>(second_encoded & 0↵
↵ x0FU);
434         block[static_cast<std::size_t>(6 + index)] =
435             static_cast<std::uint8_t>(low | (high << 4U));
436     }
437     block[2] = static_cast<std::uint8_t>(qh & LCQI_BYTE_MASK);
438     block[3] = static_cast<std::uint8_t>((qh >> LCQI_TEST_BYTE_BITS) & ↵
↵ LCQI_BYTE_MASK);
439     block[4] = static_cast<std::uint8_t>((qh >> (LCQI_TEST_BYTE_BITS * 2)) & ↵
↵ LCQI_BYTE_MASK);
440     block[5] = static_cast<std::uint8_t>((qh >> (LCQI_TEST_BYTE_BITS * 3)) & ↵
↵ LCQI_BYTE_MASK);
441     return block;
442 }
443
444 std::vector<std::uint8_t> make_q6_k_test_block() {
445     std::vector<std::uint8_t> block(
446         static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_BYTES),
447         0);
448     const std::size_t ql_offset = 0;
449     const std::size_t qh_offset = static_cast<std::size_t>(lcqi::↵
↵ LCQI_Q6_K_BLOCK_VALUES / 2);
450     const std::size_t scales_offset =
451         qh_offset + static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_VALUES / 4);
452     for (std::int32_t index = 0; index < 16; ++index) {

```

```

453     block[scales_offset + static_cast<std::size_t>(index)] = 1;
454 }
455 block[static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_BYTES - 2)] =
456     static_cast<std::uint8_t>(LCQI_TEST_F16_ONE & LCQI_BYTE_MASK);
457 block[static_cast<std::size_t>(lcqi::LCQI_Q6_K_BLOCK_BYTES - 1)] =
458     static_cast<std::uint8_t>((LCQI_TEST_F16_ONE >> LCQI_TEST_BYTE_BITS) &
459                             LCQI_BYTE_MASK);
460
461 const auto store_q6 = [&block, ql_offset, qh_offset](std::int32_t position,
462                                                     std::uint8_t encoded) {
463     const std::int32_t half = position / 128;
464     const std::int32_t local = position % 128;
465     const std::size_t ql_base = ql_offset + static_cast<std::size_t>(half *
↵ 64);
466     const std::size_t qh_base = qh_offset + static_cast<std::size_t>(half *
↵ 32);
467     const std::uint8_t low = encoded & 0x0FU;
468     const std::uint8_t high = static_cast<std::uint8_t>((encoded >> 4U) & 0
↵ x03U);
469     if (local < 32) {
470         block[ql_base + static_cast<std::size_t>(local)] =
471             static_cast<std::uint8_t>(
472                 (block[ql_base + static_cast<std::size_t>(local)] & 0xF0U)
↵ | low);
473         block[qh_base + static_cast<std::size_t>(local)] =
474             static_cast<std::uint8_t>(
475                 (block[qh_base + static_cast<std::size_t>(local)] & 0xFCU)
↵ | high);
476     } else if (local < 64) {
477         const std::size_t lane = static_cast<std::size_t>(local - 32);
478         block[ql_base + static_cast<std::size_t>(32) + lane] =
479             static_cast<std::uint8_t>(
480                 (block[ql_base + static_cast<std::size_t>(32) + lane] & 0
↵ xF0U) | low);
481         block[qh_base + lane] =
482             static_cast<std::uint8_t>((block[qh_base + lane] & 0xF3U) | (
↵ high << 2U));
483     } else if (local < 96) {
484         const std::size_t lane = static_cast<std::size_t>(local - 64);
485         block[ql_base + lane] =
486             static_cast<std::uint8_t>(
487                 (block[ql_base + lane] & 0x0FU) | (low << 4U));
488         block[qh_base + lane] =
489             static_cast<std::uint8_t>((block[qh_base + lane] & 0xCFU) | (
↵ high << 4U));
490     } else {
491         const std::size_t lane = static_cast<std::size_t>(local - 96);
492         block[ql_base + static_cast<std::size_t>(32) + lane] =
493             static_cast<std::uint8_t>(
494                 (block[ql_base + static_cast<std::size_t>(32) + lane] & 0
↵ x0FU) |
495                 (low << 4U));

```



```

496         block[qh_base + lane] =
497             static_cast<std::uint8_t>((block[qh_base + lane] & 0x3FU) | (←
    ↪ high << 6U));
498     }
499 };
500
501     store_q6(LCQI_GGML_TEST_Q6_LANE, 35);
502     store_q6(32 + LCQI_GGML_TEST_Q6_LANE, 31);
503     store_q6(64 + LCQI_GGML_TEST_Q6_LANE, 63);
504     store_q6(96 + LCQI_GGML_TEST_Q6_LANE, 0);
505     store_q6(128 + LCQI_GGML_TEST_Q6_LANE, 36);
506     return block;
507 }
508
509 void write_gguf_q4_k_fixture(const std::filesystem::path& path) {
510     std::vector<std::uint8_t> bytes;
511     bytes.push_back('G');
512     bytes.push_back('G');
513     bytes.push_back('U');
514     bytes.push_back('F');
515     append_le_u32(bytes, LCQI_GGUF_TEST_VERSION);
516     append_le_i64(bytes, LCQI_GGUF_TEST_TENSOR_COUNT);
517     append_le_i64(bytes, LCQI_GGUF_TEST_METADATA_COUNT);
518
519     append_gguf_string(bytes, "general.alignment");
520     append_le_i32(bytes, LCQI_GGUF_TYPE_UINT32);
521     append_le_u32(bytes, LCQI_GGUF_TEST_ALIGNMENT);
522     append_gguf_string(bytes, "general.architecture");
523     append_le_i32(bytes, LCQI_GGUF_TYPE_STRING);
524     append_gguf_string(bytes, "lcqi-test");
525     append_gguf_string(bytes, "tokenizer.ggml.tokens");
526     append_le_i32(bytes, LCQI_GGUF_TYPE_ARRAY);
527     append_le_i32(bytes, LCQI_GGUF_TYPE_STRING);
528     append_le_u64(bytes, 2);
529     append_gguf_string(bytes, "<bos>");
530     append_gguf_string(bytes, "hello");
531
532     append_gguf_string(bytes, "blk.0.ffn_up.weight");
533     append_le_u32(bytes, 2);
534     append_le_i64(bytes, lcqi::LCQI_QK_K_BLOCK_VALUES);
535     append_le_i64(bytes, 1);
536     append_le_i32(bytes, LCQI_GGUF_TENSOR_TYPE_Q4_K);
537     append_le_u64(bytes, LCQI_GGUF_TEST_TENSOR_OFFSET);
538     while (bytes.size() % LCQI_GGUF_TEST_ALIGNMENT != 0) {
539         bytes.push_back(0);
540     }
541     const std::vector<std::uint8_t> block = make_q4_k_test_block();
542     bytes.insert(bytes.end(), block.begin(), block.end());
543
544     std::ofstream output(path, std::ios::binary);
545     if (!output) {
546         throw std::runtime_error("cannot create GGUF fixture");

```

```

547     }
548     output.write(reinterpret_cast<const char*>(bytes.data()),
549                  static_cast<std::streamsize>(bytes.size()));
550 }
551
552 void append_gguf_uint32_metadata(std::vector<std::uint8_t>& bytes,
553                                 const std::string& key,
554                                 std::uint32_t value) {
555     append_gguf_string(bytes, key);
556     append_le_i32(bytes, LCQI_GGUF_TYPE_UINT32);
557     append_le_u32(bytes, value);
558 }
559
560 void append_gguf_float32_metadata(std::vector<std::uint8_t>& bytes,
561                                   const std::string& key,
562                                   float value) {
563     append_gguf_string(bytes, key);
564     append_le_i32(bytes, LCQI_GGUF_TYPE_FLOAT32);
565     append_le_f32(bytes, value);
566 }
567
568 void append_gguf_string_metadata(std::vector<std::uint8_t>& bytes,
569                                  const std::string& key,
570                                  const std::string& value) {
571     append_gguf_string(bytes, key);
572     append_le_i32(bytes, LCQI_GGUF_TYPE_STRING);
573     append_gguf_string(bytes, value);
574 }
575
576 void write_gguf_fixture(const std::filesystem::path& path,
577                         std::span<GgufTensorFixture> tensors,
578                         const std::vector<std::uint8_t>& metadata_bytes) {
579     std::uint64_t relative_offset = 0;
580     for (GgufTensorFixture& tensor : tensors) {
581         tensor.relative_offset = relative_offset;
582         relative_offset += tensor.bytes.size();
583         while (relative_offset % LCQI_GGUF_TEST_ALIGNMENT != 0) {
584             ++relative_offset;
585         }
586     }
587
588     std::vector<std::uint8_t> bytes;
589     bytes.push_back('G');
590     bytes.push_back('G');
591     bytes.push_back('U');
592     bytes.push_back('F');
593     append_le_u32(bytes, LCQI_GGUF_TEST_VERSION);
594     append_le_u64(bytes, static_cast<std::uint64_t>(tensors.size()));
595     append_le_u64(bytes, 14);
596     bytes.insert(bytes.end(), metadata_bytes.begin(), metadata_bytes.end());
597
598     for (const GgufTensorFixture& tensor : tensors) {

```

```

599     append_gguf_string(bytes, tensor.name);
600     append_le_u32(bytes, static_cast<std::uint32_t>(tensor.shape.size()));
601     for (const std::int64_t extent : tensor.shape) {
602         append_le_i64(bytes, extent);
603     }
604     append_le_i32(bytes, static_cast<std::int32_t>(tensor.type));
605     append_le_u64(bytes, tensor.relative_offset);
606 }
607 while (bytes.size() % LCQI_GGUF_TEST_ALIGNMENT != 0) {
608     bytes.push_back(0);
609 }
610 std::uint64_t written_relative = 0;
611 for (const GgufTensorFixture& tensor : tensors) {
612     while (written_relative < tensor.relative_offset) {
613         bytes.push_back(0);
614         ++written_relative;
615     }
616     bytes.insert(bytes.end(), tensor.bytes.begin(), tensor.bytes.end());
617     written_relative += tensor.bytes.size();
618 }
619
620 std::ofstream output(path, std::ios::binary);
621 if (!output) {
622     throw std::runtime_error("cannot create GGUF fixture");
623 }
624 output.write(reinterpret_cast<const char*>(bytes.data()),
625             static_cast<std::streamsize>(bytes.size()));
626 }
627
628 void add_gguf_f32_tensor(std::vector<GgufTensorFixture>& tensors,
629                          std::string name,
630                          std::vector<std::int64_t> shape,
631                          std::span<const float> values) {
632     tensors.push_back(GgufTensorFixture{
633         std::move(name),
634         std::move(shape),
635         lcqi::GgmlType::f32,
636         f32_payload(values),
637         0,
638     });
639 }
640
641 void write_tiny_llama_gguf_fixture(const std::filesystem::path& path) {
642     std::vector<std::uint8_t> metadata;
643     append_gguf_uint32_metadata(metadata, "general.alignment", ←
        LCQI_GGUF_TEST_ALIGNMENT);
644     append_gguf_string_metadata(metadata, "general.architecture", "llama");
645     append_gguf_string_metadata(metadata, "general.name", "lcqi tiny llama");
646     append_gguf_uint32_metadata(metadata, "llama.embedding_length", ←
        LCQI_LLAMA_TEST_HIDDEN);
647     append_gguf_uint32_metadata(metadata, "llama.block_count", ←
        LCQI_LLAMA_TEST_LAYER_COUNT);

```

```

648     append_gguf_uint32_metadata(metadata, "llama.feed_forward_length", ←
    ↪ LCQI_LLAMA_TEST_INTERMEDIATE);
649     append_gguf_uint32_metadata(metadata, "llama.attention.head_count", ←
    ↪ LCQI_LLAMA_TEST_HEADS);
650     append_gguf_uint32_metadata(metadata, "llama.attention.head_count_kv", ←
    ↪ LCQI_LLAMA_TEST_KV_HEADS);
651     append_gguf_uint32_metadata(metadata, "llama.rope.dimension_count", ←
    ↪ LCQI_LLAMA_TEST_HEAD_DIM);
652     append_gguf_float32_metadata(metadata, "llama.rope.freq_base", 10000.0F);
653     append_gguf_uint32_metadata(metadata, "llama.context_length", ←
    ↪ LCQI_LLAMA_TEST_CONTEXT);
654     append_gguf_uint32_metadata(metadata, "llama.vocab_size", ←
    ↪ LCQI_LLAMA_TEST_VOCAB);
655     append_gguf_float32_metadata(metadata, "llama.attention.←
    ↪ layer_norm_rms_epsilon", 1.0e-5F);
656     append_gguf_uint32_metadata(metadata, "tokenizer.ggml.eos_token_id", 3);
657
658     std::vector<GgufTensorFixture> tensors;
659     const std::vector<float> embedding{
660         1.0F, 0.0F, 0.0F, 0.0F,
661         0.0F, 1.0F, 0.0F, 0.0F,
662         0.0F, 2.0F, 0.0F, 0.0F,
663         0.0F, 0.0F, 1.0F, 0.0F,
664     };
665     add_gguf_f32_tensor(tensors,
666         "token_embd.weight",
667         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_VOCAB},
668         embedding);
669     add_gguf_f32_tensor(tensors,
670         "output_norm.weight",
671         {LCQI_LLAMA_TEST_HIDDEN},
672         std::vector<float>(LCQI_LLAMA_TEST_HIDDEN, 1.0F));
673     add_gguf_f32_tensor(tensors,
674         "blk.0.attn_norm.weight",
675         {LCQI_LLAMA_TEST_HIDDEN},
676         std::vector<float>(LCQI_LLAMA_TEST_HIDDEN, 1.0F));
677     add_gguf_f32_tensor(tensors,
678         "blk.0.ffn_norm.weight",
679         {LCQI_LLAMA_TEST_HIDDEN},
680         std::vector<float>(LCQI_LLAMA_TEST_HIDDEN, 1.0F));
681     const std::vector<float> hidden_to_hidden(
682         static_cast<std::size_t>(LCQI_LLAMA_TEST_HIDDEN * ←
    ↪ LCQI_LLAMA_TEST_HIDDEN),
683         0.0F);
684     std::vector<float> hidden_to_kv(
685         static_cast<std::size_t>(LCQI_LLAMA_TEST_HIDDEN * ←
    ↪ LCQI_LLAMA_TEST_HEAD_DIM),
686         0.0F);
687     hidden_to_kv[1] = LCQI_LLAMA_TEST_SMALL_ATTENTION_VALUE;
688     std::vector<float> attention_output = hidden_to_hidden;
689     attention_output[0] = LCQI_LLAMA_TEST_SMALL_OUTPUT_SCALE;
690     attention_output[10] = LCQI_LLAMA_TEST_SMALL_OUTPUT_SCALE;

```

```

691     const std::vector<float> hidden_to_intermediate(
692         static_cast<std::size_t>(LCQI_LLAMA_TEST_HIDDEN * ↵
↵ LCQI_LLAMA_TEST_INTERMEDIATE),
693         0.0F);
694     const std::vector<float> intermediate_to_hidden(
695         static_cast<std::size_t>(LCQI_LLAMA_TEST_INTERMEDIATE * ↵
↵ LCQI_LLAMA_TEST_HIDDEN),
696         0.0F);
697     add_gguf_f32_tensor(tensors,
698         "blk.0.attn.q.weight",
699         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HIDDEN},
700         hidden_to_hidden);
701     add_gguf_f32_tensor(tensors,
702         "blk.0.attn.k.weight",
703         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HEAD_DIM},
704         hidden_to_kv);
705     add_gguf_f32_tensor(tensors,
706         "blk.0.attn.v.weight",
707         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HEAD_DIM},
708         hidden_to_kv);
709     add_gguf_f32_tensor(tensors,
710         "blk.0.attn_output.weight",
711         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_HIDDEN},
712         attention_output);
713     add_gguf_f32_tensor(tensors,
714         "blk.0.ffn_gate.weight",
715         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_INTERMEDIATE},
716         hidden_to_intermediate);
717     add_gguf_f32_tensor(tensors,
718         "blk.0.ffn_up.weight",
719         {LCQI_LLAMA_TEST_HIDDEN, LCQI_LLAMA_TEST_INTERMEDIATE},
720         hidden_to_intermediate);
721     add_gguf_f32_tensor(tensors,
722         "blk.0.ffn_down.weight",
723         {LCQI_LLAMA_TEST_INTERMEDIATE, LCQI_LLAMA_TEST_HIDDEN},
724         intermediate_to_hidden);
725     write_gguf_fixture(path, tensors, metadata);
726 }
727
728 void write_tiny_gpt2_safetensors(const std::filesystem::path& path,
729     const lcqi::Gpt2ReferenceModel& model) {
730     std::vector<RawTensorFixture> tensors;
731     add_f32_tensor(tensors,
732         "transformer.wte.weight",
733         {model.config.vocab_size, model.config.hidden_size},
734         model.token_embedding);
735     add_f32_tensor(tensors,
736         "transformer.wpe.weight",
737         {model.config.max_positions, model.config.hidden_size},
738         model.position_embedding);
739     for (std::int32_t layer_id = 0; layer_id < model.config.layer_count; ++↵
↵ layer_id) {

```

```

740     const lcqi::Gpt2LayerWeightsF32& layer =
741         model.layers[static_cast<std::size_t>(layer_id)];
742     const std::string prefix = "transformer.h." + std::to_string(layer_id) ↵
↵ + ".";
743     add_f32_tensor(tensors, prefix + "ln_1.weight", {model.config.↵
↵ hidden_size}, layer.ln_1_weight);
744     add_f32_tensor(tensors, prefix + "ln_1.bias", {model.config.hidden_size↵
↵ }, layer.ln_1_bias);
745     const std::vector<float> c_attn_hf = transpose_linear_to_hf_conv1d(↵
↵ layer.c_attn);
746     add_f32_tensor(tensors,
747         prefix + "attn.c_attn.weight",
748         {model.config.hidden_size,
749         model.config.hidden_size * 3},
750         c_attn_hf);
751     add_f32_tensor(tensors,
752         prefix + "attn.c_attn.bias",
753         {model.config.hidden_size * 3},
754         layer.c_attn.bias);
755     const std::vector<float> c_proj_hf = transpose_linear_to_hf_conv1d(↵
↵ layer.c_proj);
756     add_f32_tensor(tensors,
757         prefix + "attn.c_proj.weight",
758         {model.config.hidden_size, model.config.hidden_size},
759         c_proj_hf);
760     add_f32_tensor(tensors,
761         prefix + "attn.c_proj.bias",
762         {model.config.hidden_size},
763         layer.c_proj.bias);
764     add_f32_tensor(tensors, prefix + "ln_2.weight", {model.config.↵
↵ hidden_size}, layer.ln_2_weight);
765     add_f32_tensor(tensors, prefix + "ln_2.bias", {model.config.hidden_size↵
↵ }, layer.ln_2_bias);
766     const std::vector<float> c_fc_hf = transpose_linear_to_hf_conv1d(layer.↵
↵ c_fc);
767     add_f32_tensor(tensors,
768         prefix + "mlp.c_fc.weight",
769         {model.config.hidden_size, model.config.↵
↵ intermediate_size},
770         c_fc_hf);
771     add_f32_tensor(tensors,
772         prefix + "mlp.c_fc.bias",
773         {model.config.intermediate_size},
774         layer.c_fc.bias);
775     const std::vector<float> mlp_proj_hf =
776         transpose_linear_to_hf_conv1d(layer.mlp_c_proj);
777     add_f32_tensor(tensors,
778         prefix + "mlp.c_proj.weight",
779         {model.config.intermediate_size, model.config.↵
↵ hidden_size},
780         mlp_proj_hf);
781     add_f32_tensor(tensors,

```

```

782         prefix + "mlp.c_proj.bias",
783         {model.config.hidden_size},
784         layer.mlp_c_proj.bias);
785     }
786     add_f32_tensor(tensors,
787         "transformer.ln_f.weight",
788         {model.config.hidden_size},
789         model.final_ln_weight);
790     add_f32_tensor(tensors,
791         "transformer.ln_f.bias",
792         {model.config.hidden_size},
793         model.final_ln_bias);
794     write_safetensors_raw_fixture(path, tensors);
795 }
796
797 void test_tiny_mlp() {
798     const lcqi::TinyMlpModel model = lcqi::load_model(test_model_path());
799     require(model.input_size == LCQI_TEST_INPUT_SIZE, "input size mismatch");
800     require(model.hidden_size == LCQI_TEST_HIDDEN_SIZE, "hidden size mismatch")↵
801     ↵ ;
802     require(model.output_size == LCQI_TEST_OUTPUT_SIZE, "output size mismatch")↵
803     ↵ ;
804
805     const std::vector<float> input{1.0F, 2.0F, -1.0F, 0.5F};
806     const lcqi::InferenceResult result = lcqi::run_inference(model, input);
807
808     require(result.predicted_class == LCQI_TEST_PREDICTED_CLASS, "unexpected ↵
809     ↵ predicted class");
810     require(result.logits.size() == static_cast<std::size_t>(↵
811     ↵ LCQI_TEST_OUTPUT_SIZE),
812         "unexpected logits size");
813     require_close(result.logits[0], LCQI_TEST_LOGIT_0, "logit 0 mismatch");
814     require_close(result.logits[1], LCQI_TEST_LOGIT_1, "logit 1 mismatch");
815
816     std::vector<float> hidden(static_cast<std::size_t>(LCQI_TEST_HIDDEN_SIZE), ↵
817     ↵ 0.0F);
818     lcqi::linear_i8(model.hidden, input, hidden);
819     lcqi::relu_inplace(hidden);
820     require_close(hidden[0], LCQI_TEST_HIDDEN_0, "hidden 0 mismatch");
821     require_close(hidden[1], LCQI_TEST_HIDDEN_1, "hidden 1 mismatch");
822     require_close(hidden[2], LCQI_TEST_HIDDEN_2, "hidden 2 mismatch");
823 }
824
825 void test_tokenizer_contract() {
826     const lcqi::TokenizerModel tokenizer = lcqi::load_tokenizer(↵
827     ↵ test_tokenizer_path());
828     require(tokenizer.bos_id == 0, "tokenizer BOS id mismatch");
829     require(tokenizer.eos_id == 4, "tokenizer EOS id mismatch");
830     require(tokenizer.unk_id == -1, "tokenizer UNK id mismatch");
831     require(tokenizer.chat_template_hash == "tiny-chat-template-v1",
832         "tokenizer template hash mismatch");
833 }

```



```

828     const std::vector<std::int32_t> token_ids =
829         lcqi::encode_prompt(tokenizer, "tok1 tok2 tok3");
830     const std::vector<std::int32_t> expected{0, 1, 2, 3, 4};
831     require(token_ids == expected, "tokenizer prompt ids mismatch");
832     require(lcqi::tokenizer_contract_hash(tokenizer) == ←
    ↪ LCQI_TOKENIZER_HASH_EXPECTED,
833         "tokenizer contract hash mismatch");
834 }
835
836 void test_tokenizer_rejects_unknown_without_unk() {
837     const lcqi::TokenizerModel tokenizer = lcqi::load_tokenizer(←
    ↪ test_tokenizer_path());
838     bool threw = false;
839     try {
840         static_cast<void>(lcqi::encode_prompt(tokenizer, "tok1 missing_token"))←
    ↪ ;
841     } catch (const std::runtime_error&) {
842         threw = true;
843     }
844     require(threw, "tokenizer should reject unknown token when unk id is absent"←
    ↪ );
845 }
846
847 void test_safetensors_manifest_contract() {
848     const std::filesystem::path path =
849         temp_file_path("lcqi_safetensors_manifest_contract.safetensors");
850     const std::string header =
851         R"({"__metadata__":{"format":"lcqi-fixture"},"model.embed_tokens.weight"←
    ↪ "":{"dtype":"F32","shape":[2,3],"data_offsets":[0,24]},"model.layers.0.←
    ↪ attn.q_proj.weight":{"dtype":"F16","shape":[4,3],"data_offsets"←
    ↪ :[24,48]}})";
852     write_safetensors_fixture(path, header, ←
    ↪ LCQI_SAFETENSORS_VALID_PAYLOAD_BYTES);
853
854     const lcqi::SafeTensorManifest manifest = lcqi::load_safetensors_manifest(←
    ↪ path);
855     std::filesystem::remove(path);
856
857     require(manifest.header_size == static_cast<std::uint64_t>(header.size()),
858         "safetensors header size mismatch");
859     require(manifest.data_start_offset ==
860         static_cast<std::uint64_t>(LCQI_SAFETENSORS_PREFIX_BYTES) +
861         static_cast<std::uint64_t>(header.size()),
862         "safetensors data start mismatch");
863     require(manifest.metadata.size() == 1, "safetensors metadata count mismatch"←
    ↪ );
864     require(manifest.metadata[0].first == "format" &&
865         manifest.metadata[0].second == "lcqi-fixture",
866         "safetensors metadata mismatch");
867     require(manifest.tensors.size() == 2, "safetensors tensor count mismatch");
868
869     const lcqi::SafeTensorEntry* embedding =

```



```

870     manifest.find_tensor("model.embed_tokens.weight");
871     require(embedding != nullptr, "safetensors embedding tensor missing");
872     require(embedding->dtype == "F32", "safetensors embedding dtype mismatch");
873     require(embedding->shape == std::vector<std::int64_t>({2, 3}),
874             "safetensors embedding shape mismatch");
875     require(embedding->data_begin == LCQI_SAFETENSORS_EMBED_BEGIN &&
876             embedding->data_end == LCQI_SAFETENSORS_EMBED_END,
877             "safetensors embedding offset mismatch");
878     require(embedding->byte_size() ==
879             LCQI_SAFETENSORS_EMBED_END - LCQI_SAFETENSORS_EMBED_BEGIN,
880             "safetensors embedding byte size mismatch");
881
882     const lcqi::SafeTensorEntry* q_proj =
883         manifest.find_tensor("model.layers.0.attn.q_proj.weight");
884     require(q_proj != nullptr, "safetensors q projection tensor missing");
885     require(q_proj->dtype == "F16", "safetensors q projection dtype mismatch");
886     require(q_proj->data_begin == LCQI_SAFETENSORS_QPROJ_BEGIN &&
887             q_proj->data_end == LCQI_SAFETENSORS_QPROJ_END,
888             "safetensors q projection offset mismatch");
889 }
890
891 void test_safetensors_rejects_bad_offsets() {
892     const std::filesystem::path path =
893         temp_file_path("lcqi_safetensors_bad_offsets.safetensors");
894     const std::string header =
895         R("{"tensor":{"dtype":"F32","shape":[4],"data_offsets":[0,32]}}");
896     write_safetensors_fixture(path, header, 4);
897
898     bool threw = false;
899     try {
900         static_cast<void>(lcqi::load_safetensors_manifest(path));
901     } catch (const std::runtime_error&) {
902         threw = true;
903     }
904     std::filesystem::remove(path);
905     require(threw, "safetensors parser should reject offsets beyond payload");
906 }
907
908 void test_safetensors_rejects_bad_dtype_shape_size() {
909     const std::filesystem::path path =
910         temp_file_path("lcqi_safetensors_bad_dtype_shape_size.safetensors");
911     const std::string header =
912         R("{"tensor":{"dtype":"F32","shape":[4],"data_offsets":[0,8]}}");
913     write_safetensors_fixture(path, header, 8);
914
915     bool threw = false;
916     try {
917         static_cast<void>(lcqi::load_safetensors_manifest(path));
918     } catch (const std::runtime_error&) {
919         threw = true;
920     }
921     std::filesystem::remove(path);

```

```

922     require(threw, "safetensors parser should reject dtype/shape byte mismatch" ←
    ↪ );
923 }
924
925 void test_safetensors_reads_float_tensors() {
926     const std::filesystem::path path =
927         temp_file_path("lcqi_safetensors_float_read.safetensors");
928     std::vector<std::uint8_t> f16_bytes;
929     append_le_u16(f16_bytes, LCQI_TEST_F16_ONE);
930     append_le_u16(f16_bytes, LCQI_TEST_F16_NEGATIVE_TWO);
931     std::vector<std::uint8_t> bf16_bytes;
932     append_le_u16(bf16_bytes, LCQI_TEST_BF16_ONE_POINT_FIVE);
933     append_le_u16(bf16_bytes, LCQI_TEST_BF16_NEGATIVE_HALF);
934
935     const std::array<RawTensorFixture, 3> tensors{{
936         {"f32", "F32", {2}, f32_payload(std::array<float, 2>{1.25F, -2.5F})},
937         {"f16", "F16", {2}, f16_bytes},
938         {"bf16", "BF16", {2}, bf16_bytes},
939     }};
940     write_safetensors_raw_fixture(path, tensors);
941
942     const lcqi::SafeTensorFile file = lcqi::load_safetensors_file(path);
943     std::filesystem::remove(path);
944     const std::vector<float> f32 = lcqi::read_safetensor_f32_tensor(file, "f32" ←
    ↪ );
945     const std::vector<float> f16 = lcqi::read_safetensor_f32_tensor(file, "f16" ←
    ↪ );
946     const std::vector<float> bf16 = lcqi::read_safetensor_f32_tensor(file, " ←
    ↪ bf16");
947
948     require_close(f32[0], 1.25F, "safetensors F32 value 0 mismatch");
949     require_close(f32[1], -2.5F, "safetensors F32 value 1 mismatch");
950     require_close(f16[0], 1.0F, "safetensors F16 value 0 mismatch");
951     require_close(f16[1], -2.0F, "safetensors F16 value 1 mismatch");
952     require_close(bf16[0], 1.5F, "safetensors BF16 value 0 mismatch");
953     require_close(bf16[1], -0.5F, "safetensors BF16 value 1 mismatch");
954 }
955
956 void test_gguf_manifest_reads_q4_k_tensor() {
957     const std::filesystem::path path = temp_file_path("lcqi_q4_k_fixture.gguf") ←
    ↪ ;
958     write_gguf_q4_k_fixture(path);
959
960     const lcqi::GgufManifest manifest = lcqi::load_gguf_manifest(path);
961     const lcqi::GgufTensorInfo* tensor =
962         manifest.find_tensor("blk.0.ffn_up.weight");
963     require(tensor != nullptr, "GGUF Q4_K tensor missing");
964     const std::vector<std::uint8_t> bytes =
965         lcqi::read_gguf_tensor_bytes(path, *tensor);
966     std::filesystem::remove(path);
967
968     require(manifest.version == LCQI_GGUF_TEST_VERSION, "GGUF version mismatch" ←

```

```

    ↪ );
969 require(manifest.alignment == LCQI_GGUF_TEST_ALIGNMENT, "GGUF alignment ↪
    ↪ mismatch");
970 require(manifest.metadata.size() == LCQI_GGUF_TEST_METADATA_COUNT,
971         "GGUF metadata count mismatch");
972 const lcqi::GgufMetadataEntry* tokens = manifest.find_metadata("tokenizer.↪
    ↪ ggml.tokens");
973 require(tokens != nullptr, "GGUF tokenizer tokens metadata missing");
974 require(tokens->string_values == std::vector<std::string>({"<bos>", "hello"↪
    ↪ })),
975         "GGUF tokenizer string array mismatch");
976 require(manifest.tensors.size() == LCQI_GGUF_TEST_TENSOR_COUNT,
977         "GGUF tensor count mismatch");
978 require(tensor->type == lcqi::GgmlType::q4_k, "GGUF tensor type mismatch");
979 require(tensor->shape == std::vector<std::int64_t>({lcqi::↪
    ↪ LCQI_QK_K_BLOCK_VALUES, 1}),
980         "GGUF tensor shape mismatch");
981 require(tensor->byte_size == lcqi::LCQI_Q4_K_BLOCK_BYTES,
982         "GGUF tensor byte size mismatch");
983 require(bytes == make_q4_k_test_block(), "GGUF tensor payload mismatch");
984 }
985
986 void test_q4_k_dequant_and_dot_paths() {
987     const std::vector<std::uint8_t> block = make_q4_k_test_block();
988     const std::vector<float> expected = expected_q4_k_test_values();
989     const std::vector<float> decoded =
990         lcqi::dequantize_q4_k(block, lcqi::LCQI_QK_K_BLOCK_VALUES);
991
992     require(decoded.size() == expected.size(), "Q4_K decoded size mismatch");
993     for (std::size_t index = 0; index < expected.size(); ++index) {
994         require_close(decoded[index], expected[index], "Q4_K decoded value ↪
    ↪ mismatch");
995     }
996
997     std::vector<float> input(expected.size(), 0.0F);
998     for (std::size_t index = 0; index < input.size(); ++index) {
999         input[index] = static_cast<float>(static_cast<std::int32_t>(index % 11U↪
    ↪ ) - 5) *
1000             0.125F;
1001     }
1002     float expected_dot = 0.0F;
1003     for (std::size_t index = 0; index < expected.size(); ++index) {
1004         expected_dot += expected[index] * input[index];
1005     }
1006     const float q4_f32_dot = lcqi::dot_q4_k_f32(block, input);
1007     require_close(q4_f32_dot, expected_dot, "Q4_K F32 dot mismatch");
1008
1009     const std::vector<lcqi::Q8KBlock> q8_input = lcqi::quantize_q8_k_input(↪
    ↪ input);
1010     const float q4_q8_dot = lcqi::dot_q4_k_q8(block, q8_input);
1011     require(std::fabs(q4_q8_dot - q4_f32_dot) <= LCQI_Q4K_Q8_DIFF_LIMIT,
1012             "Q4_K Q8 dot drift is too large");

```

```

1013
1014     std::vector<float> matvec_output(1, 0.0F);
1015     lcqi::matvec_q4_k_q8(block, 1, lcqi::LCQI_QK_K_BLOCK_VALUES, q8_input, ↵
↵ matvec_output);
1016     require_close(matvec_output[0], q4_q8_dot, "Q4_K Q8 matvec mismatch");
1017 }
1018
1019 void test_ggml_mixed_dequantizers() {
1020     std::vector<std::uint8_t> f32_bytes;
1021     append_le_f32(f32_bytes, 1.25F);
1022     append_le_f32(f32_bytes, -2.5F);
1023     const std::vector<float> f32 =
1024         lcqi::dequantize_ggml_tensor(lcqi::GgmlType::f32, f32_bytes, 2);
1025     require_close(f32[0], 1.25F, "GGML F32 decoded value 0 mismatch");
1026     require_close(f32[1], -2.5F, "GGML F32 decoded value 1 mismatch");
1027
1028     const std::vector<std::uint8_t> q8 = make_q8_0_test_block();
1029     const std::vector<float> q8_values =
1030         lcqi::dequantize_ggml_tensor(lcqi::GgmlType::q8_0,
1031                                     q8,
1032                                     lcqi::LCQI_Q8_0_BLOCK_VALUES);
1033     require_close(q8_values[0], -8.0F, "Q8_0 decoded value 0 mismatch");
1034     require_close(q8_values[16], 0.0F, "Q8_0 decoded value 16 mismatch");
1035     require_close(q8_values[31], 7.5F, "Q8_0 decoded value 31 mismatch");
1036
1037     const std::vector<std::uint8_t> q5 = make_q5_0_test_block();
1038     const std::vector<float> q5_values =
1039         lcqi::dequantize_ggml_tensor(lcqi::GgmlType::q5_0,
1040                                     q5,
1041                                     lcqi::LCQI_Q5_0_BLOCK_VALUES);
1042     require_close(q5_values[0], -8.0F, "Q5_0 decoded value 0 mismatch");
1043     require_close(q5_values[1], -7.5F, "Q5_0 decoded value 1 mismatch");
1044     require_close(q5_values[15], -0.5F, "Q5_0 decoded value 15 mismatch");
1045     require_close(q5_values[16], 0.0F, "Q5_0 decoded value 16 mismatch");
1046     require_close(q5_values[31], 7.5F, "Q5_0 decoded value 31 mismatch");
1047
1048     const std::vector<std::uint8_t> q6 = make_q6_k_test_block();
1049     const std::vector<float> q6_values =
1050         lcqi::dequantize_ggml_tensor(lcqi::GgmlType::q6_k,
1051                                     q6,
1052                                     lcqi::LCQI_Q6_K_BLOCK_VALUES);
1053     require_close(q6_values[LCQI_GGML_TEST_Q6_LANE], 3.0F, "Q6_K q1 mismatch");
1054     require_close(q6_values[32 + LCQI_GGML_TEST_Q6_LANE], -1.0F, "Q6_K q2 ↵
↵ mismatch");
1055     require_close(q6_values[64 + LCQI_GGML_TEST_Q6_LANE], 31.0F, "Q6_K q3 ↵
↵ mismatch");
1056     require_close(q6_values[96 + LCQI_GGML_TEST_Q6_LANE], -32.0F, "Q6_K q4 ↵
↵ mismatch");
1057     require_close(q6_values[128 + LCQI_GGML_TEST_Q6_LANE], 4.0F, "Q6_K second ↵
↵ half mismatch");
1058
1059     bool rejected_bad_size = false;

```

```

1060     try {
1061         static_cast<void>(lcqi::dequantize_ggml_tensor(
1062             lcqi::GgmlType::q8_0,
1063             std::span<const std::uint8_t>(q8.data(), q8.size() - 1U),
1064             lcqi::LCQI_Q8_0_BLOCK_VALUES));
1065     } catch (const std::runtime_error&) {
1066         rejected_bad_size = true;
1067     }
1068     require(rejected_bad_size, "GGML dequantizer accepted a truncated Q8_0 ↵
    ↵ block");
1069 }
1070
1071 void test_llama_gguf_loader_and_generation() {
1072     const std::filesystem::path path = temp_file_path("lcqi_tiny_llama_fixture.↵
    ↵ gguf");
1073     write_tiny_llama_gguf_fixture(path);
1074
1075     const lcqi::LlamaGgufLoadedModel loaded =
1076         lcqi::load_llama_gguf_reference_model(path);
1077     const std::vector<std::int32_t> prompt{1};
1078     const lcqi::LlamaGgufGenerationResult result =
1079         lcqi::llama_gguf_generate_greedy(loaded.model, prompt, 1);
1080     std::filesystem::remove(path);
1081
1082     require(loaded.model.architecture == "llama", "LLaMA GGUF architecture ↵
    ↵ mismatch");
1083     require(loaded.model.decoder.config.hidden_size == LCQI_LLAMA_TEST_HIDDEN,
1084         "LLaMA GGUF hidden size mismatch");
1085     require(loaded.model.decoder.layers.size() == LCQI_LLAMA_TEST_LAYER_COUNT,
1086         "LLaMA GGUF layer count mismatch");
1087     require(loaded.model.lm_head_tied_to_embedding,
1088         "LLaMA GGUF tiny fixture should tie lm head to embeddings");
1089     require(result.generated_ids == std::vector<std::int32_t>({1, ↵
    ↵ LCQI_LLAMA_TEST_EXPECTED_TOKEN}),
1090         "LLaMA GGUF generated ids mismatch");
1091     require(result.predicted_first_token == LCQI_LLAMA_TEST_EXPECTED_TOKEN,
1092         "LLaMA GGUF predicted token mismatch");
1093     require(loaded.report.tensors_loaded == 11, "LLaMA GGUF tensor load count ↵
    ↵ mismatch");
1094     require(loaded.report.f32_weight_bytes == loaded.report.↵
    ↵ quantized_weight_bytes,
1095         "LLaMA GGUF F32 fixture byte accounting mismatch");
1096 }
1097
1098 void test_gpt2_tiny_forward_and_generation() {
1099     const lcqi::Gpt2ReferenceModel model = lcqi::make_tiny_gpt2_reference_model↵
    ↵ ();
1100     const std::vector<std::int32_t> tokens{1, 2, 3};
1101     const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(model, tokens↵
    ↵ );
1102
1103     require(result.logits.size() == LCQI_GPT2_VOCAB_SIZE, "GPT-2 logits size ↵

```

```

↪ mismatch");
1104 require(result.predicted_token == LCQI_GPT2_PREDICTED_TOKEN,
1105         "GPT-2 predicted token mismatch");
1106 for (std::size_t index = 0; index < LCQI_GPT2_LOGITS.size(); ++index) {
1107     require_close(result.logits[index], LCQI_GPT2_LOGITS[index], "GPT-2 ↪
↪ logit mismatch");
1108 }
1109
1110 const std::vector<std::int32_t> generated =
1111     lcqi::gpt2_generate_greedy(model, tokens, 2);
1112 const std::vector<std::int32_t> expected{1, 2, 3, 3, 3};
1113 require(generated == expected, "GPT-2 greedy generation mismatch");
1114 require(static_cast<std::int32_t>(generated.size()) == ↪
↪ LCQI_GPT2_GENERATED_LENGTH,
1115         "GPT-2 generated length mismatch");
1116
1117 lcqi::Gpt2KvCache cache(model.config);
1118 lcqi::Gpt2ForwardResult cached_result;
1119 for (const std::int32_t token : tokens) {
1120     cached_result = lcqi::run_gpt2_forward_cached(model, cache, token);
1121 }
1122 require(cache.filled_tokens() == static_cast<std::int32_t>(tokens.size()),
1123         "GPT-2 KV cache filled token count mismatch");
1124 require(cache.byte_size() > 0, "GPT-2 KV cache byte size should be non-zero↪
↪ ");
1125 require(cached_result.predicted_token == result.predicted_token,
1126         "cached GPT-2 predicted token mismatch");
1127 require(cached_result.logits.size() == result.logits.size(),
1128         "cached GPT-2 logits size mismatch");
1129 for (std::size_t index = 0; index < result.logits.size(); ++index) {
1130     require_close(cached_result.logits[index],
1131                   result.logits[index],
1132                   "cached GPT-2 logit mismatch");
1133 }
1134
1135 lcqi::Gpt2CachedGreedyDecoder decoder_with_logits(model);
1136 lcqi::Gpt2ForwardResult decoder_result;
1137 for (const std::int32_t token : tokens) {
1138     decoder_result = decoder_with_logits.step_with_logits(token);
1139 }
1140 require(decoder_with_logits.filled_tokens() == static_cast<std::int32_t>(↪
↪ tokens.size()),
1141         "GPT-2 optimized decoder filled token count mismatch");
1142 require(decoder_with_logits.kv_cache_bytes() == cache.byte_size(),
1143         "GPT-2 optimized decoder KV byte size mismatch");
1144 require(decoder_result.predicted_token == result.predicted_token,
1145         "optimized cached GPT-2 predicted token mismatch");
1146 require(decoder_result.logits.size() == result.logits.size(),
1147         "optimized cached GPT-2 logits size mismatch");
1148 for (std::size_t index = 0; index < result.logits.size(); ++index) {
1149     require_close(decoder_result.logits[index],
1150                   result.logits[index],

```



```

1151         "optimized cached GPT-2 logit mismatch");
1152     }
1153
1154     lcqi::Gpt2ExecutionOptions parallel_options;
1155     parallel_options.worker_count = 2;
1156     parallel_options.parallel_min_rows = 1;
1157     lcqi::Gpt2CachedGreedyDecoder parallel_decoder(model, nullptr, ↵
↵ parallel_options);
1158     lcqi::Gpt2ForwardResult parallel_result;
1159     for (const std::int32_t token : tokens) {
1160         parallel_result = parallel_decoder.step_with_logits(token);
1161     }
1162     require(parallel_decoder.worker_count() == parallel_options.worker_count,
1163             "parallel GPT-2 decoder worker count mismatch");
1164     require(parallel_result.predicted_token == result.predicted_token,
1165             "parallel cached GPT-2 predicted token mismatch");
1166     require(parallel_result.logits.size() == result.logits.size(),
1167             "parallel cached GPT-2 logits size mismatch");
1168     for (std::size_t index = 0; index < result.logits.size(); ++index) {
1169         require_close(parallel_result.logits[index],
1170                      result.logits[index],
1171                      "parallel cached GPT-2 logit mismatch");
1172     }
1173
1174     lcqi::Gpt2CachedGreedyDecoder greedy_decoder(model);
1175     std::int32_t greedy_predicted = 0;
1176     for (const std::int32_t token : tokens) {
1177         greedy_predicted = greedy_decoder.step(token);
1178     }
1179     require(greedy_predicted == result.predicted_token,
1180             "optimized logits-free GPT-2 predicted token mismatch");
1181
1182     lcqi::Gpt2CachedGreedyDecoder prefill_decoder(model);
1183     for (std::size_t index = 0; index + 1 < tokens.size(); ++index) {
1184         prefill_decoder.advance_without_prediction(tokens[index]);
1185     }
1186     require(prefill_decoder.filled_tokens() ==
1187             static_cast<std::int32_t>(tokens.size() - 1U),
1188             "GPT-2 prefill-only decoder filled token count mismatch");
1189     const std::int32_t prefill_predicted = prefill_decoder.step(tokens.back());
1190     require(prefill_decoder.filled_tokens() == static_cast<std::int32_t>(tokens↵
↵ .size()),
1191             "GPT-2 prefill decoder final filled token count mismatch");
1192     require(prefill_predicted == result.predicted_token,
1193             "GPT-2 prefill-skipped predicted token mismatch");
1194
1195     lcqi::Gpt2HotspotProfile hotspot_profile;
1196     lcqi::Gpt2CachedGreedyDecoder profiled_decoder(model, &hotspot_profile);
1197     std::int32_t profiled_predicted = 0;
1198     for (const std::int32_t token : tokens) {
1199         profiled_predicted = profiled_decoder.step(token);
1200     }

```

```

1201     require(profiled_predicted == result.predicted_token,
1202             "profiled GPT-2 predicted token mismatch");
1203     require(hotspot_profile.decoder_steps == static_cast<std::int64_t>(tokens.↵
↵ size()),
1204             "profiled GPT-2 decoder step count mismatch");
1205     require(hotspot_profile.layer_steps ==
1206             static_cast<std::int64_t>(tokens.size()) * model.config.↵
↵ layer_count,
1207             "profiled GPT-2 layer step count mismatch");
1208     require(hotspot_profile.total_step_ms >= hotspot_profile.lm_head_ms,
1209             "profiled GPT-2 total time should include lm head time");
1210
1211     const std::vector<std::int32_t> cached_generated =
1212         lcqi::gpt2_generate_greedy_cached(model, tokens, 2);
1213     require(cached_generated == expected, "cached GPT-2 greedy generation ↵
↵ mismatch");
1214 }
1215
1216 void test_gpt2_byte_bpe_tokenizer() {
1217     const std::filesystem::path vocab_path = temp_file_path("lcqi_gpt2_vocab.↵
↵ json");
1218     const std::filesystem::path merges_path = temp_file_path("lcqi_gpt2_merges.↵
↵ txt");
1219     const std::string space_marker = "\xC4\xA0";
1220     const std::string vocab =
1221         "{ \"Hello\":7, \" \":8, \"\" + space_marker + "world\":9, \"!\":10}";
1222     const std::string merges =
1223         "#version: 0.2\n"
1224         "H e\n"
1225         "He l\n"
1226         "Hel ll\n"
1227         "Hell o\n" +
1228         space_marker + " w\n" +
1229         space_marker + "w o\n" +
1230         space_marker + "wo r\n" +
1231         space_marker + "wor l\n" +
1232         space_marker + "worl d\n";
1233     write_text_file(vocab_path, vocab);
1234     write_text_file(merges_path, merges);
1235
1236     const lcqi::Gpt2Tokenizer tokenizer =
1237         lcqi::load_gpt2_tokenizer(vocab_path, merges_path, -1, -1);
1238     std::filesystem::remove(vocab_path);
1239     std::filesystem::remove(merges_path);
1240
1241     const std::vector<std::int32_t> ids =
1242         lcqi::gpt2_encode(tokenizer, "Hello, world!");
1243     const std::vector<std::int32_t> expected{
1244         LCQI_GPT2_TOKENIZER_HELLO_ID,
1245         8,
1246         9,
1247         10,

```



```

1248     };
1249     require(ids == expected, "GPT-2 BPE ids mismatch");
1250     require(lcqi::gpt2_decode(tokenizer, ids) == "Hello, world!",
1251           "GPT-2 BPE decode mismatch");
1252 }
1253
1254 void test_gpt2_loads_hf_style_tiny_checkpoint() {
1255     const std::filesystem::path directory =
1256         temp_file_path("lcqi_gpt2_tiny_checkpoint_dir");
1257     std::filesystem::remove_all(directory);
1258     std::filesystem::create_directories(directory);
1259
1260     const lcqi::Gpt2ReferenceModel original = lcqi::↵
↵ make_tiny_gpt2_reference_model();
1261     write_text_file(
1262         directory / "config.json",
1263         R"({"model_type":"gpt2","n_embd":4,"n_head":2,"n_layer":1,"n_positions"↵
↵ :8,"n_ctx":8,"vocab_size":6,"n_inner":8,"layer_norm_epsilon":0.00001,"↵
↵ activation_function":"gelu_new","bos_token_id":0,"eos_token_id":5})");
1264     write_tiny_gpt2_safetensors(directory / "model.safetensors", original);
1265
1266     const lcqi::Gpt2ReferenceModel loaded = lcqi::load_gpt2_from_directory(↵
↵ directory);
1267     const std::vector<std::int32_t> tokens{1, 2, 3};
1268     const lcqi::Gpt2ForwardResult result = lcqi::run_gpt2_forward(loaded, ↵
↵ tokens);
1269     std::filesystem::remove_all(directory);
1270
1271     require(result.predicted_token == LCQI_GPT2_PREDICTED_TOKEN,
1272           "loaded GPT-2 predicted token mismatch");
1273     for (std::size_t index = 0; index < LCQI_GPT2_LOGITS.size(); ++index) {
1274         require_close(result.logits[index],
1275             LCQI_GPT2_LOGITS[index],
1276             "loaded GPT-2 logit mismatch");
1277     }
1278 }
1279
1280 void test_gpt2_loader_rejects_bad_shape() {
1281     const std::filesystem::path directory =
1282         temp_file_path("lcqi_gpt2_bad_shape_dir");
1283     std::filesystem::remove_all(directory);
1284     std::filesystem::create_directories(directory);
1285
1286     const lcqi::Gpt2ReferenceModel original = lcqi::↵
↵ make_tiny_gpt2_reference_model();
1287     write_text_file(
1288         directory / "config.json",
1289         R"({"model_type":"gpt2","n_embd":5,"n_head":1,"n_layer":1,"n_positions"↵
↵ :8,"n_ctx":8,"vocab_size":6,"n_inner":8,"layer_norm_epsilon":0.00001,"↵
↵ activation_function":"gelu_new","bos_token_id":0,"eos_token_id":5})");
1290     write_tiny_gpt2_safetensors(directory / "model.safetensors", original);
1291

```

```

1292     bool threw = false;
1293     try {
1294         static_cast<void>(lcqi::load_gpt2_from_directory(directory));
1295     } catch (const std::runtime_error&) {
1296         threw = true;
1297     }
1298     std::filesystem::remove_all(directory);
1299     require(threw, "GPT-2 loader should reject bad tensor shape");
1300 }
1301
1302 void test_reference_rms_norm() {
1303     const std::vector<float> input{3.0F, 4.0F};
1304     const std::vector<float> weight{1.0F, 0.5F};
1305     std::vector<float> output(2, 0.0F);
1306     lcqi::rms_norm(input, weight, 0.0F, output);
1307     require_close(output[0], LCQI_RMS_EXPECTED_0, "RMSNorm output 0 mismatch");
1308     require_close(output[1], LCQI_RMS_EXPECTED_1, "RMSNorm output 1 mismatch");
1309 }
1310
1311 void test_reference_rope() {
1312     std::vector<float> heads{1.0F, 0.0F, 0.0F, 1.0F};
1313     lcqi::apply_rope(heads, 1, 4, 1, 10000.0F);
1314     require_close(heads[0], std::cos(1.0F), "RoPE pair 0 cosine mismatch");
1315     require_close(heads[1], std::sin(1.0F), "RoPE pair 0 sine mismatch");
1316     require_close(heads[2], -std::sin(0.01F), "RoPE pair 1 negative sine ←
    ↪ mismatch");
1317     require_close(heads[3], std::cos(0.01F), "RoPE pair 1 cosine mismatch");
1318 }
1319
1320 void test_reference_kv_attention() {
1321     lcqi::DecoderConfig config;
1322     config.hidden_size = 4;
1323     config.query_heads = 2;
1324     config.kv_heads = 1;
1325     config.head_dim = 2;
1326     config.intermediate_size = 4;
1327     config.vocab_size = 4;
1328     config.max_context = 4;
1329
1330     lcqi::ReferenceKVCache cache(config, 1);
1331     const std::vector<float> key{1.0F, 0.0F};
1332     const std::vector<float> value{LCQI_ATTENTION_EXPECTED_0, ←
    ↪ LCQI_ATTENTION_EXPECTED_1};
1333     cache.append(0, 0, key, value);
1334     require(cache.filled_tokens() == 1, "KV filled token count mismatch");
1335     require_close(cache.key(0, 0, 0)[0], 1.0F, "KV key read mismatch");
1336
1337     const std::vector<float> query{1.0F, 0.0F, 0.0F, 1.0F};
1338     std::vector<float> output(4, 0.0F);
1339     lcqi::attention_decode(config, cache, 0, 0, query, output);
1340     require_close(output[0], LCQI_ATTENTION_EXPECTED_0, "attention head 0 dim 0 ←
    ↪ mismatch");

```

```

1341     require_close(output[1], LCQI_ATTENTION_EXPECTED_1, "attention head 0 dim 1↵
↵ mismatch");
1342     require_close(output[2], LCQI_ATTENTION_EXPECTED_0, "attention head 1 dim 0↵
↵ mismatch");
1343     require_close(output[3], LCQI_ATTENTION_EXPECTED_1, "attention head 1 dim 1↵
↵ mismatch");
1344 }
1345
1346 void test_reference_decoder_end_to_end() {
1347     const lcqi::ReferenceDecoderModel model = lcqi::↵
↵ make_tiny_reference_decoder_model();
1348     const std::vector<std::int32_t> tokens{1, 2, 3};
1349     const lcqi::ReferenceDecodeResult result = lcqi::run_reference_decode(model↵
↵ , tokens);
1350     require(result.logits.size() == LCQI_REFERENCE_DECODER_VOCAB_SIZE,
1351             "reference decoder logits size mismatch");
1352     require(result.predicted_token == LCQI_REFERENCE_DECODER_PREDICTED_TOKEN,
1353             "reference decoder predicted token mismatch");
1354     for (std::size_t index = 0; index < LCQI_REFERENCE_DECODER_LOGITS.size(); ↵
↵ ++index) {
1355         require_close(result.logits[index],
1356                       LCQI_REFERENCE_DECODER_LOGITS[index],
1357                       "reference decoder golden logit mismatch");
1358     }
1359 }
1360
1361 void test_reference_decoder_trace_contract() {
1362     const lcqi::ReferenceDecoderModel model = lcqi::↵
↵ make_tiny_reference_decoder_model();
1363     const std::vector<std::int32_t> tokens{1, 2, 3};
1364     const lcqi::ReferenceDecodeTraceResult trace =
1365         lcqi::run_reference_decode_with_trace(model, tokens);
1366
1367     require(trace.result.predicted_token == ↵
↵ LCQI_REFERENCE_DECODER_PREDICTED_TOKEN,
1368             "reference trace predicted token mismatch");
1369     require(!trace.checkpoints.empty(), "reference trace checkpoints missing");
1370
1371     const lcqi::ReferenceTensorCheckpoint& first = trace.checkpoints.front();
1372     require(first.name == "embedding", "first checkpoint should be embedding");
1373     require(first.token_position == LCQI_REFERENCE_TRACE_FIRST_TOKEN,
1374             "first checkpoint token position mismatch");
1375     require(first.dtype == "f32", "first checkpoint dtype mismatch");
1376     require(first.layout == "row-major", "first checkpoint layout mismatch");
1377     require(first.shape.size() == 1 &&
1378             first.shape[0] == LCQI_REFERENCE_TRACE_HIDDEN_SIZE,
1379             "first checkpoint shape mismatch");
1380     require(first.stride.size() == 1 && first.stride[0] == 1,
1381             "first checkpoint stride mismatch");
1382
1383     bool saw_logits = false;
1384     bool saw_kv_slot = false;

```

```

1385     for (const lcqi::ReferenceTensorCheckpoint& checkpoint : trace.checkpoints) {
1386         if (checkpoint.name == "kv_cache_key_slot") {
1387             saw_kv_slot = true;
1388             require(checkpoint.shape.size() == 4, "KV checkpoint rank mismatch");
1389             require(checkpoint.layout == "kv_contiguous", "KV checkpoint layout
1390 mismatch");
1391         }
1392         if (checkpoint.name == "logits") {
1393             saw_logits = true;
1394             require(checkpoint.token_position == LCQI_REFERENCE_TRACE_TOKEN_COUNT - 1,
1395 "logits checkpoint token position mismatch");
1396             require(checkpoint.shape.size() == 1 &&
1397 checkpoint.shape[0] ==
1398 static_cast<std::int32_t>(LCQI_REFERENCE_DECODER_VOCAB_SIZE),
1399 "logits checkpoint shape mismatch");
1400             require(checkpoint.values.size() == LCQI_REFERENCE_DECODER_LOGITS.
1401 size(),
1402 "logits checkpoint value size mismatch");
1403             for (std::size_t index = 0; index < LCQI_REFERENCE_DECODER_LOGITS.
1404 size(); ++index) {
1405                 require_close(checkpoint.values[index],
1406 LCQI_REFERENCE_DECODER_LOGITS[index],
1407 "trace logits checkpoint mismatch");
1408             }
1409         }
1410     }
1411     require(saw_kv_slot, "KV checkpoint missing");
1412     require(saw_logits, "logits checkpoint missing");
1413 }
1414
1415 void test_packed_i8_kernel_matches_scalar() {
1416     for (const KernelShapeCase& shape_case : LCQI_KERNEL_SHAPE_CASES) {
1417         const lcqi::QuantizedLinearLayer layer =
1418             lcqi::make_deterministic_i8_layer(shape_case.input_size,
1419 shape_case.output_size,
1420 0.03125F);
1421         const lcqi::PackedLinearI8 packed =
1422             lcqi::pack_linear_i8(layer, shape_case.output_block_size);
1423         const std::vector<float> input = lcqi::make_deterministic_input(layer.
1424 input_size);
1425         std::vector<float> scalar_output(static_cast<std::size_t>(layer.
1426 output_size), 0.0F);
1427         std::vector<float> packed_output(static_cast<std::size_t>(layer.
1428 output_size), 0.0F);
1429         lcqi::linear_i8(layer, input, scalar_output);
1430         lcqi::linear_i8_packed(packed, input, packed_output);
1431         require(lcqi::max_abs_diff(scalar_output, packed_output) <=
1432 LCQI_KERNEL_MAX_DIFF,

```

```

1426         "packed int8 kernel output differs from scalar");
1427
1428         const lcqi::PackedLinearI8 simd_packed =
1429             lcqi::pack_linear_i8(layer, LCQI_SIMD_TEST_BLOCK);
1430         if (lcqi::linear_i8_packed_avx2_available()) {
1431             std::vector<float> avx2_output(static_cast<std::size_t>(layer.↵
↵ output_size), 0.0F);
1432             lcqi::linear_i8_packed_avx2(simd_packed, input, avx2_output);
1433             require(lcqi::max_abs_diff(scalar_output, avx2_output) <= ↵
↵ LCQI_KERNEL_MAX_DIFF,
1434                 "AVX2 int8 kernel output differs from scalar");
1435         }
1436     }
1437 }
1438
1439 void test_f32_dot_kernel_matches_scalar() {
1440     std::vector<float> lhs(LCQI_F32_DOT_TEST_SIZE, 0.0F);
1441     std::vector<float> rhs(LCQI_F32_DOT_TEST_SIZE, 0.0F);
1442     for (std::size_t index = 0; index < LCQI_F32_DOT_TEST_SIZE; ++index) {
1443         lhs[index] = static_cast<float>(static_cast<std::int32_t>(index % 19U) ↵
↵ + 1) *
1444             0.03125F;
1445         rhs[index] = static_cast<float>(static_cast<std::int32_t>(index % 23U) ↵
↵ - 11) *
1446             0.0625F;
1447     }
1448
1449     const float scalar =
1450         lcqi::dot_f32_scalar_unchecked(lhs.data(), rhs.data(), lhs.size());
1451     const float dispatched = lcqi::dot_f32(lhs, rhs);
1452     require(std::fabs(scalar - dispatched) <= LCQI_F32_KERNEL_MAX_DIFF,
1453         "F32 dot kernel output differs from scalar");
1454
1455     if (lcqi::dot_f32_avx2_available()) {
1456         const float avx2 =
1457             lcqi::dot_f32_avx2_unchecked(lhs.data(), rhs.data(), lhs.size());
1458         require(std::fabs(scalar - avx2) <= LCQI_F32_KERNEL_MAX_DIFF,
1459             "AVX2 F32 dot kernel output differs from scalar");
1460     }
1461
1462     bool rejected_mismatch = false;
1463     try {
1464         (void)lcqi::dot_f32(std::span<const float>(lhs.data(), lhs.size() - 1U)↵
↵ , rhs);
1465     } catch (const std::runtime_error&) {
1466         rejected_mismatch = true;
1467     }
1468     require(rejected_mismatch, "F32 dot kernel accepted mismatched sizes");
1469 }
1470
1471 void test_f32_row_kernels_match_scalar() {
1472     std::vector<float> weights(LCQI_F32_ROW_TEST_INPUT_SIZE *

```

```

1473         LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1474         0.0F);
1475     std::vector<float> input(LCQI_F32_ROW_TEST_INPUT_SIZE, 0.0F);
1476     std::vector<float> bias(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1477     for (std::size_t index = 0; index < weights.size(); ++index) {
1478         weights[index] = static_cast<float>(static_cast<std::int32_t>(index % 29U) - 14) *
1479             0.015625F;
1480     }
1481     for (std::size_t index = 0; index < input.size(); ++index) {
1482         input[index] = static_cast<float>(static_cast<std::int32_t>(index % 13U) - 6) *
1483             0.03125F;
1484     }
1485     for (std::size_t index = 0; index < bias.size(); ++index) {
1486         bias[index] = static_cast<float>(static_cast<std::int32_t>(index % 7U) - 3) *
1487             0.125F;
1488     }
1489
1490     std::vector<float> scalar_output(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1491     std::vector<float> dispatched_output(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1492     lcqi::linear_f32_rows_scalar_unchecked(weights.data(),
1493         input.data(),
1494         bias.data(),
1495         input.size(),
1496         0,
1497         LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1498         scalar_output.data());
1499     lcqi::linear_f32_rows_unchecked(weights.data(),
1500         input.data(),
1501         bias.data(),
1502         input.size(),
1503         0,
1504         LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1505         dispatched_output.data());
1506     for (std::size_t index = 0; index < scalar_output.size(); ++index) {
1507         require(std::fabs(scalar_output[index] - dispatched_output[index]) <=
1508             LCQI_F32_KERNEL_MAX_DIFF,
1509             "F32 linear row kernel output differs from scalar");
1510     }
1511
1512     const lcqi::F32RowMax scalar_max =
1513         lcqi::max_dot_f32_rows_scalar_unchecked(weights.data(),
1514             input.data(),
1515             input.size(),
1516             0,
1517             LCQI_F32_ROW_TEST_OUTPUT_SIZE);
1518     const lcqi::F32RowMax dispatched_max =
1519         lcqi::max_dot_f32_rows_unchecked(weights.data(),
1520             input.data(),
1521             input.size(),

```

```

1522         0,
1523         LCQI_F32_ROW_TEST_OUTPUT_SIZE);
1524     require(scalar_max.row == dispatched_max.row,
1525             "F32 row max kernel selected a different row");
1526     require(std::fabs(scalar_max.value - dispatched_max.value) <= ←
    ↪ LCQI_F32_KERNEL_MAX_DIFF,
1527             "F32 row max kernel value differs from scalar");
1528
1529     if (lcqi::dot_f32_avx2_available()) {
1530         std::vector<float> avx2_output(LCQI_F32_ROW_TEST_OUTPUT_SIZE, 0.0F);
1531         lcqi::linear_f32_rows_avx2_unchecked(weights.data(),
1532                                             input.data(),
1533                                             bias.data(),
1534                                             input.size(),
1535                                             0,
1536                                             LCQI_F32_ROW_TEST_OUTPUT_SIZE,
1537                                             avx2_output.data());
1538         for (std::size_t index = 0; index < scalar_output.size(); ++index) {
1539             require(std::fabs(scalar_output[index] - avx2_output[index]) <=
1540                     LCQI_F32_KERNEL_MAX_DIFF,
1541                     "AVX2 F32 linear row kernel output differs from scalar");
1542         }
1543
1544         const lcqi::F32RowMax avx2_max =
1545             lcqi::max_dot_f32_rows_avx2_unchecked(weights.data(),
1546                                                  input.data(),
1547                                                  input.size(),
1548                                                  0,
1549                                                  LCQI_F32_ROW_TEST_OUTPUT_SIZE↪
    ↪ );
1550         require(scalar_max.row == avx2_max.row,
1551                 "AVX2 F32 row max kernel selected a different row");
1552         require(std::fabs(scalar_max.value - avx2_max.value) <= ←
    ↪ LCQI_F32_KERNEL_MAX_DIFF,
1553                 "AVX2 F32 row max kernel value differs from scalar");
1554     }
1555 }
1556
1557 } // namespace
1558
1559 int main() {
1560     try {
1561         test_tiny_mlp();
1562         test_tokenizer_contract();
1563         test_tokenizer_rejects_unknown_without_unk();
1564         test_safetensors_manifest_contract();
1565         test_safetensors_rejects_bad_offsets();
1566         test_safetensors_rejects_bad_dtype_shape_size();
1567         test_safetensors_reads_float_tensors();
1568         test_gguf_manifest_reads_q4_k_tensor();
1569         test_q4_k_dequant_and_dot_paths();
1570         test_ggml_mixed_dequantizers();

```

```
1571     test_llama_gguf_loader_and_generation();
1572     test_gpt2_tiny_forward_and_generation();
1573     test_gpt2_byte_bpe_tokenizer();
1574     test_gpt2_loads_hf_style_tiny_checkpoint();
1575     test_gpt2_loader_rejects_bad_shape();
1576     test_reference_rms_norm();
1577     test_reference_rope();
1578     test_reference_kv_attention();
1579     test_reference_decoder_end_to_end();
1580     test_reference_decoder_trace_contract();
1581     test_packed_i8_kernel_matches_scalar();
1582     test_f32_dot_kernel_matches_scalar();
1583     test_f32_row_kernels_match_scalar();
1584
1585     std::cout << "lcqi tests passed\n";
1586     return EXIT_SUCCESS;
1587 } catch (const std::exception& error) {
1588     std::cerr << "lcqi test failure: " << error.what() << "\n";
1589     return EXIT_FAILURE;
1590 }
1591 }
```


实践与代码总索引

本卷已经把实践任务、代码走读路线和完整代码清单合并到同一套内容目录。目录中不再存在一个独立的“源码卷部分”：构建入口、调用链、公共合同、tiny MLP、int8 kernel、reference decoder、GPT-2、工具脚本和测试文件都插入到相应主题之后，共同解释同一份 `books/cpu-volume-3/labs/linux_cpu_inference` 源码。

源码阅读任务

读者不需要再在实践卷和源码卷之间来回切换，也不需要先读完 16 个实践章再去末尾翻代码。实践章节提出任务，紧邻的代码走读解释完整实现，测试和 benchmark 章节给出回归证据；三者已经合并到这一本实践与代码卷里，共同构成一条可复查的工程证据链。

一、实践主题到代码走读的合并位置

- 第 1 章之后 插入“代码走读：构建入口、项目边界与调用链总图”，先把 CMake、README、模型资产、target、CLI、benchmark、trace、ledger、serving probe 和测试入口放到一张可复现地图里。
- 第 4 章之后 插入“代码走读：公共合同、Tiny MLP 与 Int8 Kernel”，把模型、推理、tokenizer、safetensors、reference decoder、GPT-2 头文件，以及 tiny MLP、基础 CLI、scalar/packed/AVX2 kernel 连起来。
- 第 10 章之后 插入“代码走读：Reference Decoder、Tokenizer、Safetensors 与 GPT-2”，把真实模型加载、byte-level BPE、safetensors、GPT-2 forward、cached KV 和 greedy generation 放在模型导入主题内部讲清楚。
- 第 16 章之后 插入“代码走读：工具、Smoke、Benchmark 与回归测试”，把 benchmark、trace、ledger、serving probe、SmolLM2 smoke、GPT-2 smoke、benchmark compare 和 CTest 回归证据集中收束。

二、最小复现命令

```
1 cmake -S books/cpu-volume-3 -B books/cpu-volume-3/build/lcqi-debug -↵  
   ↵ DCMMAKE_BUILD_TYPE=Debug  
2 cmake --build books/cpu-volume-3/build/lcqi-debug  
3 ctest --test-dir books/cpu-volume-3/build/lcqi-debug --output-on-failure  
4 make cpu3p-pdf  
5 make cpu3p-epub
```

完成这组命令后，读者应同时检查相邻实践章节的作业结果、对应代码走读、测试断言和报告字段，确认代码、测试、报告字段和章节讲解是一条证据链。

术语表

| 术语 | 实践含义 |
|-------------------|---|
| manifest | 模型资产清单，记录 tensor name、dtype、shape、offset、checksum 和 tokenizer/配置版本。 |
| shape verifier | 在 kernel 运行前检查 shape、dtype、layout、量化 descriptor 和 state edge 是否匹配。 |
| reference decoder | 以清楚正确为目标的 decoder 实现，用来生成 trace 和 golden。 |
| oracle | 比较候选实现与 reference 的判定器，可使用精确相等或误差界。 |
| trace checkpoint | 推理链路中的可复查节点，至少包含 shape、stride、dtype、layout、checksum 和摘要值。 |
| KV cache | attention 历史 K/V 状态，跨 token 存活，不能被普通 activation planner 复用。 |
| packing | 把权重改写成更适合热循环和 SIMD 的布局。 |
| backend | 执行计划落到机器的实现，例如 scalar、AVX2、AVX-512、GPU 或库调用。 |
| fallback | 后端能力不满足时显式降级到可验证路径，并记录原因。 |
| SLO | 服务目标，例如 TTFT、TPOT、queue wait、拒绝率和取消回收时间。 |